

Marks'

Basic Medical Biochemistry

A Clinical Approach

SECOND EDITION

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LIPPINCOTT WILLIAMS & WILKINS

Marks' Basic Medical Biochemistry: A Clinical Approach, 2nd Edition

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Fuel Metabolism

In order to survive, humans must meet two basic metabolic requirements: we must be able to synthesize everything our cells need that is not supplied by our diet, and we must be able to protect our internal environment from toxins and changing conditions in our external environment. In order to meet these requirements, we metabolize our dietary components through four basic types of pathways: fuel oxidative pathways, fuel storage and mobilization pathways, biosynthetic pathways, and detoxification or waste disposal pathways. Cooperation between tissues and responses to changes in our external environment are communicated through transport pathways and intercellular signaling pathways (Fig. I.1).

The foods in our diet are the fuels that supply us with energy in the form of calories. This energy is used for carrying out diverse functions such as moving, thinking, and reproducing. Thus, a number of our metabolic pathways are *fuel oxidative pathways* that convert fuels into energy that can be used for biosynthetic and mechanical work. But what is the source of energy when we are not eating—between meals, and while we sleep? How does the hunger striker in the morning headlines survive so long? We have other metabolic pathways that are *fuel storage pathways*. The fuels that we store can be mobilized during periods when we are not eating or when we need increased energy for exercise.

Our diet also must contain the compounds we cannot synthesize, as well as all the basic building blocks for compounds we do synthesize in our *biosynthetic pathways*. For example we have dietary requirements for some amino acids, but we can synthesize other amino acids from our fuels and a dietary nitrogen precursor. The compounds required in our diet for biosynthetic pathways include certain amino acids, vitamins, and essential fatty acids.

Detoxification pathways and *waste disposal pathways* are metabolic pathways devoted to removing toxins that can be present in our diets or in the air we breathe, introduced into our bodies as drugs, or generated internally from the metabolism of dietary components. Dietary components that have no value to the body, and must be disposed of, are called xenobiotics.

In general, biosynthetic pathways (including fuel storage) are referred to as *anabolic pathways*, that is, pathways that synthesize larger molecules from smaller components. The synthesis of proteins from amino acids is an example of an anabolic pathway. *Catabolic pathways* are those pathways that break down larger molecules into smaller components. Fuel oxidative pathways are examples of catabolic pathways.

In the human, the need for different cells to carry out different functions has resulted in cell and tissue specialization in metabolism. For example, our adipose tissue is a specialized site for the storage of fat and contains the metabolic pathways that allow it to carry out this function. However, adipose tissue is lacking many of the pathways that synthesize required compounds from dietary precursors. To enable our cells to cooperate in meeting our metabolic needs during changing conditions of diet, sleep, activity, and health, we need *transport pathways* into the blood and between tissues and *intercellular signaling pathways*. One means of communication is for *hormones* to carry signals to tissues about our dietary state. For example, a message that we have just had a meal, carried by the hormone insulin, signals adipose tissue to store fat.

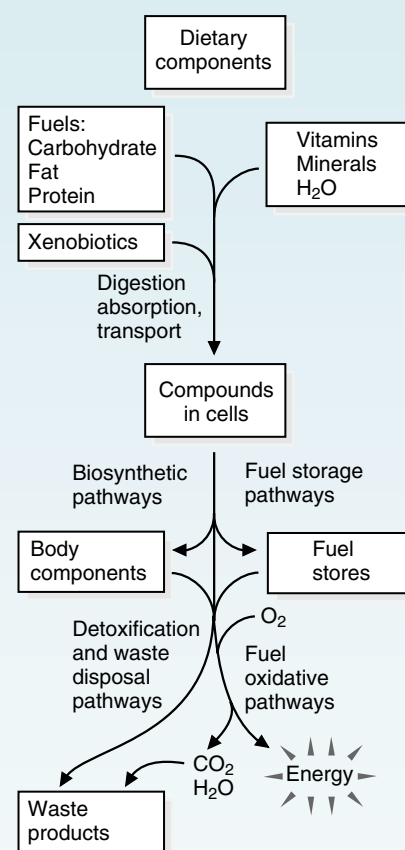


Fig. I.1. General metabolic routes for dietary components in the body. The types of Pathways are named in blue.

In the following section, we will provide an overview of various types of dietary components and examples of the pathways involved in utilizing these components. We will describe the fuels in our diet, the compounds produced by their digestion, and the basic patterns of fuel metabolism in the tissues of our bodies. We will describe how these patterns change when we eat, when we fast for a short time, and when we starve for prolonged periods. Patients with medical problems that involve an inability to deal normally with fuels will be introduced. These patients will appear repeatedly throughout the book and will be joined by other patients as we delve deeper into biochemistry.

1 Metabolic Fuels and Dietary Components

Fuel Metabolism. We obtain our fuel primarily from **carbohydrates, fats, and proteins** in our diet. As we eat, our foodstuffs are **digested** and **absorbed**. The products of digestion circulate in the blood, enter various tissues, and are eventually taken up by cells and **oxidized** to produce **energy**. To completely convert our fuels to carbon dioxide (CO_2) and water (H_2O), molecular **oxygen** (O_2) is required. We breathe to obtain this oxygen and to eliminate the **carbon dioxide** (CO_2) that is produced by the oxidation of our foodstuffs.

Fuel Stores. Any dietary fuel that exceeds the body's immediate energy needs is stored, mainly as **triacylglycerol** (fat) in adipose tissue, as **glycogen** (a carbohydrate) in muscle, liver, and other cells, and, to some extent, as **protein** in muscle. When we are fasting, between meals and overnight while we sleep, fuel is drawn from these stores and is oxidized to provide energy (Fig. 1.1).

Fuel Requirements. We require enough energy each day to drive the **basic functions** of our bodies and to support our **physical activity**. If we do not consume enough food each day to supply that much energy, the body's fuel stores supply the remainder, and we lose weight. Conversely, if we consume more food than required for the energy we expend, our body's fuel stores enlarge, and we gain weight.

Other Dietary Requirements. In addition to providing energy, the diet provides **precursors** for the **biosynthesis** of compounds necessary for cellular and tissue structure, function, and survival. Among these precursors are the **essential fatty acids** and **essential amino acids** (those that the body needs but cannot synthesize). The diet must also supply **vitamins, minerals, and water**.

Waste Disposal. Dietary components that we can utilize are referred to as **nutrients**. However, both the diet and the air we breathe contain **xenobiotic compounds**, compounds that have no use or value in the human body and may be toxic. These compounds are excreted in the urine and feces together with metabolic waste products.



Essential Nutrients

- Fuels
 - Carbohydrates
 - Fats
 - Proteins
- Required Components
 - Essential amino acids
 - Essential fatty acids
 - Vitamins
 - Minerals
 - Water

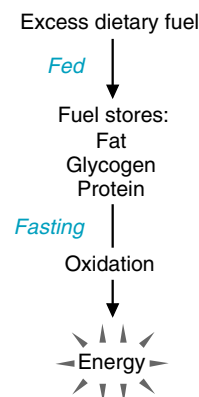


Fig. 1.1. Fate of excess dietary fuel in fed and fasting states.



THE WAITING ROOM



Percy Veere is a 59-year-old school teacher who was in good health until his wife died suddenly. Since that time, he has experienced an increasing degree of fatigue and has lost interest in many of the activities he previously enjoyed. Shortly after his wife's death, one of his married children moved far from home. Since then, Mr. Veere has had little appetite for food. When a



Percy Veere has a strong will. He is enduring a severe reactive depression after the loss of his wife. In addition, he must put up with the sometimes life-threatening antics of his hyperactive grandson, Dennis (the Menace) Veere. Yet through all of this, he will "persevere."

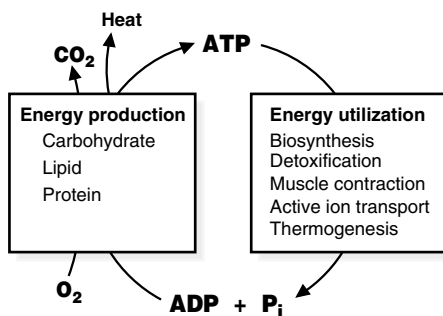


Fig. 1.2. The ATP–ADP cycle.

 Oxidative pathways are catabolic; that is, they break molecules down. In contrast, anabolic pathways build molecules up from component pieces.

neighbor found Mr. Veere sleeping in his clothes, unkempt, and somewhat confused, she called an ambulance. Mr. Veere was admitted to the hospital psychiatry unit with a diagnosis of mental depression associated with dehydration and malnutrition.

 **Otto Shape** is a 25-year-old medical student who was very athletic during high school and college, and is now “out-of-shape.” Since he started medical school, he has been gaining weight (at 5 feet 10 inches tall, he currently weighs 187 lb). He has decided to consult a physician at the student health service before the problem gets worse.

 **Ivan Applebod** is a 56-year-old accountant who has been morbidly obese for a number of years. He exhibits a pattern of central obesity, called an “apple shape,” which is caused by excess adipose tissue deposited in the abdominal area. His major recreational activities are watching TV while drinking scotch and soda and doing occasional gardening. At a company picnic, he became very “winded” while playing baseball and decided it was time for a general physical examination. At the examination, he weighed 264 lb at 5 feet 10 inches tall. His blood pressure was slightly elevated, 155 mm Hg systolic (normal = 140 mm Hg or less) and 95 mm Hg diastolic (normal = 90 mm Hg or less).

 **Ann O’Rexia** is a 23-year-old buyer for a woman’s clothing store. Despite the fact that she is 5 feet 7 inches tall and weighs 99 lb, she is convinced she is overweight. Two months ago, she started a daily exercise program that consists of 1 hour of jogging every morning and 1 hour of walking every evening. She also decided to consult a physician about a weight reduction diet.

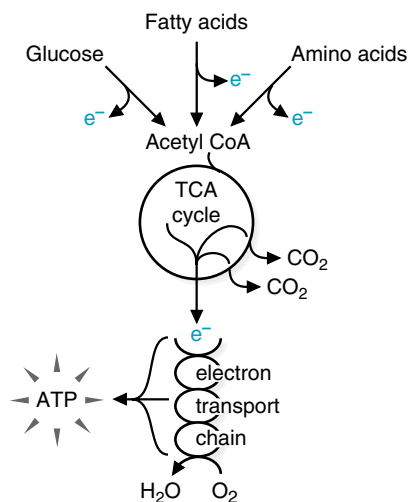


Fig. 1.3. Generation of ATP from fuel components during respiration. Glucose, fatty acids, and amino acids are oxidized to acetyl CoA, a substrate for the TCA cycle. In the TCA cycle, they are completely oxidized to CO_2 . As fuels are oxidized, electrons (e^-) are transferred to O_2 by the electron transport chain, and the energy is used to generate ATP.

I. DIETARY FUELS

The major fuels we obtain from our diet are carbohydrates, proteins, and fats. When these fuels are oxidized to CO_2 and H_2O in our cells, energy is released by the transfer of electrons to O_2 . The energy from this oxidation process generates heat and adenosine triphosphate (ATP) (Fig 1.2). Carbon dioxide travels in the blood to the lungs, where it is expired, and water is excreted in urine, sweat, and other secretions. Although the heat that is generated by fuel oxidation is used to maintain body temperature, the main purpose of fuel oxidation is to generate ATP. ATP provides the energy that drives most of the energy-consuming processes in the cell, including biosynthetic reactions, muscle contraction, and active transport across membranes. As these processes use energy, ATP is converted back to adenosine diphosphate (ADP) and inorganic phosphate (P_i). The generation and utilization of ATP is referred to as the ATP–ADP cycle.

The oxidation of fuels to generate ATP is called respiration (Fig. 1.3). Before oxidation, carbohydrates are converted principally to glucose, fat to fatty acids, and protein to amino acids. The pathways for oxidizing glucose, fatty acids, and amino acids have many features in common. They first oxidize the fuels to acetyl CoA, a precursor of the tricarboxylic acid (TCA) cycle. The TCA cycle is a series of reactions that completes the oxidation of fuels to CO_2 (see Chapter 19). Electrons lost from the fuels during oxidative reactions are transferred to O_2 by a series of proteins in the electron transport chain (see Chapter 20). The energy of electron transfer is used to convert ADP and P_i to ATP by a process known as oxidative phosphorylation.

In discussions of metabolism and nutrition, energy is often expressed in units of calories. “Calorie” in this context really means kilocalorie (kcal). Energy is also expressed in joules. One kilocalorie equals 4.18 kilojoules (kJ). Physicians tend to use units of calories, in part because that is what their patients use and understand.

A. Carbohydrates

The major carbohydrates in the human diet are starch, sucrose, lactose, fructose, and glucose. The polysaccharide starch is the storage form of carbohydrates in plants. Sucrose (table sugar) and lactose (milk sugar) are disaccharides, and fructose and glucose are monosaccharides. Digestion converts the larger carbohydrates to monosaccharides, which can be absorbed into the bloodstream. Glucose, a monosaccharide, is the predominant sugar in human blood (Fig. 1.4).

Oxidation of carbohydrates to CO_2 and H_2O in the body produces approximately 4 kcal/g (Table 1.1). In other words, every gram of carbohydrate we eat yields approximately 4 kcal of energy. Note that carbohydrate molecules contain a significant amount of oxygen and are already partially oxidized before they enter our bodies (see Fig. 1.4).

B. Proteins

Proteins are composed of amino acids that are joined to form linear chains (Fig. 1.5). In addition to carbon, hydrogen, and oxygen, proteins contain approximately 16% nitrogen by weight. The digestive process breaks down proteins to their constituent amino acids, which enter the blood. The complete oxidation of proteins to CO_2 , H_2O , and NH_4^+ in the body yields approximately 4 kcal/g.

C. Fats

Fats are lipids composed of triacylglycerols (also called triglycerides). A triacylglycerol molecule contains 3 fatty acids esterified to one glycerol moiety (Fig. 1.6).

Fats contain much less oxygen than is contained in carbohydrates or proteins. Therefore, fats are more reduced and yield more energy when oxidized. The complete oxidation of triacylglycerols to CO_2 and H_2O in the body releases approximately 9 kcal/g, more than twice the energy yield from an equivalent amount of carbohydrate or protein.



The food “calories” used in everyday speech are really “Calories,” which = kilocalories. “Calorie,” meaning kilocalorie, was originally spelled with a capital C, but the capitalization was dropped as the term became popular. Thus, a 1-calorie soft drink actually has 1 Cal (1 kcal) of energy.

Table 1.1. Caloric Content of Fuels

	kcal/g
Carbohydrate	4
Fat	9
Protein	4
Alcohol	7



Q: An analysis of **Ann O'Rexia's** diet showed she ate 100 g carbohydrate, 20 g protein, and 15 g fat each day. Approximately how many calories did she consume per day?

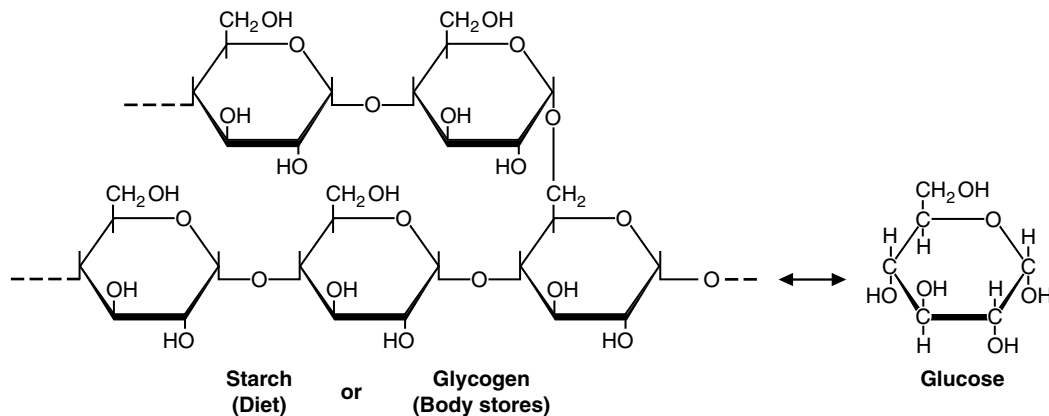


Fig. 1.4. Structure of starch and glycogen. Starch, our major dietary carbohydrate, and glycogen, the body’s storage form of glucose, have similar structures. They are polysaccharides (many sugar units) composed of glucose, which is a monosaccharide (one sugar unit). Dietary disaccharides are composed of two sugar units.



Miss O'Rexia consumed
 $100 \times 4 = 400$ kcal as carbohydrate
 $20 \times 4 = 80$ kcal as protein
 $15 \times 9 = 135$ kcal as fat

for a total of 615 kcal/day.

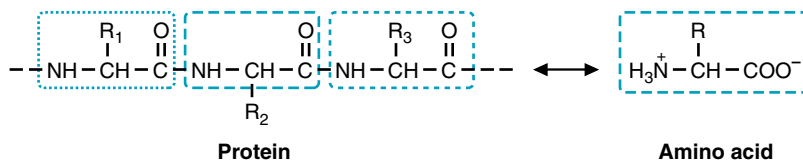


Fig. 1.5. General structure of proteins and amino acids. R = side chain. Different amino acids have different side chains. For example, R₁ might be $-\text{CH}_3$; R₂, $-\text{OH}$; R₃, $-\text{CH}_2-\text{COO}^-$.

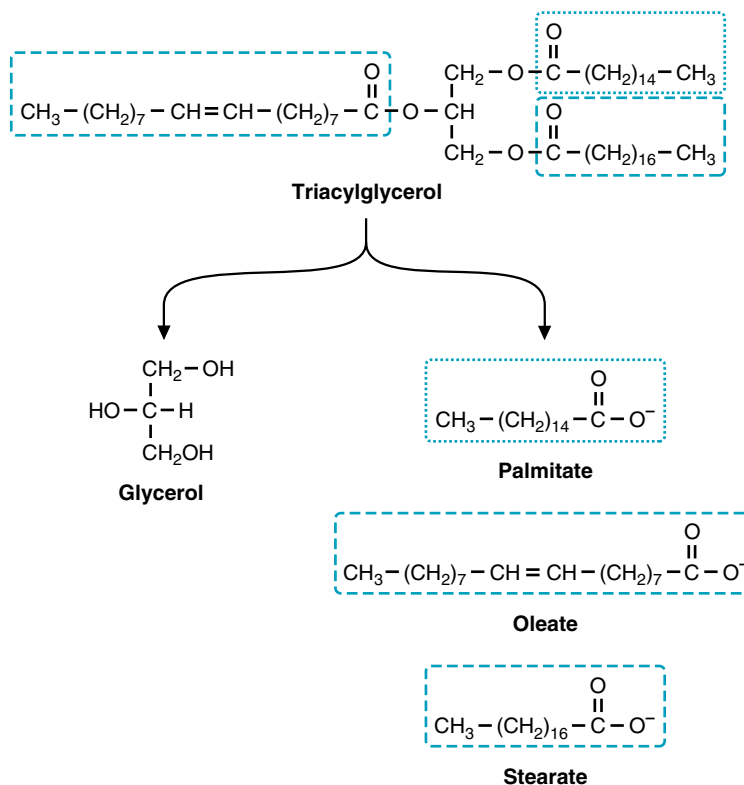


Fig. 1.6. Structure of a triacylglycerol. Palmitate and stearate are saturated fatty acids, i.e., they have no double bonds. Oleate is monounsaturated (one double bond). Polyunsaturated fatty acids have more than one double bond.



Ivan Applebod ate 585 g carbohydrate, 150 g protein, and 95 g fat each day. In addition, he drank 45 g alcohol. How many calories did he consume per day?

D. Alcohol

Many people used to believe that alcohol (ethanol, in the context of the diet) has no caloric content. In fact, ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) is oxidized to CO_2 and H_2O in the body and yields approximately 7 kcal/g—that is, more than carbohydrate but less than fat.

II. BODY FUEL STORES

Although some of us may try, it is virtually impossible to eat constantly. Fortunately, we carry supplies of fuel within our bodies (Fig. 1.7). These fuel stores are light in weight, large in quantity, and readily converted into oxidizable substances. Most of us are familiar with fat, our major fuel store, which is located in adipose tissue. Although fat is distributed throughout our bodies, it tends to increase in quantity in our hips and thighs and in our abdomens as we advance into middle age. In addition to our fat stores, we also have important, although much smaller, stores of carbohydrate in the form of glycogen located primarily in our liver and muscles. Glycogen



It is not surprising that our body fuel stores consist of the same kinds of compounds found in our diet, because the plants and animals we eat also store fuels in the form of starch or glycogen, triacylglycerols, and proteins.

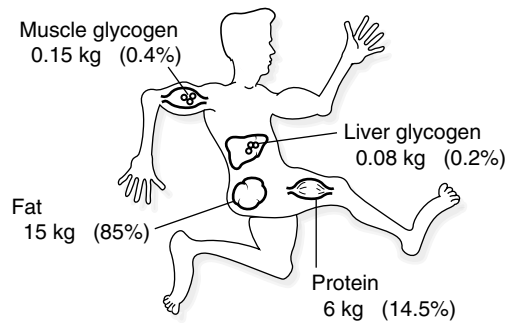


Fig. 1.7. Fuel composition of the average 70-kg man after an overnight fast (in kilograms and as percentage of total stored calories).

consists of glucose residues joined together to form a large, branched polysaccharide (see Fig. 1.4). Body protein, particularly the protein of our large muscle masses, also serves to some extent as a fuel store, and we draw on it for energy when we fast.

A. Fat

Our major fuel store is adipose triacylglycerol (triglyceride), a lipid more commonly known as fat. The average 70-kg man has approximately 15 kg stored triacylglycerol, which accounts for approximately 85% of his total stored calories (see Fig. 1.7).

Two characteristics make adipose triacylglycerol a very efficient fuel store: the fact that triacylglycerol contains more calories per gram than carbohydrate or protein (9 kcal/g versus 4 kcal/g) and the fact that adipose tissue does not contain much water. Adipose tissue contains only about 15% water, compared to tissues such as muscle that contain about 80%. Thus, the 70-kg man with 15 kg stored triacylglycerol has only about 18 kg adipose tissue.

B. Glycogen

Our stores of glycogen in liver, muscle, and other cells are relatively small in quantity but are nevertheless important. Liver glycogen is used to maintain blood glucose levels between meals. Thus, the size of this glycogen store fluctuates during the day; an average 70-kg man might have 200 g or more of liver glycogen after a meal but only 80 g after an overnight fast. Muscle glycogen supplies energy for muscle contraction during exercise. At rest, the 70-kg man has approximately 150 g of muscle glycogen. Almost all cells, including neurons, maintain a small emergency supply of glucose as glycogen.

C. Protein

Protein serves many important roles in the body; unlike fat and glycogen, it is not solely a fuel store. Muscle protein is essential for body movement. Other proteins serve as enzymes (catalysts of biochemical reactions) or as structural components of cells and tissues. Only a limited amount of body protein can be degraded, approximately 6 kg in the average 70-kg man, before our body functions are compromised.

III. DAILY ENERGY EXPENDITURE

If we want to stay in energy balance, neither gaining nor losing weight, we must, on average, consume an amount of food equal to our daily energy expenditure. The daily energy expenditure (DEE) includes the energy to support our basal metabolism (basal metabolic rate or resting metabolic rate) and our physical activity, plus the energy required to process the food we eat (diet-induced thermogenesis).



Mr. Applebod consumed
 $585 \times 4 = 2,340$ kcal as carbohydrate
 $150 \times 4 = 600$ kcal as protein
 $95 \times 9 = 855$ kcal as fat
 $45 \times 7 = 315$ kcal as alcohol
 for a total of 4,110 kcal/day.



In biochemistry and nutrition, the standard reference is often the 70-kg (154-lb) man. This standard probably was chosen because in the first half of the 20th century, when many nutritional studies were performed, young healthy medical and graduate students (who were mostly men) volunteered to serve as subjects for these experiments.



What would happen to a 70-kg man if the 135,000 kcal stored as triacylglycerols in his 18 kg of adipose tissue were stored instead as skeletal muscle glycogen? It would take approximately 34 kg glycogen to store as many calories. Glycogen, because it is a polar molecule with $-OH$ groups, binds approximately 4 times its weight in water, or 136 kg. Thus, his fuel stores would weigh 170 kg.



Daily energy expenditure =
 $RMR + \text{Physical Activity} + DIT$
 where RMR is the resting metabolic rate and DIT is diet-induced thermogenesis. BMR (basal metabolic rate) is used interchangeably with RMR in this equation.

Table 1.2. Factors Affecting BMR Expressed per kg Body Weight

Gender (males higher than females)
Body temperature (increased with fever)
Environmental temperature (increased in cold)
Thyroid status (increased in hyperthyroidism)
Pregnancy and lactation (increased)
Age (decreases with age)



What are **Ivan Applebod's** and **Ann O'Rexia's** RMR? (Compare the method for a rough estimate to values obtained with equations in Table 1.3.)



Registered dietitians use extensive tables for calculating energy requirements, based on height, weight, age, and activity level. A more accurate calculation is based on the fat-free mass (FFM), which is equal to the total body mass minus the mass of the person's adipose tissue. With FFM, the BMR is calculated using the equation $BMR = 186 + FFM \times 23.6$ kcal/kg per day. This formula eliminates differences between sexes and between aged versus young individuals that are attributable to differences in relative adiposity. However, determining FFM is relatively cumbersome—it requires weighing the patient underwater and measuring the residual lung volume.

Indirect calorimetry, a technique that measures O_2 consumption and CO_2 production, can be used when more accurate determinations are required for hospitalized patients. A portable indirect calorimeter is used to measure oxygen consumption and the respiratory quotient (RQ), which is the ratio of O_2 consumed to CO_2 produced. The RQ is 1.00 for individuals oxidizing carbohydrates, 0.83 for protein, and 0.71 for fat. From these values, the daily energy expenditure (DEE) can be determined.

A. Resting Metabolic Rate

The resting metabolic rate (RMR) is a measure of the energy required to maintain life: the functioning of the lungs, kidneys and brain, the pumping of the heart, the maintenance of ionic gradients across membranes, the reactions of biochemical pathways, and so forth. Another term used to describe basal metabolism is the basal metabolic rate (BMR). The BMR was originally defined as the energy expenditure of a person mentally and bodily at rest in a thermoneutral environment 12 to 18 hours after a meal. However, when a person is awakened and their heat production or oxygen consumption is measured, they are no longer sleeping or totally at mental rest, and their metabolic rate is called the resting metabolic rate (RMR). It is also sometimes called the resting energy expenditure (REE). The RMR and BMR differ very little in value.

The BMR, which is usually expressed in kcal/day, is affected by body size, age, sex, and other factors (Table 1.2). It is proportional to the amount of metabolically active tissue (including the major organs) and to the lean (or fat-free) body mass. Obviously, the amount of energy required for basal functions in a large person is greater than the amount required in a small person. However, the BMR is usually lower for women than for men of the same weight because women usually have more metabolically inactive adipose tissue. Body temperature also affects the BMR, which increases by 12% with each degree centigrade increase in body temperature (i.e., “feed a fever; starve a cold”). The ambient temperature affects the BMR, which increases slightly in colder climates as thermogenesis is activated. Excessive secretion of thyroid hormone (hyperthyroidism) causes the BMR to increase, whereas diminished secretion (hypothyroidism) causes it to decrease. The BMR increases during pregnancy and lactation. Growing children have a higher BMR per kilogram body weight than adults, because a greater proportion of their bodies is composed of brain, muscle, and other more metabolically active tissues. The BMR declines in aging individuals because their metabolically active tissue is shrinking and body fat is increasing. In addition, large variations exist in BMR from one adult to another, determined by genetic factors.

A rough estimate of the BMR may be obtained by assuming it is 24 kcal/day/kg body weight and multiplying by the body weight. An easy way to remember this is 1 kcal/kg/hr. This estimate works best for young individuals who are near their ideal weight. More accurate methods for calculating the BMR use empirically derived equations for different gender and age groups (Table 1.3). Even these calculations do not take into account variation among individuals.

B. Physical Activity

In addition to the RMR, the energy required for physical activity contributes to the DEE. The difference in physical activity between a student and a lumberjack is enormous, and a student who is relatively sedentary during the week may be much

Table 1.3. Equation for Predicting BMR from Body Weight (W) in kg

Males		Females	
Age Range (years)	BMR kcal/day	Age Range (years)	BMR kcal/day
0–3	$60.9W - 54$	0–3	$61.0W - 51$
3–10	$22.7W + 495$	3–10	$22.5W + 499$
10–18	$17.5W + 651$	10–18	$12.2W + 746$
18–30	$15.3W + 679$	18–30	$14.7W + 496$
30–60	$11.6W + 879$	30–60	$8.7W + 829$
>60	$13.5W + 487$	>60	$10.5W + 596$

From Energy and protein requirements: report of a Joint FAO/WHO/UNU Expert Consultation. Technical report series no. 724. Geneva World Health Organization, 1987:71. See also Schofield et al. Hum Nutr Clin Nutr 1985;39 (suppl).

Table 1.4. Typical Activities with Corresponding Hourly Activity Factors

ACTIVITY CATEGORY	Hourly Activity Factor (for Time in Activity)
Resting: sleeping, reclining	1.0
Very light: seated and standing activities, driving, laboratory work, typing, sewing, ironing, cooking, playing cards, playing a musical instrument	1.5
Light: walking on a level surface at 2.5–3 mph, garage work, electrical trades, carpentry, restaurant trades, house cleaning, golf, sailing, table tennis	2.5
Moderate: walking 3.5–4 mph, weeding and hoeing, carrying loads, cycling, skiing, tennis, dancing	5.0
Heavy: walking uphill with a load, tree felling, heavy manual digging, mountain climbing, basketball, football, soccer	7.0

Reprinted with permission from Recommended Dietary Allowances, 10th Ed. Washington, DC: National Academy Press, 1989.

The hourly activity factor is multiplied by the BMR (RMR) per hour times the number of hours engaged in the activity to give the caloric expenditure for that activity. If this is done for all of the hours in a day, the sum over 24 hours will approximately equal the daily energy expenditure.

more active during the weekend. Table 1.4 gives factors for calculating the approximate energy expenditures associated with typical activities.

A rough estimate of the energy required per day for physical activity can be made by using a value of 30% of the RMR (per day) for a very sedentary person (such as a medical student who does little but study) and a value of 60 to 70% of the RMR (per day) for a person who engages in about 2 hours of moderate exercise per day (see Table 1.4). A value of 100% or more of the RMR is used for a person who does several hours of heavy exercise per day.

C. Diet-Induced Thermogenesis

Our DEE includes a component related to the intake of food known as diet-induced thermogenesis (DIT) or the thermic effect of food (TEF). DIT was formerly called the specific dynamic action (SDA). After the ingestion of food, our metabolic rate increases because energy is required to digest, absorb, distribute, and store nutrients.

The energy required to process the types and quantities of food in the typical American diet is probably equal to approximately 10% of the kilocalories ingested. This amount is roughly equivalent to the error involved in rounding off the caloric content of carbohydrate, fat, and protein to 4, 9, and 4, respectively. Therefore, DIT is often ignored and calculations are based simply on the RMR and the energy required for physical activity.

D. Calculations of Daily Energy Expenditure

The total daily energy expenditure is usually calculated as the sum of the RMR (in kcal/day) plus the energy required for the amount of time spent in each of the various types of physical activity (see Table 1.4). An approximate value for the daily energy expenditure can be determined from the RMR and the appropriate percentage of the RMR required for physical activity (given above). For example, a very sedentary medical student would have a DEE equal to the RMR plus 30% of the RMR (or $1.3 \times \text{RMR}$) and an active person's daily expenditure could be 2 times the RMR.

E. Healthy Body Weight

Ideally, we should strive to maintain a weight consistent with good health. Overweight people are frequently defined as more than 20% above their ideal weight. But what is the ideal weight? The body mass index (BMI), calculated as



Mr. Applebod weighs 264 lb or 120 kg (264 lb divided by 2.2 lb/kg). His estimated $\text{RMR} = 24 \text{ kcal/kg/day} \times 120 = 2,880 \text{ kcal/day}$. His RMR calculated from Table 1.3 is only 2,271 kcal ($11.6 W + 879 = (11.6 \times 120) + 879$). Miss O'Rexia weighs 99 lb or 45 kg ($99/2.2 \text{ lb/kg}$). Her estimated $\text{RMR} = (24 \text{ kcal/kg/day}) \times (45 \text{ kg}) = 1,080 \text{ kcal/day}$. Her RMR from Table 1.3 is very close to this value ($14.7 W + 496 = 1,157 \text{ kcal/day}$). Thus, the rough estimate does not work well for obese patients because a disproportionately larger proportion of their body weight is metabolically inactive adipose tissue.



Based on the activities listed in Table 1.4, the average U.S. citizen is rather sedentary. Sedentary habits correlate strongly with risk for cardiovascular disease, so it is not surprising that cardiovascular disease is the major cause of death in this country.



What are reasonable estimates for Ivan Applebod's and Ann O'Rexia's daily energy expenditure?



BMI equals:

$$\text{Weight/height}^2 \text{ (kg/m}^2\text{)}$$

or

$$\frac{\text{Weight (lbs)} \times 704}{\text{height}^2 \text{ (in}^2\text{)}}$$

Where the height is measured without shoes and the weight is measured with minimal clothing.

BMI values of:

18.5 – 24.9 = desirable
 <18.5 = underweight
 25 – 29.9 = overweight
 ≥30 = obese



Are **Ivan Applebod** and **Ann O'Rexia** in a healthy weight range?

weight/height² (kg/m²), is currently the preferred method for determining whether a person's weight is in the healthy range.

In general, adults with BMI values below 18.5 are considered underweight. Those with BMIs between 18.5 and 24.9 are considered to be in the healthy weight range, between 25 and 29.9 are in the overweight or preobese range, and above 30 are in the obese range.

F. Weight Gain and Loss

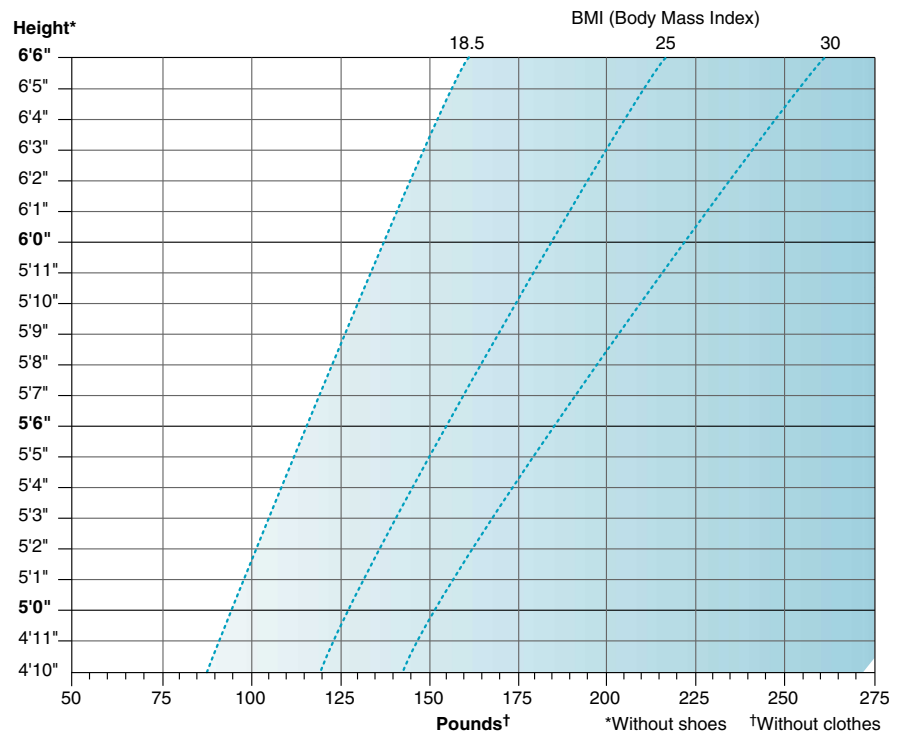
To maintain our body weight, we must stay in caloric balance. We are in caloric balance if the kilocalories in the food we eat equal our DEE. If we eat less food than we require for our DEE, our body fuel stores supply the additional calories,



To evaluate a patient's weight, physicians need standards of obesity applicable in a genetically heterogeneous population. Life insurance industry statistics have been used to develop tables giving the weight ranges, based on gender, height, and body frame size, that are associated with the greatest longevity, such as the Metropolitan Height and Weight Tables. However, these tables are considered inadequate for a number of reasons (e.g., they reflect data from upper-middle-class white groups). The BMI is the classification that is currently used clinically. It is based on two simple measurements, height without shoes and weight with minimal clothing. Patients can be shown their BMI in a nomogram and need not use calculations. The healthy weight range coincides with the mortality data derived from life insurance tables. The BMI also shows a good correlation with independent measures of body fat. The major weakness of the use of the BMI is that some very muscular individuals may be classified as obese when they are not. Other measurements to estimate body fat and other body compartments, such as weighing individuals underwater, are more difficult, expensive, and time consuming and have generally been confined to research purposes.



Mr. Applebod's BMR is 2,271 kcal/day. He is sedentary, so he only requires approximately 30% more calories for his physical activity. Therefore, his daily expenditure is approximately $2,271 + (0.3 \times 2,271)$ or $1.3 \times 2,271$ or 2,952 kcal/day. **Miss O'Rexia's** BMR is 1,157 kcal/day. She performs 2 hours of moderate exercise per day (jogging and walking), so she requires approximately 65% more calories for her physical activity. Therefore, her daily expenditure is approximately $1,157 + (0.65 \times 1,157)$ or $1.65 \times 1,157$ or 1,909 kcal/day.



If patients are above or below ideal weight (such as **Ivan Applebod** or **Ann O'Rexia**), the physician, often in consultation with a registered dietician, prescribes a diet designed to bring the weight into the ideal range.

and we lose weight. Conversely, if we eat more food than we require for our energy needs, the excess fuel is stored (mainly in our adipose tissue), and we gain weight (Fig. 1.8).

When we draw on our adipose tissue to meet our energy needs, we lose approximately 1 lb whenever we expend approximately 3,500 calories more than we consume. In other words, if we eat 1,000 calories less than we expend per day, we will lose about 2 lb/week. Because the average individual's food intake is only about 2,000 to 3,000 calories/day, eating one-third to one-half the normal amount will cause a person to lose weight rather slowly. Fad diets that promise a loss of weight much more rapid than this have no scientific merit. In fact, the rapid initial weight loss the fad dieter typically experiences is attributable largely to loss of body water. This loss of water occurs in part because muscle tissue protein and liver glycogen are degraded rapidly to supply energy during the early phase of the diet. When muscle tissue (which is approximately 80% water) and glycogen (approximately 70% water) are broken down, this water is excreted from the body.

IV. DIETARY REQUIREMENTS

In addition to supplying us with fuel and with general-purpose building blocks for biosynthesis, our diet also provides us with specific nutrients that we need to remain healthy. We must have a regular supply of vitamins and minerals and of the essential fatty acids and essential amino acids. “Essential” means that they are essential in the diet; the body cannot synthesize these compounds from other molecules and therefore must obtain them from the diet. Nutrients that the body requires in the diet only under certain conditions are called “conditionally essential.”

The Recommended Dietary Allowance (RDA) and the Adequate Intake (AI) provide quantitative estimates of nutrient requirements. The RDA for a nutrient is the average daily dietary intake level necessary to meet the requirement of nearly all (97–98%) healthy individuals in a particular gender and life stage group. Life stage group is a certain age range or physiologic status (i.e., pregnancy or lactation). The RDA is intended to serve as a goal for intake by individuals. The AI is a recommended intake value that is used when not enough data are available to establish an RDA.

A. Carbohydrates

No specific carbohydrates have been identified as dietary requirements. Carbohydrates can be synthesized from amino acids, and we can convert one type

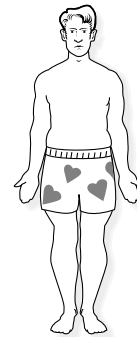


Malnutrition, the absence of an adequate intake of nutrients, occurs in the United States principally among children of families with incomes below the poverty level, the elderly, individuals whose diet is influenced by alcohol and drug usage, and those who make poor food choices. More than 13 million children in the United States live in families with incomes below the poverty level. Of these, approximately 10% have clinical malnutrition, most often anemia resulting from inadequate iron intake. A larger percentage have mild protein and energy malnutrition and exhibit growth retardation, sometimes as a result of parental neglect. Childhood malnutrition may also lead to learning failure and chronic illness later in life. A weight for age measurement is one of the best indicators of childhood malnourishment because it is easy to measure, and weight is one of the first parameters to change during malnutrition.

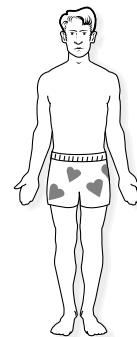
The term *kwashiorkor* refers to a disease originally seen in African children suffering from a protein deficiency. It is characterized by marked hypoalbuminemia, anemia, edema, pot belly, loss of hair, and other signs of tissue injury. The term *marasmus* is used for prolonged protein–calorie malnutrition, particularly in young children.



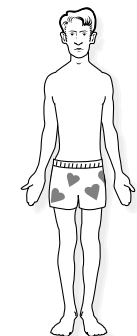
Are **Ivan Applebod** and **Ann O'Rexia** gaining or losing weight?



Positive caloric balance
Consumption > Expenditure



Caloric balance
Consumption = Expenditure



Negative caloric balance
Consumption < Expenditure

Fig. 1.8. Caloric balance.



Ivan Applebod's weight is classified as obese. His BMI is $264 \text{ lb} \times 704/70 \text{ in}^2 = 37.9$. **Ann O'Rexia** is underweight. Her BMI is $99 \text{ lb} \times 704/67 \text{ in}^2 = 15.5$.



Mr. Applebod expends about 2,952 kcal/day and consumes 4,110. By this calculation, he consumes 1,158 more kcal than he expends each day and is gaining weight. **Miss O'Rexia** expends 1,909 kcal/day while she consumes only 615. Therefore, she expends 1,294 more kcal/day than she consumes, so she is losing weight.

of carbohydrate to another. However, health problems are associated with the complete elimination of carbohydrate from the diet, partly because a low-carbohydrate diet must contain higher amounts of fat to provide us with the energy we need. High-fat diets are associated with obesity, atherosclerosis, and other health problems.

B. Essential Fatty Acids

Although most lipids required for cell structure, fuel storage, or hormone synthesis can be synthesized from carbohydrates or proteins, we need a minimal level of certain dietary lipids for optimal health. These lipids, known as essential fatty acids, are required in our diet because we cannot synthesize fatty acids with these particular arrangements of double bonds. The essential fatty acids α -linoleic and α -linolenic acid are supplied by dietary plant oils, and eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) are supplied in fish oils. They are the precursors of the eicosanoids (a set of hormone-like molecules that are secreted by cells in small quantities and have numerous important effects on neighboring cells). The eicosanoids include the prostaglandins, thromboxanes, leukotrienes, and other related compounds.

C. Protein

The RDA for protein is approximately 0.8 g high-quality protein per kilogram of ideal body weight, or approximately 60 g/day for men and 50 g/day for women. “High-quality” protein contains all of the essential amino acids in adequate amounts. Proteins of animal origin (milk, egg, and meat proteins) are high quality. The proteins in plant foods are generally of lower quality, which means they are low in one or more of the essential amino acids. Vegetarians may obtain adequate amounts of the essential amino acids by eating mixtures of vegetables that complement each other in terms of their amino acid composition.

1. ESSENTIAL AMINO ACIDS

Different amino acids are used in the body as precursors for the synthesis of proteins and other nitrogen-containing compounds. Of the 20 amino acids commonly required in the body for synthesis of protein and other compounds, nine amino acids are essential in the diet of an adult human because they cannot be synthesized in the body. These are lysine, isoleucine, leucine, threonine, valine, tryptophan, phenylalanine, methionine, and histidine.

Certain amino acids are conditionally essential, that is, required in the diet only under certain conditions. Children and pregnant women have a high rate of protein synthesis to support growth, and require some arginine in the diet, although it can be synthesized in the body. Histidine is essential in the diet of the adult in very small quantities because adults efficiently recycle histidine. The increased requirement of children and pregnant women for histidine is therefore much larger than their increased requirement of other essential amino acids. Tyrosine and cysteine are considered conditionally essential. Tyrosine is synthesized from phenylalanine, and it is required in the diet if phenylalanine intake is inadequate, or if an individual is congenitally deficient in an enzyme required to convert phenylalanine to tyrosine (the congenital disease phenylketonuria). Cysteine is synthesized by using sulfur from methionine, and it also may be required in the diet under certain conditions.

2. NITROGEN BALANCE

The proteins in the body undergo constant turnover; that is, they are constantly being degraded to amino acids and resynthesized. When a protein is degraded,



Students often use mnemonics to remember the essential amino acids. One common mnemonic is “Little TV tonight. Ha!” or **LIL** (lysine-~~i~~soleucine-~~l~~leucine) **TV** (~~t~~hreonine-~~v~~aline) **To** (~~t~~ryptophan) **PM** (~~p~~henyl- alanine-~~m~~ethionine). (**HA**) (~~h~~istidine-~~a~~rginine)!

its amino acids are released into the pool of free amino acids in the body. The amino acids from dietary proteins also enter this pool. Free amino acids can have one of three fates: they are used to make proteins, they serve as precursors for synthesis of essential nitrogen-containing compounds (e.g., heme, DNA, RNA), or they are oxidized as fuel to yield energy. When amino acids are oxidized, their nitrogen atoms are excreted in the urine principally in the form of urea. The urine also contains smaller amounts of other nitrogenous excretory products (uric acid, creatinine, and NH_4^+) derived from the degradation of amino acids and compounds synthesized from amino acids (Table 1.5). Some nitrogen is also lost in sweat, feces, and cells that slough off.

Nitrogen balance is the difference between the amount of nitrogen taken into the body each day (mainly in the form of dietary protein) and the amount of nitrogen in compounds lost (Table 1.6). If more nitrogen is ingested than excreted, a person is said to be in positive nitrogen balance. Positive nitrogen balance occurs in growing individuals (e.g., children, adolescents, and pregnant women), who are synthesizing more protein than they are breaking down. Conversely, if less nitrogen is ingested than excreted, a person is said to be in negative nitrogen balance. A negative nitrogen balance develops in a person who is eating either too little protein or protein that is deficient in one or more of the essential amino acids. Amino acids are continuously being mobilized from body proteins. If the diet is lacking an essential amino acid or if the intake of protein is too low, new protein cannot be synthesized, and the unused amino acids will be degraded, with the nitrogen appearing in the urine. If a negative nitrogen balance persists for too long, bodily function will be impaired by the net loss of critical proteins. In contrast, healthy adults are in nitrogen balance (neither positive nor negative), and the amount of nitrogen consumed in the diet equals its loss in urine, sweat, feces, and other excretions.

D. Vitamins

Vitamins are a diverse group of organic molecules required in very small quantities in the diet for health, growth, and survival (Latin *vita*, life). The absence of a vitamin from the diet or an inadequate intake results in characteristic deficiency signs and, ultimately, death. Table 1.7 lists the signs or symptoms of deficiency for each vitamin, its RDA or AI for young adults, and common food sources. The amount of each vitamin required in the diet is small (in the microgram or milligram range), compared with essential amino acid requirements (in the gram range). The vitamins are often divided into two classes, water-soluble vitamins and fat-soluble vitamins. This classification has little relationship to their function but is related to the absorption and transport of fat-soluble vitamins with lipids.

Most vitamins are used for the synthesis of coenzymes, complex organic molecules that assist enzymes in catalyzing biochemical reactions, and the deficiency symptoms reflect an inability of cells to carry out certain reactions. However, some vitamins also act as hormones. We will consider the roles played by individual vitamins as we progress through the subsequent chapters of this text.

Although the RDA or AI for each vitamin varies with age and sex, the difference is usually not very large once adolescence is reached. For example, the RDA for

Table 1.5. Major Nitrogenous Excretion Products

Urea
Creatinine
Uric acid
NH_4^+



Multiple vitamin deficiencies accompanying malnutrition are far more common in the United States than the characteristic deficiency diseases associated with diets lacking just one vitamin, because we generally eat a variety of foods. The characteristic deficiency diseases arising from single vitamin deficiencies were often identified and described in humans through observations of populations consuming a restricted diet because that was all that was available. For example, thiamine deficiency was discovered by a physician in Java, who related the symptoms of beri-beri to diets composed principally of polished rice. Today, single vitamin deficiencies usually occur as a result of conditions that interfere with the uptake or utilization of a vitamin or as a result of poor food choices or a lack of variety in the diet. For example, peripheral neuropathy associated with vitamin E deficiency can occur in children with fat malabsorption, and alcohol consumption can result in beri-beri. Vegans, individuals who consume diets lacking all animal products, can develop deficiencies in vitamin B_{12} .



In the hospital, it was learned that **Mr. Percy Veere** had lost 32 lb in the 8 months since his last visit to his family physician. On admission, his hemoglobin (the iron-containing compound in the blood, which carries O_2 from the lungs to the tissues) was 10.7 g/dL (reference range, males = 12 – 15.5), his serum iron was 38 $\mu\text{g}/\text{dL}$ (reference range, males = 42 – 135), and other hematologic indices were also abnormal. These values are indicative of an iron deficiency anemia. His serum folic acid level was 0.9 ng/mL (reference range = 3 – 20), indicating a low intake of this vitamin. His vitamin B_{12} level was 190 pg/mL (reference range = 180 – 914). A low blood vitamin B_{12} level can be caused by decreased intake, absorption, or transport, but it takes a long time to develop. His serum albumin was 3.2 g/dL (reference range = 3.5 – 5.0), which is an indicator of protein malnutrition or liver disease.

Table 1.6. Nitrogen Balance

Positive Nitrogen Balance	Growth (e.g., childhood, pregnancy)	Dietary N > Excreted N
Nitrogen Balance	Normal healthy adult	Dietary N = Excreted N
Negative Nitrogen Balance	Dietary deficiency of total protein or amino acids; catabolic stress	Dietary N < Excreted N

Table 1.7. VITAMINS ^a

Vitamin	Dietary Reference Intakes (DRI) Females (F) Males (M) (18–30 yrs old)	Some Common Food Sources	Consequences of Deficiency (Names of deficiency diseases are in bold)
Water-soluble vitamins			
Vitamin C	RDA F: 75 mg M: 90 mg UL: 2 g	Citrus fruits; potatoes; peppers, broccoli, spinach; strawberries	Scurvy: defective collagen formation leading to subcutaneous hemorrhage, aching bones, joints, and muscle in adults, rigid position and pain in infants.
Thiamin	RDA F: 1.1 mg M: 1.2 mg	Enriched cereals and breads; unrefined grains; pork; legumes, seeds, nuts	Beri-beri: (wet) Edema; anorexia, weight loss; apathy, decrease in short-term memory, confusion; irritability; muscle weakness; an enlarged heart
Riboflavin	RDA F: 1.1 mg M: 1.3 mg	Dairy products; fortified cereals; meats, poultry, fish; legumes	Ariboflavinosis: Sore throat, hyperemia, edema of oral mucosal membranes; cheilosis, angular stomatis; glossitis, magenta tongue; seborrheic dermatitis; normochromic normocytic anemia
Niacin ^b	RDA F: 14 mg NEQ M: 16 mg NEQ UL: 35 mg	Meat: chicken, beef, fish; enriched cereals or whole grains; most foods	Pellagra: Pigmented rash in areas exposed to sunlight; vomiting; constipation or diarrhea; bright red tongue; neurologic symptoms
Vitamin B ₆ (pyridoxine)	RDA F: 1.3 mg M: 1.3 mg UL: 100 mg	Chicken, fish, pork; eggs; fortified cereals, unmilled rice, oats; starchy vegetables; noncitrus fruits; peanuts, walnuts	Seborrheic dermatitis; microcytic anemia; epileptiform convulsions; depression and confusion
Folate	RDA F: 400 µg M: 400 µg	Citrus fruits; dark green vegetables; fortified cereals and breads; legumes	Impaired cell division and growth; megaloblastic anemia ; neural tube defects
Vitamin B ₁₂	RDA F: 2.4 µg M: 2.4 µg	Animal products ^c	Megaloblastic anemia Neurologic symptoms
Biotin	AI F: 30 µg M: 30 µg	Liver Egg yolk	Conjunctivitis; central nervous system abnormalities; glossitis; alopecia; dry, scaly dermatitis
Pantothenic acid	AI F: 5 mg M: 5 mg	Wide distribution in foods, especially animal tissues; whole grain cereals; legumes	Irritability and restlessness; fatigue, apathy, malaise; gastrointestinal symptoms; neurological symptoms
Choline	AI F: 550 mg M: 425 mg UL: 3.5 g	Milk; liver; eggs; peanuts	Liver damage
Fat-soluble vitamins			
Vitamin A	RDA F: 700 µg M: 900 µg UL: 3000 µg	Carrots; Dark green and leafy vegetables; sweet potatoes and squash; broccoli	Night blindness; xerophthalmia ; keratinization of epithelium in GI, respiratory and genitourinary tract, skin becomes dry and scaly
Vitamin K	RDA F: 90 µg M: 120 µg	Green leafy vegetables; cabbage family (brassica); Bacterial flora of intestine	Defective blood coagulation; hemorrhagic anemia of the newborn
Vitamin D	AI^d F: 5 µg M: 5 µg UL: 50 µg	Fortified milk; Exposure of skin to sunlight	Rickets (in children); inadequate bone mineralization (osteomalacia)
Vitamin E	RDA F: 15 mg M: 15 mg UL: 1 g	Vegetable oils, margarine; wheat germ; nuts; green leafy vegetables	Muscular dystrophy, neurologic abnormalities.

Dietary Reference Intakes (DRI): Recommended Dietary Allowance (RDA); Adequate Intake (AI); Tolerable Upper Intake Level (UL)

^aInformation for this table is from Dietary Reference Intakes for Thiamin, Riboflavin, Niacin, Vitamin B₆, Folate, Vitamin B₁₂, Pantothenic Acid, Biotin, and Choline (1998); Dietary Reference Intakes for Vitamin C, Vitamin E, Selenium, and Carotenoids (2000); Dietary Reference Intakes for Calcium, Phosphorus, Magnesium, Vitamin D, and Fluoride (1997); Dietary Reference Intakes for Vitamin A, Vitamin K, Arsenic, Boron, Chromium, Copper, Iodine, Iron, Manganese, Molybdenum, Nickel, Silicon, Vanadium, and Zinc (2001). Washington, DC: Food and Nutrition Board, Institute of Medicine, National Academy Press.^bneq = niacin equivalents. Niacin can be synthesized in the human from tryptophan, and this term takes into account a conversion factor for dietary tryptophan.^cVitamin B₁₂ is found only in animal products.^dDietary requirement assumes the absence of sunlight.

riboflavin is 0.9 mg/day for males between 9 and 13 years of age, 1.3 mg/day for males 19 to 30 years of age, still 1.3 mg/day for males older than 70 years, and 1.1 mg/day for females aged 19 to 30 years. The largest requirements occur during lactation (1.6 mg/day).

Vitamins, by definition, cannot be synthesized in the body, or are synthesized from a very specific dietary precursor in insufficient amounts. For example, we can synthesize the vitamin niacin from the essential amino acid tryptophan, but not in sufficient quantities to meet our needs. Niacin is therefore still classified as a vitamin.

Excessive intake of many vitamins, both fat-soluble and water-soluble, may cause deleterious effects. For example, high doses of vitamin A, a fat-soluble vitamin, can cause desquamation of the skin and birth defects. High doses of vitamin C cause diarrhea and gastrointestinal disturbances. One of the Reference Dietary Intakes is the Tolerable Upper Intake Level (UL), which is the highest level of daily nutrient intake that is likely to pose no risk of adverse effects to almost all individuals in the general population. As intake increases above the UL, the risk of adverse effects increases. Table 1.7 includes the UL for vitamins known to pose a risk at high levels. Intake above the UL occurs most often with dietary or pharmacologic supplements of single vitamins, and not from foods.

E. Minerals

Many minerals are required in the diet. They are generally divided into the classifications of electrolytes (inorganic ions that are dissolved in the fluid compartments of the body), minerals (required in relatively large quantities), trace minerals (required in smaller quantities), and ultratrace minerals (Table 1.8).

Sodium (Na^+), potassium (K^+), and chloride (Cl^-) are the major electrolytes (ions) in the body. They establish ion gradients across membranes, maintain water balance, and neutralize positive and negative charges on proteins and other molecules.

Calcium and phosphorus serve as structural components of bones and teeth and are thus required in relatively large quantities. Calcium (Ca^{2+}) plays many other roles in the body; for example, it is involved in hormone action and blood clotting. Phosphorus is required for the formation of ATP and of phosphorylated intermediates in metabolism. Magnesium activates many enzymes and also forms a complex with ATP. Iron is a particularly important mineral because it functions as a component of hemoglobin (the oxygen-carrying protein in the blood) and is part of many enzymes. Other minerals, such as zinc or molybdenum, are required in very small quantities (trace or ultra-trace amounts).

Sulfur is ingested principally in the amino acids cysteine and methionine. It is found in connective tissue, particularly in cartilage and skin. It has important functions in metabolism, which we will describe when we consider the action of coenzyme A, a compound used to activate carboxylic acids. Sulfur is excreted in the urine as sulfate.



A dietary deficiency of calcium can lead to osteoporosis, a disease in which bones are insufficiently mineralized and consequently are fragile and easily fractured. Osteoporosis is a particularly common problem among elderly women. Deficiency of phosphorus results in bone loss along with weakness, anorexia, malaise, and pain. Iron deficiencies lead to anemia, a decrease in the concentration of hemoglobin in the blood.



Which foods would provide Percy Veere with good sources of folate and vitamin B_{12} ?

Table 1.8. Minerals Required in the Diet

Electrolytes	Minerals	Trace Minerals	Ultratrace or Trace Minerals ^a
Sodium	Calcium	Iodine	Manganese
Potassium	Phosphorus	Selenium	Fluoride
Chloride	Magnesium	Copper	Chromium
	Iron	Zinc	Molybdenum
	Sulfur		Others?

^aThese minerals are classified as trace or as ultratrace.



Folate is found in fruits and vegetables: citrus fruits (e.g., oranges), green leafy vegetables (e.g., spinach and broccoli), fortified cereals, and legumes (e.g., peas) (see Table 1.7). Conversely, vitamin B₁₂ is found only in foods of animal origin, including meats, eggs, and milk.

Minerals, like vitamins, have adverse effects if ingested in excessive amounts. Problems associated with dietary excesses or deficiencies of minerals are described in subsequent chapters in conjunction with their normal metabolic functions.

F. Water

Water constitutes one half to four fifths of the weight of the human body. The intake of water required per day depends on the balance between the amount produced by body metabolism and the amount lost through the skin, through expired air, and in the urine and feces.

V. DIETARY GUIDELINES

Dietary guidelines or goals are recommendations for food choices that can reduce the risk of developing chronic or degenerative diseases while maintaining an adequate intake of nutrients. Many studies have shown an association between diet and exercise and decreased risk of certain diseases, including hypertension, atherosclerosis, stroke, diabetes, certain types of cancer, and osteoarthritis. Thus, the American Heart Institute and the American Cancer Institute, as well as several other groups, have developed dietary and exercise recommendations to decrease the risk of these diseases. The “Dietary Guidelines for Americans (2000)”, prepared under the joint authority of the US Department of Agriculture and the US Department of Health and Human Services, merges many of these recommendations. Recommended servings of different food groups are displayed as the food pyramid (Fig. 1.9). Issues of special concern for physicians who advise patients include the following:

A. General Recommendations

- Aim for a healthy weight and be physically active each day. For maintenance of a healthy weight, caloric intake should balance caloric expenditure. Accumulate at least 30 minutes of moderate physical activity (such as walking 2 miles) daily. A regular exercise program helps in achieving and maintaining ideal weight, cardiovascular fitness, and strength.
- Choose foods in the proportions recommended in the food pyramid, including a variety of grains and a variety of fruits and vegetables daily.
- Keep food safe to eat. For example, refrigerate leftovers promptly.

B. Vegetables, Fruits, and Grains

- Diets rich in vegetables, fruits, and grain products should be chosen. Five or more servings of vegetables and fruits should be eaten each day, particularly green and yellow vegetables and citrus fruits. Six or more daily servings of grains should be eaten (starches and other complex carbohydrates, in the form of breads, fortified cereals, rice, and pasta). In addition to energy, vegetables, fruits, and grains supply vitamins, minerals, protective substances (such as carotenoids), and fiber. Fiber, the indigestible part of plant food, has various beneficial effects, including relief of constipation.
- The consumption of refined sugar in foods and beverages should be reduced to below the American norm. Refined sugar has no nutritional value other than its caloric content, and it promotes tooth decay.

C. Fats

- Fat intake should be reduced. For those at risk of heart attacks or strokes, fat should account for no more than 30% of total dietary calories, and saturated fatty acids

Food Guide Pyramid

A Guide to Daily Food choices

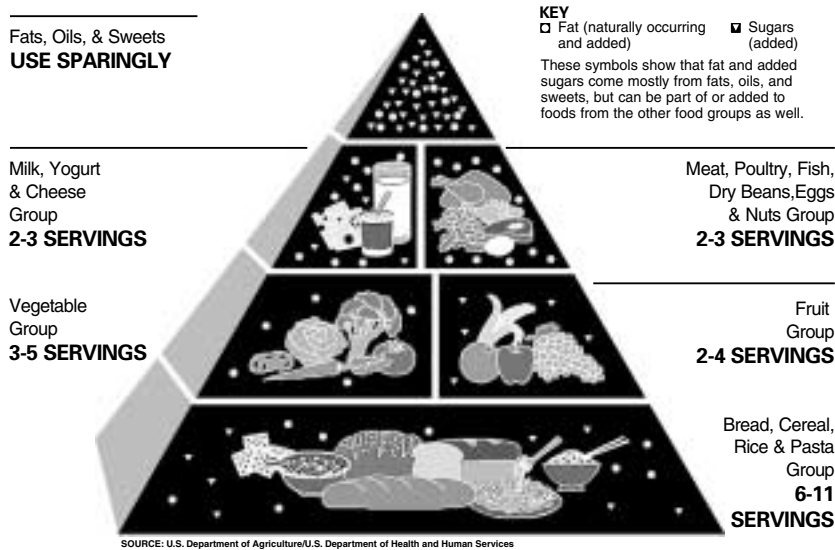


Fig. 1.9. The Food Guide Pyramid. The pyramid shows the number of servings that should be eaten each day from each food group. Within each group, a variety of foods should be eaten. Some examples of serving size: Grain products—1 slice of white bread or $\frac{1}{2}$ cup of cooked rice; Vegetable group— $\frac{1}{2}$ cup cooked vegetables; Fruit group—1 apple or banana; Milk Group—1 cup of milk or 2 oz processed cheese; Meat and Beans Group—2–3 oz cooked lean meat or fish or 1 egg or 2 tbsp peanut butter. Nutrition and Your Health: Dietary Guidelines for Americans, 2000. Washington, DC: Dietary Guidelines Committee: The U.S. Department of Agriculture and the U.S. Department of Health and Human Services.

should account for 10% or less. Foods high in saturated fat include cheese, whole milk, butter, regular ice cream, and many cuts of beef. Trans fatty acids, such as the partially hydrogenated vegetable oils used in margarine, should also be avoided.

- Cholesterol intake should be less than 300 mg/day in subjects without atherosclerotic disease and less than 200 mg/day in those with established atherosclerosis.

D. Proteins

- Protein intake for adults should be approximately 0.8 g/kg ideal body weight per day. The protein should be of high quality and should be obtained from sources low in saturated fat (e.g., fish, lean poultry, and dry beans). Vegetarians should eat a mixture of vegetable proteins that ensures the intake of adequate amounts of the essential amino acids.

E. Alcohol

- Alcohol consumption should not exceed moderate drinking. Moderation is defined as no more than one drink per day for women and no more than two drinks per day for men. A drink is defined as 1 regular beer, 5 ounces of wine (a little over $\frac{1}{2}$ cup), or 1.5 ounces of an 80-proof liquor, such as whiskey. Pregnant women should drink no alcohol.



Cholesterol is obtained from the diet and synthesized in most cells of the body. It is a component of cell membranes and the precursor of steroid hormones and of the bile salts used for fat absorption. High concentrations of cholesterol in the blood, particularly the cholesterol in lipoprotein particles called low density lipoproteins (LDL), contribute to the formation of atherosclerotic plaques. These plaques (fatty deposits on arterial walls) are associated with heart attacks and strokes. A high content of saturated fat in the diet tends to increase circulatory levels of LDL cholesterol and contributes to the development of atherosclerosis.



The ingestion of alcohol by pregnant women can result in fetal alcohol syndrome (FAS), which is marked by prenatal and postnatal growth deficiency, developmental delay, and craniofacial, limb, and cardiovascular defects.



The high intake of sodium and chloride (in table salt) of the average American diet appears to be related to the development of hypertension (high blood pressure) in individuals who are genetically predisposed to this disorder.

F. Vitamins and Minerals

- Sodium intake should be decreased in most individuals. Sodium is usually consumed as salt, NaCl. Individuals prone to salt-sensitive hypertension should eat less than 3 g sodium per day (approximately 6 g NaCl).
- Many of the required vitamins and minerals can be obtained from eating a variety of fruits, vegetables, and grains (particularly whole grains). However, calcium and iron are required in relatively high amounts. Low-fat or nonfat dairy products and dark green leafy vegetables provide good sources of calcium. Lean meats, shellfish, poultry, dark meat, cooked dry beans, and some leafy green vegetables provide good sources of iron. Vitamin B₁₂ is found only in animal sources.
- Dietary supplementation in excess of the recommended amounts (for example, megavitamin regimens) should be avoided.
- Fluoride should be present in the diet, at least during the years of tooth formation, as a protection against dental caries.

VI. XENOBIOTICS

In addition to nutrients, our diet also contains a large number of chemicals called xenobiotics, which have no nutritional value, are of no use in the body, and can be harmful if consumed in excessive amounts. These compounds occur naturally in foods, can enter the food chain as contaminants, or can be deliberately introduced as food additives.

Dietary guidelines of the American Cancer Society and the American Institute for Cancer Research make recommendations relevant to the ingestion of xenobiotic compounds, particularly carcinogens. The dietary advice that we eat a variety of food helps to protect us against the ingestion of a toxic level of any one xenobiotic compound. It is also suggested that we reduce consumption of salt-cured, smoked, and charred foods, which contain chemicals that can contribute to the development of cancer. Other guidelines encourage the ingestion of fruits and vegetables that contain protective chemicals called antioxidants.

CLINICAL COMMENTS



Otto Shape. Otto Shape sought help in reducing his weight of 187 lb (BMI of 27) to his previous level of 154 lb (BMI of 22, in the middle of the healthy range). Otto Shape was 5 feet 10 inches tall, and he calculated that his maximum healthy weight was 173 lbs. He planned on becoming a family physician, and he knew that he would be better able to counsel patients in healthy behaviors involving diet and exercise if he practiced them himself. With this information and assurances from the physician that he was otherwise in good health, Otto embarked on a weight loss program. One of his strategies involved recording all the food he ate and the portions. To analyze his diet for calories, saturated fat, and nutrients, he used the Interactive Healthy Eating Index, available online from the USDA Food and Nutrition Information Center.



Ivan Applebod. Ivan Applebod weighed 264 lb and was 70 inches tall with a heavy skeletal frame. For a male of these proportions, a BMI of 18.5 to 24.9 would correspond to a weight between 129 and 173 lb. He is currently almost 100 lb overweight, and his BMI of 37.9 is in the obese range.

Mr. Applebod's physician cautioned him that exogenous obesity (caused by overeating) represents a risk factor for atherosclerotic vascular disease, particularly when the distribution of fat is primarily "central" or in the abdominal region (apple



Physicians have an average lifespan that is longer than the general population, and generally practice healthier behaviors, especially with regard to fat consumption, exercise, alcohol consumption, and smoking. Physicians who practice healthy behaviors are more likely to counsel patients with respect to these behaviors and are better able to motivate their patients.

shape, in contrast to the pear shape, which results from adipose tissue deposited in the buttocks and hips). In addition, obesity may lead to other cardiovascular risk factors such as hypertension (high blood pressure), hyperlipidemia (high blood lipid levels), and type 2 diabetes mellitus (characterized by hyperglycemia). He already has a mild elevation in both systolic and diastolic blood pressure. Furthermore, his total serum cholesterol level was 296 mg/dL, well above the desired normal value (200 mg/dL).

Mr. Applebod was referred to the hospital's weight reduction center, where a team of physicians, dieticians, and psychologists could assist him in reaching his ideal weight range.



Ann O'Rexia. Because of her history and physical examination, Ann O'Rexia was diagnosed as having early anorexia nervosa, a behavioral disorder that involves both emotional and nutritional disturbances. Miss O'Rexia was referred to a psychiatrist with special interest in anorexia nervosa, and a program of psychotherapy and behavior modification was initiated.



Percy Veere. Percy Veere weighed 125 lb and was 71 inches tall (without shoes) with a medium frame. His BMI was 17.5, which is significantly underweight. At the time his wife died, he weighed 147 lbs. For his height, a BMI in the healthy weight range corresponds to weights between 132 and 178 lb.

Mr. Veere's malnourished state was reflected in his admission laboratory profile. The results of hematologic studies were consistent with an iron deficiency anemia complicated by low levels of folic acid and vitamin B₁₂, two vitamins that can affect the development of normal red blood cells. His low serum albumin level was caused by insufficient protein intake and a shortage of essential amino acids, which result in a reduced ability to synthesize body proteins. The psychiatrist requested a consultation with a hospital dietician to evaluate the extent of Mr. Veere's marasmus (malnutrition caused by a deficiency of both protein and total calories) as well as his vitamin and mineral deficiencies.

BIOCHEMICAL COMMENTS



Dietary Reference Intakes. Dietary Reference Intakes (DRIs) are quantitative estimates of nutrient intakes that can be used in evaluating and planning diets for healthy people. They are prepared by the Standing Committee on the Scientific Evaluation of Dietary Reference Intakes (DRI) of the Food and Nutrition Board, Institute of Medicine, and the National Academy of Science, with active input of Health Canada. The four reference intake values are the Recommended Dietary Allowance (RDA), the Estimated Average Requirement (EAR), the Adequate Intake (AI), and the Tolerable Upper Intake Level (UL). For each vitamin, the Committee has reviewed available literature on studies with humans and established criteria for adequate intake, such as prevention of certain deficiency symptoms, prevention of developmental abnormalities, or decreased risk of chronic degenerative disease. The criteria are not always the same for each life stage group. A requirement is defined as the lowest continuing intake level of a nutrient able to satisfy these criteria. The EAR is the daily intake value that is estimated to meet the requirement in half of the apparently healthy individuals in a life stage or gender group. The RDA is the EAR plus 2 standard deviations of the mean, which is the amount that should satisfy the requirement in 97 to 98% of the population. The AI level instead of an RDA is set for nutrients when there is not enough data to determine the EAR.

The Tolerable Upper Intake Level (UL) refers to the highest level of daily nutrient intake consumed over time that is likely to pose no risks of adverse effects for almost all healthy individuals in the general population. Adverse effects are defined as any significant alteration in the structure or function of the human organism. The



The prevalence of obesity in the U.S. population is increasing. In 1962, 12.8% of the population had a BMI equal to or greater than 30 and therefore were clinically obese. That number increased to 14.5% by 1980 and to 22.5% by 1998. An additional 30% were pre-obese in 1998 (BMI = 25.0 – 29.9). Therefore, more than 50% of the population is currently overweight, that is, obese or pre-obese.

Increased weight increases cardiovascular risk factors, including hypertension, diabetes, and alterations in blood lipid levels. It also increases the risk for respiratory problems, gallbladder disease, and certain types of cancer.



An example of the difference between the AI and the EAR is provided by riboflavin. Very few data exist on the nutrient requirements of very young infants. However, human milk is the sole recommended food for the first 4 to 6 months, so the AI of the vitamin riboflavin for this life stage group is based on the amount in breast milk consumed by healthy full-term infants. Conversely, the riboflavin EAR for adults is based on a number of studies in humans relating dietary intake of riboflavin to biochemical markers of riboflavin status and development of clinical deficiency symptoms.

UL does not mean that most individuals who consume more than the UL will suffer adverse health effects, but that the risk of adverse effects increases as intake increases above the UL.

Suggested References

A good, comprehensive textbook on nutrition is Shils ME, Olson JA, Shike M, Ross, AC. *Modern nutrition in health and disease*. Baltimore: Williams & Wilkins, 1999. Extensive nutrition tables, including Metropolitan Height and Weight Tables, are available in the appendices.

Recent Dietary References Intakes prepared by the Food and Nutrition Board of the National Academy of Science (1997–2001) are available in several volumes published by the National Academy Press (see Table 1.7) and may be consulted online at <http://books.nap.edu/>.

To analyze diets for calories and nutrient contents, consult food databases and resource lists made available by the USDA. The site www.nal.usda.gov/fnic provides lists of resources on food composition, such as the database U.S. Department of Agriculture, Agricultural Research Service. 2001. USDA Nutrient Database for Standard Reference, Release 14. Nutrient Data Laboratory Homepage, <http://www.nal.usda.gov/fnic/foodcomp>. This site also provides lists of resources for diet analysis, and links to the Interactive Healthy Eating Index, which is a program students can use to analyze their diets (<http://147.208.9.133>). A useful computer program for evaluating the diet of individuals, the MSU Nutriguide, can be obtained from Department of Nutrition, Michigan State University.

Dietary recommendations change frequently as new data become available. Current Dietary Recommendations are available from the following sources: Food and Nutrition Information Center, National Agricultural Library, USDA (www.fns.usda.gov); National Heart, Lung, and Blood Institute Information Center (www.nhlbi.nih.gov); American Heart Association (www.Americanheart.org); American Institute for Cancer Research (www.aicr.org); and the American Diabetes Association (www.diabetes.org). Another reliable source for nutrition information on the internet is www.navigator.tufts.edu.

A number of medical schools in the United States have received Nutrition Academic Awards from the National Institute of Heart, Blood and Lung, National Institutes of Health (www.nhlbi.nih.gov/funding/naa). These schools are developing products for medical nutrition education.



REVIEW QUESTIONS—CHAPTER 1

Directions: For each question below, select the single best answer.

1. In the process of respiration, fuels
 - (A) are stored as triacylglycerols.
 - (B) are oxidized to generate ATP.
 - (C) release energy principally as heat.
 - (D) combine with CO_2 and H_2O .
 - (E) combine with other dietary components in anabolic pathways.
2. The caloric content per gram of fuel
 - (A) is higher for carbohydrates than triacylglycerols.
 - (B) is higher for protein than for fat.
 - (C) is proportionate to the amount of oxygen in a fuel.
 - (D) is the amount of energy that can be obtained from oxidation of the fuel.
 - (E) is higher for children than adults.
3. The resting metabolic rate is
 - (A) equivalent to the caloric requirement of our major organs and resting muscle.
 - (B) generally higher per kilogram body weight in women than in men.
 - (C) generally lower per kilogram body weight in children than adults.
 - (D) decreased in a cold environment.
 - (E) approximately equivalent to the daily energy expenditure.

4. The RDA is
- (A) the average amount of a nutrient required each day to maintain normal function in 50% of the U.S. population.
 - (B) the average amount of a nutrient ingested daily by 50% of the U.S. population.
 - (C) the minimum amount of a nutrient ingested daily that prevents deficiency symptoms.
 - (D) a reasonable dietary goal for the intake of a nutrient by a healthy individual.
 - (E) based principally on data obtained with laboratory animals.
5. A 35-year old sedentary male patient weighing 120 kg was experiencing angina (chest pain) and other signs of coronary artery disease. His physician, in consultation with a registered dietician, conducted a 3-day dietary recall. The patient consumed an average of 585 g carbohydrate, 150 g protein, and 95 g fat each day. In addition, he drank 45 g alcohol. The patient
- (A) consumed between 2,500 and 3,000 kcal per day.
 - (B) had a fat intake within the range recommended in current dietary guidelines (i.e., year 2000).
 - (C) consumed 50% of his calories as alcohol.
 - (D) was deficient in protein intake.
 - (E) was in negative caloric balance.

2 The Fed or Absorptive State



Hormones are compounds that are synthesized by the endocrine glands of the body. They are secreted into the bloodstream and carry messages to different tissues concerning changes in the overall physiologic state of the body or the needs of tissues.

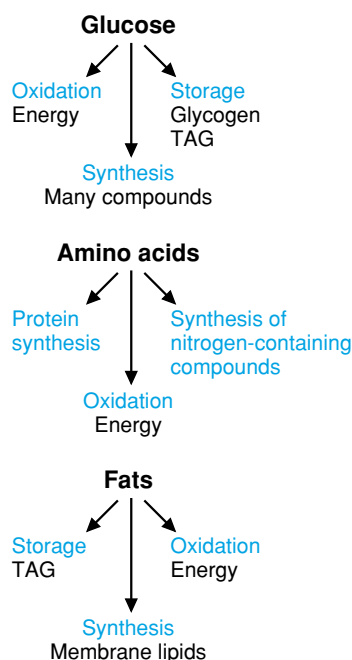


Fig. 2.1. Major fates of fuels in the fed state.

The Fed State. During a **meal**, we ingest carbohydrates, lipids, and proteins, which are subsequently **digested** and **absorbed**. Some of this food is **oxidized** to meet the immediate **energy** needs of the body. The amount consumed in **excess** of the body's energy needs is transported to the **fuel depots**, where it is stored. During the period from the start of absorption until absorption is completed, we are in the **fed**, or absorptive, state. Whether a fuel is oxidized or stored in the fed state is determined principally by the concentration of two **endocrine hormones** in the blood, **insulin** and **glucagon**.

Fate of Carbohydrates. Dietary carbohydrates are digested to monosaccharides, which are absorbed into the blood. The major monosaccharide in the blood is **glucose** (Fig 2.1). After a meal, glucose is **oxidized** by various tissues for energy, enters biosynthetic pathways, and is **stored** as **glycogen**, mainly in liver and muscle. Glucose is the major biosynthetic precursor in the body, and the carbon skeletons of most of the compounds we synthesize can be synthesized from glucose. Glucose is also converted to **triacylglycerols**. The liver packages triacylglycerols, made from glucose or from fatty acids obtained from the blood, into very low-density lipoproteins (VLDL) and releases them into the blood. The fatty acids of the VLDL are mainly stored as triacylglycerols in adipose tissue, but some may be used to meet the energy needs of cells.

Fate of Proteins. Dietary proteins are digested to **amino acids**, which are absorbed into the blood. In cells, the amino acids are converted to **proteins** or used to make various **nitrogen-containing compounds** such as neurotransmitters and heme. The carbon skeleton may also be **oxidized** for energy directly, or converted to glucose.

Fate of Fats. Triacylglycerols are the major lipids in the diet. They are digested to fatty acids and 2-monoacylglycerols, which are resynthesized into **triacylglycerols** in intestinal epithelial cells, packaged in **chylomicrons**, and secreted by way of the lymph into the blood. The **fatty acids** of the chylomicron triacylglycerols are stored mainly as triacylglycerols in **adipose** cells. They are subsequently oxidized for energy or used in biosynthetic pathways, such as synthesis of membrane lipids.



THE WAITING ROOM



Ivan Applebod returned to his doctor for a second visit. His initial efforts to lose weight had failed dismally. In fact, he now weighed 270 lb, an increase of 6 lb since his first visit 2 months ago (see Chapter 1). He reported that the recent death of his 45-year-old brother of a heart attack had made him realize that he must pay more attention to his health. Because

Mr. Applebod's brother had a history of hypercholesterolemia and because Mr. Applebod's serum total cholesterol had been significantly elevated (296 mg/dL) at his first visit, his blood lipid profile was determined, his blood glucose level was measured, and a number of other blood tests were ordered. (The blood lipid profile is a test that measures the content of the various triacylglycerol- and cholesterol-containing particles in the blood.) His blood pressure was 162 mm Hg systolic and 98 mm Hg diastolic or 162/98 mm Hg (normal = 140/90 mm Hg or less). His waist circumference was 48 inches (healthy values for men, less than 40; for women, less than 35).



The body can make fatty acids from a caloric excess of carbohydrate and protein. These fatty acids, together with the fatty acids of chylomicrons (derived from dietary fat), are deposited in adipose tissue as triacylglycerols. Thus, **Ivan Applebod's** increased adipose tissue is coming from his intake of all fuels in excess of his caloric need.

I. DIGESTION AND ABSORPTION

After a meal is consumed, foods are digested (broken down into simpler components) by a series of enzymes in the mouth, stomach, and small intestine. The products of digestion eventually are absorbed into the blood. The period during which digestion and absorption occur constitutes the fed state (Fig. 2.2)

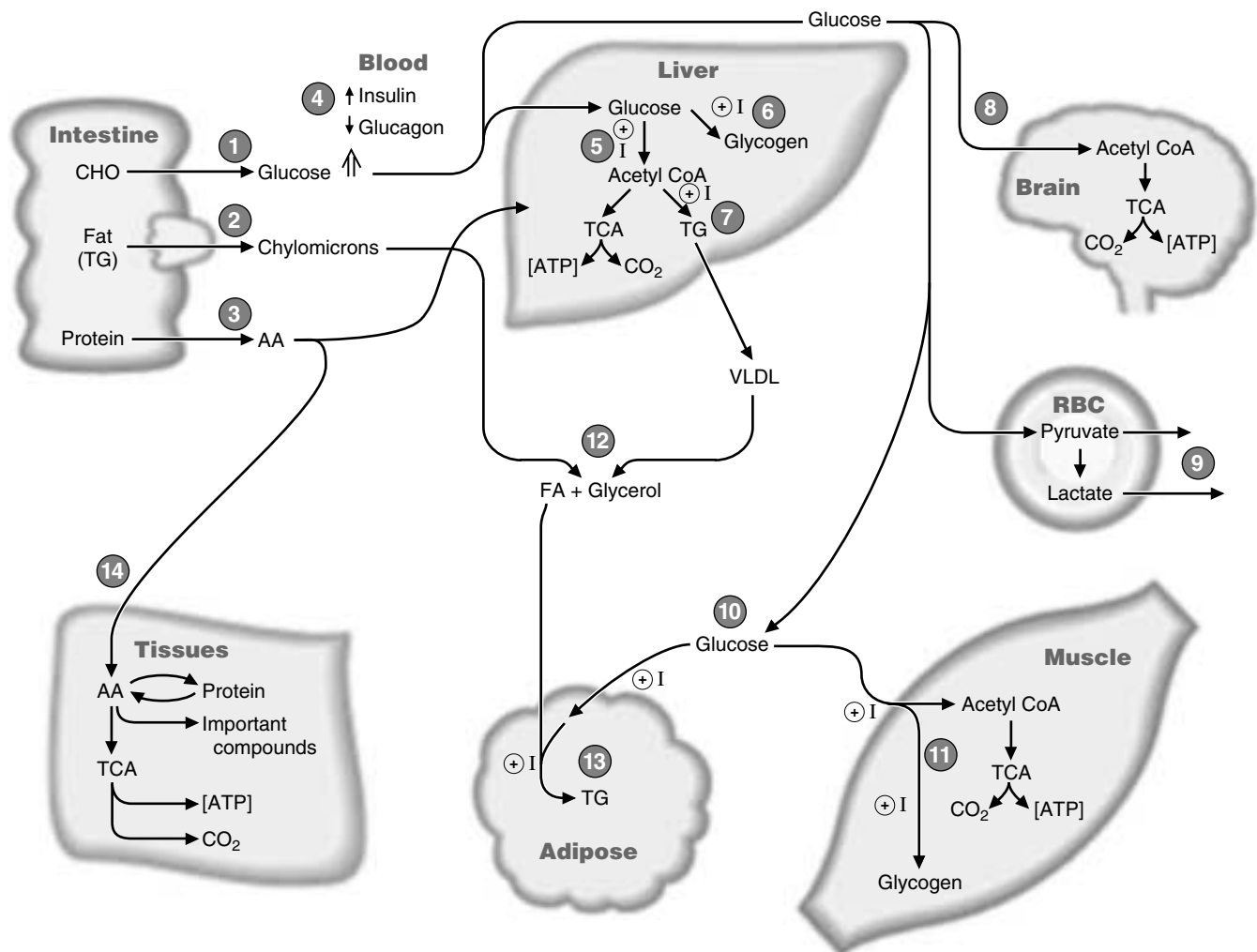


Fig. 2.2. The fed state. The circled numbers indicate the approximate order in which the processes occur. TG = triacylglycerols; FA = fatty acid; AA = amino acid; RBC = red blood cell; VLDL = very-low-density lipoprotein; I = insulin; ⊕ = stimulated by.



Digestive enzymes convert complex sugars to single sugar units for absorption. Sugars are saccharides, and the prefixes “mono” (one), “di” (two), “tri” (three), “oligo” (some), and “poly” (many) refer to the number of sugar units linked together.



Enzymes are proteins that catalyze biochemical reactions; in other words, they increase the speed at which reactions occur. Their names usually end in “ase.”



Proteins are amino acids linked through peptide bonds. Dipeptides have two amino acids, tripeptides have three amino acids, and so on. Digestive proteases are enzymes that cleave the peptide bonds between the amino acids (see Chap.1, Fig. 1.5).



Fats must be transported in the blood bound to protein or in lipoprotein complexes because they are insoluble in water. Thus, both triacylglycerols and cholesterol are found in lipoprotein complexes.



The laboratory studies ordered at the time of his second office visit show that **Ivan Applebod** has hyperglycemia, an elevation of blood glucose above normal values. At the time of this visit, his 2-hour postprandial blood glucose level was 205 mg/dL. (Two-hour postprandial refers to the glucose level measured 2 hours after a meal, when glucose should have been taken up by tissues and blood glucose returned to the fasting level, approximately 80–100 mg/dL.) His blood glucose determined after an overnight fast was 162 mg/dL. Because both of these blood glucose measurements were significantly above normal, a diagnosis of type 2 diabetes mellitus, formerly known as non-insulin-dependent diabetes mellitus (NIDDM), was made. In this disease, liver, muscle, and adipose tissue are relatively resistant to the action of insulin in promoting glucose uptake into cells and storage as glycogen and triacylglycerols. Therefore, more glucose remains in his blood.

A. Carbohydrates

Dietary carbohydrates are converted to monosaccharides. Starch, a polymer of glucose, is the major carbohydrate of the diet. It is digested by salivary α -amylase, and then by pancreatic α -amylase, which acts in the small intestine. Di-, tri-, and oligosaccharides produced by these α -amylases are cleaved to glucose by digestive enzymes located on the surface of the brush border of the intestinal epithelial cells. Dietary disaccharides also are cleaved by enzymes in this brush border. Sucrase converts the disaccharide sucrose (table sugar) to glucose and fructose, and lactase converts the disaccharide lactose (milk sugar) to glucose and galactose. Monosaccharides produced by digestion and dietary monosaccharides are absorbed by the intestinal epithelial cells and released into the hepatic portal vein, which carries them to the liver.

B. Proteins

Dietary proteins are cleaved to amino acids by proteases (see Fig. 2.2, circle 3). Pepsin acts in the stomach, and the proteolytic enzymes produced by the pancreas (trypsin, chymotrypsin, elastase, and the carboxypeptidases) act in the lumen of the small intestine. Aminopeptidases and di- and tripeptidases associated with the intestinal epithelial cells complete the conversion of dietary proteins to amino acids, which are absorbed into the intestinal epithelial cells and released into the hepatic portal vein.

C. Fats

The digestion of fats is more complex than that of carbohydrates or proteins because they are not very soluble in water. The triacylglycerols of the diet are emulsified in the intestine by bile salts, which are synthesized in the liver and stored in the gallbladder. Pancreatic lipase converts the triacylglycerols in the lumen of the intestine to fatty acids and 2-monoacylglycerols (glycerol with a fatty acid esterified at carbon 2), which interact with bile salts to form tiny microdroplets called micelles. The fatty acids and 2-monoacylglycerols are absorbed from these micelles into the intestinal epithelial cells, where they are resynthesized into triacylglycerols. The triacylglycerols are packaged with proteins, phospholipids, cholesterol, and other compounds into the lipoprotein complexes known as chylomicrons, which are secreted into the lymph and ultimately enter the bloodstream (see Fig. 2.2, circle 2).

II. CHANGES IN HORMONE LEVELS AFTER A MEAL

After a typical high carbohydrate meal, the pancreas is stimulated to release the hormone insulin, and release of the hormone glucagon is inhibited (see Fig. 2.2, circle 4). Endocrine hormones are released from endocrine glands, such as the pancreas, in response to a specific stimulus. They travel in the blood, carrying messages between tissues concerning the overall physiologic state of the body. At their target tissues, they adjust the rate of various metabolic pathways to meet the changing conditions. The endocrine hormone insulin, which is secreted from the pancreas in response to a high-carbohydrate meal, carries the message that dietary glucose is available and can be used and stored. The release of another hormone, glucagon, is suppressed by glucose and insulin. Glucagon carries the message that glucose must be generated from endogenous fuel stores. The subsequent changes in circulating hormone levels cause changes in the body's metabolic patterns, involving a number of different tissues and metabolic pathways.

III. FATE OF GLUCOSE AFTER A MEAL

A. Conversion to Glycogen, Triacylglycerols, and CO₂ in the Liver

Because glucose leaves the intestine via the hepatic portal vein, the liver is the first tissue it passes through. The liver extracts a portion of this glucose from the blood. Some of the glucose that enters hepatocytes (liver cells) is oxidized in adenosine triphosphate (ATP)-generating pathways to meet the immediate energy needs of these cells and the remainder is converted to glycogen and triacylglycerols or used for biosynthetic reactions. In the liver, insulin promotes the uptake of glucose by increasing its use as a fuel and its storage as glycogen and triacylglycerols (see Fig. 2.2, circles 5, 6, and 7).

As glucose is being oxidized to CO₂, it is first oxidized to pyruvate in the pathway of glycolysis. Pyruvate is then oxidized to acetyl CoA. The acetyl group enters the tricarboxylic acid (TCA) cycle, where it is completely oxidized to CO₂. Energy from the oxidative reactions is used to generate ATP.

Liver glycogen stores reach a maximum of approximately 200 to 300 g after a high-carbohydrate meal, whereas the body's fat stores are relatively limitless. As the glycogen stores begin to fill, the liver also begins converting some of the excess glucose it receives to triacylglycerols. Both the glycerol and the fatty acid moieties of the triacylglycerols can be synthesized from glucose. The fatty acids are also obtained preformed from the blood. The liver does not store triacylglycerols, however, but packages them along with proteins, phospholipids, and cholesterol into the lipoprotein complexes known as very-low-density lipoproteins (VLDL), which are secreted into the bloodstream. Some of the fatty acids from the VLDL are taken up by tissues for their immediate energy needs, but most are stored in adipose tissue as triacylglycerols.

B. Glucose Metabolism In Other Tissues

The glucose from the intestine that is not metabolized by the liver travels in the blood to peripheral tissues (most other tissues), where it can be oxidized for energy. Glucose is the one fuel that can be used by all tissues. Many tissues store small amounts of glucose as glycogen. Muscle has relatively large glycogen stores.

Insulin greatly stimulates the transport of glucose into the two tissues that have the largest mass in the body, muscle and adipose tissue. It has much smaller effects on the transport of glucose into other tissues.

1. BRAIN AND OTHER NEURAL TISSUES

The brain and other neural tissues are very dependent on glucose for their energy needs. They generally oxidize glucose via glycolysis and the TCA cycle completely to CO₂ and H₂O, generating ATP (see Fig. 2.2, circle 8)). Except under conditions of starvation, glucose is their only major fuel. Glucose is also a major precursor of neurotransmitters, the chemicals that convey electrical impulses (as ion gradients) between neurons. If our blood glucose drops much below normal levels, we become dizzy and light-headed. If blood glucose continues to drop, we become comatose and ultimately die. Under normal, nonstarving conditions, the brain and the rest of the nervous system require roughly 150 g glucose each day.

2. RED BLOOD CELLS

Glucose is the only fuel used by red blood cells, because they lack mitochondria. Fatty acid oxidation, amino acid oxidation, the TCA cycle, the electron transport chain, and oxidative phosphorylation (ATP generation that is dependent on oxygen



In the liver and most other tissues, glucose, fats, and other fuels are oxidized to the 2-carbon acetyl group

$$\begin{array}{c} \text{O} \\ || \\ (\text{CH}_3-\text{C}-) \end{array}$$
 of acetyl CoA. CoA,

which makes the acetyl group more reactive, is a cofactor (coenzyme A) derived from the vitamin pantothenate. The acetyl group of acetyl CoA is completely oxidized to CO₂ in the TCA cycle (see Fig 1.4). Adenosine triphosphate (ATP) is the final product of these oxidative pathways. It contains energy derived from the catabolic energy-producing oxidation reactions and transfers that energy to anabolic and other energy-requiring processes in the cell.



Fuel metabolism is often discussed as though the body consisted only of brain, skeletal and cardiac muscle, liver, adipose tissue, red blood cells, kidney, and intestinal epithelial cells ("the gut").

These are the dominant tissues in terms of overall fuel economy, and they are the tissues we describe most often. Of course, all tissues require fuels for energy, and many have very specific fuel requirements.

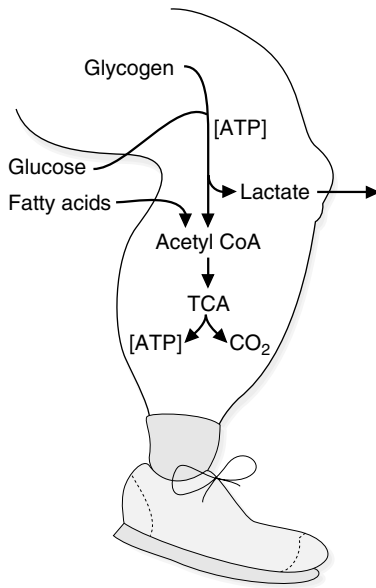


Fig. 2.3 Oxidation of fuels in exercising skeletal muscle. Exercising muscle uses more energy than resting muscle, and, therefore fuel utilization is increased to supply more ATP.



Ivan Applebod's total cholesterol level is now 315 mg/dL, slightly higher than his previous level of 296. (The currently recommended level for total serum cholesterol is 200 mg/dL or less.) His triacylglycerol level is 250 mg/dL (normal is between 60 and 160 mg/dL). These lipid levels clearly indicate that Mr. Applebod has a hyperlipidemia (high level of lipoproteins in the blood) and therefore is at risk for the future development of atherosclerosis and its consequences, such as heart attacks and strokes.

and the electron transport chain) occur principally in mitochondria. Glucose, in contrast, generates ATP from anaerobic glycolysis in the cytosol and, thus, red blood cells obtain all their energy by this process. In anaerobic glycolysis, the pyruvate formed from glucose is converted to lactate and then released into the blood (see Fig. 2.2, circle 9).

Without glucose, red blood cells could not survive. Red blood cells carry O_2 from the lungs to the tissues. Without red blood cells, most of the tissues of the body would suffer from a lack of energy because they require O_2 to completely convert their fuels to CO_2 and H_2O .

3. MUSCLE

Exercising skeletal muscles can use glucose from the blood or from their own glycogen stores, converting glucose to lactate through glycolysis or oxidizing it completely to CO_2 and H_2O . Muscle also uses other fuels from the blood, such as fatty acids (Fig. 2.3). After a meal, glucose is used by muscle to replenish the glycogen stores that were depleted during exercise. Glucose is transported into muscle cells and converted to glycogen by processes that are stimulated by insulin.

4. ADIPOSE TISSUE

Insulin stimulates the transport of glucose into adipose cells as well as into muscle cells. Adipocytes oxidize glucose for energy, and they also use glucose as the source of the glycerol moiety of the triacylglycerols they store (see Fig. 2.2, circle 10).

IV. FATE OF LIPOPROTEINS IN THE FED STATE

Two types of lipoproteins, chylomicrons and VLDL, are produced in the fed state. The major function of these lipoproteins is to provide a blood transport system for triacylglycerols, which are very insoluble in water. However, these lipoproteins also contain the lipid cholesterol, which is also somewhat insoluble in water. The triacylglycerols of chylomicrons are formed in intestinal epithelial cells from the products of digestion of dietary triacylglycerols. The triacylglycerols of VLDL are synthesized in the liver.

When these lipoproteins pass through blood vessels in adipose tissue, their triacylglycerols are degraded to fatty acids and glycerol (see Fig. 2.2, circle 12). The fatty acids enter the adipose cells and combine with a glycerol moiety that is produced from blood glucose. The resulting triacylglycerols are stored as large fat droplets in the adipose cells. The remnants of the chylomicrons are cleared from the blood by the liver. The remnants of the VLDL can be cleared by the liver, or they can form low-density lipoprotein (LDL), which is cleared by the liver or by peripheral cells.

Most of us have not even begun to reach the limits of our capacity to store triacylglycerols in adipose tissue. The ability of humans to store fat appears to be limited only by the amount of tissue we can carry without overloading the heart.

V. FATE OF AMINO ACIDS IN THE FED STATE

The amino acids derived from dietary proteins travel from the intestine to the liver in the hepatic portal vein (see Fig. 2.2, circle 3). The liver uses amino acids for the synthesis of serum proteins as well as its own proteins, and for the biosynthesis of nitrogen-containing compounds that need amino acid presursors, such as the

nonessential amino acids, heme, hormones, neurotransmitters, and purine and pyrimidine bases (e.g., adenine and cytosine in DNA). The liver also may oxidize the amino acids or convert them to glucose or ketone bodies and dispose of the nitrogen as the nontoxic compound urea.

Many of the amino acids will go into the peripheral circulation, where they can be used by other tissues for protein synthesis and various biosynthetic pathways, or oxidized for energy (see Fig. 2.2, circle 14). Proteins undergo turnover; they are constantly being synthesized and degraded. The amino acids released by protein breakdown enter the same pool of free amino acids in the blood as the amino acids from the diet. This free amino acid pool in the blood can be used by all cells to provide the right ratio of amino acids for protein synthesis or for biosynthesis of other compounds. In general, each individual biosynthetic pathway using an amino acid precursor is found in only a few tissues in the body.

VI. SUMMARY OF THE FED (ABSORPTIVE) STATE

After a meal, the fuels that we eat are oxidized to meet our immediate energy needs. Glucose is the major fuel for most tissues. Excess glucose and other fuels are stored, as glycogen mainly in muscle and liver, and as triacylglycerols in adipose tissue. Amino acids from dietary proteins are converted to body proteins or oxidized as fuels.

CLINICAL COMMENTS



Ivan Applebod. Mr. Applebod was advised that his obesity represents a risk factor for future heart attacks and strokes. He was told that his body has to maintain a larger volume of circulating blood to service his extra fat tissue. This expanded blood volume not only contributes to his elevated blood pressure (itself a risk factor for vascular disease) but also puts an increased workload on his heart. This increased load will cause his heart muscle to thicken and eventually to fail.

Mr. Applebod's increasing adipose mass has also contributed to his development of type 2 diabetes mellitus, characterized by hyperglycemia (high blood glucose levels). The mechanism behind this breakdown in his ability to maintain normal levels of blood glucose is, at least in part, a resistance by his triacylglycerol-rich adipose cells to the action of insulin.

In addition to diabetes mellitus, Mr. Applebod has a hyperlipidemia (high blood lipid level—elevated cholesterol and triacylglycerols), another risk factor for cardiovascular disease. A genetic basis for Mr. Applebod's disorder is inferred from a positive family history of hypercholesterolemia and premature coronary artery disease in a brother.

At this point, the first therapeutic steps should be nonpharmacologic. Mr. Applebod's obesity should be treated with caloric restriction and a carefully monitored program of exercise. A reduction of dietary fat and sodium would be advised in an effort to correct his hyperlipidemia and his hypertension, respectively.



Ivan Applebod's waist circumference indicates he has the android pattern of obesity (apple shape). Fat stores are distributed in the body in two different patterns, android and gynecoid. After puberty, men tend to store fat in and on their abdomens and upper body (an android pattern), whereas women tend to store fat around their breasts, hips, and thighs (a gynecoid pattern). Thus, the typical overweight male tends to have more of an apple shape than the typical overweight female, who is more pear-shaped. Abdominal fat carries a greater risk for hypertension, cardiovascular disease, hyperinsulinemia, diabetes mellitus, gallbladder disease, stroke, and cancer of the breast and endometrium. It also carries a greater risk of overall mortality. Because more men than women have the android distribution, they are more at risk for most of these conditions. But women who deposit their excess fat in a more android manner have a greater risk than women whose fat distribution is more gynecoid.

Upper body fat deposition tends to occur more by hypertrophy of the existing cells, whereas lower body fat deposition is by differentiation of new fat cells (hyperplasia). This may partly explain why many women with lower body obesity have difficulty losing weight.

BIOCHEMICAL COMMENTS



Anthropometric Measurements. Anthropometry uses measurements of body parameters to monitor normal growth and nutritional health in well-nourished individuals and to detect nutritional inadequacies or excesses. In adults, the measurements most commonly used are: height, weight,



To obtain reliable measures of skinfold thickness, procedures are carefully defined. For example, in the triceps measurement, a fold of skin in the posterior aspect of the nondominant arm midway between shoulder and elbow is grasped gently and pulled away from the underlying muscle. The skinfold thickness reading is taken at a precise time, 2 to 3 seconds after applying the caliper, because the caliper compresses the skin. Even when these procedures are performed by trained dietitians, reliable measurements are difficult to obtain.



The **waist-to-hip ratio** has been used instead of the waist circumference as a measure of abdominal obesity in an attempt to correct for differences between individuals with respect to body type or bone structure. In this measurement, the waist circumference is divided by the hip circumference (measured at the iliac crest). The average waist-to-hip ratio for men is 0.93 (with a range of 0.75–1.10), and the average for women was 0.83 (with a range of 0.70–1.00). However, the waist circumference may actually correlate better with intraabdominal fat and the associated risk factors than the waist-to-hip ratio.

triceps skinfold thickness, arm muscle circumference, and waist circumference. In infants and young children, length and head circumference are also measured.



Weight and height. Weight should be measured by using a calibrated beam or lever balance-type scale, and the patient should be in a gown or in underwear. Height for adults should be measured while the patient stands against a straight surface, without shoes, with the heels together, and with the head erect and level. The weight and height are used in calculation of the body mass index (BMI).



Skinfold thickness. Over half of the fat in the body is deposited in subcutaneous tissue under the skin, and the percentage increases with increasing weight. To provide an estimate of the amount of body fat, a standardized calipers is used to pinch a fold of the skin, usually at more than one site (e.g., the biceps, triceps, subscapular, and suprailiac areas). Obesity by this physical anthropometric technique is defined as a fatfold thickness greater than the 85th percentile for young adults; that is, 18.6 mm for males and 25.1 mm for females.



Mid-Arm Anthropometry. The arm muscle circumference (AMC), also called the mid upper arm muscle circumference (MUAMC), reflects both caloric adequacy and muscle mass and can serve as a general index of marasmic-type malnutrition. The arm circumference is measured at the midpoint of the left upper arm by a fiberglass flexible-type tape. The arm muscle circumference can be calculated from a formula that subtracts a factor related to the skinfold thickness (SFT) from the arm circumference:

$$\text{MUAMC (cm)} = \text{arm circumference (cm)} - (3.14 \times \text{SFT (mm)})/10$$

Where MUAMC is the mid upper arm muscle circumference in cm and SFT is the skinfold thickness, expressed in millimeters.

MUAMC values can be compared with reference graphs available for both sexes and all ages. Protein–calorie malnutrition and negative nitrogen balance induce muscle wasting and decrease muscle circumference.



Waist Circumference. The waist circumference is another anthropometric measurement that serves as an indicator of body composition but is used as a measure of obesity and body fat distribution (the “apple shape”), not malnutrition. It is the distance around the natural waist of a standing individual (at the umbilicus). A high-risk waistline is more than 35 inches (88 cm) for women and more than 40 inches (102 cm) for men.

Suggested References

- Garrow JS. Obesity. In: Cohen RD, Lewis B, Alberti KGMM, Denman AM, eds. The metabolic and molecular basis of acquired disease. London: Bailliere Tindall, 1990.
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REVIEW QUESTIONS—CHAPTER 2

1. During digestion of a mixed meal,
 - (A) starch and other polysaccharides are transported to the liver.
 - (B) proteins are converted to dipeptides, which enter the blood.
 - (C) dietary triacylglycerols are transported in the portal vein to the liver.
 - (D) monosaccharides are transported to adipose tissue via the lymphatic system.
 - (E) glucose levels increase in the blood.
- 2.2. After digestion of a high carbohydrate meal,
 - (A) glucagon is released from the pancreas.
 - (B) insulin stimulates the transport of glucose into the brain.
 - (C) liver and skeletal muscle use glucose as their major fuel.
 - (D) skeletal muscles convert glucose to fatty acids.
 - (E) red blood cells oxidize glucose to CO_2 .
3. Amino acids derived from digestion of dietary protein
 - (A) provide nitrogen for synthesis of nonessential amino acids in the liver.
 - (B) can be converted to glucose in most tissues.
 - (C) cannot be converted to adipose tissue fat.
 - (D) release nitrogen that is converted to urea in skeletal muscle.
 - (E) are generally converted to body proteins or excreted in the urine.
4. Elevated levels of chylomicrons were measured in the blood of a patient. What dietary therapy would be most helpful in lowering chylomicron levels?
 - (A) Decreased intake of calories
 - (B) Decreased intake of fat
 - (C) Decreased intake of cholesterol
 - (D) Decreased intake of starch
 - (E) Decreased intake of sugar
5. A male patient exhibited a BMI of 33 kg/m^2 and a waist circumference of 47 inches. What dietary therapy would you consider most helpful?
 - (A) Decreased intake of total calories, because all fuels can be converted to adipose tissue triacylglycerols
 - (B) The same amount of total calories, but substitution of carbohydrate calories for fat calories
 - (C) The same amount of total calories, but substitution of protein calories for fat calories
 - (D) A pure-fat diet, because only fatty acids synthesized by the liver can be deposited as adipose triacylglycerols
 - (E) A limited food diet, such as the ice cream and sherry diet

3 Fasting



Pathways named with the suffix “lysis” are those in which complex molecules are broken down or “lysed” into smaller units. For instance, in glycogenolysis, glycogen is lysed into glucose subunits; in glycolysis, glucose is lysed into two pyruvate molecules; in lipolysis, triacylglycerols are lysed into fatty acids and glycerol; in proteolysis, proteins are lysed into their constituent amino acids.



Gluconeogenesis means formation (genesis) of new (neo) glucose, and by definition, converts new (noncarbohydrate) precursors to glucose.



Degrees of protein–energy malnutrition (marasmus) are classified according to BMI.

Protein–energy Malnutrition	BMI (kg/cm ²)
I	17.0–18.4
II	16.0–16.9
III	<16.0

Percy Veere has grade I protein–energy malnutrition. At his height of 71 inches, his body weight would have to be above 132 lb to achieve a BMI greater than 18.5. **Ann O’Rexia** has grade III malnutrition. At 66 inches, she needs a body weight greater than 114 lb to achieve a BMI of 18.5.

The Fasting State. Fasting begins approximately 2 to 4 hours after a meal, when blood glucose levels return to basal levels, and continues until blood glucose levels begin to rise after the start of the next meal. Within about 1 hour after a meal, blood glucose levels begin to fall. Consequently, **insulin** levels decline, and **glucagon** levels rise. These changes in hormone levels trigger the **release of fuels** from the body stores. **Liver glycogen** is **degraded** by the process of **glycogenolysis**, which supplies glucose to the blood. **Adipose triacylglycerols** are **mobilized** by the process of **lipolysis**, which releases fatty acids and glycerol into the blood. Use of **fatty acids** as a fuel increases with the length of the fast; they are the **major fuel** oxidized during overnight fasting.

Fuel Oxidation. During fasting, **glucose** continues to be **oxidized** by **glucose-dependent tissues** such as the brain and red blood cells, and **fatty acids** are oxidized by tissues such as muscle and liver. Muscle and most other tissues oxidize fatty acids completely to CO_2 and H_2O . However, the **liver** partially oxidizes fatty acids to smaller molecules called **ketone bodies**, which are released into the blood. Muscle, kidney, and certain other tissues derive energy from completely oxidizing ketone bodies in the tricarboxylic acid (TCA) cycle.

Maintenance of Blood Glucose. As fasting progresses, the **liver produces glucose** not only by **glycogenolysis** (the release of glucose from glycogen), but also by a second process called **gluconeogenesis** (the synthesis of glucose from non-carbohydrate compounds). The major **sources of carbon** for gluconeogenesis are **lactate**, **glycerol**, and **amino acids**. When the carbons of the amino acids are converted to glucose by the liver, their **nitrogen** is converted to **urea**.

Starvation. When we fast for 3 or more days, we are in the starved state. **Muscle** continues to burn fatty acids but **decreases** its use of **ketone bodies**. As a result, the concentration of ketone bodies rises in the blood to a level at which the **brain** begins to **oxidize** them for energy. The brain then needs less glucose, so the liver decreases its rate of gluconeogenesis. Consequently, less **protein** in muscle and other tissues is degraded to supply amino acids for gluconeogenesis. Protein sparing preserves vital functions for as long as possible. Because of these changes in the fuel utilization patterns of various tissues, humans can survive for extended periods without ingesting food.



THE WAITING ROOM



Percy Veere had been admitted to the hospital with a diagnosis of mental depression associated with malnutrition (see Chap. 1). At the time of admission, his body weight of 125 lb gave him a body mass index (BMI) of 17.5 (healthy range, 18.5–24.9). His serum albumin was 10% below the low end of the normal range, and he exhibited signs of iron and vitamin deficiencies.

Additional tests were made to help evaluate Mr. Veere's degree of malnutrition and his progress toward recovery. His arm circumference and triceps skinfold were measured, and his mid upper arm muscle circumference was calculated (see Chap. 2, Anthropometric Measurements). His serum transferrin, as well as his serum albumin, were measured. Fasting blood glucose and serum ketone body concentration were determined on blood samples drawn the next day before breakfast. A 24-hour urine specimen was collected to determine ketone body excretion and creatinine excretion for calculation of the creatinine–height index, a measure of protein depletion from skeletal muscle.



Ann O'Rexia was receiving psychological counseling for anorexia nervosa, but with little success (see Chap. 1). She saw her gynecologist because she had not had a menstrual period for 5 months. She also complained of becoming easily fatigued. The physician recognized that Ann's body weight of 85 lb was now less than 65% of her ideal weight. (Her BMI was now 13.7.) The physician recommended immediate hospitalization. The admission diagnosis was severe malnutrition secondary to anorexia nervosa. Clinical findings included decreased body core temperature, blood pressure, and pulse (adaptive responses to malnutrition). Her physician ordered measurements of blood glucose and ketone body levels and made a spot check for ketone bodies in the urine as well as ordering tests to assess the functioning of her heart and kidneys.

I. THE FASTING STATE

Blood glucose levels peak approximately 1 hour after eating and then decrease as tissues oxidize glucose or convert it to storage forms of fuel. By 2 hours after a meal, the level returns to the fasting range (between 80 and 100 mg/dL). This decrease in blood glucose causes the pancreas to decrease its secretion of insulin, and the serum insulin level decreases. The liver responds to this hormonal signal by starting to degrade its glycogen stores and release glucose into the blood.

If we eat another meal within a few hours, we return to the fed state. However, if we continue to fast for a 12-hour period, we enter the basal state (also known as the postabsorptive state). A person is generally considered to be in the basal state after an overnight fast, when no food has been eaten since dinner the previous evening. By this time, the serum insulin level is low and glucagon is rising. Figure 3.1 illustrates the main features of the basal state.

A. Blood Glucose and the Role of the Liver during Fasting

The liver maintains blood glucose levels during fasting, and its role is thus critical. Glucose is the major fuel for tissues such as the brain and neural tissue, and the sole fuel for red blood cells. Most neurons lack enzymes required for oxidation of fatty acids, but can use ketone bodies to a limited extent. Red blood cells lack mitochondria, which contain the enzymes of fatty acid and ketone body oxidation, and can use only glucose as a fuel. Therefore, it is imperative that blood glucose not decrease too rapidly nor fall too low.

Initially, liver glycogen stores are degraded to supply glucose to the blood, but these stores are limited. Although liver glycogen levels may increase to 200 to 300 g after a meal, only approximately 80 g remain after an overnight fast. Fortunately, the liver has another mechanism for producing blood glucose, known as gluconeogenesis. In gluconeogenesis, lactate, glycerol, and amino acids are used as carbon sources to synthesize glucose. As fasting continues, gluconeogenesis progressively adds to the glucose produced by glycogenolysis in the liver.



Percy Veere had not eaten much on his first day of hospitalization. His fasting blood glucose determined on the morning of his second day of hospitalization was 72 mg/dL (normal, overnight fasting = 80–100 mg/dL). Thus, in spite of his malnutrition and his overnight fast, his blood glucose was being maintained at nearly normal levels through gluconeogenesis using amino acid precursors. If his blood glucose had decreased below 50 to 60 mg/dL during fasting, his brain would have been unable to absorb glucose fast enough to obtain the glucose needed for energy and neurotransmitter synthesis, resulting in coma and eventual death. Although many other tissues, such as the red blood cell, are also totally or partially dependent on glucose for energy, they are able to function at lower concentrations of blood glucose than the brain.

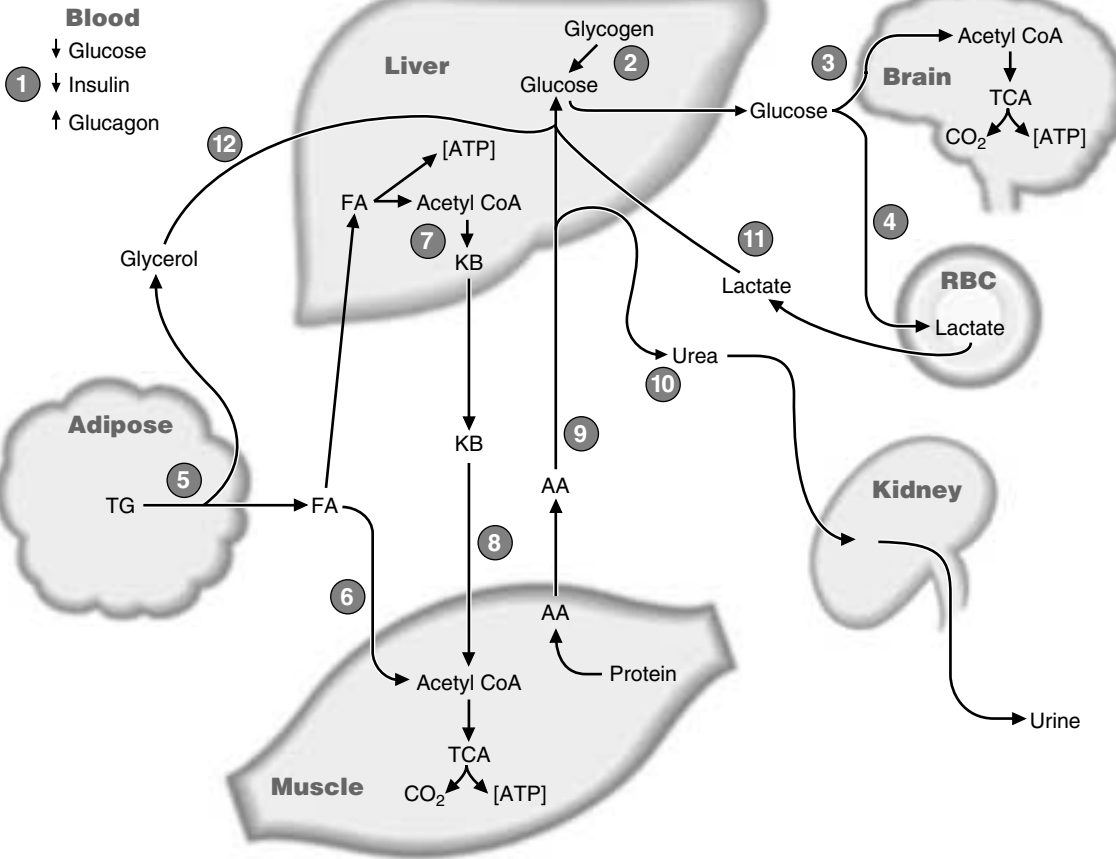


Fig. 3.1. Basal state. This state occurs after an overnight (12-hour) fast. The circled numbers serve as a guide indicating the approximate order in which the processes begin to occur. KB = ketone bodies. Other abbreviations are defined in Figure 2.1.

Lactate is a product of glycolysis in red blood cells and exercising muscle, glycerol is obtained from lipolysis of adipose triacylglycerols, and amino acids are generated by the breakdown of protein. Because our muscle mass is so large, most of the amino acid is supplied from degradation of muscle protein. These compounds travel in the blood to the liver, where they are converted to glucose by gluconeogenesis. Because the nitrogen of the amino acids can form ammonia, which is toxic to the body, the liver converts this nitrogen to urea. Urea has two amino groups for just one carbon ($\text{NH}_2\text{-CO-NH}_2$). It is a very soluble, nontoxic compound that can be readily excreted by the kidneys and thus is an efficient means for disposing of excess ammonia.

As fasting progresses, gluconeogenesis becomes increasingly more important as a source of blood glucose. After a day or so of fasting, liver glycogen stores are depleted and gluconeogenesis is the only source of blood glucose.

B. Role of Adipose Tissue During Fasting

Adipose triacylglycerols are the major source of energy during fasting. They supply fatty acids, which are quantitatively the major fuel for the human body. Fatty acids are not only oxidized directly by various tissues of the body; they are also partially oxidized in the liver to 4-carbon products called ketone bodies. Ketone bodies are subsequently oxidized as a fuel by other tissues.

As blood insulin levels decrease and blood glucagon levels rise, adipose triacylglycerols are mobilized by a process known as lipolysis. They are converted to fatty acids and glycerol, which enter the blood.



On the second day of hospitalization, **Percy Veere's** serum ketone body level was 110 μM . (Normal valve after a 12-hour fast, is approximately 70 μM .) No ketone bodies were detectable in his urine. At this stage of protein-energy malnutrition, Mr. Veere still has remaining fat stores. After 12 hours of fasting, most of his tissues are using fatty acids as a major fuel, and the liver is beginning to produce ketone bodies from fatty acids. As these ketone bodies increase in the blood, their use as a fuel will increase.

It is important to realize that most fatty acids cannot provide carbon for gluconeogenesis. Thus, of the vast store of food energy in adipose tissue triacylglycerols, only the small glycerol portion travels to the liver to enter the gluconeogenic pathway.

Fatty acids serve as a fuel for muscle, kidney, and most other tissues. They are oxidized to acetyl CoA, and subsequently to CO_2 and H_2O in the TCA cycle, producing energy in the form of adenosine triphosphate (ATP). In addition to the ATP required to maintain cellular integrity, muscle uses ATP for contraction, and the kidney uses it for urinary transport processes.

Most of the fatty acids that enter the liver are converted to ketone bodies rather than being completely oxidized to CO_2 . The process of conversion of fatty acids to acetyl CoA produces a considerable amount of energy (ATP), which drives the reactions of the liver under these conditions. The acetyl CoA is converted to the ketone bodies acetoacetate and β -hydroxybutyrate, which are released into the blood (Fig. 3.2).

The liver lacks an enzyme required for ketone body oxidation. However, ketone bodies can be further oxidized by most other cells with mitochondria, such as muscle and kidney. In these tissues, acetoacetate and β -hydroxybutyrate are converted to acetyl CoA and then oxidized in the TCA cycle, with subsequent generation of ATP.

C. Summary of the Metabolic Changes during a Brief Fast

In the initial stages of fasting, stored fuels are used for energy (see Fig. 3.1). The liver plays a key role by maintaining blood glucose levels in the range of 80 to 100 mg/dL, first by glycogenolysis and subsequently by gluconeogenesis. Lactate, glycerol, and amino acids serve as carbon sources for gluconeogenesis. Amino acids are supplied by muscle. Their nitrogen is converted in the liver to urea, which is excreted by the kidneys.

Fatty acids, which are released from adipose tissue by the process of lipolysis, serve as the body's major fuel during fasting. The liver oxidizes most of its fatty acids only partially, converting them to ketone bodies, which are released into the blood. Thus, during the initial stages of fasting, blood levels of fatty acids and ketone bodies begin to increase. Muscle uses fatty acids, ketone bodies, and (when exercising and while supplies last) glucose from muscle glycogen. Many other tissues use either fatty acids or ketone bodies. However, red blood cells, the brain, and other neural tissues use mainly glucose. The metabolic capacities of different tissues with respect to pathways of fuel metabolism are summarized in Table 3.1.

II. METABOLIC CHANGES DURING PROLONGED FASTING

If the pattern of fuel utilization that occurs during a brief fast were to persist for an extended period, the body's protein would be quite rapidly consumed to the point at which critical functions would be compromised. Fortunately, metabolic changes occur during prolonged fasting that conserve (spare) muscle protein by causing muscle protein turnover to decrease. Figure 3.3 shows the main features of metabolism during prolonged fasting (starvation).

B. Role of Liver During Prolonged Fasting

After 3 to 5 days of fasting, when the body enters the starved state, muscle decreases its use of ketone bodies and depends mainly on fatty acids for its fuel. The

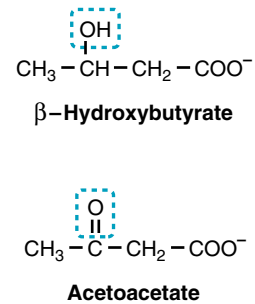


Fig. 3.2. β -Hydroxybutyrate and acetoacetate. These compounds are ketone bodies formed in the liver. A third ketone body, acetone ($\text{CH}_3\text{-CO-CH}_3$), is produced by nonenzymatic decarboxylation of acetoacetate. However, acetone is expired in the breath and not metabolized to a significant extent in the body.



The liver synthesizes a number of serum proteins and releases them into the blood. These proteins decrease in the blood during protein malnutrition. Two of these serum proteins, albumin and transferrin (an iron-binding transport protein), are often measured to assess the state of protein malnutrition. Serum albumin is the traditional standard of protein malnutrition. Neither measurement is specific for protein malnutrition. Serum albumin and transferrin levels decrease with hepatic disease, certain renal diseases, surgery, and a number of other conditions, in addition to protein malnutrition. Serum transferrin levels also decrease in iron deficiency. **Percy Veere's** values were below the normal range for both of these proteins, indicating that his muscle mass is unable to supply sufficient amino acids to sustain both synthesis of serum proteins by the liver and gluconeogenesis.

Process	Liver	Adipose Tissue	Kidney Cortex	Muscle	Brain	RBC
TCA cycle (acetyl CoA $\rightarrow$ CO ₂ + H ₂ O)	+++	++	+++	+++	+++	--
β -Oxidation of fatty acids	+++	--	++	+++	--	--
Ketone body formation	+++	--	+	--	--	--
Ketone body utilization	--	+	+	+++	+++ (prolonged starvation)	--
Glycolysis (glucose $\rightarrow$ CO ₂ + H ₂ O)	+++	++	++	+++	+++	--
Lactate production (glucose $\rightarrow$ lactate)	+	+	---	+++ exercise	+	+++
Glycogen metabolism (synthesis and degradation)	+++	+	+	+++	+	--
Gluconeogenesis (lactate, amino acids, glycerol $\rightarrow$ glucose)	+++	--	+	--	--	--
Urea cycle (ammonia $\rightarrow$ urea)	+++	--	--	--	--	--
Lipogenesis (glucose $\rightarrow$ fatty acids)	+++	+	--	--	--	--

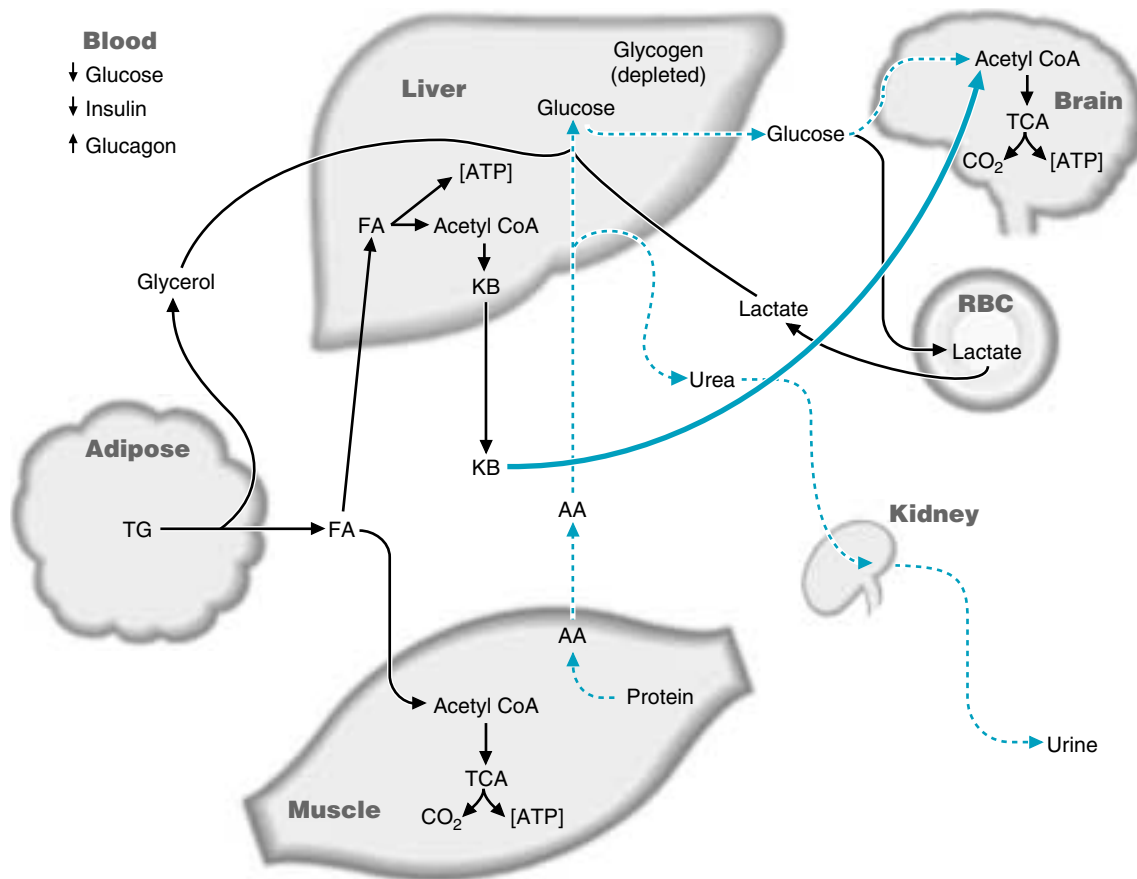


Fig. 3.3. Starved state. Abbreviations are defined in Figures 2.1 and 3.1. Dashed lines indicate processes that have decreased, and the heavy solid line indicates a process that has increased relative to the fasting state.

liver, however, continues to convert fatty acids to ketone bodies. The result is that the concentration of ketone bodies rises in the blood (Fig. 3.4). The brain begins to take up these ketone bodies from the blood and to oxidize them for energy. Therefore, the brain needs less glucose than it did after an overnight fast (Table 3.2).

Glucose is still required, however, as an energy source for red blood cells, and the brain continues to use a limited amount of glucose, which it oxidizes for energy and uses as a source of carbon for the synthesis of neurotransmitters. Overall, however, glucose is “spared” (conserved). Less glucose is used by the body, and, therefore, the liver needs to produce less glucose per hour during prolonged fasting than during shorter periods of fasting.

Because the stores of glycogen in the liver are depleted by approximately 30 hours of fasting, gluconeogenesis is the only process by which the liver can supply glucose to the blood if fasting continues. The amino acid pool, produced by the breakdown of protein, continues to serve as a major source of carbon for gluconeogenesis. A fraction of this amino acid pool is also being used for biosynthetic functions (e.g., synthesis of heme and neurotransmitters) and new protein synthesis, processes that must continue during fasting. However, as a result of the decreased rate of gluconeogenesis during prolonged fasting, protein is “spared”; less protein is degraded to supply amino acids for gluconeogenesis.

While converting amino acid carbon to glucose in gluconeogenesis, the liver also converts the nitrogen of these amino acids to urea. Consequently, because glucose production decreases during prolonged fasting compared with early fasting, urea production also decreases (Fig. 3.5).

B. Role of Adipose Tissue During Prolonged Fasting

During prolonged fasting, adipose tissue continues to break down its triacylglycerol stores, providing fatty acids and glycerol to the blood. These fatty acids serve as the major source of fuel for the body. The glycerol is converted to glucose, whereas the fatty acids are oxidized to CO_2 and H_2O by tissues such as muscle. In the liver, fatty acids are converted to ketone bodies that are oxidized by many tissues, including the brain.

A number of factors determine how long we can fast and still survive. The amount of adipose tissue is one factor, because adipose tissue supplies the body with its major source of fuel. However, body protein levels can also determine the length of time we can fast. Glucose is still used during prolonged fasting (starvation), but in greatly reduced amounts. Although we degrade protein to supply amino acids for gluconeogenesis at a slower rate during starvation than during the first days of a fast, we are still losing protein that serves vital functions for our tissues. Protein can become so depleted that the heart, kidney, and other vital tissues stop functioning, or we can develop an infection and not have adequate reserves to mount an immune response. In addition to fuel problems, we are also deprived of the vitamin and mineral precursors of coenzymes and other compounds necessary for tissue function. Because of either a lack of ATP or a decreased intake of electrolytes, the electrolyte composition of the blood or cells could become incompatible with life. Ultimately, we die of starvation.

Table 3.2. Metabolic Changes during Prolonged Fasting Compared with Fasting 24 Hours

Muscle	↓	Use of ketone bodies
Brain	↑	Use of ketone bodies
Liver	↓	Gluconeogenesis
Muscle	↓	Protein degradation
Liver	↓	Production of urea



Ann O'Rexia's admission laboratory studies showed a blood glucose level of 65 mg/dL (normal fasting blood glucose = 80 – 100 mg/dL).

Her serum ketone body concentration was 4,200 μM (normal = $\sim 70 \mu\text{M}$). The Ketostix (Bayer Diagnostics, Mishawaka, IN) urine test was moderately positive, indicating that ketone bodies were present in the urine. In her starved state, ketone body use by her brain is helping to conserve protein in her muscles and vital organs.

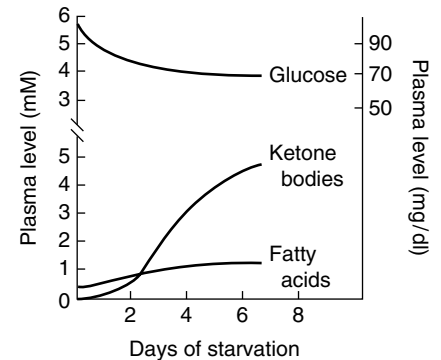


Fig. 3.4. Changes in the concentration of fuels in the blood during prolonged fasting.



Death by starvation occurs with loss of roughly 40% of body weight, when approximately 30 to 50% of body protein has been lost, or 70 to 95% of body fat stores. Generally, this occurs at BMIs of approximately 13 for men, and 11 for women.

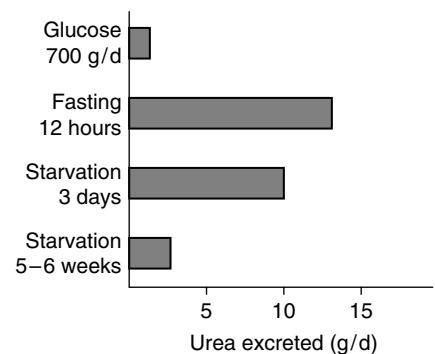


Fig. 3.5. Changes in urea excretion during fasting. Urea production is very low in a person consuming only glucose. It increases during fasting as muscle protein is broken down to supply amino acids for gluconeogenesis. However, as fasting progresses, urea synthesis decreases. Because the brain meets some of its energy needs by oxidizing ketone bodies after 3 to 5 days of fasting, gluconeogenesis decreases, sparing protein in muscle and other tissues.



Creatinine–Height Index. The most widely used biochemical marker for estimating body muscle mass is the 24-hour urinary creatinine excretion. Creatinine is a degradation product formed in active muscle at a constant rate, in proportion to the amount of muscle tissue present in a patient. In a protein-malnourished individual, urinary creatinine will decrease in proportion to the decrease in muscle mass. To assess depletion of muscle mass, creatinine excreted is expressed relative to the height, the creatinine–height index (CHI). The amount of creatinine (in milligrams) excreted by the subject in 24 hours is divided by the amount of creatinine excreted by a normal, healthy subject of the same height and sex. The resulting ratio is multiplied by 100 to express it as a percentage. **Percy Veere's** CHI was 85% (80–90% of normal indicates a mild deficit; 60–80% indicates a moderate deficit; less than 60% of normal indicates a severe deficit of muscle mass).



Fig. 3.6. Photograph of a patient with anorexia nervosa. From a MedCom slide, 1970.

CLINICAL COMMENTS



Percy Veere. As a result of his severely suppressed appetite for food, Percy Veere has developed a mild degree of protein–calorie malnutrition. When prolonged, this type of protein malnutrition can cause changes in the villi of the small intestine that reduce its absorptive capacity for what little food is ingested.

Despite his insufficient intake of dietary carbohydrates, Mr. Veere's blood glucose level is 72 mg/dL, close to the lower limit (80 mg/dL) of the normal range for a well-nourished, healthy person after a 12-hour fast. This is the finding you would expect; it reflects the liver's capacity to maintain adequate levels of blood glucose by means of gluconeogenesis, even during prolonged and moderately severe caloric restriction. Amino acids from degradation of protein, principally in skeletal muscle, supply most of the precursors for gluconeogenesis.

Percy Veere has several indicators of his protein malnutrition: his serum albumin and transferrin levels are below normal, his mid-upper-arm muscle circumference (MUAMC) is at the 12th percentile, and his creatinine–height index (CHI) was at 85%. The low levels of serum proteins reflect a low dietary protein intake, and possibly diminished capacity to absorb dietary amino acids. Consequently, amino acids are being mobilized from degradation of protein in muscle and other tissues to supply precursors for new protein synthesis as well as gluconeogenesis. The result is a loss of muscle mass, indicated by the MUAMC and the CHI, and decreased levels of serum proteins.

Fatty acids mobilized from adipose tissue are the major source of energy for most tissues. Because he is eating, and not in total starvation, his ketone bodies were only moderately elevated in the blood (110 μ M vs. normal of 70 μ M) and did not appear in the urine.

After several psychological counseling sessions, and the promise of an extended visit from his grandchild, Mr. Veere resumed his normal eating pattern.



Ann O'Rexia. Ann O'Rexia has anorexia nervosa, a chronic disabling disease in which poorly understood psychological and biologic factors lead to disturbances in the patient's body image. These patients typically pursue thinness in spite of the presence of severe emaciation and a "skeletal appearance" (Fig. 3.6). They generally have an intense fear of being overweight and deny the seriousness of their low body weight.

Amenorrhea (lack of menses) usually develops during anorexia nervosa and other conditions when a woman's body fat content falls to approximately 22% of her total body weight. The immediate cause of amenorrhea is a reduced production of the gonadotropic protein hormones (luteinizing hormone and follicle-stimulating hormone) by the anterior pituitary; the connection between this hormonal change and body fat content is not yet understood.

Ms. O'Rexia is suffering from the consequences of prolonged and severe protein and caloric restriction. Fatty acids, released from adipose tissue by lipolysis, are being converted to ketone bodies in the liver, and the level of ketone bodies in the blood is extremely elevated (4,200 μ M vs. normal of 70 μ M). The fact that her kidneys are excreting ketone bodies is reflected in the moderately positive urine test for ketone bodies noted on admission.

Although Ms. O'Rexia's blood glucose is below the normal fasting range (65 mg/dL vs. normal of 80 mg/dL), she is experiencing only a moderate degree of hypoglycemia (low blood glucose) despite her severe, near starvation diet. Her blood glucose level reflects the ability of the brain to use ketone bodies as a fuel when they are elevated in the blood, thereby decreasing the amount of glucose that must be synthesized from amino acids provided by protein degradation.

Ms. O'Rexia's BMI showed that she was close to death through starvation. She was therefore hospitalized and placed on enteral nutrition (nutrients provided through tube feeding). The general therapeutic plan, outlined in Chapter 1, of nutritional restitution and identification and treatment of those emotional factors leading to the patient's anorectic behavior was continued. She was coaxed into eating small amounts of food while hospitalized.

BIOCHEMICAL COMMENTS



Clinical Use of Metabolite Measurements in Blood and Urine.

When a patient develops a metabolic problem, it is difficult to examine cells to determine the cause. To obtain tissue for metabolic studies, biopsies must be performed. These procedures can be difficult, dangerous, or even impossible, depending on the tissue. Cost is an additional problem. However, both blood and urine can be obtained readily from patients, and measurements of substances in the blood and urine can help in diagnosing a patient's problem. Concentrations of substances that are higher or lower than normal indicate which tissues are malfunctioning. For example, if blood urea nitrogen (BUN) levels are low, a problem centered in the liver might be suspected because urea is produced in the liver. Conversely, high blood levels of urea suggest that the kidney is not excreting this compound normally. Decreased urinary and blood levels of creatinine indicate diminished production of creatinine by skeletal muscle. However, high blood creatinine levels could indicate an inability of the kidney to excrete creatinine, resulting from renal disease. If high levels of ketone bodies are found in the blood or urine, the patient's metabolic pattern is that of the starved state. If the high levels of ketone bodies are coupled with elevated levels of blood glucose, the problem is most likely a deficiency of insulin; that is, the patient probably has type 1, formerly called insulin-dependent, diabetes mellitus. Without insulin, fuels are mobilized from tissues rather than being stored.

These relatively easy and inexpensive tests on blood and urine can be used to determine which tissues need to be studied more extensively to diagnose and treat the patient's problem. A solid understanding of fuel metabolism helps in the interpretation of these simple tests.

Suggested References

Owen OE, Tappy L, Mozzoli MA, Smalley KJ. Acute starvation. In: Cohen RD, Lewis B, Alberti KGMM, Denman AM (eds). *The Metabolic and Molecular Basis of Acquired Disease*. London: Bailliere Tindall, 1990.

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REVIEW QUESTIONS—CHAPTER 3

You will need some information from Chapters 1 and 2, as well as Chapter 3, to answer these questions.

1. By 24 hours after a meal,
 - (A) gluconeogenesis in the liver is the major source of blood glucose.
 - (B) muscle glycogenolysis provides glucose to the blood.
 - (C) muscles convert amino acids to blood glucose.
 - (D) fatty acids released from adipose tissue provide carbon for synthesis of glucose.
 - (E) ketone bodies provide carbon for gluconeogenesis.

2. The liver is the only tissue that
 - (A) contains significant glycogen stores.
 - (B) oxidizes fatty acids during overnight fasting.
 - (C) oxidizes ketone bodies during overnight fasting.
 - (D) converts ammonia to urea.
 - (E) converts glucose to lactate.
3. In a well-nourished individual, as the length of fasting increases from overnight to 1 week,
 - (A) blood glucose levels decrease by approximately 50%.
 - (B) red blood cells switch to using ketone bodies.
 - (C) muscles decrease their use of ketone bodies, which increase in the blood.
 - (D) the brain begins to use fatty acids as a major fuel.
 - (E) adipose tissue triacylglycerols are nearly depleted.
4. A hospitalized patient had low levels of serum albumin and high levels of blood ammonia. His CHI was 98%. His BMI was 20.5. Blood urea nitrogen was not elevated, consistent with normal kidney function. The diagnosis most consistent with these findings is
 - (A) loss of hepatic function (e.g., alcohol-induced cirrhosis).
 - (B) anorexia nervosa.
 - (C) kwashiorkor (protein malnutrition).
 - (D) marasmus (protein–energy malnutrition).
 - (E) decreased absorption of amino acids by intestinal epithelial cells (e.g., celiac disease).
5. Otto Shape, an overweight medical student (see Chapter 1), discovered that he could not exercise enough during his summer clerkship rotations to lose 2 to 3 lb per week. He decided to lose weight by eating only 300 kcal/day of a dietary supplement that provided half the calories as carbohydrate and half as protein. In addition, he consumed a multivitamin supplement. During the first 3 days on this diet,
 - (A) his protein intake met the RDA for protein.
 - (B) his carbohydrate intake met the fuel needs of his brain.
 - (C) both his adipose mass and his muscle mass decreased.
 - (D) he remained in nitrogen balance.
 - (E) he developed severe hypoglycemia.

Chemical and Biological Foundations of Biochemistry

The discipline of biochemistry developed as chemists began to study the molecules of cells, tissues, and body fluids and physicians began to look for the molecular basis of various diseases. Today, the practice of medicine depends on understanding the roles and interactions of the enormous number of different chemicals enabling our bodies to function. The task is less overwhelming if one knows the properties, nomenclature and function of classes of compounds, such as carbohydrates and enzymes. The intent of this section is to review some of this information in a context relevant to medicine. Students enter medical school with different scientific backgrounds and some of the information in this section will, therefore, be familiar to many students.

We begin by discussing the relationship of metabolic acids and buffers to blood pH in Chapter 4. Chapter 5 focuses on the nomenclature, structure, and some of the properties of the major classes of compounds found in the human body. The structure of a molecule determines its function and its fate, and the common name of a compound can often tell you something about its structure.

Proteins are linear chains of amino acids that fold into complex 3-dimensional structures. They function in the maintenance of cellular and tissue structure and the transport and movement of molecules. Some proteins are enzymes, which are catalysts that enormously increase the rate of chemical reactions in the body. Chapters 6 and 7 describe the amino acids and their interactions within proteins that provide proteins with a flexible and functional 3-dimensional structure. Chapters 8 and 9 describe the properties, functions, and regulation of enzymes.

Our proteins and other compounds function within a specialized environment defined by their location in cells or body fluids. Their ability to function is partially dependent on membranes that restrict the free movement of molecules. Chapter 10 includes a brief review of the components of cells, their organization into subcellular organelles, and the manner in which various types of molecules move into cells and between compartments within a cell.

In a complex organism like the human, different cell types carry out different functions. This specialization of function requires cells to communicate with each other. One of the ways they communicate is through secretion of chemical messengers that carry a signal to another cell. In Chapter 11, we consider some of the principles of cell signaling and describe some of the chemical messenger systems.

Both in this book and in medical practice, you will need to interconvert different units used for the weight and size of compounds and for their concentration in blood and other fluids. Table 1 provides definitions of some of the units used for these interconversions.



The nomenclature used to describe patients may include the name of a class of compounds. For example a patient with diabetes mellitus who has hyperglycemia has hyper (high) concentrations of carbohydrates (glyc) in her blood (emia).



From a biochemist's point of view, most metabolic diseases are caused by enzymes and other proteins that malfunction and the pharmacological drugs used to treat these diseases correct that malfunction. For example, individuals with atherosclerosis, who have elevated blood cholesterol levels, are treated with a drug that inhibits an enzyme in the pathway for cholesterol synthesis. Even a bacterial infection can be considered a disease of protein function, if one considers the bacterial toxins that are proteins, the enzymes in our cells affected by these toxins, and the proteins involved in the immune response when we try to destroy these bacteria.



Q: Di Abietes had an elevated blood glucose level of 684 mg/dL. What is the molar concentration of glucose in Di's blood? (Hint: the molecular weight of glucose ($C_6H_{12}O_6$) is 180 grams per mole.)



mg/dL is the common way clinicians in the United States express blood glucose concentration. A concentration of 684 mg/dL is 684 mg per 100 mL of blood, or 6,480 mg per one liter (L), or 6.48 g/L. If 6.84 g/L is divided by 180 grams per mole, one obtains a value of 0.038 mol/L, which is 0.038 M, or 38 mM.

Table 1 Common Units Expressed in Equivalent Ways

1 M	1 mol/L	molecular weight in g/L
1 mM	1 millimol/L	10^{-3} mol/L
1 mM	1 micromol/L	10^{-6} mol/L
1 μ M	1 nanomol/L	10^{-9} mol/L
1 mL	1 milliliter	10^{-3} L
1 mg%	1 mg/100 mL	10^{-3} g/100 mL
1 mg/dL	1 mg/100 mL	10^{-3} g/100 mL
1 mEq/L	1 milliequivalent/L	mM X valence of ion
1 kg	1000 g	2.2 lbs (pounds)
1 cm	10^{-2} m	0.394 inches

4 Water, Acids, Bases, and Buffers

Approximately 60% of our body is water. It acts as a solvent for the substances we need, such as K^+ , glucose, adenosine triphosphate (ATP), and proteins. It is important for the transport of molecules and heat. Many of the compounds produced in the body and dissolved in water contain chemical groups that act as acids or bases, releasing or accepting hydrogen ions. The hydrogen ion content and the amount of body water are controlled to maintain a constant environment for the cells called homeostasis (same state) (Fig. 4.1). Significant deviations from a constant environment, such as acidosis or dehydration, may be life-threatening. This chapter describes the role of water in the body and the buffer systems used by the body to protect itself from acids and bases produced from metabolism.

Water. Water is distributed between intracellular and extracellular compartments, the latter comprising interstitial fluids, blood, and lymph. Because water is a **dipolar molecule** with an uneven distribution of electrons between the hydrogen and oxygen atoms, it forms hydrogen bonds with other polar molecules and acts as a **solvent**.

The pH of water. Water dissociates to a slight extent to form hydrogen (H^+) and hydroxyl (OH^-) ions. The concentration of hydrogen ions determines the acidity of the solution, which is expressed in terms of **pH**. The pH of a solution is the negative log of its hydrogen ion concentration.

Acids and Bases. An **acid** is a substance that can **release hydrogen ions** (protons), and a **base** is a substance that can **accept hydrogen ions**. When dissolved in water, almost all the molecules of a **strong acid** dissociate and release their hydrogen ions, but only a small percentage of the total molecules of a weak acid dissociate. A **weak acid** has a characteristic **dissociation constant, K_a** . The relationship between the pH of a solution, the K_a of an acid, and the extent of its dissociation are given by the **Henderson-Hasselbalch equation**.

Buffers. A buffer is a mixture of an **undissociated acid** and its **conjugate base** (the form of the acid having lost its proton). It causes a solution to resist changes in pH when either H^+ or OH^- is added. A buffer has its greatest buffering capacity in the pH range near its **pK_a** (the negative log of its K_a). Two factors determine the effectiveness of a buffer, its pK_a relative to the pH of the solution and its concentration.

Metabolic Acids and Bases. Normal metabolism generates CO_2 , **metabolic acids** (e.g., **lactic acid** and **ketone bodies**) and **inorganic acids** (e.g., **sulfuric acid**). The major source of acid is CO_2 , which reacts with water to produce **carbonic acid**. To maintain the pH of body fluids in a range compatible with life, the body has **buffers** such as **bicarbonate**, **phosphate**, and **hemoglobin** (see Fig 4.1). Ultimately, **respiratory mechanisms** remove carbonic acid through the **expiration of CO_2** , and the **kidneys excrete acid** as **ammonium ion** (NH_4^+) and other ions.

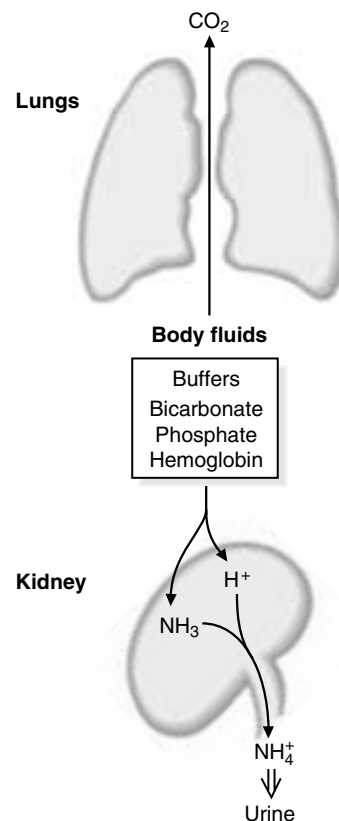


Fig. 4.1. Maintenance of body pH. The body produces approximately 13 to 22 moles of acid per day from normal metabolism. The body protects itself against this acidity by buffers that maintain a neutral pH and by the expiration of CO_2 through the lungs and the excretion of NH_4^+ and other ions through the kidney.



Di Abietes has a ketoacidosis. When the amount of insulin she injects is inadequate, she remains in a condition similar to a fasting state even though she ingests food (see Chapters 2 and 3). Her liver continues to metabolize fatty acids to the ketone bodies acetoacetic acid and β -hydroxybutyric acid. These compounds are weak acids that dissociate to produce anions (acetoacetate and β -hydroxybutyrate, respectively) and hydrogen ions, thereby lowering her blood and cellular pH below the normal range.



Percy Veere is a 59-year-old school teacher who persevered through a period of malnutrition associated with mental depression precipitated by the death of his wife (see Chapters 1 and 3). Since his recovery, he has been looking forward to an extended visit from his grandson.



THE WAITING ROOM



Dianne (Di) Abietes is a 26-year-old woman who was diagnosed with type 1 diabetes mellitus at the age of 12 years. She has an absolute insulin deficiency resulting from autoimmune destruction of the β -cells of her pancreas. As a result, she depends on daily injections of insulin to prevent severe elevations of glucose and ketone bodies in her blood.

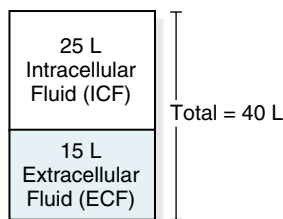
When **Di Abietes** could not be aroused from an afternoon nap, her roommate called an ambulance, and Di was brought to the emergency room of the hospital in a coma. Her roommate reported that Di had been feeling nauseated and drowsy and had been vomiting for 24 hours. Di is clinically dehydrated, and her blood pressure is low. Her respirations are deep and rapid, and her pulse rate is rapid. Her breath has the “fruity” odor of acetone.

Blood samples are drawn for measurement of her arterial blood pH, arterial partial pressure of carbon dioxide (PaCO_2), serum glucose, and serum bicarbonate (HCO_3^-). In addition, serum and urine are tested for the presence of ketone bodies, and Di is treated with intravenous normal saline and insulin. The laboratory reports that her blood pH is 7.08 (reference range = 7.36–7.44) and that ketone bodies are present in both blood and urine. Her blood glucose level is 648 mg/dL (reference range = 80–110 after an overnight fast, and no higher than 200 in a casual glucose sample taken without regard to the time of a last meal).



Dennis “the Menace” Veere, age 3 years, was brought to the emergency department by his grandfather, **Percy Veere**. While Dennis was visiting his grandfather, he climbed up on a chair and took a half-full 500-tablet bottle of 325-mg aspirin (acetylsalicylic acid) tablets from the kitchen counter. Mr. Veere discovered Dennis with a mouthful of aspirin, which he removed, but he could not tell how many tablets Dennis had already swallowed. When they arrived at the emergency room, the child appeared bright and alert, but Mr. Veere was hyperventilating.

A. Total body water



B. Extracellular fluid



Fig. 4.2. Fluid compartments in the body based on an average 70-kg man.

I. WATER

Water is the solvent of life. It bathes our cells, dissolves and transports compounds in the blood, provides a medium for movement of molecules into and throughout cellular compartments, separates charged molecules, dissipates heat, and participates in chemical reactions. Most compounds in the body, including proteins, must interact with an aqueous medium function. In spite of the variation in the amount of water we ingest each day and produce from metabolism, our body maintains a nearly constant amount of water that is approximately 60% of our body weight (Fig. 4.2).

A. Fluid Compartments in the Body

Total body water is roughly 50 to 60% of body weight in adults and 75% of body weight in children. Because fat has relatively little water associated with it, obese people tend to have a lower percentage of body water than thin people, women tend to have a lower percentage than men, and older people have a lower percentage than younger people.

Approximately 40% of the total body water is intracellular and 60% extracellular. The extracellular water includes the fluid in plasma (blood after the cells have been removed) and interstitial water (the fluid in the tissue spaces, lying between

cells). Transcellular water is a small, specialized portion of extracellular water that includes gastrointestinal secretions, urine, sweat, and fluid that has leaked through capillary walls because of such processes as increased hydrostatic pressure or inflammation.

B. Hydrogen Bonds in Water

The dipolar nature of the water (H_2O) molecule allows it to form hydrogen bonds, a property that is responsible for the role of water as a solvent. In H_2O , the oxygen atom has two unshared electrons that form an electron dense cloud around it. This cloud lies above and below the plane formed by the water molecule (Fig. 4.3). In the covalent bond formed between the hydrogen and oxygen atoms, the shared electrons are attracted toward the oxygen atom, thus giving the oxygen atom a partial negative charge and the hydrogen atom a partial positive charge. As a result, the oxygen side of the molecule is much more electronegative than the hydrogen side, and the molecule is dipolar.

Both the hydrogen and oxygen atoms of the water molecule form hydrogen bonds and participate in hydration shells. A hydrogen bond is a weak noncovalent interaction between the hydrogen of one molecule and the more electronegative atom of an acceptor molecule. The oxygen of water can form hydrogen bonds with two other water molecules, so that each water molecule is hydrogen-bonded to approximately four close neighboring water molecules in a fluid three-dimensional lattice (see Fig. 4.3).

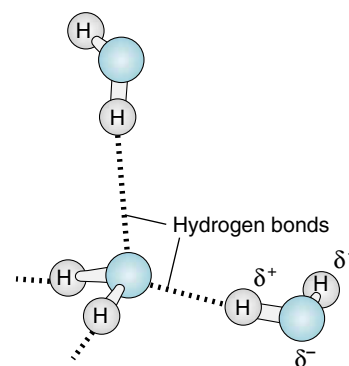


Fig. 4.3. Hydrogen bonds between water molecules. The oxygen atoms are shown in blue.

1. WATER AS A SOLVENT

Polar organic molecules and inorganic salts can readily dissolve in water because water also forms hydrogen bonds and electrostatic interactions with these molecules. Organic molecules containing a high proportion of electronegative atoms (generally oxygen or nitrogen) are soluble in water because these atoms participate in hydrogen bonding with water molecules (Fig. 4.4). Chloride (Cl^-), bicarbonate (HCO_3^-), and other anions are surrounded by a hydration shell of water molecules arranged with their hydrogen atoms closest to the anion. In a similar fashion, the oxygen atom of water molecules interacts with inorganic cations such as Na^+ and K^+ to surround them with a hydration shell.

Although hydrogen bonds are strong enough to dissolve polar molecules in water and to separate charges, they are weak enough to allow movement of water and solutes. The strength of the hydrogen bond between two water molecules is only approximately 4 kcal, roughly 1/20th of the strength of the covalent O–H bond in the water molecule. Thus, the extensive water lattice is dynamic and has many strained bonds that are continuously breaking and reforming. The average hydrogen bond between water molecules lasts only about 10 psec (1 picosecond is 10^{-12} sec), and each water molecule in the hydration shell of an ion stays only 2.4 nsec (1 nanosecond = 10^{-9} sec). As a result, hydrogen bonds between water molecules and polar solutes continuously dissociate and reform, thereby permitting solutes to move through water and water to pass through channels in cellular membranes.

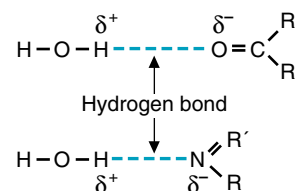


Fig. 4.4. Hydrogen bonds between water and polar molecules. R denotes additional atoms.

2. WATER AND THERMAL REGULATION

The structure of water also allows it to resist temperature change. Its heat of fusion is high, so a large drop in temperature is needed to convert liquid water to the solid state of ice. The thermal conductivity of water is also high, thereby facilitating heat dissipation from high energy-using areas such as the brain into the blood and the total body water pool. Its heat capacity and heat of vaporization are remarkably high; as liquid water is converted to a gas and evaporates from the

Table 4.1. Distribution of ions in body fluids.

	ECF* mmol/L	ICF
Cations		
Na ⁺	145	12
K ⁺	4	150
Anions		
Cl ⁻	105	5
HCO ₃ ⁻	25	12
Inorganic Phosphate	2	100

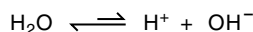
*The content of inorganic ions is very similar in plasma and interstitial fluid, the two components of the extracellular fluid. (ECF, extracellular fluid; ICF, intracellular fluid).



In the emergency room, **Di Abietes** was rehydrated with intravenous saline, which is a solution of 0.9% NaCl. Why was saline used instead of water?



Di Abietes has an osmotic diuresis. Because her blood levels of glucose and ketone bodies are so high, these compounds are passing from the blood into the glomerular filtrate in the kidneys and then into the urine. As a consequence of the high osmolality of the glomerular filtrate, much more water is being excreted in the urine than usual. Thus, Di has polyuria (increased urine volume). As a result of water lost from the blood into the urine, water passes from inside cells into the interstitial space and into the blood, resulting in an intracellular dehydration. The dehydrated cells in the brain are unable to carry out their normal functions. As a result, Di is in a coma.

**Fig. 4.5.** The dissociation of water.

The laboratory reported that **Di Abietes'** blood pH was 7.08 (reference range = 7.37–7.43). What was the [H⁺] in her blood compared with the concentration at a normal pH of 7.4?

Equation 4.1. Definition of pH.

$$\text{pH} = -\log [\text{H}^+]$$

skin, we feel a cooling effect. Water responds to the input of heat by decreasing the extent of hydrogen bonding and to cooling by increasing the bonding between water molecules.

C. Electrolytes

Both extracellular fluid (ECF) and intracellular fluid (ICF) contain electrolytes, a general term applied to bicarbonate and inorganic anions and cations. The electrolytes are unevenly distributed between compartments; Na⁺ and Cl⁻ are the major electrolytes in the ECF (plasma and interstitial fluid), and K⁺ and phosphates such as HPO₄⁻² are the major electrolytes in cells (Table 4.1). This distribution is maintained principally by energy-requiring transporters that pump Na⁺ out of cells in exchange for K⁺ (see Chapter 10).

D. Osmolality and Water Movement

Water distributes between the different fluid compartments according to the concentration of solutes, or osmolality, of each compartment. The osmolality of a fluid is proportionate to the total concentration of all dissolved molecules, including ions, organic metabolites, and proteins (usually expressed as milliosmoles (mOsm)/kg water). The semipermeable cellular membrane that separates the extracellular and intracellular compartments contains a number of ion channels through which water can freely move, but other molecules cannot. Likewise, water can freely move through the capillaries separating the interstitial fluid and the plasma. As a result, water will move from a compartment with a low concentration of solutes (lower osmolality) to one with a higher concentration to achieve an equal osmolality on both sides of the membrane. The force it would take to keep the same amount of water on both sides of the membrane is called the osmotic pressure.

As water is lost from one fluid compartment, it is replaced with water from another to maintain a nearly constant osmolality. The blood contains a high content of dissolved negatively charged proteins and the electrolytes needed to balance these charges. As water is passed from the blood into the urine to balance the excretion of ions, the blood volume is replenished with water from interstitial fluid. When the osmolality of the blood and interstitial fluid is too high, water moves out of the cells. The loss of cellular water also can occur in hyperglycemia, because the high concentration of glucose increases the osmolality of the blood.

II. ACIDS AND BASES

Acids are compounds that donate a hydrogen ion (H⁺) to a solution, and bases are compounds (such as the OH⁻ ion) that accept hydrogen ions. Water itself dissociates to a slight extent, generating hydrogen ions (H⁺), which are also called protons, and hydroxide ions (OH⁻) (Fig. 4.5). The hydrogen ions are extensively hydrated in water to form species such as H₃O⁺, but nevertheless are usually represented as simply H⁺. Water itself is neutral, neither acidic nor basic.

A. The pH of Water

The extent of dissociation by water molecules into H⁺ and OH⁻ is very slight, and the hydrogen ion concentration of pure water is only 0.0000001 M, or 10⁻⁷ mol/L. The concentration of hydrogen ions in a solution is usually denoted by the term *pH*, which is the negative log₁₀ of the hydrogen ion concentration expressed in mol/L (Equation 4.1). Therefore, the pH of pure water is 7.

The dissociation constant for water, K_d, expresses the relationship between the hydrogen ion concentration [H⁺], the hydroxide ion concentration [OH⁻],

and the concentration of water $[H_2O]$ at equilibrium (Equation 4.2). Because water dissociates to such a small extent, $[H_2O]$ is essentially constant at 55.5 M. Multiplication of the K_d for water (approximately 1.8×10^{-16} M) by 55.5 M gives a value of approximately 10^{-14} (M)², which is called the ion product of water (K_w) (Equation 4.3). Because K_w , the product of $[H^+]$ and $[OH^-]$, is always constant, a decrease of $[H^+]$ must be accompanied by a proportionate increase of $[OH^-]$.

A pH of 7 is termed neutral because $[H^+]$ and $[OH^-]$ are equal. Acidic solutions have a greater hydrogen ion concentration and a lower hydroxide ion concentration than pure water (pH < 7.0), and basic solutions have a lower hydrogen ion concentration and a greater hydroxide ion concentration (pH > 7.0).

B. Strong and Weak Acids

During metabolism, the body produces a number of acids that increase the hydrogen ion concentration of the blood or other body fluids and tend to lower the pH (Table 4.2). These metabolically important acids can be classified as weak acids or strong acids by their degree of dissociation into a hydrogen ion and a base (the anion component). Inorganic acids such as sulfuric acid (H_2SO_4) and hydrochloric acid (HCl) are strong acids that dissociate completely in solution (Fig. 4.6). Organic acids containing carboxylic acid groups (e.g., the ketone bodies acetoacetic acid and β -hydroxybutyric acid) are weak acids that dissociate only to a limited extent in water. In general, a weak acid (HA), called the conjugate acid, dissociates into a hydrogen ion and an anionic component (A^-), called the conjugate base. The name of an undissociated acid usually ends in “ic acid” (e.g., acetoacetic acid) and the name of the dissociated anionic component ends in “ate” (e.g., acetoacetate).

The tendency of the acid (HA) to dissociate and donate a hydrogen ion to solution is denoted by its K_a , the equilibrium constant for dissociation of a weak acid (Equation 4.4). The higher the K_a , the greater is the tendency to dissociate a proton.

In the Henderson-Hasselbalch equation, the formula for the dissociation constant of a weak acid is converted to a convenient logarithmic equation (Equation 4.5). The term pK_a represents the negative log of K_a . If the pK_a for a weak acid is known, this

Equation 4.2. Dissociation of water.

$$K_d = \frac{[H^+][OH^-]}{[H_2O]}$$

Equation 4.3. The ion product of water.

$$K_w = [H^+][OH^-] = 1 \times 10^{-14}$$

Equation 4.4. The K_a acid

For the reaction
 $HA \leftrightarrow A^- + H^+$

$$K_a = \frac{[H^+][A^-]}{[HA]}$$



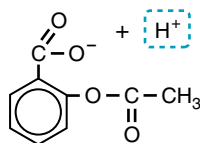
0.9% NaCl is 0.9 g NaCl/100 mL, equivalent to 9 g/L. NaCl has a molecular weight of 58 g/mole, so the concentration of NaCl in isotonic saline is 0.155 M, or 155 mM. If all of the NaCl were dissociated into Na^+ and Cl^- ions, the osmolality would be 310 mOsm/kg water. Because NaCl is not completely dissociated and some of the hydration shells surround undissociated NaCl molecules, the osmolality of isotonic saline is approximately 290 mOsm/kg H_2O . The osmolality of plasma, interstitial fluids, and ICF is also approximately 290 mOsm/kg water, so that no large shifts of water or swelling occur when isotonic saline is given intravenously.

Equation 4.5. The Henderson-Hasselbalch equation

$$pH = pK_a + \log \frac{[A^-]}{[HA]}$$



Dennis Veere has ingested an unknown number of acetylsalicylic acid (aspirin) tablets. Acetylsalicylic acid is rapidly converted to salicylic acid in the body. The initial effect of aspirin is to produce a respiratory alkalosis caused by a stimulation of the “metabolic” central respiratory control center in the hypothalamus. This increases the rate of breathing and the expiration of CO_2 . This is followed by a complex metabolic acidosis caused partly by the dissociation of salicylic acid (salicylic acid $\leftrightarrow$ salicylate⁻ + H^+ , $pK_a \approx 3.5$).



Acetylsalicylate

Salicylate also interferes with mitochondrial ATP production, resulting in increased generation of CO_2 and accumulation of lactate and other organic acids in the blood. Subsequently, salicylate may impair renal function, resulting in the accumulation of strong acids of metabolic origin, such as sulfuric acid and phosphoric acid. Usually, children who ingest toxic amounts of aspirin are acidotic by the time they arrive in the emergency room.



From inspection, you can tell that her $[H^+]$ is greater than normal, but less than 10 times higher. A 10-fold change in $[H^+]$ changes the pH by 1 unit. For Di, the pH of 7.08 = $-\log [H^+]$, and therefore her $[H^+]$ is $1 \times 10^{-7.08}$. To calculate her $[H^+]$, express -7.08 as $-8 + 0.92$. The antilog to the base 10 of 0.92 is 8.3. Thus, her $[H^+]$ is 8.3×10^{-8} compared to 4.0×10^{-8} at pH 7.4, or slightly more than double the normal value.

Table 4.2. Acids in the Blood of a Healthy Individual.

Acid	Anion	pK _a	Major Sources
Strong acid			
Sulfuric acid (H ₂ SO ₄)	Sulfate SO ₄ ⁻²	Completely Dissociated	Dietary sulfate and S-containing amino acids
Weak acid			
Carbonic acid (R-COOH)	Bicarbonate (R-COO ⁻)	3.80	CO ₂ from TCA cycle
Lactic acid (R-COOH)	Lactate (R-COO ⁻)	3.73	Anaerobic glycolysis
Pyruvic acid (R-COOH)	Pyruvate (R-COO ⁻)	2.39	Glycolysis
Citric acid (R-3COOH)	Citrate (R-3COO ⁻)	3.13; 4.76; 6.40	TCA cycle and diet (e.g., citrus fruits)
Acetoacetic acid (R-COOH)	Acetoacetate (R-COO ⁻)	3.62	Fatty acid oxidation to ketone bodies
β-Hydroxybutyric acid (R-COOH)	β-Hydroxybutyrate (R-COO ⁻)	4.41	Fatty acid oxidation to ketone bodies
Acetic acid (R-COOH)	Acetate (R-COO ⁻)	4.76	Ethanol metabolism
Dihydrogen phosphate (H ₂ PO ₄ ⁻)	Monohydrogen phosphate (HPO ₄ ⁻²)	6.8	Dietary organic phosphates
Ammonium ion (NH ₄ ⁺)	Ammonia (NH ₃)	9.25	Dietary nitrogen-containing compounds

equation can be used to calculate the ratio of the unprotonated to the protonated form at any pH. From this equation, you can see that a weak acid is 50% dissociated at a pH equal to its pK_a.

Most of the metabolic carboxylic acids have pK_as between 2 and 5, depending on the other groups on the molecule (see Table 4.2). The pK_a reflects the strength of an acid. Acids with a pK_a of 2 are stronger acids than those with a pK_a of 5 because, at any pH, a greater proportion is dissociated.

III. BUFFERS

Buffers, which consist of a weak acid and its conjugate base, cause a solution to resist changes in pH when hydrogen ions or hydroxide ions are added. In Figure 4.7, the pH of a solution of the weak acid acetic acid is graphed as a function of the

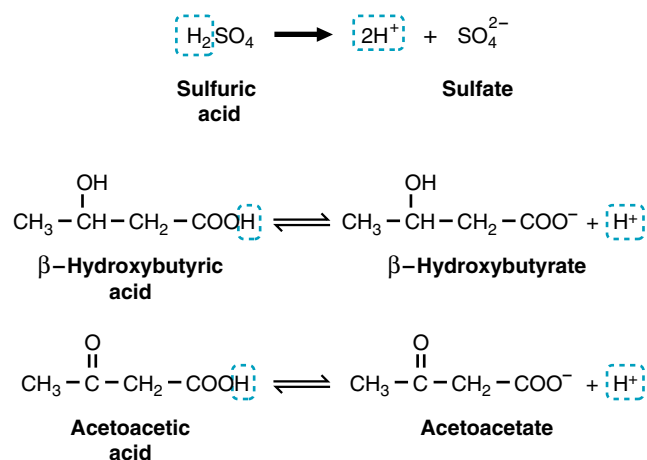


Fig. 4.6. Dissociation of acids. Sulfuric acid is a strong acid that dissociates into H⁺ ions and sulfate. The ketone bodies acetoacetic acid and β-hydroxybutyric acid are weak acids that partially dissociate into H⁺ and their conjugate bases.

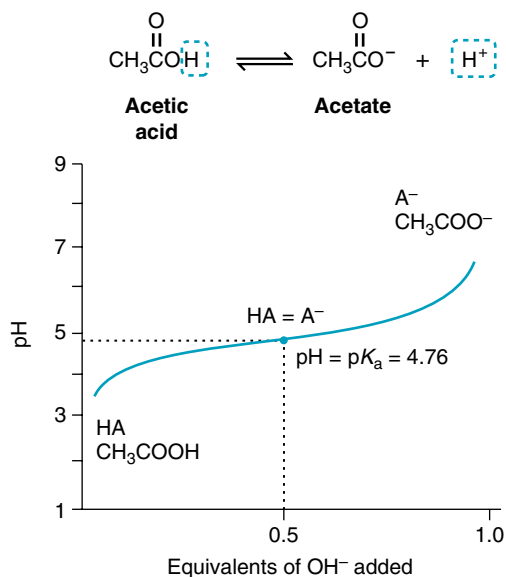


Fig. 4.7. The titration curve for acetic acid.

amount of OH^- that has been added. The OH^- is expressed as equivalents of total acetic acid present in the dissociated and undissociated forms. At the midpoint of this curve, 0.5 equivalents of OH^- have been added, and half of the conjugate acid has dissociated so that $[\text{A}^-]$ equals $[\text{HA}]$. This midpoint is expressed in the Henderson-Hasselbalch equation as the pK_a , defined as the pH at which 50% dissociation occurs. As you add more OH^- ions and move to the right on the curve, more of the conjugate acid molecules (HA) dissociate to generate H^+ ions, which combine with the added OH^- ions to form water. Consequently, only a small increase in pH results. If you add hydrogen ions to the buffer at its pK_a (moving to the left of the midpoint in Fig. 4.7), conjugate base molecules (A^-) combine with the added hydrogen ions to form HA, and almost no decrease in pH occurs.

As can be seen from Figure 4.7, a buffer can only compensate for an influx or removal of hydrogen ions within approximately 1 pH unit of its pK_a . As the pH of a buffered solution changes from the pK_a to one pH unit below the pK_a , the ratio of $[\text{A}^-]$ to HA changes from 1:1 to 1:10. If more hydrogen ions were added, the pH would fall rapidly because relatively little conjugate base remains. Likewise, at 1 pH unit above the pK_a of a buffer, relatively little undissociated acid remains. More concentrated buffers are more effective simply because they contain a greater total number of buffer molecules per unit volume that can dissociate or recombine with hydrogen ions.

IV. METABOLIC ACIDS AND BUFFERS

An average rate of metabolic activity produces roughly 22,000 mEq acid per day. If all of this acid were dissolved at one time in unbuffered body fluids, their pH would be less than 1. However, the pH of the blood is normally maintained between 7.36 and 7.44, and intracellular pH at approximately 7.1 (between 6.9 and 7.4). The widest range of extracellular pH over which the metabolic functions of the liver, the beating of the heart, and conduction of neural impulses can be maintained is 6.8 to 7.8. Thus, until the acid produced from metabolism can be excreted as CO_2 in expired air and as ions in the urine, it needs to be buffered in the body fluids. The major buffer systems in the body are: the bicarbonate–carbonic acid buffer system, which operates principally in extracellular fluid; the hemoglobin buffer system in red blood cells; the phosphate buffer system in all types of cells; and the protein buffer system of cells and plasma.

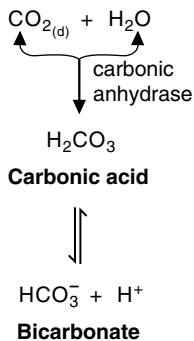


Fig. 4.8. The bicarbonate buffer system.



The pK_a for dissociation of bicarbonate anion (HCO_3^-) into H^+ and carbonate (CO_3^{2-}) is 9.8; therefore, only trace amounts of carbonate exist in body fluids.

Equation 4.6: The Henderson-Hasselbalch equation for the bicarbonate buffer system

$$\begin{aligned}
 \text{pH} &= \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]} \\
 \text{pH} &= 6.1 + \log \frac{[\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3] + [\text{CO}_2(d)]} \\
 \text{pH} &= 6.1 + \log \frac{[\text{HCO}_3^-]}{0.03\text{Paco}_2}
 \end{aligned}$$

where $[\text{CO}_2(d)]$ is the concentration of dissolved CO_2 , the $[\text{HCO}_3^-]$ is expressed as mEq/mL, and Paco_2 as mm Hg. The constant of 0.03 is a solubility constant that also corrects for the use of different units.

A. The Bicarbonate Buffer System

The major source of metabolic acid in the body is the gas CO_2 , produced principally from fuel oxidation in the TCA cycle. Under normal metabolic conditions, the body generates more than 13 moles of CO_2 per day (approximately 0.5–1 kg). CO_2 dissolves in water and reacts with water to produce carbonic acid, H_2CO_3 , a reaction accelerated by the enzyme carbonic anhydrase (Fig. 4.8). Carbonic acid is a weak acid that partially dissociates into H^+ and bicarbonate anion, HCO_3^- .

Carbonic acid is both the major acid produced by the body, and its own buffer. The pK_a of carbonic acid itself is only 3.8, so at the blood pH of 7.4 it is almost completely dissociated and theoretically unable to buffer and generate bicarbonate. However, carbonic acid can be replenished from CO_2 in body fluids and air because the concentration of dissolved CO_2 in body fluids is approximately 500 times greater than that of carbonic acid. As base is added and H^+ is removed, H_2CO_3 dissociates into hydrogen and bicarbonate ions, and dissolved CO_2 reacts with H_2O to replenish the H_2CO_3 (see Fig. 4.8). Dissolved CO_2 is in equilibrium with the CO_2 in air in the alveoli of the lungs, and thus the availability of CO_2 can be increased or decreased by an adjustment in the rate of breathing and the amount of CO_2 expired. The pK_a for the bicarbonate buffer system in the body thus combines K_h (the hydration constant for the reaction of water and CO_2 to form H_2CO_3) with the chemical pK_a to obtain the value of 6.1 used in the Henderson-Hasselbalch equation (Equation 4.6). To use the terms for blood components measured in the emergency room, the dissolved CO_2 is expressed as a fraction of the partial pressure of CO_2 in arterial blood, Paco_2 .

B. Bicarbonate and Hemoglobin in the Red Blood Cell

The bicarbonate buffer system and hemoglobin in red blood cells cooperate in buffering the blood and transporting CO_2 to the lungs. Most of the CO_2 produced from tissue metabolism in the TCA cycle diffuses into the interstitial fluid and the blood plasma and then into red blood cells (Fig. 4.9, circle 1). Although no carbonic anhydrase can be found in blood plasma or interstitial fluid, the red blood cells contain high amounts of this enzyme, and CO_2 is rapidly converted to carbonic acid (H_2CO_3) within these cells (circle 2). As the carbonic acid dissociates (circle 3), the H^+ released is also buffered by combination with hemoglobin (Hb, circle 4). The side chain of the amino acid histidine in hemoglobin has a pK_a of 6.7 and is thus able to accept a proton. The bicarbonate anion is transported out of the red blood cell into the blood in exchange for chloride anion, and thus bicarbonate is relatively high in the plasma (circle 5) (see Table 4.1).

As the red blood cell approaches the lungs, the direction of the equilibrium reverses. CO_2 is released from the red blood cell, causing more carbonic acid to dissociate into



The partial pressure of CO_2 (Paco_2) in **Di Abietes'** arterial blood was 28 mm Hg (reference range = 37–43), and her serum bicarbonate level was 8 mEq/L (reference range = 24–28). Elevated levels of ketone bodies had produced a ketoacidosis, and **Di Abietes** was exhaling increased amounts of CO_2 by breathing deeply and frequently (Kussmaul's breathing) to compensate. Why does this occur? Ketone bodies are weak acids that partially dissociate, increasing H^+ levels in the blood and the interstitial fluid surrounding the "metabolic" respiratory center in the hypothalamus that controls the rate of breathing. As a result of the decrease in pH, her respiratory rate increased, causing a fall in the partial pressure of arterial CO_2 (Paco_2). As bicarbonate and increased protons combined to form carbonic acid and produce more CO_2 , her bicarbonate concentration decreased ($\text{HCO}_3^- + \text{H}^+ \rightarrow \text{H}_2\text{CO}_3 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$). As shown by Di's low arterial blood pH of 7.08, the Kussmaul's breathing was unable to fully compensate for the high rate of acidic ketone body production.

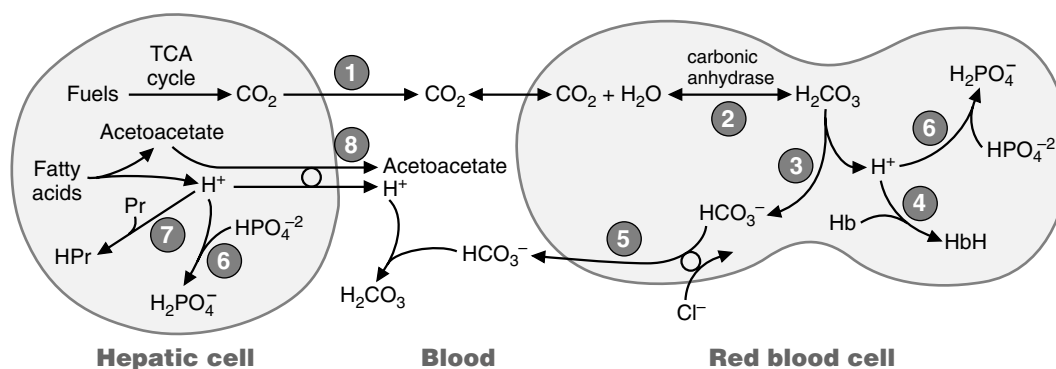


Fig. 4.9. Buffering systems of the body. CO_2 produced from cellular metabolism is converted to bicarbonate and H^+ in the red blood cells. Within the red blood cell, the H^+ is buffered by hemoglobin (Hb) and phosphate (HPO_4^{2-}). The bicarbonate is transported into the blood to buffer H^+ generated by the production of other metabolic acids, such as the ketone body acetoacetic acid. Other proteins (Pr) also serve as intracellular buffers.

CO_2 and water and more hydrogen ions to combine with bicarbonate. Hemoglobin loses some of its hydrogen ions, a feature that allows it to bind oxygen more readily (see Chapter 7). Thus, the bicarbonate buffer system is intimately linked to the delivery of oxygen to tissues.

The respiratory center within the hypothalamus, which controls the rate of breathing, is sensitive to changes in pH. As the pH falls, individuals breathe more rapidly and expire more CO_2 . As the pH rises, they breathe more shallowly. Thus, the rate of breathing contributes to regulation of pH through its effects on the dissolved CO_2 content of the blood.

Bicarbonate and carbonic acid, which diffuse through the capillary wall from the blood into interstitial fluid, provide a major buffer for both plasma and interstitial fluid. However, blood differs from interstitial fluid in that the blood contains a high content of extracellular proteins, such as albumin, which contribute to its buffering capacity through amino acid side chains that are able to accept and release protons. The protein content of interstitial fluid is too low to serve as an effective buffer.

C. Intracellular pH

Phosphate anions and proteins are the major buffers involved in maintaining a constant pH of intracellular fluids. The inorganic phosphate anion H_2PO_4^- dissociates to generate H^+ and the conjugate base, HPO_4^{2-} with a pK_a of 7.2 (see Fig. 4.9, circle 6). Thus, phosphate anions play a major role as an intracellular buffer in the red blood cell and in other types of cells, where their concentration is much higher than in blood and interstitial fluid (See Table 4.1, extracellular fluid). Organic phosphate anions, such as glucose 6-phosphate and ATP, also act as buffers. Intracellular fluid contains a high content of proteins that contain histidine and other amino acids that can accept protons, in a fashion similar to hemoglobin (see Fig. 4.9, circle 7).

The transport of hydrogen ions out of the cell is also important for maintenance of a constant intracellular pH. Metabolism produces a number of other acids in addition to CO_2 . For example, the metabolic acids acetoacetic acid and β -hydroxybutyric acid are produced from fatty acid oxidation to ketone bodies in the liver, and lactic acid is produced by glycolysis in muscle and other tissues. The pK_a for most metabolic carboxylic acids is below 5, so these acids are completely dissociated at the pH of blood and cellular fluid. Metabolic anions are transported out of the cell together with H^+ (see Fig. 4.9, circle 8). If the cell becomes too acidic, more H^+ is transported out in exchange for Na^+ ions by a different transporter. If the cell becomes too alkaline, more bicarbonate is transported out in exchange for Cl^- ions.



Both **Di Abietes** and **Percy Veere** were hyperventilating when they arrived at the emergency room: Ms. Abietes in response to her primary metabolic acidosis and Mr. Veere in response to anxiety. Hyperventilation raised blood pH in both cases: In Ms. Abietes case it partially countered the acidosis and in Mr. Veere's case it produced a respiratory alkalosis (abnormally high arterial blood pH).



Phosphoric acid (H_3PO_4) dissociates into H^+ ions and dihydrogen phosphate (H_2PO_4^-) with a pK_a of 2.15. The dissociation of H_2PO_4^- into H^+ and monohydrogen ions (HPO_4^{2-}) occurs with a pK_a of 7.20. Monohydrogen phosphate dissociates into H^+ and phosphate anions (PO_4^{3-}) with a pK_a of 12.4.

D. Urinary Hydrogen, Ammonium, and Phosphate Ions

The nonvolatile acid that is produced from body metabolism cannot be excreted as expired CO_2 and is excreted in the urine. Most of the nonvolatile acid hydrogen ion is excreted as undissociated acid that generally buffers the urinary pH between 5.5 and 7.0. A pH of 5.0 is the minimum urinary pH. The acid secretion includes inorganic acids such as phosphate and ammonium ions, as well as uric acid, dicarboxylic acids, and tricarboxylic acids such as citric acid (see Table 4.2). One of the major sources of nonvolatile acid in the body is sulfuric acid (H_2SO_4). Sulfuric acid is generated from the sulfate-containing compounds ingested in foods and from metabolism of the sulfur-containing amino acids, cysteine and methionine. It is a strong acid that is dissociated into H^+ and sulfate anion (SO_4^{-2}) in the blood and urine (see Fig. 4.6). Urinary excretion of H_2PO_4^- helps to remove acid. To maintain metabolic homeostasis, we must excrete the same amount of phosphate in the urine that we ingest with food as phosphate anions or organic phosphates such as phospholipids. Whether the phosphate is present in the urine as H_2PO_4^- or HPO_4^{-2} depends on the urinary pH and the pH of blood.

Ammonium ions are major contributors to buffering urinary pH, but not blood pH. Ammonia (NH_3) is a base that combines with protons to produce ammonium (NH_4^+) ions ($\text{NH}_3 + \text{H}^+ \leftrightarrow \text{NH}_4^+$), a reaction that occurs with a pK_a of 9.25. Ammonia is produced from amino acid catabolism or absorbed through the intestine, and kept at very low concentrations in the blood because it is toxic to neural tissues. Cells in the kidney generate NH_4^+ and excrete it into the urine in proportion to the acidity (proton concentration) of the blood. As the renal tubular cells transport H^+ into the urine, they return bicarbonate anions to the blood.



Ammonium ions are acids that dissociate to form the conjugate base, ammonia, and hydrogen ions. What is the form present in blood? In urine?



Diabetes mellitus is diagnosed by the concentration of plasma glucose (plasma is blood from which the red blood cells have been removed by centrifugation). Because plasma glucose normally increases after a meal, the normal reference ranges and the diagnostic level are defined relative to the time of consumption of food, or to consumption of a specified amount of glucose during an oral glucose tolerance test. After an overnight fast, values for the fasting plasma glucose below 110 mg/dL are considered normal and above 126 mg/dL define diabetes mellitus. Fasting plasma glucose values greater or equal to 110 and less than 126 mg/dL define an intermediate condition termed impaired fasting glucose. Normal casual plasma glucose levels (casual is defined as any time of day without regard to the time since a last meal) should not be above 200 mg/dL. A 2-hour postprandial (after a meal or after an oral glucose load) plasma glucose level between 140 and 199 mg/dL defines a condition known as impaired glucose tolerance. A level above 200 mg/dL defines “overt” diabetes mellitus.

E. Hydrochloric Acid

Hydrochloric acid (HCl), also called gastric acid, is secreted by parietal cells of the stomach into the stomach lumen, where the strong acidity denatures ingested proteins so they can be degraded by digestive enzymes. When the stomach contents are released into the lumen of the small intestine, gastric acid is neutralized by bicarbonate secreted from pancreatic cells and by cells in the intestinal lining.

CLINICAL COMMENTS



Dianne Abietes. Di Abietes has type 1 diabetes mellitus (formerly called juvenile or insulin-dependent diabetes mellitus, IDDM). She maintains her insulin level through two daily subcutaneous (under the skin) injections of insulin. If her blood insulin levels fall too low, free fatty acids leave her adipocytes (fat cells) and are converted by the liver to the ketone bodies acetoacetic acid and β -hydroxybutyric acid. As these acids accumulate in the blood, a metabolic acidosis known as diabetic ketoacidosis (DKA) develops. Until insulin is administered to reverse this trend, several compensatory mechanisms operate to minimize the extent of the acidosis. One of these mechanisms is a stimulation of the respiratory center in the hypothalamus induced by the acidosis, which leads to deeper and more frequent respiration (Kussmaul’s respiration). CO_2 is expired more rapidly than normal, and the blood pH rises. The results of the laboratory studies performed on **Di Abietes** in the emergency room were consistent with a moderately severe DKA. Her arterial blood pH and serum bicarbonate were low, and ketone bodies were present in her blood and urine (normally, ketone bodies are not present in the urine). In addition, her serum glucose level was 648 mg/dL (reference range = 80 – 110 fasting and no higher than 200 in a random glucose sample). Her hyperglycemia, which induces an osmotic diuresis, contributed to her dehydration and the hyperosmolality of her body fluids.

Treatment was initiated with intravenous saline solutions to replace fluids lost with the osmotic diuresis and hyperventilation. The osmotic diuresis resulted from increased urinary water volume to dilute the large amounts of glucose and ketone bodies excreted in the urine. Hyperventilation increased the water of respiration lost with expired air. A loading dose of regular insulin was given as an intravenous bolus followed by additional insulin each hour as needed. The patient's metabolic response to the treatment was monitored closely.



Dennis Veere. Dennis remained alert in the emergency room. While awaiting the report of his initial serum salicylate level, his stomach was lavaged, and several white tablets were found in the stomach aspirate. He was examined repeatedly and showed none of the early symptoms of salicylate toxicity, such as respiratory stimulation, upper abdominal distress, nausea, or headache.

His serum salicylate level was reported as 92 $\mu\text{g/mL}$ (the usual level in an adult receiving a therapeutic dosage of 4–5 g/day is 120–350 $\mu\text{g/mL}$, and a level of 800 $\mu\text{g/mL}$ is considered potentially lethal). He was admitted for overnight observation and continued to do well. A serum salicylate level the following morning was 24 $\mu\text{g/mL}$. He was discharged later that day.



Percy Veere. At the emergency room, Mr. Veere complained of light-headedness and “pins and needles” (paresthesias) in his hands and around his lips. These symptoms resulted from an increase in respiratory drive mediated in this case through the “behavioral” rather than the “metabolic” central respiratory control system (as seen in **Di Abietes** when she was in diabetic ketoacidosis). The behavioral system was activated in Mr. Veere's case by his anxiety over his grandson's potential poisoning. His alveolar hyperventilation caused Mr. Veere's PaCO_2 to decrease below the normal range of 37 to 43 mm Hg. The alkalemia caused his neurologic symptoms. After being reassured that his grandson would be fine, Mr. Veere was asked to breathe slowly into a small paper bag placed around his nose and mouth, allowing him to reinhale the CO_2 being exhaled through hyperventilation. Within 20 minutes, his symptoms disappeared.

BIOCHEMICAL COMMENTS



Body water and dehydration. Dehydration, or loss of water, occurs when salt and water intake is less than the combined rates of renal plus extrarenal volume loss (Fig. 4.10). In a true hypovolemic state, total body water, functional ECF volume, and ICF volume are decreased. One of the causes of hypovolemia is an intake of water that is inadequate to resupply the daily excretion volume (maintenance of fluid homeostasis). The amount of water lost by the kidneys is determined by the amount of water necessary to dilute the ions, acids, and other solutes excreted. Both urinary solute excretion and water loss in expired air, which amount to almost 400 mL/day, occur during fasting as well as during periods of normal food intake. Thus, people who are lost in the desert or shipwrecked continue to lose water in air and urine, in addition to their water loss through the skin and sweat glands. Comatose patients and patients who are debilitated and unable to swallow also continue to lose water and become dehydrated. Dysphagia (loss of appetite), such as **Percy Veere** experienced during his depression (see Chapters 1 and 3), can result in dehydration because food intake is normally a source of fluid. Cerebral injuries that cause a loss of thirst also can lead to dehydration.

Dehydration also can occur secondary to excessive water loss through the kidneys, lungs, and skin. Excessive renal loss could be the result of impaired kidney function, or impaired response by the hormones that regulate water balance (e.g.,



The pK_a for dissociation of ammonium ions is 9.25. Thus, the undissociated conjugate acid form, NH_4^+ , predominates at the physiologic pH of 7.4 and urinary pH of 5.5 to 7.0. Ammonia is an unusual conjugate base because it is not an anion.

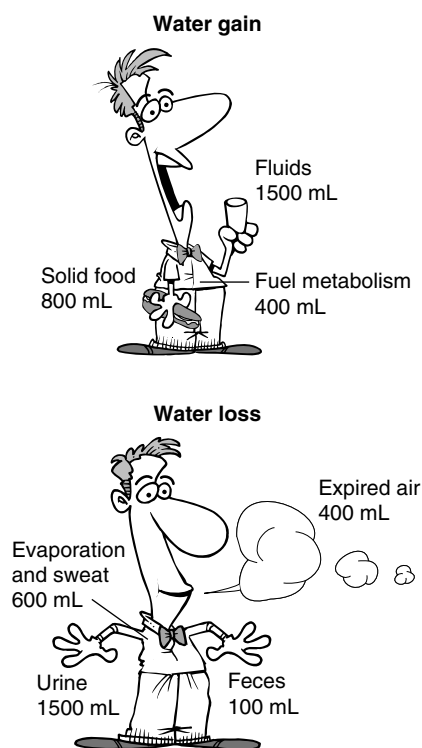


Fig. 4.10. Body fluid homeostasis (constant body water balance). Intake is influenced by availability of fluids and food, thirst, hunger, and the ability to swallow. The rates of breathing and evaporation and urinary volume influence water loss. The body adjusts the volume of urinary excretion to compensate for variations in other types of water loss and for variations in intake. The hormones aldosterone and antidiuretic hormone (ADH) help to monitor blood volume and osmolality through mechanisms regulating thirst and sodium and water balance.

antidiuretic hormone [ADH] and aldosterone). However, in individuals such as **Di Abietes** with normal kidney function, dehydration can occur as a response to water dilution of the high concentrations of ketone bodies, glucose, and other solutes excreted in the urine (osmotic diuresis). If the urinary water loss is associated with an acidosis, hyperventilation occurs, increasing the amount of water lost in expired air. Water loss in expired air (pulmonary water loss) could occur also as a result of a tracheotomy or cerebral injury. An excessive loss of water and electrolytes through the skin can result from extensive burns.

Gastrointestinal water losses also can result in dehydration. We secrete approximately 8 to 10 L fluid per day into our intestinal lumen. Normally, more than 90% of this fluid is reabsorbed in the intestines. The percentage reabsorbed can be decreased by vomiting, diarrhea, tube drainage of gastric contents, or loss of water into tissues around the gut via bowel fistulas.

Suggested References

Many physiology textbooks have good in-depth descriptions of acid-base balance. A very useful reference for medical students is: Brensilver J, Goldberger, E. *A Primer of Water, Electrolyte, and Acid-Base Syndromes*. 8th Ed. Philadelphia: F.A. Davis, 1996 .

For additional information on toxic effects of aspirin, see Roberts LJ II, Morrow JD. Analgesic-antipyretic and antiinflammatory agents and drugs employed in the treatment of gout. In: Hardman JG, Limbird LE, Goodman AG, eds. *Goodman and Gilman's The Pharmacological Basis of Therapeutics*. New York: McGraw-Hill, 2001:697–701.



REVIEW QUESTIONS—CHAPTER 4

1. A decrease of blood pH from 7.5 to 6.5 would be accompanied by which of the following changes in ion concentration?
 - (A) A 10-fold increase in hydrogen ion concentration.
 - (B) A 10-fold increase in hydroxyl ion concentration.
 - (C) An increase in hydrogen ion concentration by a factor of 7.5/6.5.
 - (D) A decrease in hydrogen ion concentration by a factor of 6.5/7.5.
 - (E) A shift in concentration of buffer anions, with no change in hydrogen ion concentration.
2. Which of the following describes a universal property of buffers?
 - (A) Buffers are usually composed of a mixture of strong acids and strong bases.
 - (B) Buffers work best at the pH at which they are completely dissociated.
 - (C) Buffers work best at the pH at which they are 50% dissociated.
 - (D) Buffers work best at one pH unit lower than the pK_a .
 - (E) Buffers work equally well at all concentrations.
3. A patient with an enteropathy (intestinal disease) produced large amounts of ammonia (NH_3) from bacterial overgrowth in the intestine. The ammonia was absorbed through the intestine into the portal vein and entered the circulation. Which of the following is a likely consequence of his ammonia absorption?
 - (A) A decrease of blood pH
 - (B) Conversion of ammonia to ammonium ion in the blood
 - (C) A decreased concentration of bicarbonate in the blood
 - (D) Kussmaul's respiration
 - (E) Increased expiration of CO_2

4. Which of the following physiologic/pathologic conditions is most likely to result in an alkalosis, provided that the body could not fully compensate?
- (A) Production of lactic acid by muscles during exercise
 - (B) Production of ketone bodies by a patient with diabetes mellitus
 - (C) Repeated vomiting of stomach contents, including HCl
 - (D) Diarrhea with loss of the bicarbonate anions secreted into the intestine
 - (E) An infection resulting in a fever and hypercatabolism
5. Laboratory tests on the urine of a patient identified the presence of methylmalonate ($^-\text{OOC}-\text{CH}(\text{CH}_3)-\text{COO}^-$). Methylmalonate
- (A) is a strong acid.
 - (B) is the conjugate base of a weak acid.
 - (C) is 100% dissociated at its pK_a .
 - (D) is 50% dissociated at the pH of the blood.
 - (E) is a major intracellular buffer.

5 Structures of the Major Compounds of the Body



$\text{CH}_3\text{CH}_2\text{OH}$ Ethanol
 CH_3OH Methanol

The names of chemical groups are often incorporated into the common name of a compound, and denote important differences in chemical structure. For example, in the name *ethanol*, the “eth” denotes the ethyl group (CH_3CH_2-), the “ol” denotes the alcohol group (OH), and the “an” denotes the single bonds between the carbon atoms. Methanol contains a methyl group (CH_3) instead of the ethyl group. Methanol (also called wood alcohol) is much more toxic to humans than ethanol, the alcohol in alcoholic beverages. Ingestion of methanol results in visual disturbances, bradycardia, coma, and seizures.



Hydrophobic molecules are transported in the blood bound to carrier proteins or incorporated into lipoprotein complexes that have a hydrophobic lipid core and a more polar surface.

The body contains compounds of great structural diversity, ranging from relatively simple sugars and amino acids to enormously complex polymers such as proteins and nucleic acids. Many of these compounds have common structural features related to their names, their solubility in water, the pathways in which they participate, or their physiologic function. Thus, learning the terminology used to describe individual compounds and classes of compounds can greatly facilitate learning biochemistry.

In this chapter, we describe the major classes of carbohydrates and lipids and some of the classes of nitrogen-containing compounds. The structures of amino acids, proteins, the nucleic acids, and vitamins are covered in more detail in subsequent chapters.

Functional Groups on Molecules. Organic molecules are composed principally of carbon and hydrogen. However, their unique characteristics are related to structures termed functional groups involving **oxygen, nitrogen, phosphorus, or sulfur**.

Carbohydrates. Carbohydrates, commonly known as sugars, can be classified by their carbonyl group (**aldo-** or **ketosugars**), the number of carbons they contain (e.g., **pentoses**, **hexoses**), or the positions of the hydroxyl groups on their asymmetric carbon atoms (**D-** or **L-sugars**, **stereoisomers**, or **epimers**). They can also be categorized according to their substituents (e.g., **amino sugars**), or the number of monosaccharides (such as glucose) joined through **glycosidic bonds** (**disaccharides**, **oligosaccharides**, and **polysaccharides**). **Glycoproteins** and **proteoglycans** have sugars attached to their protein components.

Lipids. Lipids are a group of structurally diverse compounds defined by their **hydrophobicity**; they are not very soluble in water. The major lipids of the human body are the **fatty acids**, which are esterified to **glycerol** to form **triacylglycerols** (triglycerides) or **phosphoacylglycerols** (phosphoglycerides). In the **sphingolipids**, a fatty acid is attached to sphingosine, which is derived from serine and another fatty acid. **Glycolipids** contain sugars attached to a lipid hydroxyl group. Specific **polyunsaturated fatty acids** are precursors of **eicosanoids**. The lipid **cholesterol** is a component of membranes, and the precursor of other compounds that contain the steroid nucleus, such as the **bile salts** and **steroid hormones**. Cholesterol is one of the compounds synthesized from a 5-carbon precursor called the **isoprene** unit.

Nitrogen-containing compounds. Nitrogen in **amino groups** or **heterocyclic ring** structures often carries a positive charge at neutral pH. **Amino acids** contain a carboxyl group, an amino group, and one or more additional carbons. **Purines**, **pyrimidines**, and **pyridines** have heterocyclic nitrogen-containing ring structures. **Nucleosides** comprise one of these ring structures attached to a sugar. The addition of a phosphate produces a **nucleotide**.



THE WAITING ROOM



Di Abietes recovered from her bout of diabetic ketoacidosis and was discharged from the hospital (see Chapter 4). She has returned for a follow-up visit as an outpatient. She reports that she has been compliant with her recommended diet and that she faithfully gives herself insulin by subcutaneous injection twice daily. Her serum glucose levels are monitored in the hospital laboratory approximately every 2 weeks, and she self-monitors her blood glucose levels every other day.



Lotta Topaigne is a 47-year-old woman who came to the physician's office complaining of a severe throbbing pain in the right great toe that began 8 hours earlier. The toe has suffered no trauma but appears red and swollen. It is warmer than the surrounding tissue and is exquisitely tender to even light pressure. Ms. Topaigne is unable to voluntarily flex or extend the joints of the digit, and passive motion of the joints causes great pain.

I. FUNCTIONAL GROUPS ON BIOLOGIC COMPOUNDS

A. Biologic Compounds

The organic molecules of the body consist principally of carbon, hydrogen, oxygen, nitrogen, sulfur, and phosphorus joined by covalent bonds. The key element is carbon, which forms four covalent bonds with other atoms. Carbon atoms are joined through double or single bonds to form the carbon backbone for structures of varying size and complexity (Fig. 5.1). Groups containing 1, 2, 3, 4, and 5 carbons plus hydrogen are referred to as methyl, ethyl, propionyl, butyl, and pentanyl groups, respectively. If the carbon chain is branched, the prefix “iso” is used. If the compound contains a double bond, “ene” is sometimes incorporated into the name. Carbon structures that are straight or branched with single or double bonds, but do not contain a ring, are called aliphatic.

Carbon-containing rings are found in a number of biologic compounds. One of the most common is the six-membered carbon-containing benzene ring, sometimes called a phenyl group (see Fig. 5.1). This ring has three double bonds, but the electrons are shared equally by all six carbons and delocalized in planes above and below the ring. Compounds containing the benzene ring, or a similar ring structure with benzene-like properties, are called aromatic.

B. Functional Groups

Biochemical molecules are defined both by their carbon skeleton and by structures called functional groups that usually involve bonds between carbon and oxygen, carbon and nitrogen, carbon and sulfur, and carbon and phosphate groups (Fig. 5.2). In carbon–carbon and carbon–hydrogen bonds, the electrons are shared equally between atoms, and the bonds are nonpolar and relatively unreactive. In carbon–oxygen and carbon–nitrogen bonds, the electrons are shared unequally, and the bonds are polar and more reactive. Thus, the properties of the functional groups usually determine the types of reactions that occur and the physiologic role of the molecule.

Functional group names are often incorporated into the common name of a compound. For example, a **ketone** might have a name that ends in “one” like **acetone**, and the name of a compound that contains a hydroxyl (alcohol or OH group) might end in “ol” (e.g., **ethanol**). The acyl group is the portion of the molecule that provides



Di Abietes had a metabolic acidosis resulting from an increased hepatic production of ketone bodies. Her response to therapy was followed with screening tests for ketone bodies in her urine that employed a paper strip containing nitroprusside, a compound that reacts with keto groups. Her blood glucose was measured with an enzymatic assay that is specific for the sugar D-glucose and will not react with other sugars.

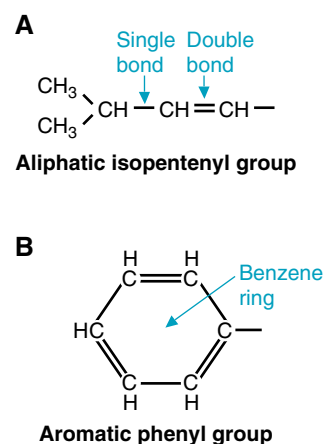
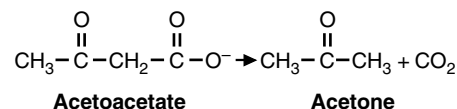
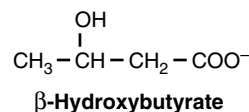


Fig. 5.1. Examples of aliphatic and aromatic compounds. **A.** An **isoprene** group, which is an aliphatic group. The “iso” prefix denotes branching, and the “ene” denotes a double bond. **B.** A benzene ring (or phenyl group), which is an aromatic group.



The ketone bodies synthesized in the liver are β -hydroxybutyrate and acetoacetate. A third ketone body, acetone, is formed by the nonenzymatic decarboxylation of acetoacetate.



Acetone is volatile and accounts for the sweet mousy odor in the breath of patients such as **Di Abietes** when they have a ketoacidosis. What functional groups are present in each of these ketone bodies?

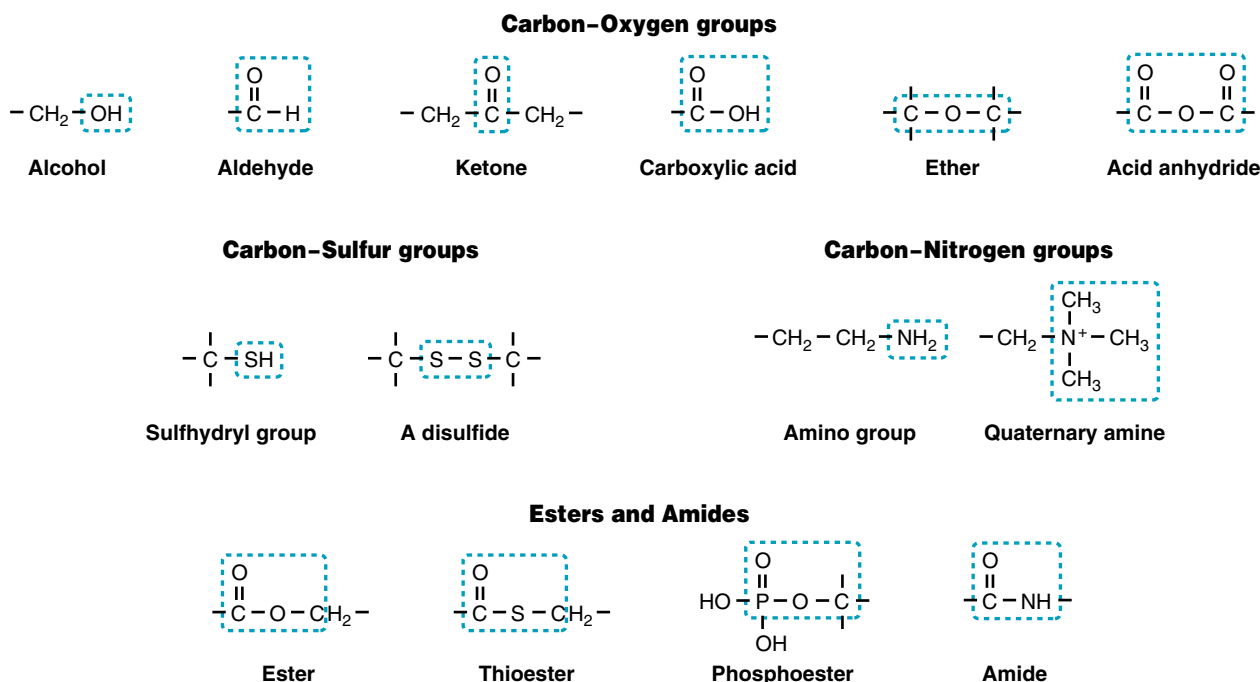
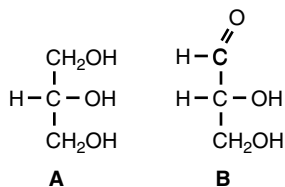


Fig. 5.2. Major types of functional groups found in biochemical compounds of the human body.

Q:



Which compound is **glycerol**, and which is **glyceraldehyde**?

Q:

Judging from the structures shown in the question on page 55 which compound is more oxidized, β -hydroxybutyrate or acetoacetate? Which is more reduced?

A:

β -hydroxybutyrate and acetoacetate are **carboxylates** (dissociated carboxylic acids). Acetoacetate and acetone contain **keto/ketone** groups. Because β -hydroxybutyrate contains an alcohol (**hydroxyl**) group and not a keto group, the general name of ketone bodies for these compounds is really a misnomer.

the carbonyl (—C=O) group in an ester or amide linkage. It is denoted in a name by an “yl” ending. For example, the fat stores of the body are **triacylglycerols**. Three acyl (fatty acid) groups are esterified to **glycerol**, a compound containing three alcohol groups. In the remainder of this chapter, we will bold the portions of names of compounds that refer to a class of compounds or a structural feature.

1. OXIDIZED AND REDUCED GROUPS

The carbon–carbon and carbon–oxygen groups are described as “oxidized” or “reduced” according to the number of electrons around the carbon atom. Oxidation is the loss of electrons and results in the loss of hydrogen atoms together with one or two electrons, or the gain of an oxygen atom or hydroxyl group. Reduction is the gain of electrons and results in the gain of hydrogen atoms or loss of an oxygen atom. Thus, the carbon becomes progressively more oxidized (and less reduced) as we go from an alcohol to an aldehyde or a ketone to a carboxyl group (see Fig. 5.2). Carbon–carbon double bonds are more oxidized (and less reduced) than carbon–carbon single bonds.

2. GROUPS THAT CARRY A CHARGE

Acidic groups contain a proton that can dissociate, usually leaving the remainder of the molecule as an anion with a negative charge (see Chapter 4). In biomolecules, the major anionic substituents are **carboxylate** groups, **phosphate** groups, or **sulfate** groups (the “ate” suffix denotes a negative charge) (Fig. 5.3). Phosphate groups attached to metabolites are often abbreviated as P with a circle around it, or just as “P”, as in glucose-6-P.

Compounds containing nitrogen are usually basic and can acquire a positive charge (Fig. 5.4). Nitrogen has five electrons in its valence shell. If only three of these electrons form covalent bonds with other atoms, the nitrogen has no charge. If the remaining two electrons form a bond with a hydrogen ion or a carbon atom, the nitrogen carries a positive charge. Amines consist of nitrogen attached through single bonds to hydrogen atoms and to one or more carbon atoms. Primary amines,

such as **dopamine**, have one carbon–nitrogen bond. These amines are weak acids with a pK_a of approximately 9, so that at pH 7.4 they carry a positive charge. Secondary, tertiary, and quaternary amines have 2, 3, and 4 nitrogen–carbon bonds, respectively (see Fig. 5.4).

C. Polarity of Bonds and Partial Charges

Polar bonds are covalent bonds in which the electron cloud is denser around one atom (the atom with the greater electronegativity) than the other. Oxygen is more electronegative than carbon, and a carbon–oxygen bond is therefore polar, with the oxygen atom carrying a partial negative charge and the carbon atom carrying a partial positive charge (Fig. 5.5). In nonpolar carbon–carbon bonds and carbon–hydrogen bonds, the two electrons in the covalent bond are shared almost equally. Nitrogen, when it has only three covalent bonds, also carries a partial negative charge relative to carbon, and the carbon–nitrogen bond is polarized. Sulfur can carry a slight partial negative charge.

1. SOLUBILITY

Water is a dipolar molecule in which the oxygen atom carries a partial negative charge and the hydrogen atoms carry partial positive charges (see Chapter 4). For molecules to be soluble in water, they must contain charged or polar groups that can associate with the partial positive and negative charges of water. Thus, the solubility of organic molecules in water is determined by both the proportion of polar to nonpolar groups attached to the carbon–hydrogen skeleton and to their relative positions in the molecule. Polar groups or molecules are called **hydrophilic** (water-loving), and nonpolar groups or molecules are **hydrophobic** (water-fearing). Sugars such as glucose 6-phosphate, for example, contain so many polar groups that they are very hydrophilic and almost infinitely water-soluble (Fig. 5.6). The water molecules interacting with a polar or ionic compound form a hydration shell around the compound.

Compounds that have large nonpolar regions are relatively water insoluble. They tend to cluster together in an aqueous environment and form weak associations through van der Waals forces, often termed **hydrophobic bonds**. Hydrophobic compounds are essentially pushed together as the water molecules maximize the number of energetically favorable hydrogen bonds they can form with each other in the water lattice. Thus, lipids form droplets or separate layers in an aqueous environment (e.g., vegetable oils in a salad dressing).

2. REACTIVITY

Another consequence of bond polarity is that atoms that carry a partial (or full) negative charge are attracted to atoms that carry a partial (or full) positive charge and vice versa. These partial or full charges dictate the course of biochemical reactions—which follow the same principles of electrophilic and nucleophilic attack characteristic of organic reactions in general. The partial positive charge on the carboxyl carbon attracts more negatively charged groups and accounts for many of the reactions of carboxylic acids. An ester is formed when a carboxylic acid and an alcohol combine, splitting out water (Fig. 5.7). Similarly, a thioester is formed when an acid combines with a sulfhydryl group, and an amide is formed when an acid combines with an amine. Similar reactions result in the formation of a phosphoester from phosphoric acid and an alcohol and in the formation of an anhydride from two acids.

D. Nomenclature

Biochemists use two systems for the identification of the carbons in a chain. In the first system, the carbons in a compound are numbered, starting with the carbon in

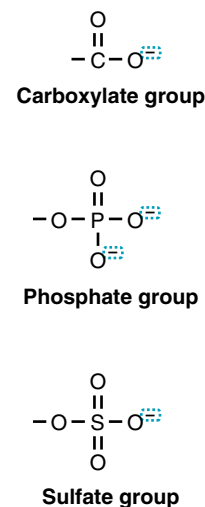
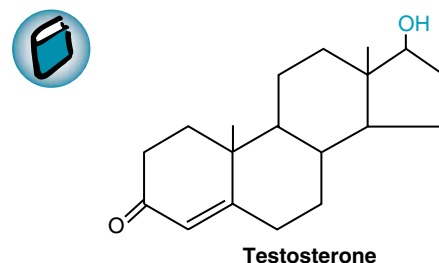


Fig. 5.3. Examples of anions formed by dissociation of acidic groups. At physiologic pH, carboxylic acids, phosphoric acid, and sulfuric acid are dissociated into hydrogen ions and negatively charged anions.



In medicine and biochemistry, the common or trivial names of compounds are used rather than the systematic nomenclature favored by chemists. Sometimes such names reflect functional groups, classes of compounds or the source from which the compound was first isolated. The compound **testosterone** provides an example. It was first isolated from monkey testis, the “ster” in its name denotes the steroid ring structure, and the “one” denotes a ketone group.

A: “A” contains three **alcohol** groups and is called **glycerol**. Compound B contains an **aldehyde group**, and is called **glyceraldehyde**.

A: Acetoacetate is more oxidized than β -hydroxybutyrate. The carbon in the keto group contains one less hydrogen than the carbon to which the OH group is attached. It has lost an electron.

the most oxidized group (e.g., the carboxyl group). In the second system, the carbons are given Greek letters, starting with the carbon next to the most oxidized group. Hence, the compound shown in Figure 5.8 is known as 3-hydroxybutyrate or β -hydroxybutyrate.

II. CARBOHYDRATES

A. Monosaccharides

Simple monosaccharides consist of a linear chain of three or more carbon atoms, one of which forms a carbonyl group through a double bond with oxygen (Fig. 5.9). The other carbons of an unmodified monosaccharide contain hydroxyl groups, resulting in the general formula for an unmodified sugar of $C_nH_{2n}O_n$. The suffix “ose” is used for the names of sugars. If the carbonyl group is an aldehyde, the sugar is an aldose; if the carbonyl group is a ketone, the sugar is a ketose. Monosaccharides are also classified according to their number of carbons: Sugars containing 3, 4, 5, 6, and 7 carbons are called trioses, tetroses, pentoses, hexoses, and heptoses, respectively. Fructose is therefore a ketohexose (see Fig. 5.9), and glucose is an aldohexose (see Fig. 5.6).

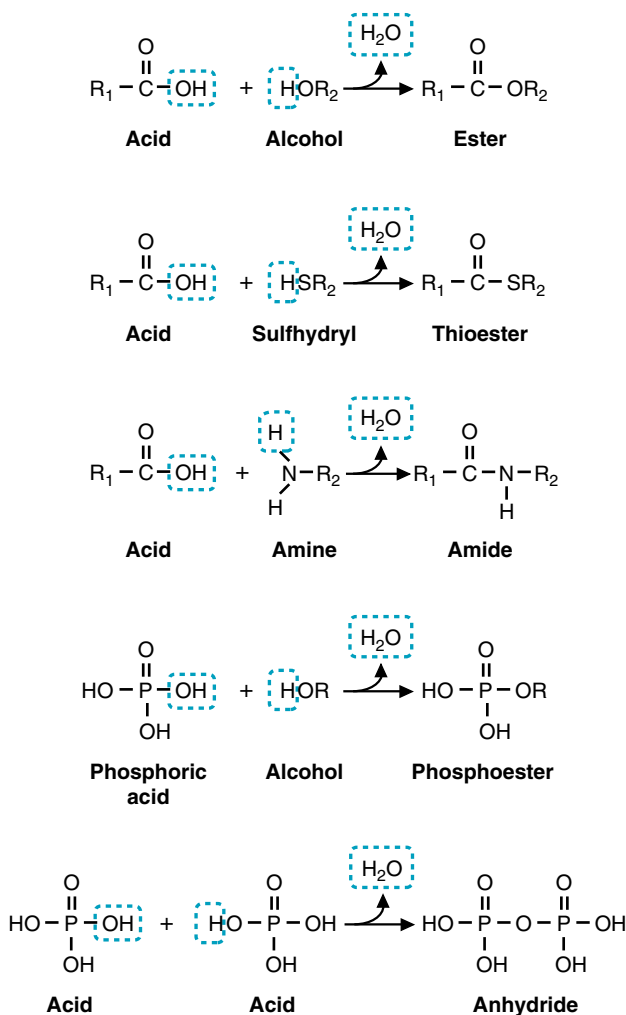
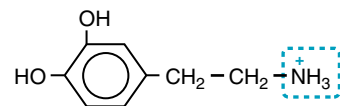
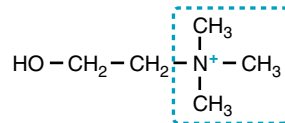


Fig. 5.7. Formation of esters, thioesters, amides, phosphoesters and anhydrides.



Dopamine (a primary amine)



Choline (a quaternary amine)

Fig. 5.4. Examples of amines. At physiologic pH, many amines carry positive charges.

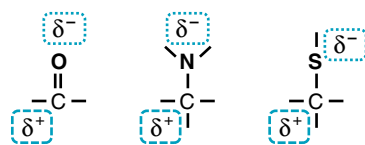
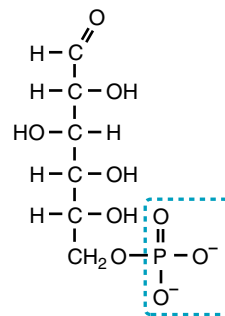


Fig. 5.5. Partial charges on carbon-oxygen, carbon-nitrogen, and carbon-sulfur bonds.



Glucose 6-phosphate

Fig. 5.6. Glucose 6-phosphate, a very polar and water-soluble molecule.

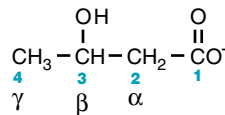


Fig. 5.8. Two systems for identifying the carbon atoms in a compound. This compound is called 3-hydroxybutyrate or β -hydroxybutyrate.

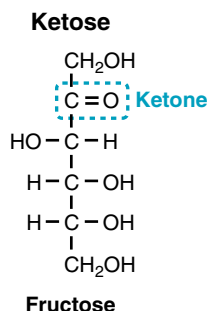


Fig. 5.9. Fructose is a ketohexose.



The stereospecificity of D-glucose is still frequently denoted in medicine by the use of its old name, dextrose. A solution used for intravenous infusions in patients is a 5% (5 g/100 mL) solution of dextrose.



Are D-mannose and D-galactose stereoisomers? Are they epimers of each other? (see Fig. 5.12)

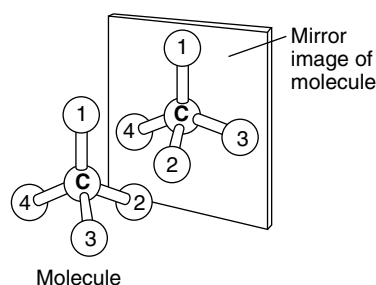
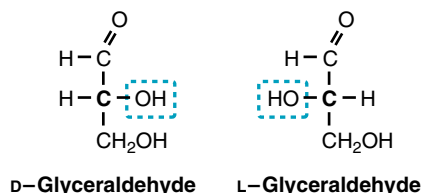


Fig. 5.10. D- and L-Glyceraldehyde. The carbon in the center contains four different substituent groups arranged around it in a tetrahedron. A different arrangement creates an isomer that is a nonsuperimposable mirror image. If you rotate the mirror image structure so that groups 1 and 2 align, group 3 will be in the position of group 4, and group 4 will be in position 3.

1. D- AND L-SUGARS

A carbon atom that contains four different chemical groups forms an asymmetric (or chiral) center (Fig. 5.10). The groups attached to the asymmetric carbon atom can be arranged to form two different isomers that are mirror images of each other and not superimposable. Monosaccharide stereoisomers are designated D or L based on whether the position of the hydroxyl group furthest from the carbonyl carbon matches D or L glyceraldehyde (Fig. 5.11). Although a more sophisticated system of nomenclature using the designations of (R) and (S) is generally used to describe the positions of groups on complex molecules such as drugs, the D and L designation is still used in medicine for describing sugars and amino acids. Because glucose (the major sugar in human blood) and most other sugars in human tissues belong to the D series, sugars are assumed to be D unless L is specifically added to the name.

2. STEREOISOMERS AND EPIMERS

Stereoisomers have the same chemical formula but differ in the position of the hydroxyl group on one or more of their asymmetric carbons (Fig. 5.12). A sugar with n asymmetric centers has 2^n stereoisomers unless it has a plane of symmetry. Epimers are stereoisomers that differ in the position of the hydroxyl group at only one of their asymmetric carbons. D-glucose and D-galactose are epimers of each other, differing only at position 4, and can be interconverted in human cells by enzymes called epimerases. D-mannose and D-glucose are also epimers of each other.

3. RING STRUCTURES

Monosaccharides exist in solution mainly as ring structures in which the carbonyl (aldehyde or ketone) group has reacted with a hydroxyl group in the same molecule to form a five- or six-membered ring (Fig. 5.13). The oxygen that was on the hydroxyl group is now part of the ring, and the original carbonyl carbon, which now contains a $-\text{OH}$ group, has become the anomeric carbon atom. An hydroxyl group on the anomeric carbon drawn down below the ring is in the α -position; drawn up above the ring, it is in the β position. In the actual three-dimensional structure, the ring is not planar but usually takes a “chair” conformation in which the hydroxyl groups are located at a maximal distance from each other.

In solution, the hydroxyl group on the anomeric carbon spontaneously (non-enzymatically) changes from the α to the β position through a process called mutarotation. When the ring opens, the straight chain aldehyde or ketone is formed. When the ring closes, the hydroxyl group may be in either the α or the β position (Fig. 5.14). This process occurs more rapidly in the presence of cellular enzymes called mutarotases. However, if the anomeric carbon forms a bond with another molecule, that bond is fixed in the α or β position, and the sugar cannot mutarotate. Enzymes are specific for α or β bonds between sugars and other molecules, and react with only one type.

4. SUBSTITUTED SUGARS

Sugars frequently contain phosphate groups, amino groups, sulfate groups or N-acetyl groups. Most of the free monosaccharides within cells are phosphorylated at their terminal carbons, which prevents their transport out of the cell (see glucose 6-phosphate in Fig. 5.6). Amino sugars such as galactosamine and glucosamine contain an amino group instead of a hydroxyl group on one of the carbon atoms, usually carbon 2 (Fig. 5.15). Frequently this amino group has been acetylated to form an N-acetylated sugar. In complex molecules termed proteoglycans, many of

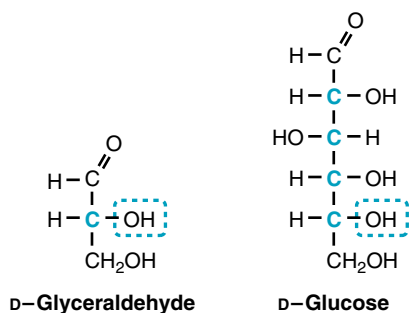


Fig. 5.11. D-Glyceraldehyde and D-glucose. These sugars have the same configuration at the asymmetric carbon atom farthest from the carbonyl group. Both belong to the D series. Asymmetric carbons are shown in blue.

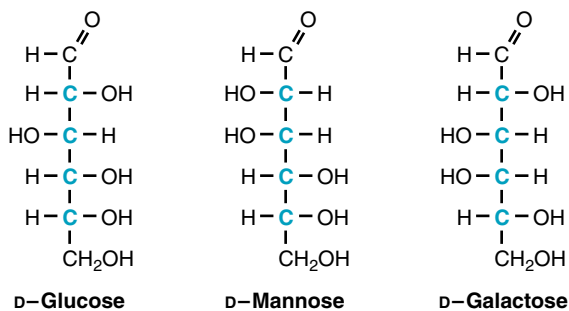


Fig. 5.12. Examples of stereoisomers. These compounds have the same chemical formula ($C_6H_{12}O_6$) but differ in the positions of the hydroxyl groups on their asymmetric carbons (in blue).

A: They are stereoisomers, but not epimers of each other. They have the same chemical formula, but differ in the position of two hydroxyl groups.

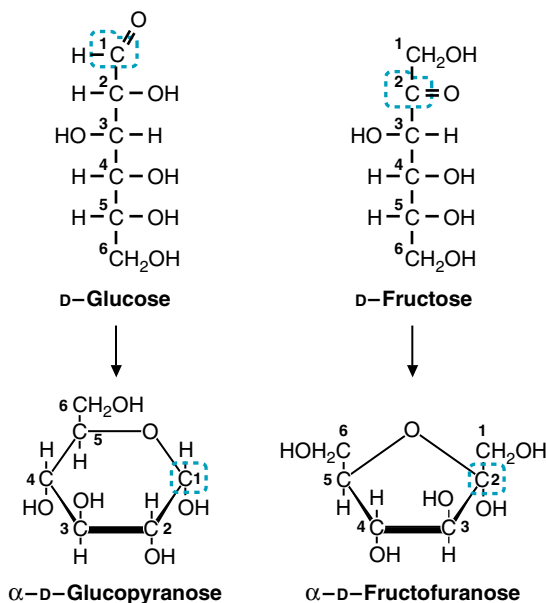


Fig. 5.13. Pyranose and furanose rings formed from glucose and fructose. The anomeric carbons are highlighted.

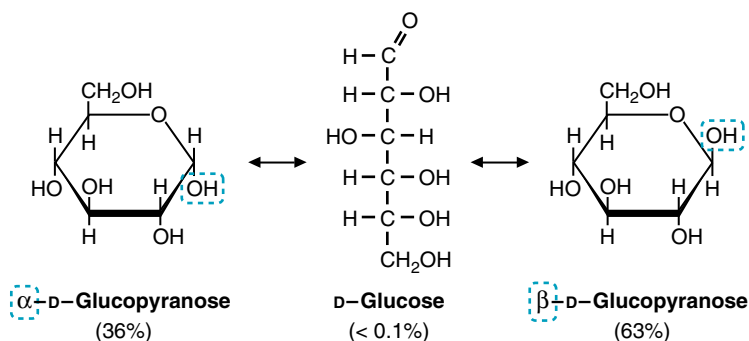


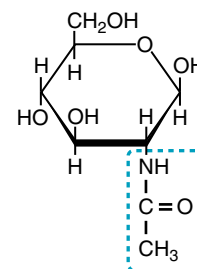
Fig. 5.14. Mutarotation of glucose in solution, with percentages of each form at equilibrium.

the *N*-acetylated sugars also contain negatively charged sulfate groups attached to a hydroxyl group on the sugar.

5. OXIDIZED AND REDUCED SUGARS

Sugars can be oxidized at the aldehyde carbon to form an acid. Technically the compound is no longer a sugar, and the ending on its name is changed from “-ose” to “onic acid” or “onate” (e.g., gluconic acid, Fig. 5.16). If the carbon containing the terminal hydroxyl group is oxidized, the sugar is called a uronic acid (e.g., glucuronic acid).

If the aldehyde of a sugar is reduced, all of the carbon atoms contain alcohol (hydroxyl) groups, and the sugar is a polyol (e.g., sorbitol)(see Fig. 5.16). If one of



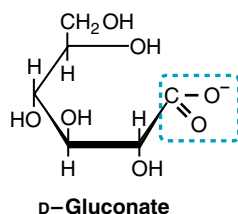
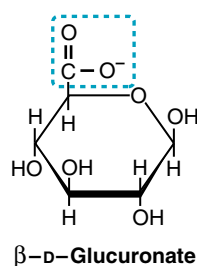
N-Acetyl- β -D-glucosamine

Fig. 5.15. An *N*-acetylated amino sugar. The *N*-denotes the amino group to which the acetyl group is attached, shown in the blue box.



Proteoglycans contain many long unbranched polysaccharide chains attached to a core protein. The polysaccharide chains, called glycosaminoglycans, are composed of repeating disaccharide units containing oxidized acid sugars (such as glucuronic acid), sulfated sugars, and *N*-acetylated amino sugars. The large number of negative charges causes the glycosaminoglycan chains to radiate out from the protein so that the overall structure resembles a bottlebrush. The proteoglycans are essential parts of the extracellular matrix, the aqueous humor of the eye, secretions of mucus-producing cells, and cartilage.

Oxidized sugars



Reduced sugars

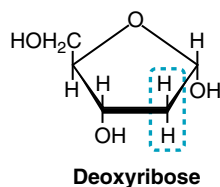
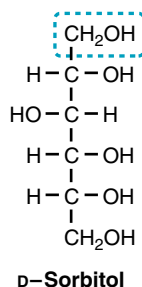


Fig. 5.16. Oxidized and reduced sugars. The affected group is shown in blue. Gluconic acid (D-gluconate) is formed by oxidation of the glucose aldehyde carbon. Glucuronic acid is formed by oxidation of the glucose terminal OH group. Sorbitol, a sugar alcohol, is formed by reduction of the glucose aldehyde group. Deoxyribose is formed by reduction of ribose.

the hydroxyl groups of a sugar is reduced so that the carbon contains only hydrogen, the sugar is a deoxysugar, such as the **deoxyribose** in DNA.

B. Glycosides

1. N- AND O-GLYCOSIDIC BONDS

The hydroxyl group on the anomeric carbon of a monosaccharide can react with an $-OH$ or an $-NH$ group of another compound to form a **glycosidic bond**. The linkage may be either α or β , depending on the position of the atom attached to the anomeric carbon of the sugar. *N*-glycosidic bonds are found in **nucleosides** and **nucleotides**. For example, in the adenosine moiety of ATP, the nitrogenous base adenine is linked to the sugar ribose through a β -*N*-glycosidic bond (Fig. 5.17). In contrast, *O*-glycosidic bonds, such as those found in lactose, join sugars to each other or attach sugars to the hydroxyl group of an amino acid on a protein.

2. DISACCHARIDES, OLIGOSACCHARIDES, AND POLYSACCHARIDES

A disaccharide contains two monosaccharides joined by an *O*-glycosidic bond. Lactose, which is the sugar in milk, consists of galactose and glucose linked through a $\beta(1 \rightarrow 4)$ bond formed between the β $-OH$ group of the anomeric carbon of galactose and the hydroxyl group on carbon 4 of glucose (see Fig. 5.17). Oligosaccharides contain from 3 to roughly 12 monosaccharides linked together. They are often found

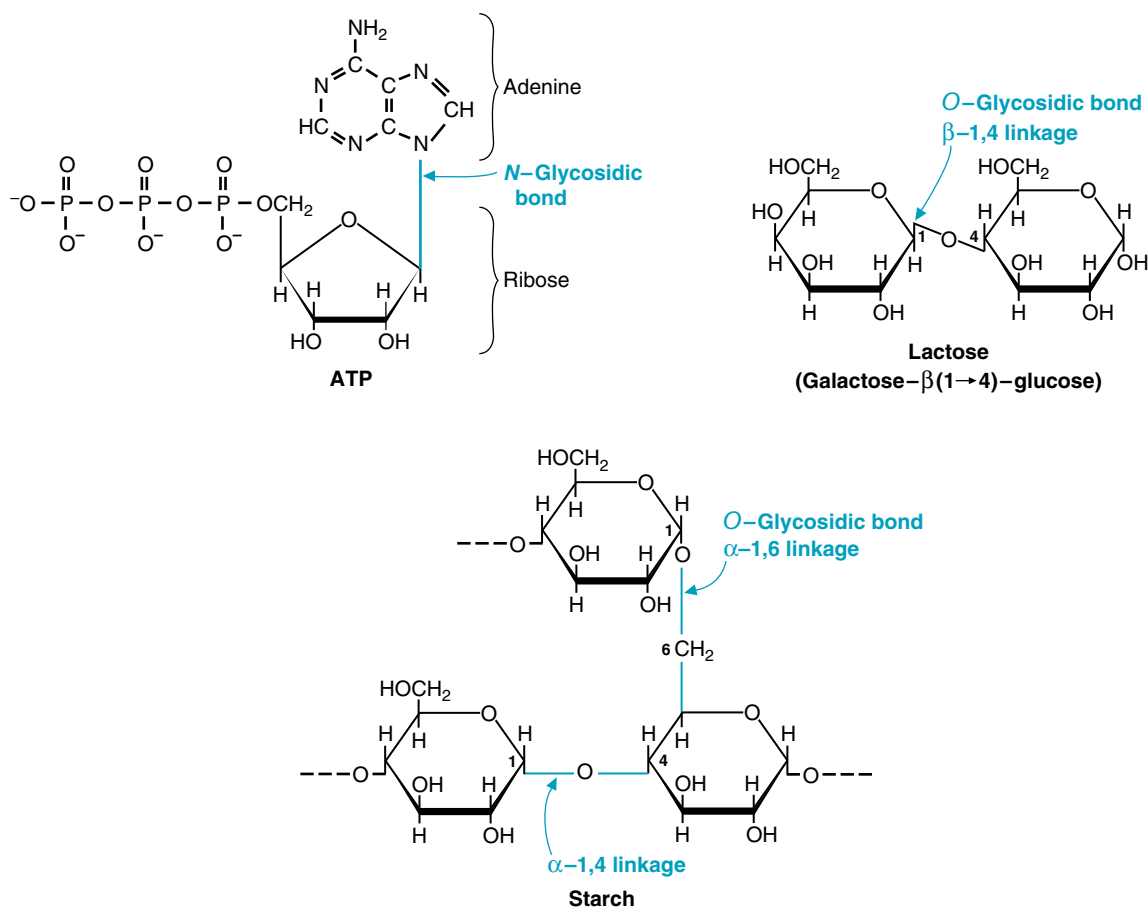


Fig. 5.17. *N*- and *O*-glycosidic bonds. Adenosine triphosphate (ATP) contains a β , *N*-glycosidic bond. Lactose contains an *O*-glycosidic $\beta(1 \rightarrow 4)$ bond. Glycogen contains α -1,4 and α -1,6 *O*-glycosidic bonds.

attached through *N*- or *O*-glycosidic bonds to proteins to form **glycoproteins** (see Chapter 6). Polysaccharides contain tens to thousands of monosaccharides joined by glycosidic bonds to form linear chains or branched structures. Amylopectin (a form of starch) and glycogen (the storage form of glucose in human cells) are branched polymers of glucosyl residues linked through $\alpha(1\rightarrow 4)$ and $\alpha(1\rightarrow 6)$ bonds.

III. LIPIDS

A. Fatty Acids

Fatty acids are usually straight aliphatic chains with a methyl group at one end (called the ω -carbon) and a carboxyl group at the other end (Fig. 5.18). Most fatty acids in the human have an even number of carbon atoms, usually between 16 and 20. Saturated fatty acids have single bonds between the carbons in the chain, and unsaturated fatty acids contain one or more double bonds. The most common saturated fatty acids present in the cell are palmitic acid (C16) and stearic acid (C18). Although these two fatty acids are generally called by their common names, shorter fatty acids are often called by the Latin word for the number of carbons, such as octanoic acid (8 carbons) and decanoic acid (10 carbons).

Monounsaturated fatty acids contain one double bond, and polyunsaturated fatty acids contain two or more double bonds (see Fig. 5.18). The position of a double bond is designated by the number of the carbon in the double bond that is closest to the carboxyl group. For example, oleic acid, which contains 18 carbons and a double bond between position 9 and 10, is designated 18:1, Δ^9 . The number 18 denotes the number of carbon atoms, 1 (one) denotes the number of double bonds, and Δ^9



The melting point of a fatty acid increases with chain length and decreases with the degree of saturation. Thus, fatty acids with many double bonds, such as those in vegetable oils, are liquid at room temperature and saturated fatty acids, such as those in butterfat, are solids. Lipids with lower melting points are more fluid at body temperature and contribute to the fluidity of our cellular membranes.

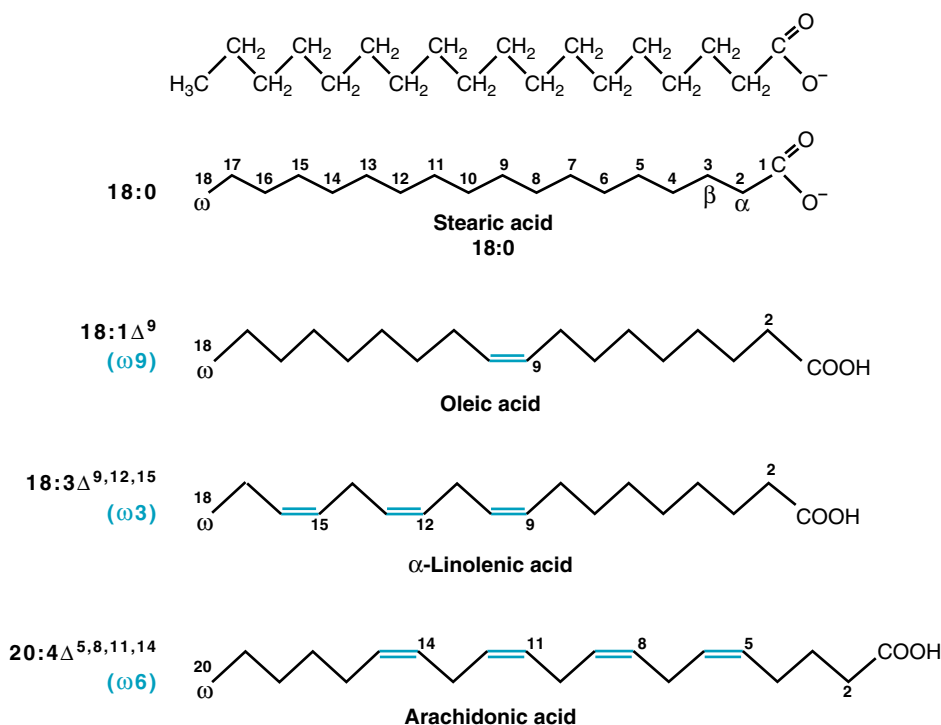


Fig. 5.18. Saturated fatty acids and unsaturated fatty acids. In stearic acid, the saturated fatty acid at the top of the figure, all the atoms are shown. A more common way of depicting the same structure is shown below. The carbons are either numbered starting with the carboxyl group or given Greek letters starting with the carbon next to the carboxyl group. The methyl (or ω) carbon at the end of the chain is always called the ω -carbon regardless of the chain length. 18:0 refers to the number of carbon atoms (18) and the number of double bonds (0). In the unsaturated fatty acids shown, not all of the carbons are numbered, but note that the double bonds are *cis* and spaced at three-carbon intervals. Both $\omega 3$ and $\omega 6$ fatty acids are required in the diet.



The eicosanoids are a group of hormone-like compounds produced by many cells in the body. They are derived from polyunsaturated fatty acids such as arachidonic acid that contain 20 carbons (eicosa) and have 3, 4, or 5 double bonds. The prostaglandins, thromboxanes, and leukotrienes belong to this group of compounds.



Palmitoleic acid, oleic acid, and arachidonic acid are the most common unsaturated fatty acids in the cell. Palmitoleic acid is a 16:1, Δ^9 fatty acid. How would you name it as an ω fatty acid?

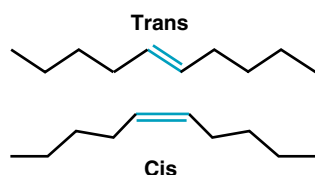


Fig. 5.19. *Cis* and *trans* double bonds in fatty acid side chains. Note that the *cis* double bond causes the chain to bend.

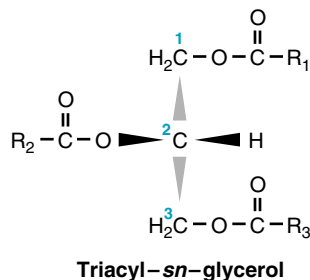


Fig. 5.20. A triacylglycerol. Note that carbons 1 and 3 of the glycerol moiety are not identical. The broad end of each arrowhead is closer to the reader than the narrow, pointed end.

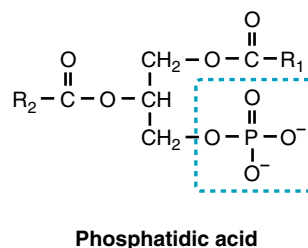
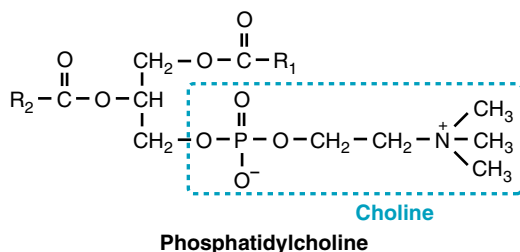


Fig. 5.21. Phosphoacylglycerols. Phospholipids found in membranes, such as phosphatidylcholine, have a polar group attached to the phosphate.

denotes the position of the double bond between the 9th and 10th carbon atoms. Oleic acid can also be designated 18:1(9), without the Δ . Fatty acids are also classified by the distance of the double bond closest to the ω end (the methyl group at the end farthest from the carboxyl group). Thus oleic acid is an ω 9 fatty acid, and linolenic acid is an ω 3 fatty acid. Arachidonic acid, a polyunsaturated fatty acid with 20 carbons and 4 double bonds, is an ω 6 fatty acid that is completely described as 20:4, $\Delta^{5,8,11,14}$.

The double bonds in most naturally occurring fatty acids are in the *cis* configuration (Fig. 5.19). The designation *cis* means that the hydrogens are on the same side of the double bond and the acyl chains on the other side. In *trans* fatty acids, the acyl chains are on opposite sides of the double bond. Margarine and the fat used in preparing French fries are probably the major sources of *trans* fatty acids found in humans. *Trans* fatty acids are produced by the chemical hydrogenation of polyunsaturated fatty acids in vegetable oils and are not a natural food product.

C. Acylglycerols

An **acylglycerol** comprises glycerol with one or more fatty acids (the **acyl** group) attached through ester linkages (Fig. 5.20). Monoacylglycerols, diacylglycerols, and triacylglycerols contain 1, 2, or 3 fatty acids esterified to glycerol, respectively. **Triacylglycerols** rarely contain the same fatty acid at all three positions and are therefore called mixed triacylglycerols. Unsaturated fatty acids, when present, are most often esterified to carbon 2. In the three-dimensional configuration of glycerol, carbons 1 and 3 are not identical, and enzymes are specific for one or the other carbon.

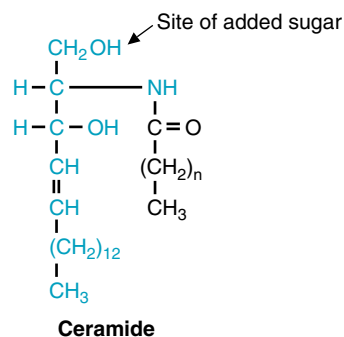
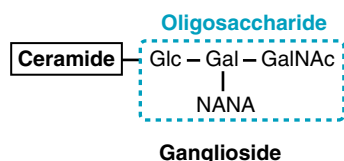
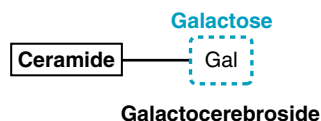
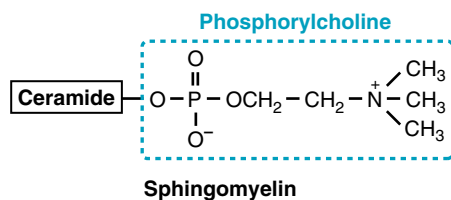
D. Phosphoacylglycerols

Phosphoacylglycerols contain fatty acids esterified to position 1 and 2 of glycerol and a phosphate (alone or with a substituent) attached to carbon 3. If only a phosphate group is attached to carbon 3, the compound is **phosphatidic acid** (see Fig. 5.21). Phosphatidic acid is a precursor for the synthesis of the other phosphoacylglycerols.

Phosphatidylcholine is one of the major phosphoacylglycerols found in membranes (see Fig. 5.21). The **amine** is positively charged at neutral pH, and the phosphate negatively charged. Thus, the molecule is amphipathic; it contains large polar and nonpolar regions. Phosphatidylcholine is also called lecithin. Removal of a fatty acyl group from a phosphoacyl glycerol leads to a lyso-lipid. For example, removing the fatty acyl group from lecithin forms lysolecithin.

E. Sphingolipids

Sphingolipids do not have a glycerol backbone; they are formed from sphingosine. (Fig. 5.22). Sphingosine is derived from **serine** and a specific fatty acid, palmitate. **Ceramides** are **amides** formed from sphingosine by attaching a fatty acid to the



A:

A: Palmitoleic acid is an $\omega 7$ fatty acid. It has one double bond between the 9th and 10th carbons. It has 16 carbons, like palmitic acid, so the double bond must be at the 7th carbon from the ω end.

Fig. 5.22. Sphingolipids, derivatives of ceramide. The structure of ceramide is shown at the bottom of the figure. The portion of ceramide shown in blue is sphingosine. The NH and OH were contributed by serine. Different groups are added to the hydroxyl group of ceramide to form sphingomyelin, galactocerebrosides, and gangliosides. NANA = N-acetylneuraminic acid, also called sialic acid; Glc = glucose; Gal = galactose; GalNAc = N-acetylgalactosamine.

amino group. Various sphingolipids are then formed by attaching different groups to the hydroxyl group on ceramide. As reflected in the names for **cerebrosides** and **gangliosides**, these sphingolipids contain sugars attached to the hydroxyl group of ceramide through glycosidic bonds. They are glycolipids (more specifically, glycosphingolipids). Sphingomyelin, which contains a phosphorylcholine group attached to ceramide, is a component of cell membranes and the myelin sheath around neurons.

F. Steroids

Steroids contain a four-ring structure called the steroid nucleus (Fig. 5.23). Cholesterol is the steroid precursor in human cells from which all of the steroid hormones are synthesized by modifications to the ring or C20 side chain. Although cholesterol is not very water soluble, it is converted to amphipathic water-soluble bile salts such as cholic acid. Bile salts line the surfaces of lipid droplets called micelles in the lumen of the intestine, where they keep the droplets emulsified in the aqueous environment.

Cholesterol is one of the compounds synthesized in the human from branched 5-carbon units with one double bond called an isoprenyl unit (see Fig. 5.1A). Isoprenyl

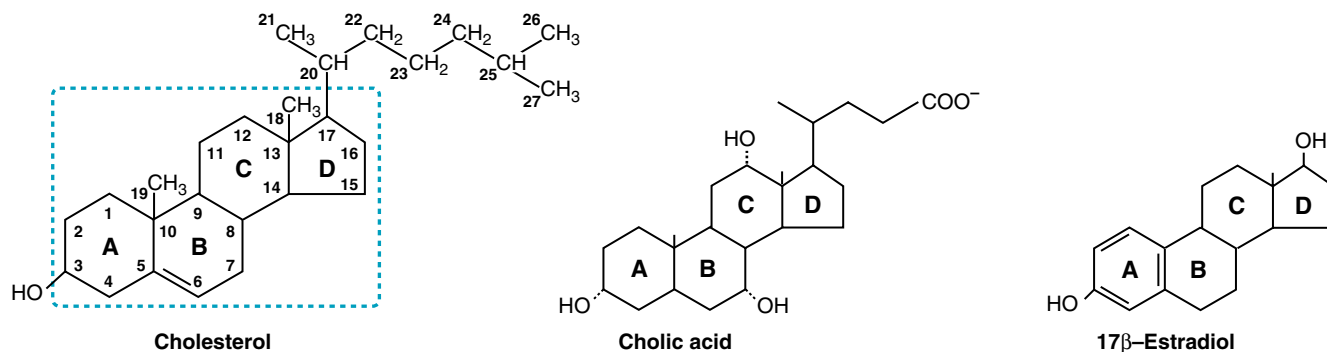


Fig. 5.23. Cholesterol and its derivatives. The steroid nucleus is shown in blue. The bile salt, cholic acid, and the steroid hormone 17β-estradiol are derived from cholesterol and contain the steroid ring structure.



What structural features account for the differences in the solubility of **cholesterol**, **estradiol** and **cholic acid** in the body? (see Fig. 5.23)

units are combined in long chains to form other structures, such as the side chains of coenzyme Q in humans and vitamin A in plants.

IV. NITROGEN-CONTAINING COMPOUNDS

Nitrogen, as described in Section IB2, is an electronegative atom with two unshared electrons in its outer valence shell. At neutral pH, the nitrogen in amino groups is usually bonded to four other atoms and carries a positive charge. However, the presence of nitrogen atom in an organic compound will increase its solubility in water, whether the nitrogen is charged or uncharged.

A. Amino Acids

Amino acids are compounds that contain an amino group and a carboxylic acid group. In proteins, the amino acids are always L-α amino acids (the amino group is attached to the α carbon in the L-configuration) (Fig. 5.24). These same amino acids also serve as precursors of nitrogen-containing compounds in the body, such as phosphatidylcholine (see Fig. 5.21) and are the basis of most human amino acid metabolism. However, our metabolic reactions occasionally produce an amino acid that has a β or γ amino group, such as the neurotransmitter γ-aminobutyric acid (see Fig. 5.24). However, only α amino acids are incorporated into proteins.

B. Nitrogen-Containing Ring Structures

1. PURINES, PYRIMIDINES AND PYRIDINES

Nitrogen is also a component of ring structures referred to as heterocyclic rings or nitrogenous bases. The three most common types of nitrogen-containing rings in the



Although D-amino acids are not usually incorporated into proteins in living organisms, they serve many other functions in bacteria, such as synthesis of cross-links in cell walls.

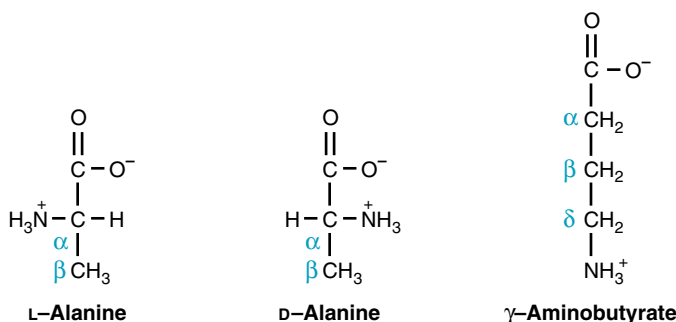


Fig. 5.24. The structure of amino acids.

body are **purines** (e.g., adenine), **pyrimidines** (e.g., thymine), and **pyridines** (e.g., the vitamins nicotinic acid, also called niacin, and pyridoxine, also called vitamin B₆) (Fig. 5.25). The suffix “**ine**” denotes the presence of nitrogen (**amine**) in the ring. The pyrimidine uracil is an exception to this general type of nomenclature. The utility of these nitrogen-containing ring structures lies in the ability of the nitrogen to form hydrogen bonds and to accept and donate electrons while still part of the ring. In contrast, the unsubstituted aromatic benzene ring, in which electrons are distributed equally among all six carbons (see Fig. 5.1), is nonpolar, hydrophobic, and relatively unreactive.

2. NUCLEOSIDES AND NUCLEOTIDES

Nitrogenous bases form nucleosides and nucleotides. A nucleoside consists of a nitrogenous base joined to a sugar, usually ribose or deoxyribose, through an *N*-glycosidic bond (see Fig. 5.17). If phosphate groups are attached to the sugar, the compound becomes a nucleotide. In the name of the nucleotide adenosine triphosphate (ATP), the addition of the ribose is indicated by the name change from adenine to **adenosine** (for the glycosidic bond). Monophosphate, diphosphate, or triphosphate are added to the name to indicate the presence of 1, 2, or 3 phosphate groups in the nucleotide. The structures of the nucleotides that serve as precursors of DNA and RNA are discussed in more detail in Section Three, Chapter 12.

3. TAUTOMERS

In many of the nitrogen-containing rings, the hydrogen can shift to produce a tautomer, a compound in which the hydrogen and double bonds have changed position (i.e., $\text{--N=C--OH} \rightarrow \text{--NH--C=O}$) (Fig. 5.26). Tautomers are considered the same compound, and the structure may be represented either way. Generally one tautomeric form is more reactive than the other. For example, in the two tautomeric forms of uric acid, a proton can dissociate from the enol form to produce urate.

V. FREE RADICALS

Radicals are compounds that have a single electron, usually in an outer orbital. Free radicals are radicals that exist independently in solution or in a lipid environment.

Q: Compounds that cannot be oxidized as fuels in the human are often excreted in the urine. Chemical modifications often occur in the liver, kidney, or other tissues that inactivate or detoxify the chemicals, make them more water-soluble, or otherwise target such molecules for excretion. Uric acid, the basis of **Lotta Topaigne's** pain, is excreted in the urine (see Fig. 5.26). Judging from the similarity in structure, do you think it is derived from the degradation of purines, pyrimidines, or pyridines?

A: Cholesterol is composed almost entirely of CH_2 groups and is therefore water-insoluble. Estradiol is likewise relatively water-insoluble. However, cholic acid contains a hydrophilic carboxyl group, and three hydroxyl groups. As shown by the dashed lines, the three hydroxyl groups all lie on one side of the molecule, thus creating a hydrophilic surface.

Lotta Topaigne's gout is caused by depositions of monosodium urate crystals in the joint of her big toe. At a blood pH of 7.4, all of the uric acid has dissociated a proton to form urate, which is not very water-soluble and forms crystals of the Na^+ salt. In the more acidic urine generated by the kidney, the acidic form, uric acid, may precipitate to form kidney stones.

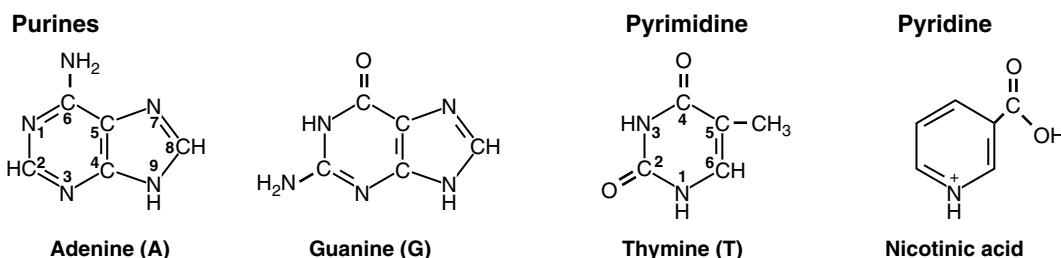


Fig. 5.25. The nitrogenous bases.

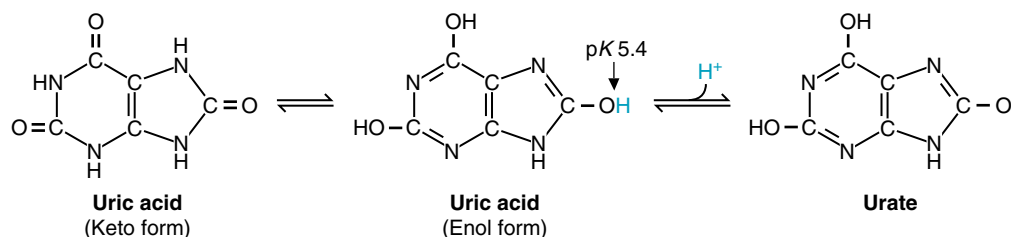


Fig. 5.26. Tautomers of uric acid. The tautomeric form affects the reactivity. The enol form dissociates a proton to form urate.



Free radicals are not just esoteric reactants; they are the agents of cell death and destruction. They are involved in all chronic disease states (e.g., coronary artery disease, diabetes mellitus, arthritis, and emphysema) as well as acute injury (e.g., radiation, strokes, myocardial infarction, and spinal cord injury). Through free radical defense mechanisms in our cells, we can often restrict the damage attributed to the “normal” aging process.



A: Uric acid contains two rings and is similar to the purines adenine and guanine (adenine is shown in Fig. 5.25). In fact, it is the urinary excretion product formed from the oxidation of these two purine bases. It is not very soluble in water, particularly if the pH is near the pK_a of its acidic OH group. If present in excess amounts, Na^+ urate tends to precipitate in joints, causing the severe pain of gout experienced by **Ms. Topaigne**.



The reducing sugar test. The reducing sugar test was used for detection of sugar in the urine long before specific enzymatic assays for glucose and galactose became available. In this test, the aldehyde group of a sugar is oxidized as it donates electrons to copper; the copper becomes reduced and produces a blue color. In alkaline solution, keto sugars (e.g., fructose) also react in this test because they form tautomers that are aldehydes. Ring structures of sugars will also react, but only if the ring can open (i.e., it is not attached to another compound through a glycosidic bond). Until a specific test for fructose becomes available, a congenital disease resulting in the presence of fructose in the urine is indicated by a positive reducing sugar test and negative results in the specific enzymatic assays for glucose or galactose.

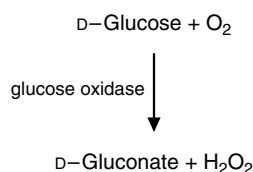


Fig. 5.27. Glucose oxidase.

Although many enzymes generate radicals as intermediates in reactions, these are not usually released into the cell to become free radicals.

Many of the compounds in the body are capable of being converted to free radicals by natural events that remove one of their electrons, or by radiation. Radiation, for example, dissociates water into the hydrogen atom and the hydroxyl radical:



In contrast, water normally dissociates into a proton and the negatively charged hydroxyl ion. The hydroxyl radical forms organic radicals by taking one electron (as $\text{H}\cdot$) from a compound such as an unsaturated membrane lipid, which then has a single unpaired electron and is a new radical.

Compounds that are radicals may be written with, or without, the radical showing. For example, nitrogen dioxide, a potent, reactive, toxic radical present in smog and cigarette smoke, may be designated in medical and lay literature as NO_2 rather than $\text{NO}_2\cdot$. Superoxide, a radical produced in the cell and that is the source of much destruction, is correctly written as the superoxide anion, O_2^- . However, to emphasize its free radical nature, the same compound is sometimes written as $\text{O}_2^{\cdot-}$. If a compound is designated as a radical in the medical literature, you can be certain that it is a reactive radical, and that its radical nature is important for the pathophysiology under discussion. (Reactive oxygen and nitrogen-containing free radicals are discussed in more detail in Chapter 24).

CLINICAL COMMENTS



Dianne Abietes. The severity of clinical signs and symptoms in patients with diabetic ketoacidosis (DKA), such as **Di Abietes**, is directly correlated with the concentration of ketone bodies in the blood. Direct quantitative methods for measuring acetoacetate and β -hydroxybutyrate are not routinely available. As a result, clinicians usually rely on semiquantitative reagent strips (Ketostix, Bayer Diagnostics, Mishawaha, IN) or tablets (Acetest, Bayer Diagnostics, Mishawaha, IN) to estimate the level of acetoacetate in the blood and the urine. The nitroprusside on the strips and in the tablets reacts with acetoacetate and to a lesser degree with acetone (both of which have ketone groups), but does not react with β -hydroxybutyrate (which does not have a ketone group). β -hydroxybutyrate is the predominant ketone body present in the blood of a patient in DKA, and its concentration could decline at a disproportionately rapid rate compared with that of acetoacetate and acetone. Therefore, tests employing the nitroprusside reaction to monitor the success of therapy in such a patient may be misleading.

In contrast to the difficulty of ketone body measurements, diabetic patients can self-monitor blood glucose levels at home, thereby markedly decreasing the time and expense of the many blood glucose determinations they need. Blood obtained from a finger prick is placed on the pad of a plastic strip. The strip has been impregnated with an enzyme (usually the bacterial enzyme glucose oxidase) that specifically converts the glucose in the blood to a compound (hydrogen peroxide, H_2O_2) that reacts with a dye to produce a color (Fig. 5.27). The intensity of the color, which is directly proportionate to the concentration of glucose in the patient's blood, is read on an instrument called a blood glucose monitor.



Lotta Topaigne. Ms. Topaigne has acute gouty arthritis (podagra) involving her right great toe. Polarized light microscopy of the fluid aspirated from the joint space showed crystals of monosodium urate phagocytosed by white blood cells. The presence of the relatively insoluble urate crystals within the joint space activates an inflammatory cascade leading to the classic components of joint inflammation (pain, redness, warmth, swelling, and limitation of joint motion).

BIOCHEMICAL COMMENTS

**Chlorinated Aromatic Hydrocarbon Environmental**

Toxins. As a result of human endeavor, toxic compounds containing chlorinated benzene rings have been widely distributed in the environment.

The pesticide DDT and the class of chemicals called dioxins provide examples of chlorinated aromatic hydrocarbons and structurally related compounds that are very hydrophobic and poorly biodegraded (Fig. 5.28). As a consequence of their persistence and lipophilicity, these chemicals are concentrated in the adipose tissue of fish, fish-eating birds, and carnivorous mammals, including humans.

DDT, a chlorinated biphenyl, was widely used in the United States as an herbicide between the 1940s and 1960s (see Fig. 5.28). Although it has not been used in this country since 1972, the chlorinated benzene rings are resistant to biodegradation, and U.S. soil and water are still contaminated with small amounts. DDT is still used in other parts of the world. Because this highly lipophilic molecule is stored in the fat of animals, organisms accumulate progressively greater amounts of DDT at each successive stage of the food chain. Fish-eating birds, one of the organisms at the top of the food chain, have declined in population because of the effect of DDT on the thickness of their eggshells. DDT is not nearly as toxic in the human, although long-term exposure or exposure to high doses may cause reversible neurologic symptoms, hepatotoxic effects, or cancer.

Dioxins, specifically chlorinated dibenzo-p-dioxins (CDDs), constitute another class of environmental toxins that are currently of great concern (see Fig. 5.28). They have been measured at what is termed background levels in the blood, adipose tissue, and breast milk of all humans tested. CDDs are formed as a byproduct during the production of other chlorinated compounds and herbicides, and from the chlorine bleaching process used by pulp and paper mills. They are released during the incineration of industrial, municipal, and domestic waste, and during the combustion of fossil fuels, and are found in cigarette smoke and the exhaust from gasoline and diesel fuels. They can also be formed from the combustion of organic matter during forest fires. They enter the atmosphere as particulate matter, are vaporized, and can spread large distances to enter soil and water.

As humans at the top of the food chain, we have acquired our background levels of dioxins principally through the consumption of food, primarily meat, dairy products,



Most of this chapter has dealt with the names and structures of compounds that are nutrients for the human or metabolites that can be produced from reactions in the human body. However, our health is also affected by naturally occurring and man-made xenobiotic compounds (compounds that have no nutrient value and are not produced in the human) that we ingest, inhale, or absorb through our skin. DDT and dioxins provide examples of chlorinated aromatic hydrocarbons, an important class of manmade chemicals present in the environment.



The accumulation of DDT in adipose tissue may be protective in the human because it decreases the amount of DDT available to pass through nonpolar lipid membranes to reach neurons in the brain, or pass through placental membranes to reach the fetus. Eventually we convert DDT to more polar metabolites that are excreted in the urine. However, some may pass with lipid into the breast milk of nursing mothers.

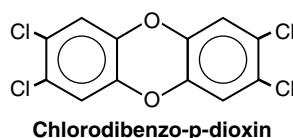
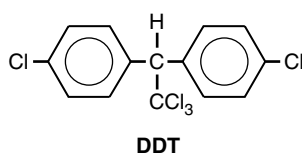


Fig. 5.28. Environmental toxins. Dichlorodiphenyl trichlorethane (DDT) is a member of a class of aromatic hydrocarbons that contain two chlorinated benzene (phenyl) rings joined by a chlorinated ethane molecule. Chlorodibenzo-p-dioxin's (CDDs) are a related class of more than 75 chlorinated hydrocarbons that all contain a dibenzo-p-dioxin (DD) molecule comprising two benzene rings joined via two oxygen bridges at adjacent carbons on each of the benzene rings. 2,3,7,8 Tetrachlorodibenzo-p-dioxin, shown above, is one of the most toxic and the most extensively studied. Chlorinated dibenzofurans (CDF) are structurally and toxicologically related.



Most of what is known about the toxicity of dioxins in the human comes from individuals exposed incidentally or chronically to higher levels (e.g., industrial accidents or presence in areas sprayed with Agent Orange or other herbicides contaminated with dioxins.). The lowest dose effects are probably associated with thymic atrophy and decreased immune response, chloracne and related skin lesions, and neoplasia (cancer). Dioxins can cross into the placenta to cause developmental and reproductive effects, decreased prenatal growth, and prenatal mortality.

and fish. Once in the human body, dioxins are stored in human fat and adipose tissue, and have an average half-life of approximately 5 to 15 years. They are unreactive, poorly degraded, and not readily converted to more water-soluble compounds that can be excreted in the urine. They are slowly excreted in the bile and feces, and together with lipids enter the breast milk of nursing mothers.

Suggested References

Nomenclature of chemical compounds. Most undergraduate organic textbooks provide a more detailed account of the nomenclature used for organic molecules, including the R,S nomenclature for chiral centers.

Environmental toxins. Good literature reviews for many environmental toxins are published by the U.S. Department of Health and Human Services; Public Health Service Agency for Toxic Substances and Disease Registry, including: Toxicological Profile for Chlorinated Dibenzo-p-dioxins, 1998, and the Toxicological Profile for 4,4'-DDT, 4,4'-DDE, 4,4'-DDD (Update), 1994.



REVIEW QUESTIONS—CHAPTER 5

Directions: Select the single best answer for each of the questions below. Base your answers on your knowledge of nomenclature. You need not recognize any of the structures shown to answer the questions.

- Which of the following is a universal characteristic of water-soluble organic compounds?
 - They are composed of carbon and hydrogen atoms.
 - They must contain a group that has a full negative charge.
 - They must contain a group that has a full positive charge.
 - They contain polar groups that can hydrogen bond with water.
 - They contain aromatic groups.
- $\text{CH}_2\text{OH}-\text{CH}_2-\text{CH}_2-\text{COO}^-$

A patient was admitted to the hospital emergency room in a coma. Laboratory tests found high levels of the compound shown above in her blood. On the basis of its structure (and your knowledge of the nomenclature of functional groups), you identify the compound as

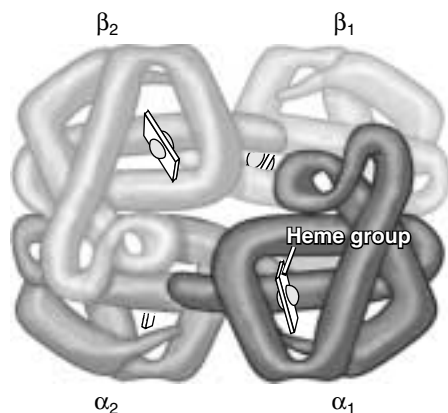
 - methanol (wood alcohol).
 - ethanol (alcohol).
 - ethylene glycol (antifreeze).
 - β -hydroxybutyrate (a ketone body).
 - γ -hydroxybutyrate (the “date rape” drug).
- A patient was diagnosed with a deficiency of the lysosomal enzyme α -glycosidase. The name of the deficient enzyme suggests that it hydrolyzes a glycosidic bond, which is a bond formed
 - through multiple hydrogen bonds between two sugar molecules.
 - between the anomeric carbon of a sugar and an O–H (or N) of another molecule.
 - between two anomeric carbons in polysaccharides.
 - internally between the anomeric carbon of a monosaccharide and its own 5th carbon hydroxyl group.
 - between the carbon containing the aldol or keto group and the α carbon.

4. A patient was diagnosed with a hypertriglyceridemia. This condition is named for the high blood levels of lipids composed of
- (A) 3 fatty acyl groups attached to a glycerol backbone.
 - (B) a glycerol lipid containing a phosphorylcholine group.
 - (C) a sphingolipid containing three fatty acyl groups.
 - (D) three glycerol moieties attached to a fatty acid.
 - (E) three glyceraldehyde moieties attached to a fatty acid.
5. A patient was diagnosed with a sphingolipidoses, which are congenital diseases involving the inability to degrade sphingolipids. All sphingolipids have in common
- (A) a glycerol backbone.
 - (B) ceramide.
 - (C) phosphorylcholine.
 - (D) N-acetylneuraminic acid (NANA).
 - (E) a steroid ring structure to which sphingosine is attached.

6 Amino Acids in Proteins



The genetic code is the sequence of three bases (nucleotides) in DNA containing the information for the linear sequence of amino acids in a polypeptide chain (its primary structure). A gene is the portion of DNA that encodes a functional product, such as a polypeptide chain. Mutations, which are changes in the nucleotides in a gene, result in a change in the products of that gene that may be inherited. The genetically inherited disease sickle cell anemia is, for example, caused by a mutation in the gene encoding one of the subunits of hemoglobin. Hemoglobin is the protein present in red blood cells that reversibly binds O_2 and transports it to tissues. The adult hemoglobin protein comprises four polypeptide chains, 2α and 2β . The α and β subunits differ in primary structure (i.e., they have different sequences of amino acids and are encoded by different genes). Sickle cell anemia is caused by a mutation in DNA that changes just one amino acid in the hemoglobin β chains from a glutamic acid to a valine.



Proteins have many functions in the body. They serve as transporters of hydrophobic compounds in the blood, as cell adhesion molecules that attach cells to each other and to the extracellular matrix, as hormones that carry signals from one group of cells to another, as ion channels through lipid membranes, and as enzymes that increase the rate of biochemical reactions. The unique characteristics of a protein are dictated by its linear sequence of amino acids, termed its primary structure. The primary structure of a protein determines how it can fold and how it interacts with other molecules in the cell to perform its function. The primary structures of all of the diverse human proteins are synthesized from 20 amino acids arranged in a linear sequence determined by the genetic code.

General properties of amino acids. Each of the amino acids used for protein synthesis has the same general structure (Fig. 6.1). It contains a **carboxylic acid** group, an **amino group** attached to the α -carbon in an **L configuration**, a **hydrogen atom**, and a chemical group called a **side chain** that is different for each amino acid. In solution, the free amino acids exist as **zwitterions**, ions in which the amino group is positively charged and the carboxylate group is negatively charged. In proteins, these amino acids are joined into linear polymers called **polypeptide chains** through **peptide bonds** between the carboxylic acid group of one amino acid and the amino group of the next amino acid.

Classification of amino acids according to chemical properties of the side chains. The chemical properties of the side chain determine the types of bonds and interactions each amino acid in a polypeptide chain can make with other molecules. Thus, amino acids are often grouped by polarity of the side chain (**charged**, **nonpolar hydrophobic**, or **uncharged polar**) or by structural features (**aliphatic**, **cyclic**, or **aromatic**). The side chains of the **nonpolar hydrophobic** amino acids (alanine, valine, leucine, isoleucine, phenylalanine, and methionine) cluster together to exclude water in the **hydrophobic effect**. The **uncharged polar** amino acids (serine, threonine, tyrosine, asparagine, and glutamine) participate in **hydrogen bonding**. Cysteine, which contains a sulfhydryl group, forms **disulfide bonds**. The negatively charged **acidic** amino acids (aspartate and glutamate) form **ionic (electrostatic) bonds** with positively charged molecules, such as the **basic** amino acids (lysine, arginine, and histidine). The charge on the amino acid at a particular pH is determined by the pK_a of each group that has a dissociable proton.

Amino acid substitutions in the primary structure. Mutations in the genetic code result in proteins with an altered **primary structure**. Mutations resulting in single amino acid substitutions can affect the functioning of a protein or can confer an advantage specific to a tissue or a set of circumstances. Many proteins such as **hemoglobin** exist in the human population as **polymorphisms** (genetically determined variations in primary structure.)

Within the same individual, the primary structure of many proteins varies with the stage of development and is present in **fetal** and **adult isoforms**, such as fetal and adult hemoglobin. The primary structure of some proteins, such as **creatine kinase**, can also vary between tissues (**tissue-specific isozymes**) or between intracellular locations in the same tissue. **Electrophoretic separation** of tissue-specific isozymes has been useful in medicine as a means of identifying the tissue site of injury.

Modified amino acids. In addition to the amino acids encoded by DNA that form the primary structure of proteins, many proteins contain specific amino acids that have been modified by **phosphorylation**, **oxidation**, **carboxylation**, or other reactions. When these reactions are enzyme-catalyzed, they are referred to as **post-translational modifications**.

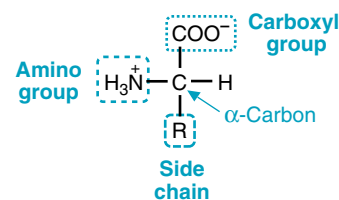


Fig. 6.1. General structure of the amino acids found in proteins.



THE WAITING ROOM



Will Sichel is a 17-year-old boy who came to the hospital emergency room with severe pain in his lower back, abdomen, and legs, which began after a 2-day history of nausea and vomiting caused by gastroenteritis. He was diagnosed as having sickle cell disease at age 3 years and has been admitted to the hospital on numerous occasions for similar vaso-occlusive sickle cell crises.

On admission, the patient's hemoglobin level in peripheral venous blood was 7.8 g/dL (reference range = 12–16 g/dL). The hematocrit or packed cell volume (the percentage of the total volume of blood made up by red blood cells) was 23.4% (reference range = 41–53%). His serum total bilirubin level (a pigment derived from hemoglobin degradation) was 2.3 mg/dL (reference range = 0.2–1.0 mg/dL). An x-ray of his abdomen showed radiopaque stones in his gallbladder. With chronic hemolysis (red blood cell destruction), the amount of heme degraded to bilirubin is increased. These stones are the result of the chronic excretion of excessive amounts of bilirubin from the liver into the bile, leading to bilirubinate crystal deposition in the gallbladder lumen.



Cal Kulis is an 18-year-old boy who was brought to the hospital by his mother because of the sudden onset of severe pain in the left flank radiating around his left side toward his pubic area. His urine was reddish-brown in color, and his urinalysis showed the presence of many red blood cells. When his urine was acidified with acetic acid, clusters of flat hexagonal transparent crystals of cystine were noted. An x-ray of his abdomen showed radiopaque calculi (stones) in both kidneys. There was no family history of kidney stone disease.



Di Abietes, who has type 1 diabetes mellitus, was giving herself subcutaneous injections of insulin regular NPH beef insulin twice daily after her disease was first diagnosed (see Chapters 4 and 5). Subsequently, her physician switched her to synthetic human insulin. At this visit, her physician changed her insulin therapy and has written a prescription for Humalog mix 75/25 (Eli Lilly, Indianapolis, IN), a mixture of Humalog in protamine suspension and unbound Humalog insulin (25%).



Ann Jeina is a 54-year-old woman who is 68 inches tall and weighs 198 lb. She has a history of high blood pressure and elevated serum cholesterol levels. After a heated argument with a neighbor, Mrs. Jeina experienced a “tight pressure-like band of pain” across her chest, associated with shortness of breath, sweating, and a sense of light-headedness.

After 5 hours of intermittent chest pain, she went to the hospital emergency room, where her electrocardiogram showed changes consistent with an acute infarction of the anterior wall of her heart. She was admitted to the cardiac care unit. Blood was sent to the laboratory for various tests, including the total creatine kinase (CK) level and the MB (“muscle–brain”) fraction of CK in the blood.



The term *calculus* is used to describe any abnormal concretion (concrete-like precipitate) of mineral salts. These almost always form within the cavity of a hollow organ, such as the kidney (kidney or renal stones) or the lumen of a duct (e.g., common bile duct stones).



The term *angina* describes a crushing or compressive pain. The term *angina pectoris* is used when this pain is located in the center of the chest, often radiating to the neck or arms. The most common mechanism for the latter symptom is a decreased supply of oxygen to the heart muscle caused by atherosclerotic coronary artery disease, which results in obstruction of the vessels that supply arterial blood to cardiac muscle.

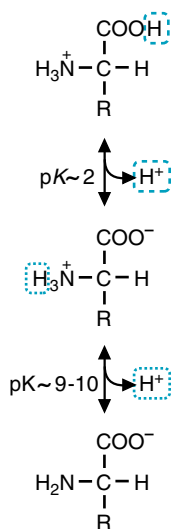


Fig. 6.2. Dissociation of the α -carboxyl and α -amino groups of amino acids. At physiologic pH (~ 7), a form in which both the α -carboxyl and α -amino groups are charged predominates. Some amino acids also have ionizable groups on their side chains.



The L-stereospecificity is often incorporated into the names of drugs derived from amino acids either as “L” or “levo” for the direction in which L amino acids rotate polarized light. For example, patients with Parkinson’s disease are treated with L-DOPA (a derivative of the amino acid L-tyrosine), and hypothyroid patients are treated with levothyroxine (a different derivative of L-tyrosine).

I. GENERAL STRUCTURE OF THE AMINO ACIDS

Twenty different amino acids are commonly found in proteins. They are all α -amino acids, amino acids in which the amino group is attached to the α -carbon (the carbon atom next to the carboxylate group) (see Fig 6.1). The α -carbon has two additional substituents, a hydrogen atom and an additional chemical group called a side chain ($-R$). The side chain is different for each amino acid.

At a physiologic pH of 7.4, the amino group on these amino acids carries a positive charge, and the carboxylic acid group is negatively charged (Fig. 6.2). The pK_a of the primary carboxylic acid groups for all of the amino acids is approximately 2 (1.8–2.4). At pH values much lower than the pK_a (higher hydrogen ion concentrations), all of the carboxylic acid groups are protonated. At the pK_a , 50% of the molecules are dissociated into carboxylate anions and protons, and at a pH of 7.4, more than 99% of the molecules are dissociated (see Chapter 4). The pK_a for all of the α -amino groups is approximately 9.5 (8.8–11.0), so that at the lower pH of 7.4, most of the amino groups are fully protonated and carry a positive charge. The form of an amino acid that has both a positive and a negative charge is called a zwitterion. Because these charged chemical groups can form hydrogen bonds with water molecules, all of these amino acids are water-soluble at physiologic pH.

In all of the amino acids but glycine, the α -carbon is an asymmetric carbon atom that has four different substituents and can exist in either the D or L configuration (Fig. 6.3). The amino acids in mammalian proteins are all L-amino acids represented with the amino group to the left if the carboxyl group is at the top of the structure. These same amino acids serve as precursors of nitrogen-containing compounds synthesized in the body, and thus human amino acid metabolism is also centered on L-amino acids. The amino acid glycine is neither D nor L because the α -carbon atom contains two hydrogen atoms.

The chemical properties of the amino acids give each protein its unique characteristics. Proteins are composed of one or more linear polypeptide chains containing

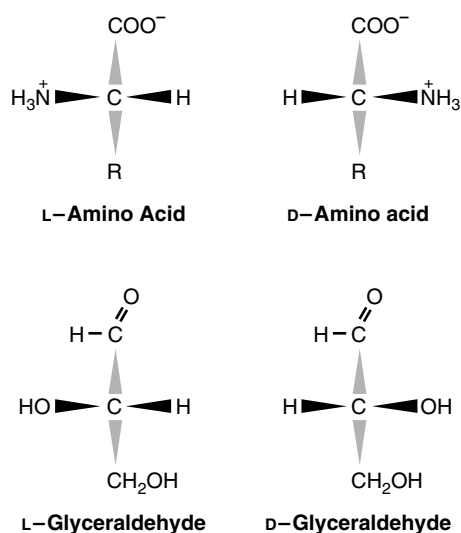


Fig. 6.3. L- and D-amino acids. The L forms are the only ones found in human proteins. Bonds coming out of the paper are shown by black arrows; those going in, by shaded arrows. The α -amino groups and H-atoms come toward the reader, and the α -carboxyl and side chains go away. The L and D forms are mirror images that cannot be superimposed by rotating the molecule. The reference for the L and D forms are the stereoisomers of glyceraldehyde.

hundreds of amino acids. The sequence of amino acids, termed the primary structure, is determined by the genetic code for the protein. In the polypeptide chains, amino acids are joined through peptide bonds between the carboxylic acid of one amino acid and the amino group of the adjacent amino acid (Fig. 6.4). Thus, the amino group, the α -carbon, and the carboxyl groups form the peptide backbone, and the side chains of the amino acids extend outward from this backbone. The side chains interact with the peptide backbone of other regions of the chain or with the side chains of other amino acids in the protein to form hydrophobic regions, electrostatic bonds, hydrogen bonds, or disulfide bonds. These interactions dictate the folding pattern of the molecule. The three-dimensional folding of the protein forms distinct regions called binding sites that are lined with amino acid side chains that interact specifically with another molecule termed a ligand (such as the heme in hemoglobin). Thus, the chemical properties of the side chains determine how the protein folds, how it binds specific ligands, and how it interacts with its environment (such as the aqueous medium of the cytoplasm).

The names of the different amino acids have been given three-letter and one-letter abbreviations (Table 6.1). The three-letter abbreviations use the first two letters in the name plus the third letter of the name or the letter of a characteristic sound, such as trp for tryptophan. The one-letter abbreviations use the first letter of the name of the most frequent amino acid in proteins (such as an “A” for alanine). If the first letter has already been assigned, the letter of a characteristic sound is used (such as an “R” for arginine). Single-letter abbreviations are usually used to denote the amino acids in a polypeptide sequence.

II. CLASSIFICATION OF AMINO ACID SIDE CHAINS

In Figure 6.5, the 20 amino acids used for protein synthesis are grouped into different classifications according to the polarity and structural features of the side chains. These groupings can be helpful in describing common functional roles or metabolic pathways of the amino acids. However, some amino acid side chains fit into a number of different classifications and are therefore grouped differently in different textbooks. Two of the characteristics of the side chain that are useful for classification are its pK_a and its hydropathic index, shown in Table 6.2. The hydropathic index is a scale used to denote the hydrophobicity of the side chain; the more positive the hydropathic index, the greater the tendency to cluster with other nonpolar molecules and exclude water in the hydrophobic effect. The more negative the hydropathic index of an amino acid, the more hydrophilic is its side chain.

A. Nonpolar, Aliphatic Amino Acids

Glycine is the simplest amino acid, and it really does not fit well into any classification because its side chain is only a hydrogen atom. Alanine and the branched chain amino acids (valine, leucine, and isoleucine) have bulky, nonpolar, aliphatic side chains. The high degree of hydrophobicity of the branched chain amino acid side chains is denoted by their high hydropathic index (see Table 6.2). Electrons are shared equally between the carbon and hydrogen atoms in these side chains, so that they cannot hydrogen bond with water. Within proteins, these amino acid side chains will cluster together to form hydrophobic cores. Their association is also promoted by van der Waals forces between the positively charged nucleus of one atom and the electron cloud of another. This force is effective over short distances when many atoms pack closely together.

The roles of proline and glycine in amino acid structure differ from those of the nonpolar amino acids. The amino acid proline contains a ring involving its α -carbon and its α -amino group, which are part of the peptide backbone. This rigid ring

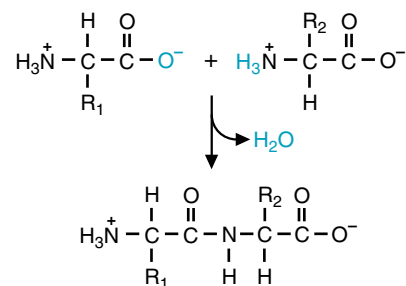


Fig. 6.4. Peptide bonds. Amino acids in a polypeptide chain are joined through peptide bonds between the carboxyl group of one amino acid and the amino group of the next amino acid in the sequence.



According to some biochemists, “W” stands for the characteristic sound in “twyptophan” (see Table 6.1).

Table 6.1. Abbreviations for the Amino Acids

Name	Abbreviations*	
	Three-Letter	One-Letter
Alanine	Ala	A
Arginine	Arg	R
Asparagine	Asn	N
Aspartate	Asp	D
Cysteine	Cys	C
Glutamate	Glu	E
Glutamine	Gln	Q
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Leucine	Leu	L
Lysine	Lys	K
Methionine	Met	M
Phenylalanine	Phe	F
Proline	Pro	P
Serine	Ser	S
Threonine	Thr	T
Tryptophan	Trp	W
Tyrosine	Tyr	Y
Valine	Val	V

*Three-letter abbreviations are generally used. One-letter abbreviations are used mainly to list the amino acid sequences of long protein chains.



The proteolytic digestive enzyme chymotrypsin cleaves the peptide bonds formed by the carboxyl groups of large, bulky uncharged amino acids. Which amino acids fall into this category?

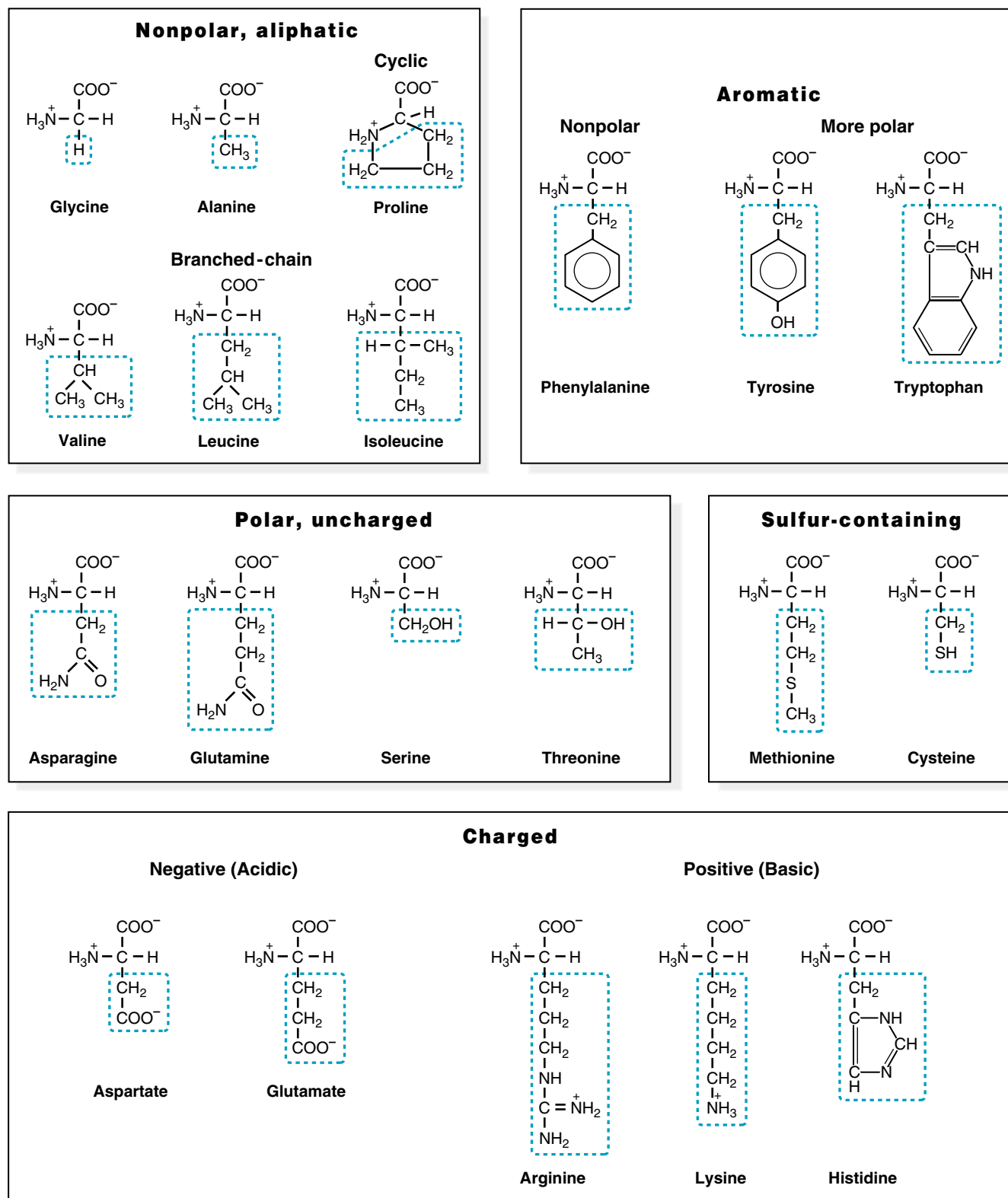


Fig. 6.5. The side chains of the amino acids. The side chains are highlighted. The amino acids are grouped by the polarity and structural features of their side chains. These groupings are not absolute, however. Tyrosine and tryptophan, often listed with the nonpolar amino acids, are more polar than other aromatic amino acids because of their phenolic and indole rings, respectively.

Table 6.2. Properties of the Common Amino Acids

Amino Acid	pK _{a1} * (Carboxyl)	pK _{a2} (Amino)	pK _{aR} (R Group)	Hydropathy Index**
Nonpolar aliphatic				
Glycine	2.4	9.8		−0.4
Proline	2.0	11.0		−1.6
Alanine	2.3	9.7		1.8
Leucine	2.4	9.6		3.8
Valine	2.3	9.6		4.2
Isoleucine	2.4	9.7		4.5
Aromatic				
Phenylalanine	1.8	9.1		2.8
Tyrosine	2.2	9.1	10.5	−1.3
Tryptophan	2.4	9.4		−0.9
Polar uncharged				
Threonine	2.1	9.6	13.6	−0.7
Serine	2.2	9.2	13.6	−0.8
Asparagine	2.0	8.8		−3.5
Glutamine	2.2	9.1		−3.5
Sulfur-containing				
Cysteine	2.0	10.3	8.4	2.5
Methionine	2.3	9.2		1.9
Charged negative				
Aspartate	1.9	9.6	3.9	−3.5
Glutamate	2.2	9.7	4.1	−3.5
Charged positive				
Histidine	1.8	9.3	6.0	−3.2
Lysine	2.2	9.0	10.5	−3.9
Arginine	2.2	9.0	12.5	−4.5
Average	2.2	9.5		

*When these amino acids reside in proteins, the pK_a for the side chains may vary to some extent from the value for the free amino acid, depending on the local environment of the amino acid in the three-dimensional structure of the protein.

**The hydropathy index is a measure of the hydrophobicity of the amino acid (the higher the number, the more hydrophobic). Values based on Kyte J, Doolittle RF. J Mol Biol 1982;157:105B132.

causes a kink in the peptide backbone that prevents it from forming its usual configuration. Because the side chain of glycine is so small compared with that of other amino acids, it causes the least amount of steric hindrance in a protein (i.e., it does not significantly impinge on the space occupied by other atoms or chemical groups). Therefore, glycine is often found in bends or in the tightly packed chains of fibrous proteins.

B. Aromatic Amino Acids

The aromatic amino acids have been grouped together because they all contain ring structures with similar properties, but their polarity differs a great deal. The aromatic ring is a six-membered carbon–hydrogen ring with three conjugated double bonds (the benzene ring or phenyl group). The substituents on this ring determine whether the amino acid side chain engages in polar or hydrophobic interactions. In the amino acid phenylalanine, the ring contains no substituents, and the electrons are shared equally between the carbons in the ring, resulting in a very nonpolar hydrophobic structure in which the rings can stack on each other (Fig. 6.6). In tyrosine, a hydroxyl group on the phenyl ring engages in hydrogen bonds, and the side chain is therefore more polar and more hydrophilic. The more complex ring structure in tryptophan is an indole ring with a nitrogen that can engage in hydrogen bonds. Tryptophan is therefore also more polar than phenylalanine.

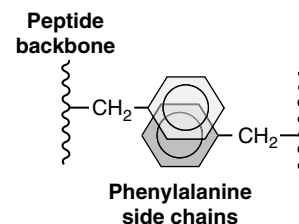
C. Aliphatic, Polar, Uncharged Amino Acids

Amino acids with side chains that contain an amide group (asparagine and glutamine) or a hydroxyl group (serine and threonine) can be classified as aliphatic, polar, uncharged amino acids. Asparagine and glutamine are amides of the amino



Chymotrypsin's highest activity is toward peptide bonds formed by the carboxyl groups of aromatic amino acids (phenylalanine, tyrosine, tryptophan). The side chains of these amino acids are all large and uncharged. One of the chymotrypsin isozymes also exhibits activity toward leucine and methionine, which are similar in polarity.

A. Hydrophobic interaction



B. Hydrogen bonds

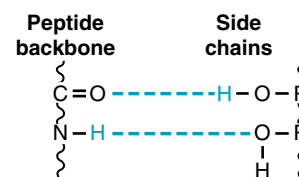


Fig. 6.6. Hydrophobic and hydrogen bonds. **A.** Strong hydrophobic interactions occur with the stacking of aromatic groups in phenylalanine side chains. **B.** Examples of hydrogen bonds in which a hydrogen atom is shared by a nitrogen in the peptide backbone and an oxygen atom in an amino acid side chain or between an oxygen in the peptide backbone and an oxygen in an amino acid side chain.



Cal Kulis passed a renal stone shortly after admission, with immediate relief of flank pain. Stone analysis showed its major component to be cystine. Normally, amino acids are filtered by the renal glomerular capillaries into the tubular urine but are almost entirely reabsorbed from this fluid back into the blood via transport proteins in the proximal tubular cells of the kidney. Cal Kulis has cystinuria, a genetically inherited amino acid substitution in the transport protein that normally reabsorbs cystine, arginine, and lysine from the kidney lumen back into the renal tubular cells. Therefore, his urine contained high amounts of these amino acids. Cystine, which is less soluble than other amino acids, precipitates in the urine to form renal stones (calculi).



Will Sichel has sickle cell anemia caused by a point mutation in his DNA that changes the sixth amino acid in the β -globin chain of hemoglobin from glutamate to valine. What difference would you expect to find in the chemical bonds formed by these two amino acids?

acids aspartate and glutamate. The hydroxyl groups and the amide groups in the side chains allow these amino acids to form hydrogen bonds with water, with each other and the peptide backbone, or with other polar compounds in the binding sites of the proteins (see Fig. 6.6). As a consequence of their hydrophilicity, these amino acids are frequently found on the surface of water-soluble globular proteins. Cysteine, which is sometimes included in this class of amino acids, has been separated into the class of sulfur-containing amino acids.

D. Sulfur-Containing Amino Acids

Both cysteine and methionine contain sulfur. The side chain of cysteine contains a sulfhydryl group that has a pK_a of approximately 8.4 for dissociation of its hydrogen, so cysteine is predominantly undissociated and uncharged at the physiologic pH of 7.4. The free cysteine molecule in solution can form a covalent disulfide bond with another cysteine molecule through spontaneous (nonenzymatic) oxidation of their sulfhydryl groups. The resultant amino acid, cystine, is present in blood and tissues, and is not very water-soluble. In proteins, the formation of a cystine disulfide bond between two appropriately positioned cysteine sulfhydryl groups often plays an important role in holding two polypeptide chains or two different regions of a chain together (Fig. 6.7). Methionine, although it contains a sulfur group, is a nonpolar amino acid with a large bulky side chain that is hydrophobic. It does not contain a sulfhydryl group and cannot form disulfide bonds. Its important and central role in metabolism is related to its ability to transfer the methyl group attached to the sulfur atom to other compounds.

E. The Acidic and Basic Amino Acids

The amino acids aspartate and glutamate have carboxylic acid groups that carry a negative charge at physiologic pH (see Fig. 6.5). The basic amino acids histidine,

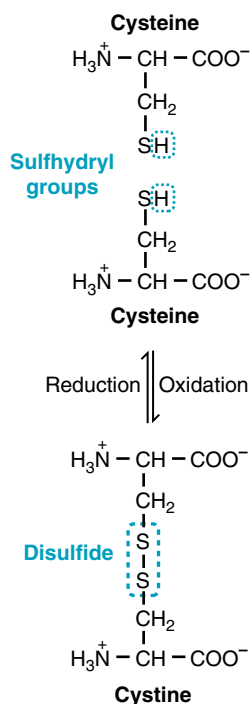


Fig. 6.7. A disulfide bond. Covalent disulfide bonds may be formed between two molecules of cysteine or between two cysteine residues in a protein. The disulfide compound is called cystine. The hydrogens of the cysteine sulfhydryl groups are removed during oxidation.

lysine, and arginine have side chains containing nitrogen that can be protonated and positively charged at physiologic and lower pH values. Histidine has a nitrogen-containing imidazole ring for a side chain, lysine has a primary amino group on the 6th or ϵ carbon (from the sequence α , β , γ , δ , ϵ), and arginine has a guanidinium group.

The positive charges on the basic amino acids enables them to form ionic bonds (electrostatic bonds) with negatively charged groups, such as the side chains of acidic amino acids or the phosphate groups of coenzymes (Fig. 6.8). In addition, lysine and arginine side chains often form ionic bonds with negatively charged compounds bound to the protein binding sites, such as the phosphate groups in ATP. The acidic and basic amino acid side chains also participate in hydrogen bonding and the formation of salt bridges (such as the binding of an inorganic ion such as Na^+ between two partially or fully negatively charged groups).

The charge on these amino acids at physiologic pH is a function of their pK_a s for dissociation of protons from the α -carboxylic acid groups, the α -amino groups, and the side chains. The titration curve of histidine illustrates the changes in amino acid structure occurring as the pH of the solution is changed from less than 1 to 14 by the addition of hydroxide ions (Fig. 6.9). At low pH, all groups carry protons; amino groups have a positive charge, and carboxylic acid groups have zero charge. As the pH is increased by the addition of alkali (OH^-), the proton dissociates from the carboxylic acid group, and its charge changes from zero to negative with a pK_a of approximately 2, the pH at which 50% of the protons have dissociated.

The histidine side chain is an imidazole ring with a pK_a of approximately 6 that changes from a predominantly protonated positively charged ring to an uncharged ring at this pH. The amino group on the α -carbon titrates at a much higher pH (between 9 and 10), and the charge changes from positive to zero as the pH rises. The pH at which the net charge on the molecules in solution is zero is called the isoelectric point (pI). At this pH, the molecules will not migrate in an electric field toward either a positive pole (cathode) or a negative pole (anode), because the number of negative charges on each molecule is equal to the number of positive charges.

Amino acid side chains change from uncharged to negatively charged, or positively charged to uncharged as they release protons (Fig. 6.10). The acidic amino acids lose a proton from their carboxylic acid side chains at a pH of roughly 4, and are thus negatively charged at pH 7.4. Cysteine and tyrosine lose protons at their pK_a s (~ 8.4 and 10.5 , respectively), so their side chains are uncharged at physiologic pH. Histidine, lysine, and arginine side chains change from positively charged to neutral at their pK_a s. The side chains of the two basic amino acids, arginine and lysine, have pK_a values above 10, so that the positively charged form always predominates at physiologic pH. The side chain of histidine ($\text{pK}_a \sim 6.0$) dissociates near physiologic pH, so only a portion of the histidine side chains carry a positive charge (see Fig. 6.9).

In proteins, only the amino acid side chains and the amino group at the amino terminal and carboxyl group at the carboxyl terminal have dissociable protons. All of the other carboxylic acid and amino groups on the α -carbons are joined in peptide bonds that have no dissociable protons. The amino acid side chains might have very different pK_a s than those of the free amino acids if they are involved in hydrogen or ionic bonds with other amino acid side chains. The pK_a of the imidazole group of histidine, for example, is often shifted to a higher value between 6 and 7 so that it adds and releases a proton in the physiologic pH range.

III. VARIATIONS IN PRIMARY STRUCTURE

Although almost every amino acid in the primary structure of a protein contributes to its conformation (three-dimensional structure), the primary structure of a protein

A: Glutamate carries a negative charge on its side chain at physiologic pH and thus can engage in ionic bonds or hydrogen bonds with water or other side chains. Valine is a hydrophobic amino acid and therefore tends to interact with other hydrophobic side chains to exclude water. (The effect of this substitution on hemoglobin structure is described in more detail in Chapter 7.)

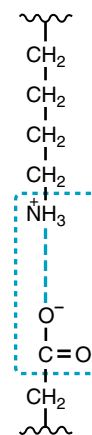


Fig. 6.8. Electrostatic interaction between the positively charged side chain of lysine and the negatively charged side chain of aspartate.

B: Electrophoresis is a technique used to separate proteins on the basis of charge that has been extremely useful in medicine to identify proteins with different amino acid composition. The net charge on a protein at a certain pH is a summation of all of the positive and negative charges on all of the ionizable amino acid side chains plus the N-terminal amino and C-terminal carboxyl groups. Theoretically, the net charge of a protein at any pH could be determined from its amino acid composition by calculating the concentration of positively and negatively charged groups from the Henderson-Hasselbalch equation (see Chapter 4). However, hydrogen bonds and ionic bonds between amino acid side chains in the protein make this calculation unrealistic.

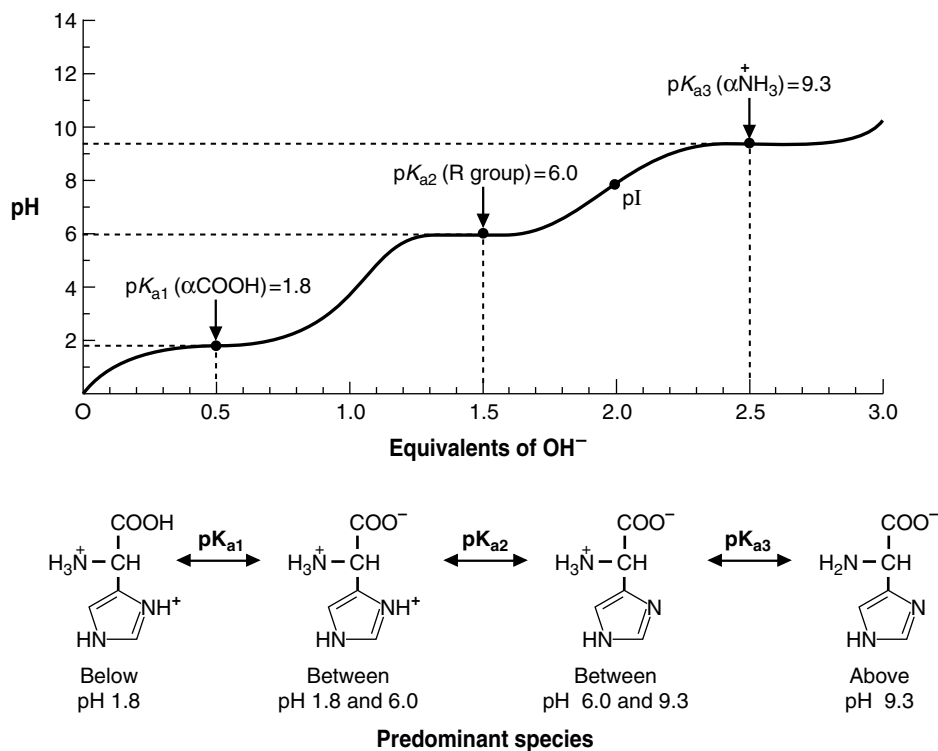


Fig. 6.9. Titration curve of histidine. The ionic species that predominates in each region is shown below the graph. pI is the isoelectric point (at which there is no net charge on the molecule).

	Form that predominates below the pK _a	pK _a	Form that predominates above the pK _a
Aspartate	$\text{---CH}_2\text{---COOH}$	3.9	$\text{---CH}_2\text{---COO}^- + \text{H}^+$
Glutamate	$\text{---CH}_2\text{---CH}_2\text{---COOH}$	4.1	$\text{---CH}_2\text{---CH}_2\text{---COO}^- + \text{H}^+$
Histidine	$\text{---CH}_2\text{---Imidazole}^+$	6.0	$\text{---CH}_2\text{---Imidazole} + \text{H}^+$
Cysteine	$\text{---CH}_2\text{SH}$	8.4	$\text{---CH}_2\text{S}^- + \text{H}^+$
Tyrosine	$\text{---C}_6\text{H}_4\text{---OH}$	10.5	$\text{---C}_6\text{H}_4\text{---O}^- + \text{H}^+$
Lysine	$\text{---CH}_2\text{---CH}_2\text{---CH}_2\text{---CH}_2\text{---NH}_3^+$	10.5	$\text{---CH}_2\text{---CH}_2\text{---CH}_2\text{---CH}_2\text{---NH}_2 + \text{H}^+$
Arginine	$\text{---CH}_2\text{---CH}_2\text{---CH}_2\text{---NH---C(=NH}_2^+)(\text{NH}_2)$	12.5	$\text{---CH}_2\text{---CH}_2\text{---CH}_2\text{---NH---C(=NH)(NH}_2) + \text{H}^+$

Fig. 6.10. Dissociation of the side chains of the amino acids. As the pH increases, the charge on the side chain goes from 0 to + or from + to 0. The pK_a is the pH at which half the molecules of an amino acid in solution have side chains that are charged. Half are uncharged.

can vary to some degree between species. Even within the human species, the amino acid sequence of a normal functional protein can vary somewhat among individuals, tissues of the same individual, and the stage of development. These variations in the primary structure of a functional protein are tolerated if they are confined to noncritical regions (called variant regions), if they are conservative substitutions (replace one amino acid with one of similar structure), or if they confer an advantage. If many different amino acid residues are tolerated at a position, the region is called hypervariable. In contrast, the regions that form binding sites or are critical for forming a functional three-dimensional structure are usually invariant regions that have exactly the same amino acid sequence from individual to individual, tissue to tissue, or species to species.

A. Polymorphism in Protein Structure

Within the human population, the primary structure of a protein may vary slightly among individuals. The variations generally arise from mutations in DNA that are passed to the next generation. The mutations can result from the substitution of one base for another in the DNA sequence of nucleotides (a point mutation), from deletion or insertions of bases into DNA, or from larger changes (see Chap.14). For many alleles, the variation has a distinct phenotypic consequence that contributes to our individual characteristics, produces an obvious dysfunction (a congenital or genetically inherited disease), or increases susceptibility to certain diseases. A defective protein may differ from the most common allele by as little as a single amino acid that is a nonconservative substitution (replacement of one amino acid with another of a different polarity or very different size) in an invariant region. Such mutations might affect the ability of the protein to carry out its function, catalyze a particular reaction, reach the appropriate site in a cell, or be degraded. For other proteins, the variations appear to have no significance.

Variants of an allele that occur with a significant frequency in the population are referred to as polymorphisms. Thus far in studies of the human genome, almost one third of the genetic loci appear to be polymorphic. When a particular variation of an allele, or polymorphism, increases in the general population to a frequency of over



Is the substitution of a glutamate for a valine in sickle cell hemoglobin a conservative replacement? What about the substitution of an aspartate for a glutamate?



For the most part, human chromosomes occur as homologous pairs, with each member of a pair containing the same genetic information. One member of the pair is inherited from the mother and one from the father. Genes are arranged linearly along each chromosome. A genetic locus is a specific position or location on a chromosome. Alleles are alternate versions of a gene at a given locus. For each locus (site), we have two alleles of each gene, one from our mother and one from our father. If both alleles of a gene are identical, the individual is homozygous for this gene; if the alleles are different, he is heterozygous for this gene. **Will Sichel** has two identical alleles for the sickle variant of the β -globin gene that results in substitution of a valine for a glutamate residue at the sixth position of the β -globin chain. He is, therefore, homozygous and has sickle cell anemia. Individuals with one normal gene and one sickle cell allele are heterozygous. They are carriers of the disease and have sickle cell trait.



Will Sichel's hemoglobin, HbS, comprises two normal α chains and 2 β -globin chains with the sickle cell variant ($\alpha_2\beta_2^S$). The change in amino acid composition from a glutamate to a valine in the β chain allows sickle hemoglobin to be separated from normal adult hemoglobin (HbA, or $\alpha_2\beta_2^A$) by electrophoresis. In electrophoresis, an aliquot of blood or other solution containing proteins is applied to a support, such as paper or a gel. When an electrical field is applied, proteins migrate a distance toward the anode (negative pole) or cathode (positive pole) that reflects small differences in their net charge. Electrophoresis of **Will Sichel's** blood shows that he is homozygous for the sickle cell variant, HbS, and has increased amounts of fetal hemoglobin, HbF. Individuals with sickle cell trait are heterozygous and have both HbA and HbS, plus small amounts of HbF ($\alpha_2\gamma_2$).

In heterozygous individuals with sickle cell trait, the sickle cell allele provides some protection against malaria. Malaria is caused by the parasite *Plasmodium falciparum*, which spends part of its life cycle in red blood cells. The infected red blood cells of individuals with normal hemoglobin (HbA) develop protrusions that attach to the lining of capillaries. This attachment occludes the vessels and prevents oxygen from reaching cells in the affected region, resulting in cell death. In heterozygous individuals, HbS in infected cells aggregates into long fibers that cause the cell to become distorted. These distorted cells containing the parasite are preferentially recognized by the spleen and are rapidly destroyed, thus ending the life of the parasite.

In **Will Sichel** and other homozygous individuals with sickle cell anemia, the red blood cells sickle more frequently, especially under conditions of low oxygen tension (see Chapter 7). The result is a vaso-occlusive crisis in which the sickled cells clog capillaries and prevent oxygen from reaching cells (ischemia), thereby causing pain. The enhanced destruction of the sickled cells by the spleen results in anemia. Consequently, the sickle cell allele is of little advantage to homozygous individuals.

Because heterozygous individuals occur more frequently in a population than homozygous individuals, a selective advantage in a heterozygous state can outweigh a disadvantage in a homozygous state, causing the mutation to become a stable polymorphism in a population. As a consequence, the frequency of sickle cell anemia in parts of equatorial Africa in which malaria was endemic in the past is 1 in 25 births. Migration from Africa accounts for the high frequency of sickle cell anemia among blacks in the United States, which is approximately 1/400 at birth.



The substitution of a glutamate for a valine is a nonconservative replacement because a negatively charged amino acid is substituted for a hydrophobic branched chain aliphatic amino acid. However, the substitution of an aspartate for a glutamate is a conservative replacement because the two amino acids have the same polarity and nearly the same size.



Homologous families of proteins are proteins that have the same ancestral protein and arose from the same gene. The term *homologs* includes both orthologs and paralogs. Orthologs are genes from different species that have evolved from a common ancestral gene as different species developed (for example, human and pork insulin). By contrast, paralogs are genes related by duplication within the genome of a single species (e.g., myoglobin and hemoglobin). Normally, orthologs retain the same function in the course of evolution, whereas paralogs evolve new functions that may, or may not, be related to the original one.

1%, it is considered stable. The sickle cell allele is an example of a point mutation that is stable in the human population. Its persistence is probably attributable to selective pressure for the heterozygous mutant phenotype, which confers some protection against malaria.

B. Protein Families and Superfamilies.

A homologous family of proteins is composed of proteins related to the same ancestral protein. Groups of proteins with similar, but not identical, structure and function that have evolved from the same gene after the gene was duplicated are called paralogs and considered members of the same protein family. Once a gene has duplicated, one gene can continue to perform original function, and the second copy can mutate into a protein with another function or another type of regulation. This process is called divergent evolution. Very large families of homologous proteins are called a superfamily, which is subdivided by name into families of proteins with the most similarity in structure.

The paralogs of a protein family are considered different proteins and have different names because they have different functions. They are all present in the same individual. Myoglobin and the different chains of hemoglobin, for example, are paralogs and members of the same globin family that have similar, but not identical, structures and functions. Myoglobin, an intracellular heme protein present in most cells that stores and transports O_2 to mitochondria, is a single polypeptide chain containing one heme oxygen-binding site. In contrast, hemoglobin is composed of four globin chains, each with a heme oxygen-binding site that is present in red blood cells and transports O_2 from the lungs to tissues. The gene for myoglobin is assumed to have evolved from gene duplication of the α -chain for hemoglobin, which evolved from duplication of the β -chain. Figure 6.11 compares a region of the structure of myoglobin and the α and β chains of hemoglobin. Among these three proteins, only 15 invariant (identical) residues are present, but many of the other amino acid residues are conservative substitutions.



To compare the primary structure of two homologous polypeptide chains, the sequences are written left to right from the amino terminal to the carboxyl terminal. The sequences are aligned with computer programs that maximize the identity of amino acids and minimize the differences caused by segments that are present in one protein and not in the other.

The primary structure of human globin proteins

	1	5	10	15
Myoglobin	gly-----	leu-ser-asp-gly	glu-trp-gln-leu-val	leu-asn-val-trp-gly-lys-val-
β chain Hemoglobin	val-his-leu-thr-pro	glu-glu-lys-ser-ala	val-thr-ala-leu-trp-gly-lys-val-	
α chain hemoglobin	val-----	leu-ser-pro-ala-asp-lys	thr-asn-val-lys-ala-ala-trp-gly-lys-val-	
ζ chain Hemoglobin	met-ser-leu-thr-lys	thr-glu-arg-thr-ile-ile	val-ser-met-trp-ala-lys-ile-	
γ chain Hemoglobin	met-gly-his-phe-thr	glu-glu-asp-lys-ala-thr	ile-thr-ser-leu-trp-gly-lys-val-	

Fig 6.11. The primary structures of a region in human globin proteins. Gaps in the structure, indicated with dashes, are introduced to maximize the alignment between proteins in structure comparisons. They are assumed to coincide with mutations that caused a deletion. Regions of sequence similarity (identity and conservative substitution) are indicated in blue. Within these regions, there are smaller regions of invariant residues that are exactly the same from protein to protein. Myoglobin is a single polypeptide chain. The α and β chains are part of hemoglobin A ($\alpha_2\beta_2$). The ζ chain is part of embryonic hemoglobin ($\zeta_2\epsilon_2$). The γ chain is part of fetal hemoglobin (HbF), $\alpha_2\gamma_2$.

C. Tissue and Developmental Variations in Protein Structure

Within the same individual, different isoforms or isozymes of a protein may be synthesized during different stages of fetal and embryonic development, may be present in different tissues, or may reside in different intracellular locations. Isoforms of a protein all have the same function. If they are isozymes (isoforms of enzymes), they catalyze the same reactions. However, isoforms have somewhat different properties and amino acid structure.

1. DEVELOPMENTAL VARIATION

Hemoglobin isoforms provide an example of variation during development. Hemoglobin is expressed as the fetal isozyme HbF during the last trimester of pregnancy until after birth, when it is replaced with HbA. HbF is composed of two hemoglobin α and 2 hemoglobin γ polypeptide chains, in contrast to the adult hemoglobin, hemoglobin A, which is 2 α and 2 β chains. During the embryonic stages of development, chains with a different amino acid composition, the embryonic ϵ and ζ chains, are produced (see Fig. 6.11). These differences are believed to arise evolutionarily from mutation of a duplicated α gene to produce ζ , and mutation of a duplicate α gene to produce ϵ . The fetal and embryonic forms of hemoglobin have a much higher affinity for O_2 than the adult forms, and thus confer an advantage at the low O_2 tensions to which the fetus is exposed. At different stages of development, the globin genes specific for that stage are expressed and translated.

2. TISSUE-SPECIFIC ISOFORMS

Proteins that differ somewhat in primary structure and properties from tissue to tissue, but retain essentially the same function, are called tissue-specific isoforms or isozymes. The enzyme creatine kinase is an example of a protein that exists as tissue-specific isozymes, each composed of two subunits with 60 to 72% sequence homology. Of the two creatine kinases that bind to the muscle sarcomere, the M form is produced in skeletal muscle, and the B polypeptide chains are produced in the brain. The protein comprises two subunits, and skeletal muscle therefore produces an MM creatine kinase, and the brain produces a BB form. The heart produces both types of chains and therefore forms a heterodimer, MB, as well as an MM dimer. Two more creatine kinase isozymes are found in mitochondria, a heart mitochondrial creatine kinase, and the “universal” isoform found in other tissues. (In general, most proteins present in both the mitochondria and cytosol will be present as different isoforms.) The advantage conferred on different tissues by having their own isoform of creatine kinase is unknown. However, tissue-specific isozymes such as MB creatine kinase are useful in diagnosing sites of tissue injury and cell death.

The structure of proteins involved in the response to hormones has been studied in greater depth than many other types of proteins, and most of these proteins are present as several tissue-specific isoforms that help different tissues respond differently to the same hormone. One of these proteins present in cell membranes is adenylyl cyclase, an enzyme that catalyzes the synthesis of intracellular 3',5' cyclic adenosine monophosphate (cAMP) (Fig. 6.12). In human tissues, at least nine different isoforms of adenylyl cyclase are coded by different genes in different tissues. Although they have an overall sequence homology of 50%, the two intracellular regions involved in the synthesis of cAMP are an invariant consensus sequence with a 93% identity. The different isoforms help cells respond differently to the same hormone.



The term *isozyme* was originally defined as meaning enzymes with a different primary structure, catalyzing the same reaction, encoded by different genes. However, the term is now used more broadly. Some isozymes are now known to arise from alternate splicing of pre-mRNA to form different mRNAs (mRNA is the final nucleic acid template used for protein synthesis.)



A myocardial infarction (heart attack) is caused by an atheromatous obstruction or a severe spasm in a coronary artery that prevents the flow of blood to an area of heart muscle. Thus, heart cells in this region suffer from a lack of oxygen and blood-borne fuel. Because the cells cannot generate ATP, the membranes become damaged, and enzymes leak from the cells into the blood.

Creatine kinase (CK or CPK) is one of these enzymes. The protein is composed of two subunits, which may be either of the muscle (M) or the brain (B) type. The MB form, containing one M and one B subunit, is found primarily in cardiac muscle. It can be separated electrophoretically from other CK isozymes and the amount in the blood used to determine if a myocardial infarction has occurred. On admission to the hospital, **Ann Jeina's** total CK was 182 units/L (reference range 5–174 U/L). Her MB fraction was 6.8% (reference range 5% or less of the total CK). Although these values are only slightly elevated, they are typical of the phase immediately following a myocardial infarction. Additional information was provided by myoglobin and troponin T (Tn-T) measurements.



Variations in protein structure among species have been used to develop a phylogenetic tree of ancestral relationships between species showing the progress of evolution. According to evolutionary theory, related species evolved from a common ancestor, and proteins all can be grouped into clusters of proteins (called orthologous proteins) that evolved from a common ancestral protein. The more similar the amino acid sequence of an orthologous protein from two different species, the closer is the relationship between those species.



Although bovine (beef) insulin is identical to human insulin in those amino acid residues essential for activity, the amino acid residues that are in the variable regions can act as antigens and stimulate the formation of antibodies against bovine insulin. Consequently, recombinant DNA techniques have been used for the synthesis of insulin, identical in structure to native insulin, such as Humulin (intermediate-acting insulin) (see Chapter 17 for information on recombinant DNA technology.) Although **Di Abietes** had not yet experienced an allergic response to her beef insulin, the cost of Humulin is now sufficiently low that most patients are being changed from beef insulin to the synthetic forms of human insulin.



The 20 DNA-encoded amino acids form the linear sequence of amino acids known as the primary structure of the protein, and determine the folding pattern for the three-dimensional conformation of the protein. Posttranslational modifications usually occur once the protein has already folded into its characteristic pattern. They are like accessories (jewelry, ties, etc.) that provide additional variations to your basic wardrobe.

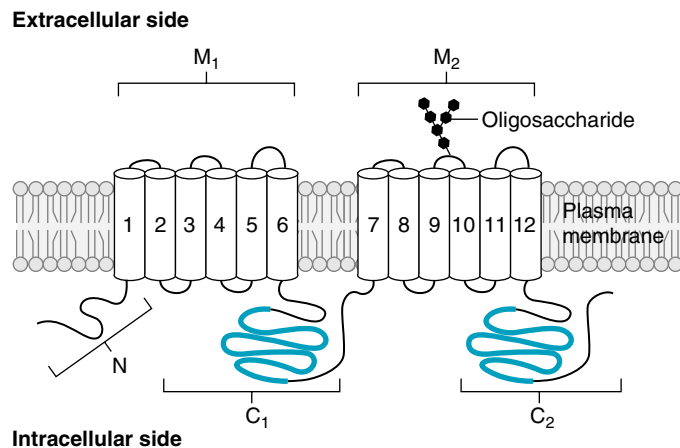


Fig 6.12. Invariant regions in the isoforms of adenylyl cyclase. The invariant regions are on the cytosolic side of the membrane in the C1 and C2 loops shown in blue. These amino acid residues participate in the catalytic function of the enzyme, synthesis of 3',5'-cyclic AMP. The protein also has several helical regions that span the membrane (M1 helices and M2 helices), represented as tubes. An oligosaccharide chain is attached to an extracellular domain. N is the amino terminus. (Copyrighted from Taussig R, Gilman AG. *J Biol Chem* Mammalian membrane-bound adenylyl cyclases 1995;270:1–4.)

D. Species Variations in the Primary Structure of Insulin

Species variations in primary structure are also important in medicine, as illustrated by the comparison of human, beef, and pork insulin. Insulin is one of the hormones that are highly conserved between species, with very few amino acid substitutions and none in the regions that affect activity. Insulin is a polypeptide hormone of 51 amino acids that is composed of two polypeptide chains (Fig. 6.13). It is synthesized as a single polypeptide chain, but is cleaved in three places before secretion to form the C peptide and the active insulin molecule containing the A and B chains. The folding of the A and B chains into the correct three-dimensional structure is promoted by the presence of one intrachain and two interchain disulfide bonds formed by cysteine residues. The invariant residues consist of the cysteine residues engaged in disulfide bonds and the residues that form the surface of the insulin molecule that binds to the insulin receptor. The amino acid substitutions in bovine and porcine insulin (shown in blue in Fig. 6.13.) are not in amino acids that affect its activity. Consequently, bovine and pork insulin were used for many years for the treatment of diabetes mellitus. However, even with only a few different amino acids, some patients developed an immune response to these insulins.

IV. MODIFIED AMINO ACIDS

After synthesis of a protein has been completed, a few amino acid residues in the primary sequence may be further modified in enzyme-catalyzed reactions that add a chemical group, oxidize, or otherwise modify specific amino acids in the protein. Because protein synthesis occurs by a process known as translation, these changes are called posttranslational modification. More than 100 different posttranslationally modified amino acid residues have been found in human proteins. These modifications change the structure of one or more specific amino acids on a protein in a way that may serve a regulatory function, target or anchor the protein in membranes, enhance a protein's association with other proteins, or target it for degradation (Fig. 6.14).

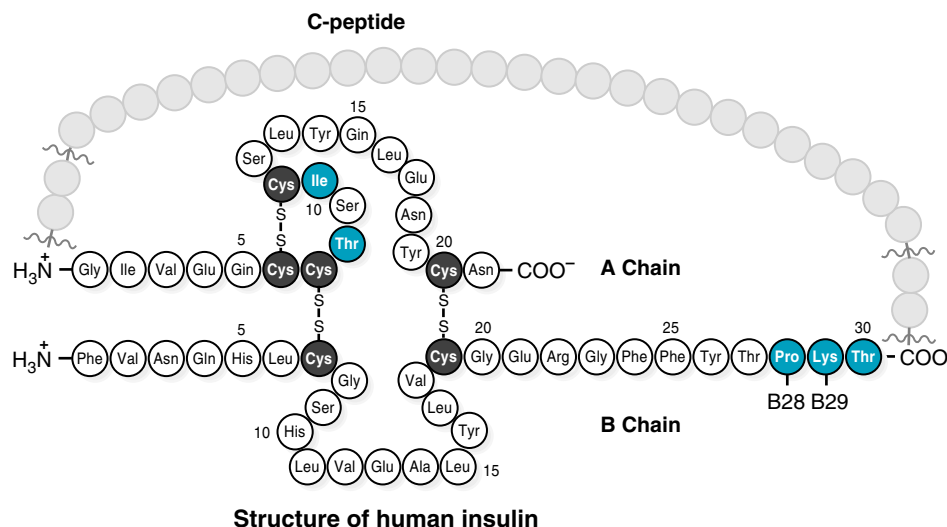


Fig. 6.13. The primary structure of human insulin. The substituted amino acids in bovine (beef) and porcine (pork) insulin are shown in blue. Threonine 30 at the carboxy terminal of the B chain is replaced by alanine in both beef and pork insulin. In beef insulin, threonine 8 on the A chain is also replaced with alanine, and isoleucine 10 with valine. The cysteine residues, which form the disulfide bonds holding the chains together, are invariant. In the bioengineered insulin Humalog (lispro insulin), the position of proline at B28 and lysine at B29 is switched. Insulin is synthesized as a longer precursor molecule, proinsulin, which is one polypeptide chain. Proinsulin is converted to insulin by proteolytic cleavage of certain peptide bonds (squiggly lines in the figure). The cleavage removes a few amino acids and the 31-amino acid C-peptide that connects the A and B chains. The active insulin molecule, thus, has two nonidentical chains.

B. Glycosylation

Oligosaccharides (small carbohydrate chains) are bound to proteins by either *N*-linkages or *O*-linkages (see Fig. 6.14). *N*-linked oligosaccharides are found attached to cell surface proteins, where they protect the cell from proteolysis or an immune attack. In contrast, an *O*-glycosidic link is a common way of attaching oligosaccharides to the serine or threonine hydroxyl groups in secreted proteins. The intracellular polysaccharide glycogen is attached to a protein through an *O*-glycosidic linkage to a tyrosine.

C. Fatty Acylation or Prenylation

Many membrane proteins contain a covalently attached lipid group that interacts hydrophobically with lipids in the membrane. Palmitoyl groups (C16) are often attached to plasma membrane proteins, and the myristoyl group (C14) is often attached to proteins in the lipid membranes of intracellular vesicles (see Fig. 6.14). The farnesyl (C15) or geranylgeranyl group (C20) are synthesized from the five-carbon isoprene unit (isopentenyl pyrophosphate, see Fig. 5.1A) and are therefore called isoprenoids. These are attached in ether linkage to a specific cysteine residue of certain membrane proteins, particularly proteins involved in regulation.

D. Regulatory Modifications

Phosphorylation, acetylation, and adenosine diphosphate (ADP)-ribosylation of specific amino acid residues in a polypeptide can alter bonding by that residue and change the activity of the protein (see Fig. 6.14). Phosphorylation of an OH group on serine, threonine, or tyrosine by a protein kinase (an enzyme that transfers a phosphate group from ATP to a protein) introduces a large, bulky, negatively charged group that can alter the activity of a protein. Reversible acetylation occurring on lysine residues of histone proteins in the chromosome changes their



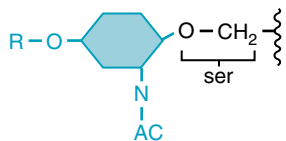
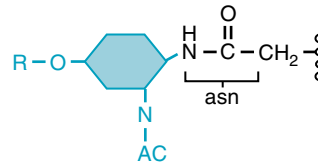
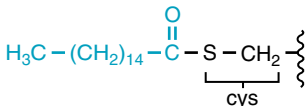
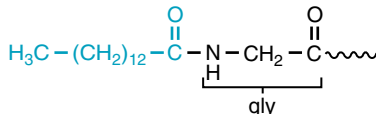
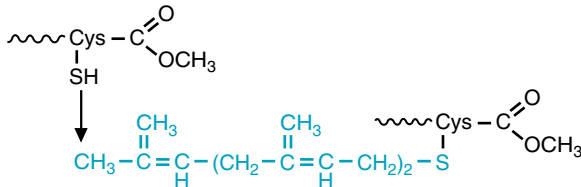
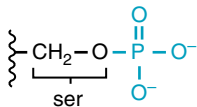
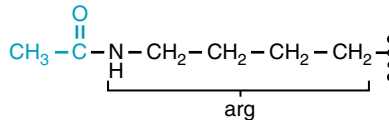
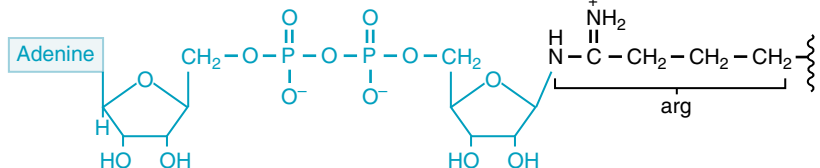
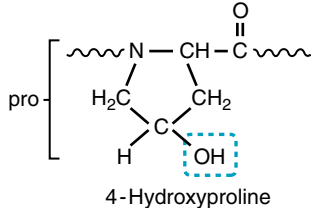
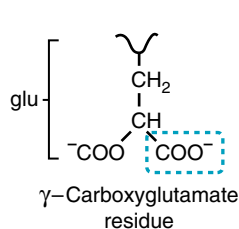
Carbohydrate addition**O-glycosylation:** OH of ser, thr, tyr,**N-glycosylation:** NH₂ of asn**Lipid addition****Palmitoylation:** Internal SH of cys**Myristoylation:** NH of N-terminal gly**Prenylation:** SH of cys**Regulation****Phosphorylation:** OH of ser, thr, tyr**Acetylation:** NH₂ of lys, terminus**ADP-ribosylation:** N of arg, gln; S of cys**Modified amino acids****Oxidation:** pro, lys**Carboxylation:** glu

Fig. 6.14. Posttranslational modifications of amino acids in proteins. Some of the common amino acid modifications and the sites of attachment are illustrated. The added group is shown in blue. Because these modifications are enzyme-catalyzed, only a specific amino acid in the primary sequence is altered. R-O-hexagon refers to an oligosaccharide. In *N*-glycosylation, the attached sugar is usually N-acetylglucosamine (N-Ac).

interaction with the negatively charged phosphate groups of DNA. ADP-ribosylation is the transfer of an ADP-ribose from NAD^+ to an arginine, glutamine, or a cysteine residue on a target protein in the membrane (primarily in leukocytes, skeletal muscles, brain, and testes). This modification may regulate the activity of these proteins

E. Other Amino Acid Posttranslational Modifications

A number of other posttranslational modifications of amino acid side chains alter the activity of the protein in the cell (see Fig 6.14). Carboxylation of the γ carbon of glutamate (carbon 4) in certain blood clotting proteins is important for attaching the clot to a surface. Calcium ions mediate this attachment by binding to the two negatively charged carboxyl groups of γ -glutamate and two additional negatively charged groups provided by phospholipids in the cell membrane. Collagen, an abundant fibrous extracellular protein, contains the oxidized amino acid hydroxyproline. The addition of the hydroxyl group to the proline side chain provides an extra polar group that can engage in hydrogen bonding between the polypeptide strands of the fibrous protein.

F. Selenocysteine

The unusual amino acid selenocysteine is found in a few enzymes and is required for their activity (Fig. 6.15). Its synthesis is not a posttranslational modification, however, but a modification to serine that occurs while serine is bound to a unique tRNA. The selenocysteine is then inserted into the protein as it is being synthesized.



A number of pathogenic bacteria produce bacterial toxins that are ADP-ribosyl transferases (NAD^+ -glycohydrolases). These enzymes hydrolyze the N-glycosidic bond of NAD^+ and transfer the ADP-ribose portion to a specific amino acid residue on a protein in the affected human cell. Cholera A-B toxin, a pertussis toxin, and a diphtheria toxin are all ADP-ribosyl transferases.

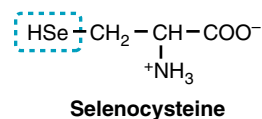


Fig. 6.15. Selenocysteine.

CLINICAL COMMENTS



Will Sichel. Will Sichel was treated for 3 days with parenteral (intravascular) narcotics, hydration, and nasal inhalation of oxygen for his vaso-occlusive crisis. The diffuse severe pains of sickle cell crises result from occlusion of small vessels in a variety of tissues, thereby causing damage to cells from ischemia (low blood flow) or hypoxia (low levels of oxygen). Vaso-occlusion occurs when HbS molecules in red blood cells polymerize in the capillaries, where the partial pressure of O_2 ($p\text{O}_2$) is low. This polymerization causes the red blood cells to change from a biconcave disc to a sickle shape that cannot deform to pass through the narrow capillary lumen. The cells aggregate in the capillaries and occlude blood flow.

In addition, **Will Sichel** was treated with hydroxyurea therapy, which increases the production of red blood cells containing fetal hemoglobin. HbF molecules cannot participate in sickling. **Will Sichel's** acute symptoms gradually subsided. Had his severe pain persisted, partial exchange blood transfusions would have been considered because no other effective therapy is currently available. Patients with sickle cell anemia periodically experience sickle cell crises, and Will's physician urged him to seek medical help whenever symptoms reappeared.



Cal Kulis. Mr. Kulis has cystinuria, a relatively rare disorder, with a prevalence that ranges between 1 in 2,500 to 1 in 15,000 births, depending on the population studied. It is a genetically determined disease with a complex recessive mode of inheritance resulting from allelic mutations. These mutations lead to a reduction in the activity of renal tubular cell transport proteins that normally carry cystine from the tubular lumen into the renal tubular cells. The transport of the basic amino acids (lysine, arginine, and ornithine, an amino acid

found in the urea cycle but not in proteins) is also often compromised, and they appear in the urine.

Because cystine is produced by oxidation of cysteine, conservative treatment of cystinuria includes decreasing the amount of cysteine within the body and, hence, the amount of cystine eventually filtered by the kidneys. Reduction of cysteine levels is accomplished by restricting dietary methionine, which contributes its sulfur to the pathway for cysteine formation. To increase the amount of cystine that remains in solution, the volume of fluid ingested daily is increased. Crystallization of cystine is further prevented by chronically alkalinizing the urine. Finally, drugs may be administered to enhance the conversion of urinary cystine to more soluble compounds. If these conservative measures fail to prevent continued cystine stone formation, existing stones may be removed by a surgical technique that involves sonic fracture. The fragmented stones may then pass spontaneously or may be more easily extracted surgically because of their smaller size.



The switch in position of amino acids in lispro does not affect the action of this synthetic insulin on cells because it is not in a critical invariant region, but it does affect the ability of insulin to bind zinc. Normally, human insulin is secreted from the pancreas as a zinc hexamer in which six insulin molecules are bound to the zinc atom. When zinc insulin is injected, the binding to zinc slows the absorption from the subcutaneous (under the skin) injection site. Lispro cannot bind zinc to form a hexamer, and thus it is absorbed much more quickly than other insulins.



Di Abietes. Di Abietes' treatment was first changed to daily injections of Humulin instead of beef insulin. Humulin is now mass-produced by recombinant DNA techniques that insert the human DNA sequences for the insulin A and B chains into the *Escherichia coli* or yeast genome (see Chapter 17). The insulin chains that are produced are then extracted from the media and treated to form the appropriate disulfide bonds between the chains. As costs have fallen for production of the synthetic human insulins, they have replaced pork insulin and the highly antigenic beef insulin.

Di's physician then recommended that she take Humalog, an insulin preparation containing lispro, an ultra-fast-acting bioengineered insulin analog in which lysine at position B29 has been moved to B28 and proline at B28 has been moved to B29 (hence, lispro) (see Fig. 6.13). With lispro, Di will be able to time her injections minutes before her consumption of carbohydrate-containing meals, rather than having to remember to give herself an insulin injection 1 hour before a meal.



Ann Jeina. Mrs. Jeina continued to be monitored in the cardiac care unit. At admission, her CK levels were elevated (182 units/L compared with a reference range of 38–174 U/L), and the MB fraction was high at 6.8% of the total (reference = <5% of the total). Her total CK continued to rise (228 units/L 12 hours after admission and 266 units/L at 24 hours), as did her MB fraction (8% at 12 hours and 10.8% at 24 hours). Within 2 hours of the onset of an acute myocardial infarction, the MB form of CK begins leaking from heart cells that were injured by the ischemic process. These rising serum levels of the MB fraction (and, therefore, of the total CK) reach their peak 12 to 36 hours later and usually return to normal within 3 to 5 days from the onset of the infarction (Fig. 6.16). In addition to the CK measurements, her blood levels were also analyzed for myoglobin and the heart isoform of troponin-T, a protein involved in muscle contraction (see Chapter 7).

BIOCHEMICAL COMMENTS



Enzyme and Protein Databases. Large databases of protein structure have been assembled to collate data from various laboratories around the world. The National Library of Medicine maintains a website, PubMed, which catalogues the medical literature, and a search engine that allows you to search for reference articles by topic or author

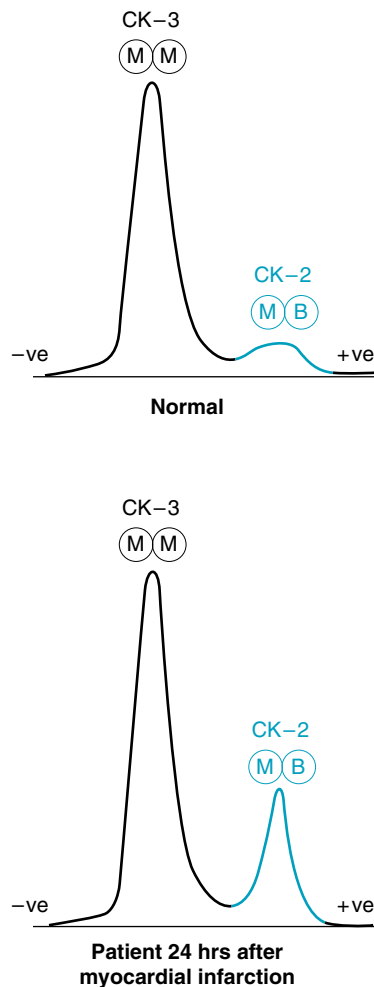
Creatine kinase isozymes in blood

Fig. 6.16. Electrophoretic separation of serum creatine kinase enzymes from a normal healthy adult and from a patient who had a myocardial infarction 24 hours previously. Creatine kinase catalyzes the reversible transfer of a phosphate from ATP to creatine to form phosphocreatine and ADP. The reaction is an important part of energy metabolism in heart muscle, skeletal muscle, and brain. Three different forms of the dimer exist: BB (or CK-1) found in brain, MB (or CK-2) found only in heart, and MM (or CK-3), found only in skeletal and heart muscle (cathode, -ve; anode, +ve).

(<http://www.ncbi.nlm.nih.gov/pubmed>). From the menu, you can directly enter a protein database or a structure database. These databases are linked, so that you can type in the name of a protein, such as human hemoglobin chain A, obtain a list of contributors to sequence data, and retrieve a complete amino acid sequence for many proteins. You can link to PubMed to find recent articles about the protein, or you can link to the structure database. A program called Cn3D can be downloaded from this site that allows you to view the three-dimensional versions of the protein structures.

These databases are only a few of more than 500 biologic databases that have been assembled to collate and exchange biologic information in the areas of DNA, RNA, genomics, gene mapping, and protein structure. The first issue of the *Journal of Nucleic Acid Research* each year provides a description of currently available

biologic databases. Their goal is to provide information that can relate a particular DNA sequence or mutation to the protein involved, to its function, and to the pathologic consequences of a particular amino acid substitution, by comparing proteins that have similar functional elements.

Will these databases be of any use to the practicing physician or to the practice-oriented medical student? Very few students will ever want to do protein modeling. However, students and physicians may wish to use the literature search in PubMed as part of their approach to evidence-based medicine. They also may wish to use it to track definitions or fundamental knowledge about particular topics. Thus, a move has begun to link biomedical and basic science textbooks to PubMed.

Suggested Reference

Tatusov RL, Koonin EV, Lipman, DJ. A genomic perspective on protein families. *Science* 1997;278:631–637.



REVIEW QUESTIONS—CHAPTER 6

- In a polypeptide at physiologic pH, hydrogen bonding may occur between
 - the side chains of a leucine residue and a lysine residue.
 - the side chains of an aspartyl residue and a glutamyl residue.
 - the terminal α -amino group and the terminal α -carboxyl group.
 - the amide group in the peptide bond and an aspartyl side chain.
 - the SH groups of two cysteine residues.
- Which of the following shows the linear sequence of atoms joined by covalent bonds in a peptide backbone?
 - $\text{--N--C--C--N--C--C--N--C--C--}$
 - $\text{--N--C--O--N--C--O--N--C--O--}$
 - $\text{--N--C--C--O--N--C--C--O--N--C--C--O--}$
 - $\text{--N--H--C--C--N--H--C--C--N--H--C--C--}$
 - $\text{--N--H--C--O--H--N--H--C--O--H--N--H--C--C--}$
- Di Abietes'** different preparations of insulin contain some insulin complexed with protamine that is absorbed slowly after injection. Protamine is a protein preparation from rainbow trout sperm containing arginine-rich peptides that bind insulin. Which of the following provides the best explanation for complex formation between protamine and insulin?
 - Arginine is a basic amino acid that binds to negatively charged amino acid side chains in insulin.
 - Arginine is a basic amino acid that binds to the α -carboxylic acid groups at the N-terminals of insulin chains.
 - Arginine is a large bulky hydrophobic amino acid that complexes with leucine and phenylalanine in insulin.
 - Arginine forms disulfide bonds with the cysteine residues that hold the A and B chains together.
 - Arginine has a side chain that forms peptide bonds with the carboxyl terminals of the insulin chains.
- Protein kinases phosphorylate proteins only at certain hydroxyl groups on amino acid side chains. Which of the following groups of amino acids all contain side chain hydroxyl groups?
 - aspartate, glutamate, and serine
 - serine, threonine, and tyrosine
 - threonine, phenylalanine, and arginine
 - lysine, arginine, and proline
 - alanine, asparagine, and serine

5. In a single individual, the primary structures of enzymes catalyzing the same reaction
- (A) are exactly the same from cell type to cell type, although the amount of enzyme may differ.
 - (B) stay the same throughout the lifetime of that individual.
 - (C) are identical if the enzymes are paralogs.
 - (D) are identical to all members of the homologous family.
 - (E) may differ between different cellular compartments of the same cell.

7 Structure–Function Relationships in Proteins



Diseases can be caused by changes in protein structure that affect the protein's ability to bind other molecules and carry out its function. They also can be caused by conformational changes in proteins that affect their solubility and degradability. In amyloidosis AL, immunoglobulin chains form an insoluble protein aggregate called amyloid in organs and tissues. Alzheimer's disease and familial amyloid polyneuropathy are neurodegenerative diseases characterized by the deposition of amyloid. Prion diseases result from misfolding and aggregation of a normal cellular protein. Even in sickle cell anemia, the mutation in hemoglobin principally affects the quaternary structure of hemoglobin and its solubility, and not its ability to bind oxygen.

A multitude of different proteins can be formed from only 20 common amino acids because these amino acids can be linked together in an enormous variety of sequences determined by the genetic code. The sequence of amino acids, its **primary structure**, determines the way a protein folds into a unique three-dimensional structure, which is its **native conformation**. Once folded, the three-dimensional structure of a protein forms binding sites for other molecules, thereby dictating the function of the protein in the body. In addition to creating binding sites, a protein must fold in such a way that it is flexible, stable, able to function in the correct site in the cell, and capable of being degraded by cellular enzymes.

Levels of protein structure. Protein structure is described in terms of four different levels: **primary**, **secondary**, **tertiary**, and **quaternary** (Fig. 7.1). The **primary structure** of a protein is the linear sequence of amino acids in the polypeptide chain. **Secondary structure** consists of local regions of polypeptide chains formed into structures generally stabilized by hydrogen bonds, such as the regular structures called α -helices and β -sheets. The **rigidity of the peptide backbone** determines the types of secondary structure that can occur. The **tertiary structure** involves folding of the secondary structural elements into an overall three-dimensional conformation. In **globular proteins** such as **myoglobin**, the tertiary structure generally forms a densely packed **hydrophobic core** with polar amino acid side chains on the outside. Some proteins exhibit **quaternary structure**, the combination of two or more **subunits**, each composed of a polypeptide chain.

Domains and folds. The tertiary structure of a globular protein is made up of structural **domains**, regions of structure that are recognized as separate and linked to other domains in a simple way. Within a domain, a combination of secondary

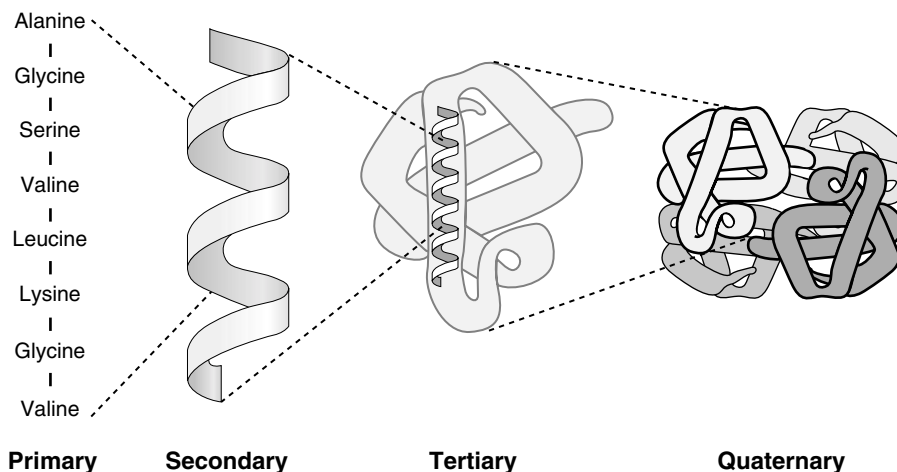


Fig. 7.1. Levels of structure in a protein.

structural elements forms a fold, such as the **nucleotide binding fold**, or an **actin fold**. Folds are defined by their similarity in a number of different proteins.

Quarternary structure. Assembly of globular polypeptide subunits into a multi-subunit complex can provide the opportunity for **cooperative binding** of ligands (e.g., O_2 binding to hemoglobin), form **binding sites** for complex molecules (e.g., antigen binding to immunoglobulin), and increase **stability** of the protein. The polypeptide chains of **fibrous proteins** such as **collagen** are aligned along an axis, have repeating elements, and are extensively linked to each other through hydrogen bonds.

Ligand binding. Proteins form binding sites for a specific molecule, called a **ligand** (e.g., ATP or O_2) or for another protein. The affinity of a binding site for its ligand is quantitatively characterized by an **association** or **affinity constant**, K_a , (or its **dissociation constant**, K_d).

Folding of proteins. The primary structure of a protein dictates the way that it folds into its tertiary structure, which is a **stable conformation** that is identical to the shape of other molecules of the same protein (that is, its **native conformation**.)

Chaperonins act as templates to overcome the kinetic barrier to reaching a stable conformation. **Prion proteins** cause neurodegenerative diseases by acting as a template for misfolding. Heat, acid, and other agents cause proteins to **denature**, that is, to unfold or refold and lose their native three-dimensional conformation.



THE WAITING ROOM



Will Sichel, who has sickle cell anemia, was readmitted to the hospital with symptoms indicating that he was experiencing another sickle cell crisis (see Chapter 6).



Anne Jeina is a 54-year-old woman who arrived in the hospital 4 days ago, approximately 5 hours after she began to feel chest pain (see Chapter 6). In the emergency room, the physician had drawn blood for the measurement of myoglobin, CK-MB (creatine kinase, muscle-brain fraction) and cTN-T (cardiac troponin T subunit). The results from these tests had supported the diagnosis of an acute MI (myocardial infarction), and Mrs. Jeina was hospitalized.



Amy Lloyd is a 62-year-old woman who presents with weakness, fatigue, an enlarged tongue (macroglossia), and edema. She has signs and symptoms of cardiac failure, including electrocardiographic abnormalities. Initial laboratory studies showed a serum creatinine of 1.9 mg/dL (reference range [females] = 0.5–1.1) indicating mild renal failure. A urinalysis indicated the presence of a moderate proteinuria and numerous white blood cells in the urinary sediment. She was subsequently diagnosed with amyloidosis/AL secondary to a plasma cell dyscrasia.



Di Abietes returned to her physician's office for a routine visit to monitor her treatment (see Chapters 4, 5, and 6.) Her physician drew blood for an HbA_{1c} (pronounced hemoglobin A-1- c) determination. The laboratory reported a value of 8.5%, compared with to a normal reference range of 5.8 to 7.2%.



Amyloidosis is a term encompassing many diseases that share a common feature—the extracellular deposition of pathologic insoluble fibrillar proteins called amyloid in organs and tissues. In **Amy Lloyd's** disease, amyloidosis/AL, the amyloid is derived from immunoglobulin light chains (AL = amyloidosis, light-chain related).

I. GENERAL CHARACTERISTICS OF THREE-DIMENSIONAL STRUCTURE

The overall conformation of a protein, the particular position of the amino acid side chains in three-dimensional space, gives a protein its function.

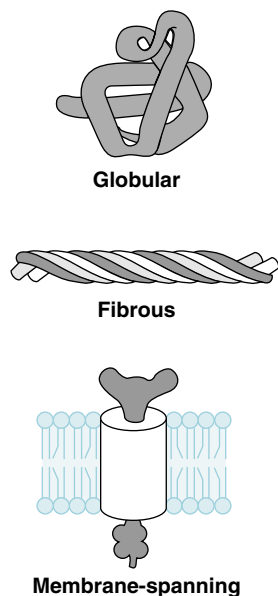


Fig. 7.2. General shapes of proteins.



Requirements of a Protein Structure

- Function
- Binding specificity
- Flexibility
- Solubility or lipophilicity
- Stability
- Degradability

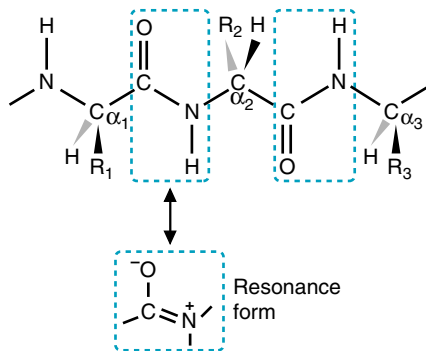


Fig. 7.3. The peptide backbone. Because of the resonance nature of the peptide bond, the C and N of the peptide bonds form a series of rigid planes. Rotation within allowed torsion angles can occur around the bonds attached to the α -carbon. The side chains are *trans* to each other, and alternate above and below the peptide chain. The actual peptide bond is a hybrid between the resonance forms shown, resulting in a partial negative charge on the carbonyl oxygen, a partial positive charge on the nitrogen, and partial double bond character for the peptide bond itself.

A. Descriptions of Protein Structure

Proteins are generally grouped into major structural classifications: globular proteins, fibrous proteins, transmembrane proteins, and DNA-binding proteins (Fig. 7.2). Globular proteins are usually soluble in aqueous medium and resemble irregular balls. The fibrous proteins are geometrically linear, arranged around a single axis, and have a repeating unit structure. Another general classification, transmembrane proteins, consists of proteins that have one or more regions aligned to cross the lipid membrane. DNA-binding proteins, usually classified separately, are considered in Chapter 16.

The structure of these proteins is often described according to levels called primary, secondary, tertiary, and quaternary structure (see Fig. 7.1). The primary structure is the linear sequence of amino acid residues joined through peptide bonds to form a polypeptide chain. The secondary structure refers to recurring structures (such as the regular structure of the α -helix) that form in short localized regions of the polypeptide chain. The overall three-dimensional conformation of a protein is its tertiary structure. The quaternary structure is the association of polypeptide subunits in a geometrically specific manner.

B. Requirements of the Three-Dimensional Structure

The overall three-dimensional structure of a protein must meet certain requirements to enable the protein to function in the cell or extracellular medium of the body. The first requirement is the creation of a binding site that is specific for just one molecule, or a group of molecules with similar structural properties. The specific binding sites of a protein usually define its role. The three-dimensional structure also must exhibit the degrees of flexibility and rigidity appropriate to its function. Some rigidity is essential for the creation of binding sites and for a stable structure (i.e., a protein that just flopped all over the place could not accomplish anything). However, flexibility and mobility in structure enables the protein to fold as it is synthesized, and to adapt as it binds other proteins and small molecules. The three-dimensional structure must have an external surface appropriate for its environment (e.g., cytoplasmic proteins need to keep polar amino acids on the surface to remain soluble in an aqueous environment). In addition, the conformation must also be stable, with little tendency to undergo refolding into a form that cannot fulfill its function or that precipitates in the cell. Finally, the protein must have a structure that can be degraded when it is damaged or no longer needed in the cell. Almost every region in the sequence of amino acids, the primary structure, participates in fulfilling one or more of these requirements through the chemical properties of the peptide bonds and the individual amino acid side chains.

II. THE THREE-DIMENSIONAL STRUCTURE OF THE PEPTIDE BACKBONE

The amino acids in a polypeptide chain are sequentially joined by peptide bonds between the carboxyl group of one amino acid and amide group of the next amino acid in the sequence (Fig. 7.3). Usually the peptide bond assumes a *trans* configuration in which successive α -carbons and their R groups are located on opposite sides of the peptide bond.

The polypeptide backbone can only bend in a very restricted way. The peptide bond itself is a hybrid of two resonance structures, one of which has double bond character, so that the carboxyl and amide groups that form the bond must, therefore, remain planar (see Fig. 7.3.). As a consequence, the peptide backbone consists of a sequence of rigid planes formed by the peptide groups (see Fig. 7.3). However, rotation within certain allowed angles (torsion angles) can occur around the bond between the α -carbon and the α -amino group and around the bond between the α -carbon and the carbonyl group. This rotation is subject to steric constraints that maximize the

distance between atoms in the different amino acid side chains and forbid torsion (rotation) angles that place the side chain atoms too close to each other. These folding constraints, which depend on the specific amino acids present, limit the secondary and tertiary structures that can be formed from the polypeptide chain.

III. SECONDARY STRUCTURE

Regions within polypeptide chains form recurring, localized structures known as secondary structures. The two regular secondary structures called the α -helix and the α -sheet contain repeating elements formed by hydrogen bonding between atoms of the peptide bonds. Other regions of the polypeptide chain form nonregular non-repetitive secondary structures, such as loops and coils.

A. The α -Helix

The α -helix is a common secondary structural element of globular proteins, membrane-spanning domains, and DNA-binding proteins. It has a stable rigid conformation that maximizes hydrogen bonding while staying within the allowed rotation angles of the polypeptide backbone. The peptide backbone of the α -helix is formed by strong hydrogen bonds between each carbonyl oxygen atom and the amide hydrogen (N-H) of an amino acid residue located four residues further down the chain (Fig. 7.4). Thus, each peptide bond is connected by hydrogen bonds to the peptide bond four amino acid residues ahead of it and four amino acid residues behind it in the amino acid sequence. The core of the helix is tightly packed, thereby maximizing association energies between atoms. The *trans* side chains of the amino acids project backward and outward from the helix, thereby avoiding steric hindrance with the polypeptide backbone and with each other (Fig. 7.5).

B. β -Sheets

β -Sheets are a second type of regular secondary structure that maximizes hydrogen bonding between the peptide backbones while maintaining the allowed torsion angles. In β -sheets, the hydrogen bonding usually occurs between regions of separate neighboring polypeptide strands aligned parallel to each other (Fig. 7.6). Thus, the carbonyl oxygen of one peptide bond is hydrogen-bonded to the nitrogen of a peptide bond on an adjacent strand. (This pattern contrasts with the α -helix in which the peptide backbone hydrogen bonds are within the same strand.) Optimal hydrogen bonding occurs when the sheet is bent (pleated) to form β -pleated sheets.

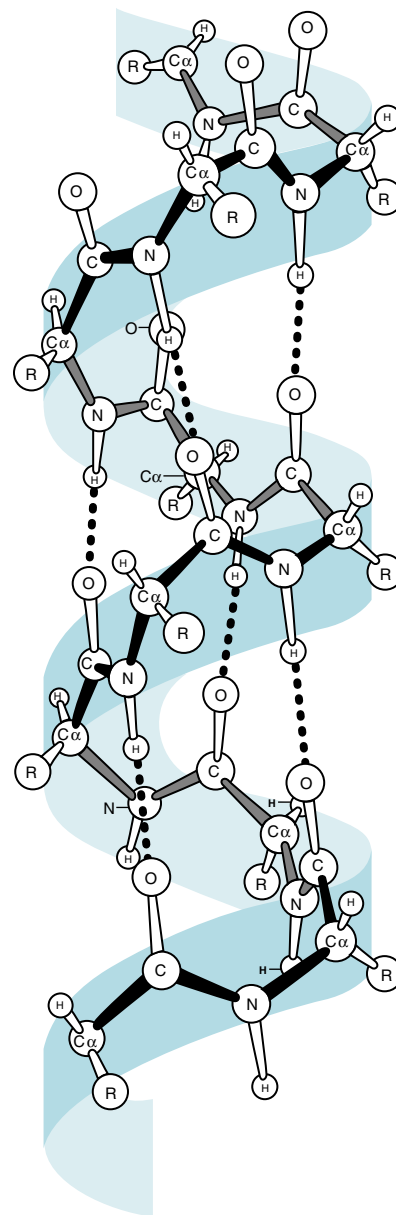


Fig. 7.4. The α -helix. Each oxygen of a carbonyl group of a peptide bond forms a hydrogen bond with the hydrogen atom attached to a nitrogen atom in a peptide bond four amino acids further along the chain. The result is a highly compact and rigid structure.

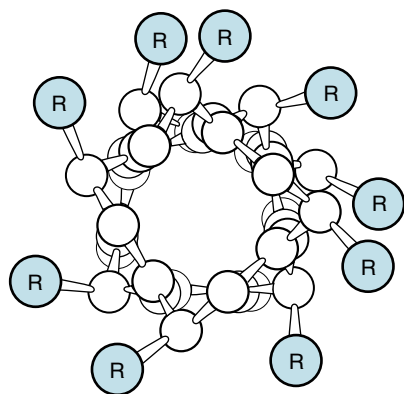


Fig. 7.5. A view down the axis of an α -helix. The side chains (R) jut out from the helix. Steric hindrance occurs if they come within their van der Waals radii of each other, and a stable helix cannot form.

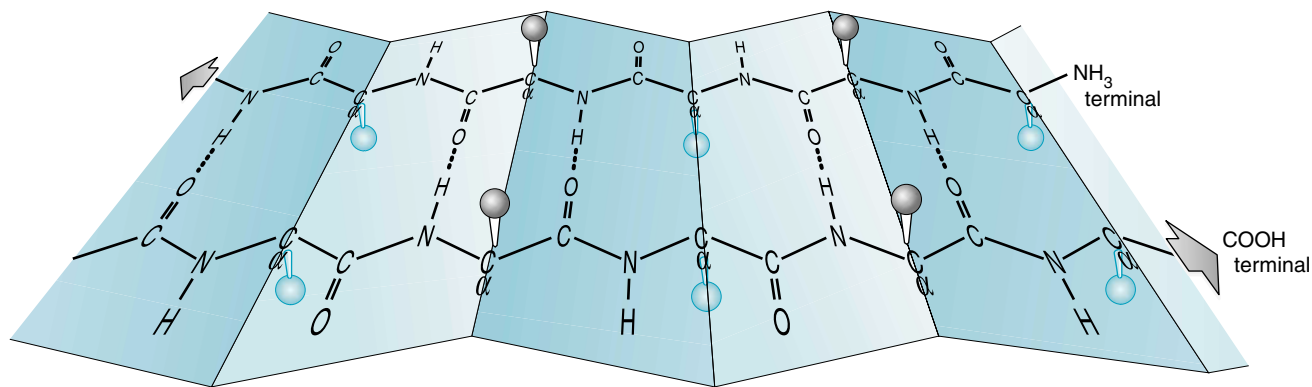


Fig. 7.6. A β -pleated sheet. In this case, the chains are oriented in opposite directions (antiparallel). The large arrows show the direction of the carboxy terminal. The amino acid side chains (R) in one strand are *trans* to each other, and alternate above and below the plane of the sheet, which can have a hydrophobic face and a polar face that engages in hydrogen bonding.



A renal biopsy used in the diagnosis of **Amy Lloyd's** disease showed amorphous deposits in the glomeruli. When stained with Congo red dye, these deposits appeared red with ordinary light microscopy and exhibited apple green fluorescence when viewed in polarized light. This staining is characteristic of the amyloid fibril structure, which is composed of repeated β -sheets aligned orthogonally (perpendicular) to the axis of the fiber.

A number of diseases involve deposition of a characteristic amyloid fiber. However, in each of these diseases, the amyloid is derived from a different protein that has changed its conformation (three-dimensional structure) to that of the amyloid repeated β -sheet structure. Once amyloid deposition begins, it seems to proceed rapidly, as if the fibril itself were promoting formation and deposition of more fibrils (a phenomenon called "seeding"). The different clinical presentations in each of these diseases results from differences in the function of the native protein and the site of amyloid deposition.

The β -pleated sheet is described as parallel if the polypeptide strands run in the same direction (as defined by their amino and carboxy terminals) and anti-parallel if they run in opposite directions. Antiparallel strands are often the same polypeptide chain folded back on itself, with simple hairpin turns or long runs of polypeptide chain connecting the strands. The amino acid side chains of each polypeptide strand alternate between extending above and below the plane of the β -sheet (see Fig. 7.6). Parallel sheets tend to have hydrophobic residues on both sides of the sheets; antiparallel sheets usually have a hydrophobic side and a hydrophilic side. Frequently, sheets twist in one direction.

C. Nonrepetitive Secondary Structures

α -Helices and β -pleated sheets are patterns of regular structure with a repeating element, the turn of a helix or a pleat. In contrast, bends, loops, and turns are nonregular secondary structures that do not have a repeating element. They are characterized by an abrupt change of direction and are often found on the protein surface. For example, β -turns are short regions usually involving four successive amino acid residues. They often connect strands of antiparallel β -sheets (Fig. 7.7.). The surface of large globular proteins usually has at least one omega loop, a structure with a neck like the capital Greek letter omega, Ω .

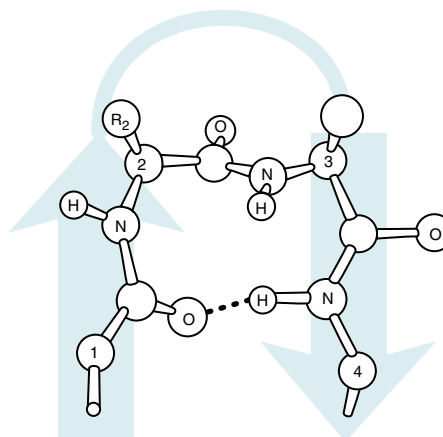


Fig. 7.7. β -turn. The four amino acid residues that form the β -turn (also called a hairpin loop) are held together by hydrogen bonds, which make this an extremely stable structure.

D. Patterns of Secondary Structure

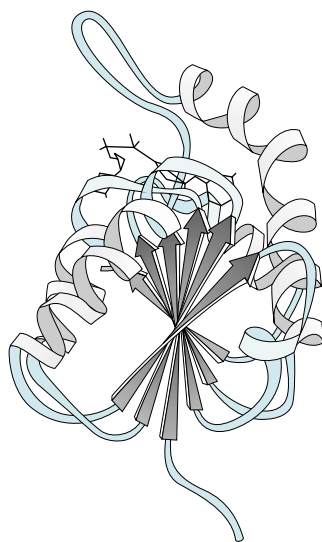
Figure 7.8 is a three-dimensional drawing of a globular domain in the soluble enzyme lactate dehydrogenase (LDH). It illustrates the combination of secondary structural elements to form patterns. This LDH domain is typical of globular proteins, which average approximately 31% α -helical structure and approximately 28% β -pleated sheets (with a wide range of variation.) The helices of globular domains have an average span of approximately 12 residues, corresponding to approximately three to four helical turns, although many are much longer. The β -sheets, represented in diagrams by an arrow for each strand, are an average of six residues long and six strands wide (2–15 strands). Like the β -sheet in the lactate dehydrogenase domain, they generally twist to the right, rather than lie flat (see Fig. 7.8). Most globular domains, such as this LDH domain, also contain motifs. Motifs are relatively small arrangements of secondary structure recognized in many different proteins. For example, certain of the β -strands are connected with α -helices to form the $\beta\alpha\beta\alpha\beta$ structural motif.

The remaining polypeptide segments connecting the helices and β -sheets are said to have a coil or loop conformation (see Fig. 7.8). Although some of the connecting segments recognized in many proteins have been given names (like the Ω -loops), other segments such as those in this LDH domain appear disordered or irregular. These nonregular regions, generally called coils, should never be referred to as “random coils.” They are neither truly disordered nor random; they are stabilized through specific hydrogen bonds dictated by the primary sequence of the protein and do not vary from one molecule of the protein to another of the same protein.

The nonregular coils, loops, and other segments are usually more flexible than the relatively rigid helices and β -pleated sheets. They often form hinge regions that allow segments of the polypeptide chain to move as a compound binds or to move as the protein folds around another molecule.



Although it is usually assumed that proteins can have truly disordered regions, the more that is learned about protein structure, the less disordered these regions seem. Even regions that look truly disordered may form a specific binding site for another molecule (e.g., the intracellular C-terminal of transmembrane proteins). Eventually all regions of all proteins may be classified as a particular named pattern.



Lactate dehydrogenase domain 1

Fig. 7.8. Ribbon drawing showing the arrangement of secondary structures into a three-dimensional pattern in domain 1 of lactate dehydrogenase. The individual polypeptide strands in the six-stranded β -sheet are shown with arrows. Different strands are connected by helices and by nonrepetitive structures (turns, coils and loops), shown in blue. This domain is the nucleotide binding fold. NAD^+ is bound to a site created by the helices (upper left of figure.) (Modified from Richardson JS. *Adv Protein Chem. The anatomy and taxonomy of protein structure* 1981;34:167).



Flexibility is one of the most important features of protein structure.

Although every portion of every amino acid in a protein is engaged in bonding to the rest of the protein, to water, or to a ligand, proteins do not have rigid structures. The three-dimensional structure is flexible and dynamic, with rapidly fluctuating movement in the exact position of amino acid side chains and domains. These fluctuations are like wiggling or shaking that can occur without unfolding. They allow ions and water to diffuse through the structure and provide alternate conformations for ligand binding.

IV. TERTIARY STRUCTURE

The tertiary structure of a protein is the folding pattern of the secondary structural elements into a three-dimensional conformation, as shown for the LDH domain in Figure 7.8. As illustrated with examples below, this three-dimensional structure is designed to serve all aspects of the protein's function. It creates specific and flexible binding sites for ligands (the compound that binds), illustrated with actin and myoglobin. The tertiary structure also maintains residues on the surface appropriate for the protein's cellular location, polar residues for cytosolic proteins, and hydrophobic residues for transmembrane proteins (illustrated with the β_2 -adrenergic receptor).

A. Domains in the Tertiary Structure

The tertiary structure of large complex proteins is often described in terms of physically independent regions called structural domains. You can usually identify domains from visual examination of a three-dimensional figure of a protein, such as the three-dimensional figure of G-actin shown in Fig. 7.9. Each domain is formed from a continuous sequence of amino acids in the polypeptide chain that are folded into a three-dimensional structure independently of the rest of the protein, and two domains are connected through a simpler structure like a loop (e.g., the hinge region of Fig. 7.9). The structural features of each domain can be discussed independently of another domain in the same protein, and the structural features of one domain may not match that of other domains in the same protein.

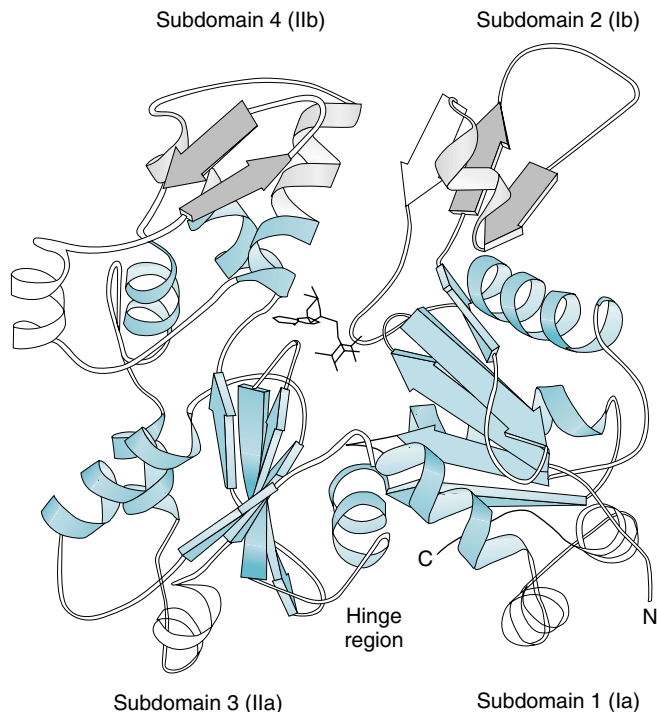


Fig. 7.9. G-Actin. ATP binds in the center of the cleft. The two domains that form the cleft are further subdivided into subdomains 1–4. The overall structure is found in many ATP-binding proteins and is called the actin fold. The conformations of the regions shown in blue are nearly superimposable among the proteins that contain the actin fold. (From Kabsch W, Holmes KC. The actin fold. *FASEB J* 1995;9:167–174.)

B. Folds in Globular Proteins

Folds are relatively large patterns of three-dimensional structure that have been recognized in many proteins, including proteins from different branches of the phylogenetic tree. A characteristic activity is associated with each fold, such as ATP binding and hydrolysis (the actin fold) or NAD^+ binding (the nucleotide-binding fold). These three examples of fold patterns are discussed below.

1. THE ACTIN FOLD

In the three-dimensional drawing of G-actin shown in Figure 7.9, all four subdomains contribute to a folding pattern called the actin fold, named for the first protein in which it was described. ATP is bound into the middle of the cleft of the actin fold by amino acid residues contributed by domains on both sides; thus, ATP binding promotes a conformational change that closes the cleft. Once bound, ATP is cleaved to ADP and phosphate.

The actin fold is found in proteins as diverse as actin, which polymerizes to form the cytoskeleton, heat shock protein 70 (hsp70), which uses ATP energy in changing the conformation of other proteins and hexokinase, which catalyzes phosphorylation of glucose (see Chapter 8 for further discussion of hexokinase.) Although these proteins have very little sequence identity, three-dimensional drawings of their actin folds are almost superimposable. The amount of sequence identity they do have is consistent with their membership in the same fold family and establishes that they are all homologs of the same ancestral protein. In all of these proteins, ATP binding results in large conformational changes that contribute to the function of the protein.

2. THE NUCLEOTIDE BINDING FOLD

A fold also can be formed by one domain. In the example of secondary structures provided by lactate dehydrogenase (see Fig. 7.8), domain 1 alone forms the nucleotide binding fold. This fold is a binding site for NAD^+ or, in other proteins, molecules with a generally similar structure (e.g., riboflavin). However, many proteins that bind NAD^+ or NADP^+ contain a very different fold from a separate fold family. These two different NAD^+ binding folds arise from different ancestral lines and have different structures, but have similar properties and function. They are believed to be the product of convergent evolution.

C. The Solubility of Globular Proteins in an Aqueous Environment

Most globular proteins are soluble in the cell. In general, the core of a globular domain has a high content of amino acids with nonpolar side chains (val, leu, ile, met, and phe), out of contact with the aqueous medium. This hydrophobic core is densely packed to maximize attractive van der Waals forces, which exert themselves over short distances. The charged polar amino acid side chains (arg, his, lys, asp, and glu) are generally located on the surface of the protein, where they form ion pairs (salt bridges) or are in contact with aqueous solvent. Charged side chains often bind inorganic ions (e.g., K^+ , PO_4^{3-} or Cl^-) to decrease repulsion between like charges. When charged amino acids are located on the interior, they are generally involved in forming specific binding sites. The polar uncharged amino acid side chains of ser, thr, asn, gln, tyr, and trp are also usually found on the surface of the protein, but may occur in the interior, hydrogen bonded to other side chains. Although cystine disulfide bonds (the bond formed by two cysteine sulfhydryl groups) are sometimes involved in the formation of tertiary structure, they are generally not needed.



More than 1000 folds have now been recognized, and it is predicted that there are only a few thousand different folds for all the proteins that have ever existed. One of the values of classifying proteins into fold patterns is that a specific function, such as ATP binding and hydrolysis, is associated with each fold. Currently, we know the primary structure of many proteins involved in inherited diseases without knowing the function of the protein. However, the function of a domain in a protein can be predicted if its amino acid sequence can be matched to that of other proteins in a fold family with a known function. Classifying proteins into fold families also may be useful for drug design. The structure of a bacterial protein targeted by a drug can be compared with that of human proteins in the same fold family so that an antibacterial drug can be designed that more specifically targets the bacterial protein.



G-actin subunits polymerize to form F-actin, which forms the cytoskeleton (see Chapter 10). The conformational change caused by ATP binding to the G-actin subunits promotes their addition to the growing ends of the F-actin polymers. Hydrolysis of bound ATP to ADP by the G-actin subunit promotes dissociation.



Many features of the adrenaline (β_2 -adrenergic) receptor are typical of hormone receptors. Each extracellular loop is actually a structural domain, but several loops together form the binding site for the hormone. The binding site is sometimes referred to as a binding domain (a functional domain), even though it is not formed from a continuous segment of the polypeptide chain. The amino terminus (residues 1–34) extends out of the membrane and has branched high mannose oligosaccharides linked through N-glycosidic bonds to the amide of asparagine. It is anchored in the lipid plasma membrane by a palmitoyl group (shown as a squiggle in Fig. 7.10) that forms a thioester with the SH residue of a cysteine. The COOH terminus, which extends into the cytoplasm, has a number of serine and threonine phosphorylation sites (shown as blue circles in Fig. 7.10).

D. Tertiary Structure of Transmembrane Proteins

Transmembrane proteins, such as the β_2 -adrenergic receptor, contain membrane-spanning domains and intracellular and extracellular domains on either side of the membrane (Fig. 7.10). Many ion channel proteins, transport proteins, neurotransmitter receptors, and hormone receptors contain similar membrane-spanning segments that are α -helices with hydrophobic residues exposed to the lipid bilayer. These rigid helices are connected by loops containing hydrophilic amino acid side chains that extend into the aqueous medium on both sides of the membrane. In the β_2 -adrenergic receptor, the helices clump together so that the extracellular loops form a surface that acts as a binding site for the hormone adrenaline (epinephrine), our fight or flight hormone. The binding site is sometimes referred to as a binding domain (a functional domain), even though it is not formed from a continuous segment of the polypeptide chain. Once adrenaline binds to the receptor, a conformational change in the arrangement of rigid helical structures is transmitted to the intracellular domains that form a binding site for another signaling protein, a heterotrimeric G protein (a guanosine triphosphate [GTP]-binding protein composed of three different subunits). Thus, receptors require both rigidity and flexibility to transmit signals across the cell membrane.

As discussed in Chapter 6, transmembrane proteins usually have a number of posttranslational modifications that provide additional chemical groups to fulfill requirements of the three-dimensional structure. The amino terminus (residues 1–34) extends out of the membrane and has branched high mannose oligosaccharides linked through N-glycosidic bonds to the amide of asparagine (see Fig. 7.10). It is anchored in the lipid plasma membrane by a palmitoyl group that forms a thioester with the SH residue of a cysteine. The COOH terminus, which extends

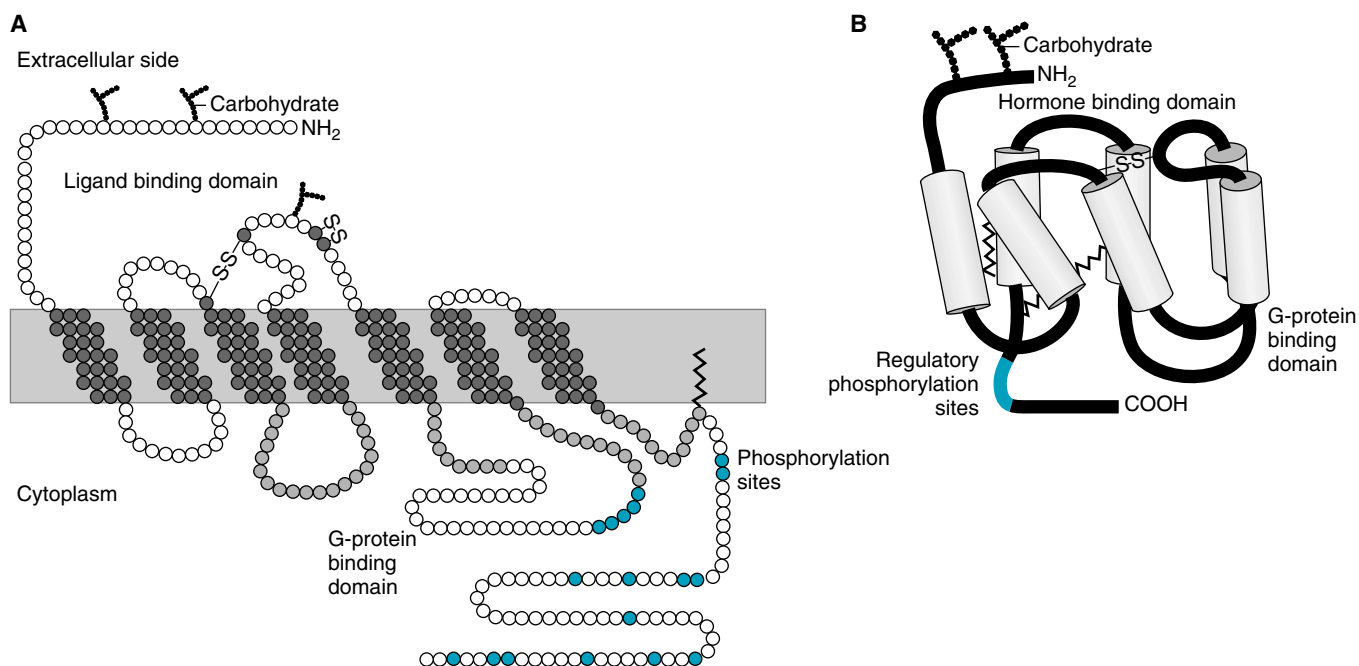


Fig. 7.10. β_2 -Adrenergic receptor. The receptor has seven α -helix domains that span the membrane and is therefore a member of the heptahelical class of receptors. **A.** The transmembrane domains are drawn in an extended form. **B.** The seven transmembrane helices (shown as tubes) form a cylindrical structure. Loops connecting helices form the hormone binding site on the external side of the plasma membrane, and a binding site for a G-protein is on the intracellular side. The protein also contains oligosaccharide chains, palmitoyl groups, and phosphorylation sites (shown in blue; see Chapter 6).

into the cytoplasm, has a number of serine and threonine phosphorylation sites (shown as blue circles) that regulate receptor activity.

V. QUATERNARY STRUCTURE

The quaternary structure of a protein refers to the association of individual polypeptide chain subunits in a geometrically and stoichiometrically specific manner. Many proteins function in the cell as dimers, tetramers, or oligomers, proteins in which two, four, or more subunits, respectively, have combined to make one functional protein. The subunits of a particular protein always combine in the same number and in the same way, because the binding between the subunits is dictated by the tertiary structure, which is dictated by the primary structure, which is determined by the genetic code.

A number of different terms are used to describe subunit structure. The prefixes “homo” or “hetero” are used to describe identical or different subunits, respectively, of 2, 3, or 4 subunit proteins (e.g., heterotrimeric G proteins have three different subunits). A protomer is a unit structure composed of nonidentical subunits. In contrast, F-actin is an oligomer, a multisubunit protein composed of identical G-actin subunits. “Multimer” is sometimes used as a more generic term to designate a complex with many subunits of more than one type.

The contact regions between the subunits of globular proteins resemble the interior of a single subunit protein; they contain closely packed nonpolar side chains, hydrogen bonds involving the polypeptide backbones and their side chains, and occasional ionic bonds or salt bridges. The subunits of globular proteins are very rarely held together by interchain disulfide bonds, and never by other covalent bonds. In contrast, fibrous and other structural proteins may be extensively linked to other proteins through covalent bonds.

Assembly into a multisubunit structure increases the stability of a protein. The increase in size increases the number of possible interactions between amino acid residues and therefore makes it more difficult for a protein to unfold and refold. As a result, many soluble proteins are composed of two or four identical or nearly identical subunits with an average size of approximately 200 amino acids.

A multisubunit structure has many advantages besides increased stability. It may enable the protein to exhibit cooperativity between subunits in binding ligands (illustrated later with hemoglobin) or to form binding sites with a high affinity for large molecules (illustrated with antigen binding to the immunoglobulin molecule, IgG). An additional advantage of a multisubunit structure is that the different subunits can have different activities and cooperate in a common function. Examples of enzymes that have regulatory subunits or exist as multiprotein complexes are provided in Chapter 9.



Insulin is composed of two nonidentical polypeptide chains attached to each other through disulfide bonds between the chains (see Chapter 6, Fig. 6.13).

The subunits of globular proteins are generally not held together by disulfide bonds, but regions of the same chain may be connected by disulfide bonds that form as the chain folds. Insulin actually fits this generalization because it is synthesized as a single polypeptide chain, which forms the disulfide bonds. Subsequently, a proteolytic enzyme in secretory vesicles clips the polypeptide chain into two nonidentical subunits. Generally, each subunit of most protomers and oligomers is synthesized as a separate polypeptide chain. In fibrous proteins, which have a regular, sometimes repeating sequence of amino acids, interchain binding serves different functions. In collagen, for example, extensive interchain binding provides great tensile strength. The structure of collagen is discussed in Chapter 49 on connective tissue.



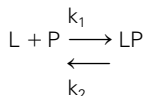
Creatine phosphokinase (CK), one of the proteins measured to follow Ann Jeina's myocardial infarction

(see Chapter 6) is present in cells as dimers (two subunits). The dimers may be homodimers (two identical subunits of either the M [muscle] isozyme or the B [brain] isozyme), or heterodimers (MB). The MB isozyme is produced only by the heart and readily released from injured cardiomyocytes into the blood (see Chapter 6).

Serum levels of lactate dehydrogenase (LDH) were formerly used to diagnose an acute MI. LDH is present in cells as a tetramer of four identical, or nearly identical, subunits. Each subunit is a separate polypeptide chain with a molecular weight of 35 kD (approximately 35,000 g/mole). These subunits are present as two isoforms, the H isoform (for heart) and the M isoform (for skeletal muscle). Although the heart produces principally the H₄ form (four H subunits combined into a tetramer) and skeletal muscles produce principally the M₄ isoform, the heart, skeletal muscle, and other tissues produce several intermediate combinations (e.g., H₃M, H₂M₂). These tetrameric isoforms all have similar activities, but the individual monomeric subunits are inactive. Measurements of LDH isozymes in the serum are no longer used for diagnosis of a recent MI because the enzyme is large, released slowly, and the isozyme pattern is not as specific for the heart as is CK.

Equation 7.1. The association constant, K_a for a binding site on a protein.

Consider a reaction in which a ligand (L) binds to a protein (P) to form a ligand–protein complex (LP) with a rate constant of k_1 . LP dissociates with a rate constant of k_2 :



then,

$$K_{\text{eq}} = \frac{k_1}{k_2} = \frac{[\text{LP}]}{[\text{L}][\text{P}]} = \frac{K_a}{K_d} = 1$$

The equilibrium constant, K_{eq} , is equal to the association constant (K_a) or $1/K_d$, the dissociation constant. Unless otherwise given, the concentrations of L, P, and LP are expressed as mol/L, and K_a has the units of $(\text{mol/L})^{-1}$.

VI. QUANTITATION OF LIGAND BINDING

In the examples of tertiary structure discussed above, the folding of a protein created a three-dimensional binding site for a ligand (NAD^+ for the lactate dehydrogenase domain 1, ATP for hexokinase, or adrenaline for the β_2 adrenergic receptor). The binding affinity of a protein for a ligand is quantitatively described by its association constant, K_a , which is the equilibrium constant for the binding reaction of a ligand (L) with a protein (P) (Eq. 7.1).

K_a is equal to the rate constant (k_1) for association of the ligand with its binding site divided by the rate constant (k_2) for dissociation of the ligand–protein complex (LP). K_d , the dissociation constant for ligand–protein binding, is the reciprocal of K_a . The tighter the binding of the ligand to the protein, the higher is the K_a and the lower is the K_d . The K_a is useful for comparing proteins produced by different alleles, or for describing the affinity of a receptor for different drugs.

VII. STRUCTURE–FUNCTION RELATIONSHIPS IN MYOGLOBIN AND HEMOGLOBIN

Myoglobin and hemoglobin are two oxygen-binding proteins with a very similar primary structure (Fig. 7.11). However, myoglobin is a globular protein composed of a single polypeptide chain that has one O_2 binding site. Hemoglobin is a tetramer composed of two different types of subunits (2 α and 2 β polypeptide chains, referred to as two $\alpha\beta$ protomers). Each subunit has a strong sequence homology to myoglobin and contains an O_2 binding site. A comparison between myoglobin and hemoglobin illustrates some of the advantages of a multisubunit quaternary structure.

The tetrameric structure of hemoglobin facilitates saturation with O_2 in the lungs and release of O_2 as it travels through the capillary beds (Fig. 7.12). When the amount of oxygen bound to myoglobin or hemoglobin is plotted against the partial pressure of oxygen (pO_2), a hyperbolic curve is obtained for myoglobin, whereas that for hemoglobin is sigmoidal. These curves show that when the pO_2 is high, as in the lungs, both myoglobin and hemoglobin are saturated with oxygen. However,

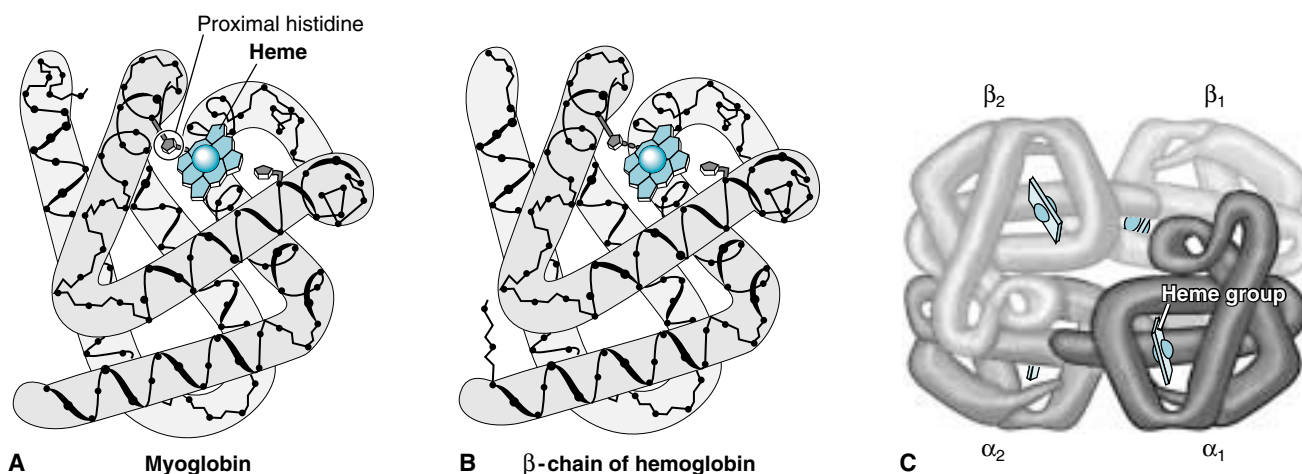


Fig. 7.11. Myoglobin and hemoglobin. Myoglobin consists of a single polypeptide chain, which is similar in structure to the α and β subunits of hemoglobin. In all of the subunits, heme is tightly bound in a hydrophobic binding pocket. The proximal histidine extends down from a helix to bind to the Fe atom. The oxygen binds between the distal histidine and the heme. Panel C displays the quaternary structure of hemoglobin (From Frescht A. *Structure and Mechanism in Protein Science*. New York: WH Freeman and Company, 1999. Used with permission.)

at the lower levels of pO_2 in oxygen-using tissues, hemoglobin cannot bind oxygen as well as myoglobin (i.e., its percent saturation is much lower). Myoglobin, which is present in heart and skeletal muscle, can bind the O_2 released by hemoglobin, which it stores to meet the demands of contraction. As O_2 is used in the muscle cell for generation of ATP during contraction, it is released from myoglobin and picked up by cytochrome oxidase, a heme-containing enzyme in the electron transport chain that has an even higher affinity for oxygen than myoglobin.

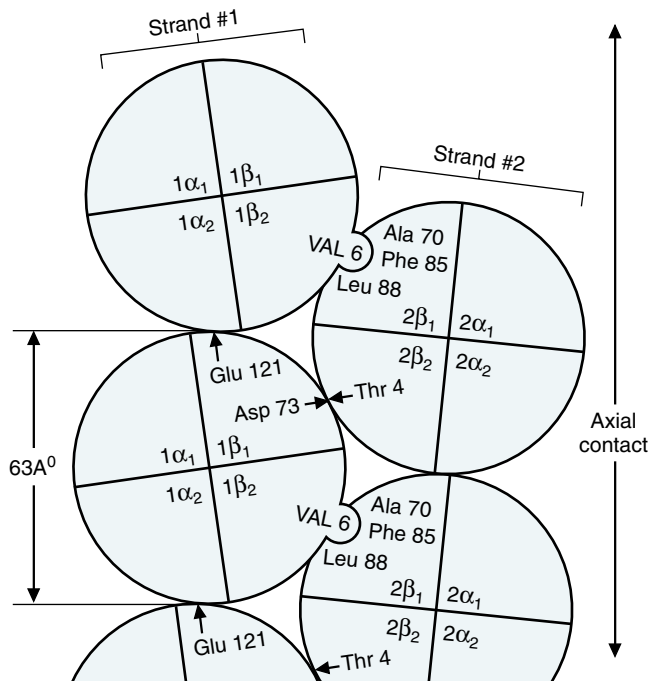
A. Oxygen Binding and Heme

The tertiary structure of myoglobin consists of eight α -helices connected by short coils, a structure that is known as the globin fold (see Fig. 7.11). This structure is unusual for a globular protein in that it has no β -sheets. The helices create a hydrophobic O_2 binding pocket containing tightly bound heme with an iron atom (Fe^{2+}) in its center.

Heme consists of a planar porphyrin ring composed of four pyrrole rings that lie with their nitrogen atoms in the center, binding an Fe^{2+} atom (Fig. 7.13).



Sickle cell anemia is really a disease caused by the wrong quaternary structure. The painful vasoocclusive crises experienced by **Will Sichel** are caused by the polymerization of HbS molecules into long fibers that distort the shape of the red blood cells into sickle cells. The substitution of a hydrophobic valine for a glutamate in the β_2 chain creates a knob on the surface of deoxygenated hemoglobin that fits into a hydrophobic binding pocket on the β_1 subunit of a different hemoglobin molecule. A third hemoglobin molecule, which binds to the first and second hemoglobin molecules through aligned polar interactions, binds a fourth hemoglobin molecule through its valine knob. Thus the polymerization continues until long fibers are formed.



Polymerization of the hemoglobin molecules is highly dependent on the concentration of HbS and is promoted by the conformation of the deoxygenated molecules. At 100% oxygen saturation, even high concentrations of HbS will not polymerize. A red blood cell spends the longest amount of time at the lower oxygen concentrations of the venous capillary bed, where polymerization is most likely initiated.



Myoglobin is readily released from skeletal muscle or cardiac tissue when the cell is damaged. It has a small molecular weight, 17,000 kDa, and is not complexed to other proteins in the cell. (Da is the abbreviation for Dalton, which is a unit of mass approximately equal to one H atom. Thus, a molecular weight of 17,000 kDa is equal to approximately 17,000 g/mole.) Large injuries to skeletal muscle that result from physical crushing or lack of ATP production result in cellular swelling and the release of myoglobin and other proteins into the blood. Myoglobin passes into the urine and turns the urine red because the heme (which is red) remains covalently attached to the protein. During an acute MI, myoglobin is one of the first proteins released into the blood from damaged cardiac tissue. (However, the amount released is not high enough to cause myoglobinuria.) Laboratory measurements of serum myoglobin are used for early diagnosis in patients such as **Ann Jeina**. Because myoglobin is not present in skeletal muscle and the heart as tissue-specific isozymes, and the amount released from the heart is much smaller than the amount that can be released from a large skeletal muscle injury, myoglobin measurements are not specific for an MI.

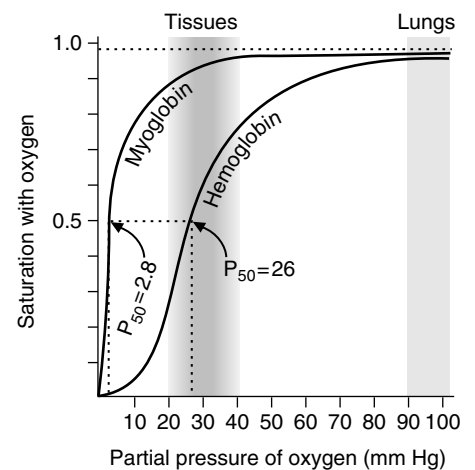


Fig. 7.12. Oxygen saturation curves for myoglobin and hemoglobin. Note that the curve for myoglobin is hyperbolic, whereas that for hemoglobin is sigmoidal. The effect of the tetrameric structure is to inhibit O_2 binding at low O_2 concentrations. P_{50} is the partial pressure of O_2 (pO_2) at which the protein is half-saturated with O_2 . P_{50} for myoglobin is 2.8, and that for hemoglobin is 26.

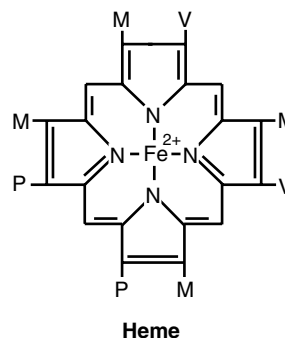


Fig. 7.13. Heme. The Fe is bound to four nitrogen atoms in the center of the heme porphyrin ring. Methyl (M, CH_3), vinyl (V, $-\text{CH}=\text{CH}_2$), and propionate (P, $\text{CH}_2\text{CH}_2\text{COO}^-$) side chains extend out from the four pyrrole rings that constitute the porphyrin ring.

Negatively charged propionate groups on the porphyrin ring interact with arginine and histidine side chains from the hemoglobin, and the hydrophobic methyl and vinyl groups that extend out from the porphyrin ring interact with hydrophobic amino acid side chains from hemoglobin. All together, there are approximately 16 different interactions between myoglobin amino acids and different groups in the porphyrin ring.

Organic ligands that are tightly bound, such as the heme of myoglobin, are called prosthetic groups. A protein with its attached prosthetic group is called a holoprotein; without the prosthetic group, it is called an apoprotein. The tightly bound prosthetic group is an intrinsic part of the protein and does not dissociate until the protein is degraded.

Within the binding pocket of myoglobin, O_2 binds directly to the Fe^{2+} atom on one side of the planar porphyrin ring (Fig. 7.14). The Fe^{2+} atom is able to chelate six different ligands; four of the ligand positions are in a plane and taken by the central nitrogens in the planar porphyrin ring. Two ligand positions are perpendicular to this plane. One of these positions is taken by the nitrogen atom on a histidine, called the proximal histidine, which extends down from a myoglobin helix. The other position is taken by O_2 .

The proximal histidine of myoglobin and hemoglobin is sterically repelled by the heme porphyrin ring. Thus, when the histidine binds to the Fe^{2+} in the middle of the

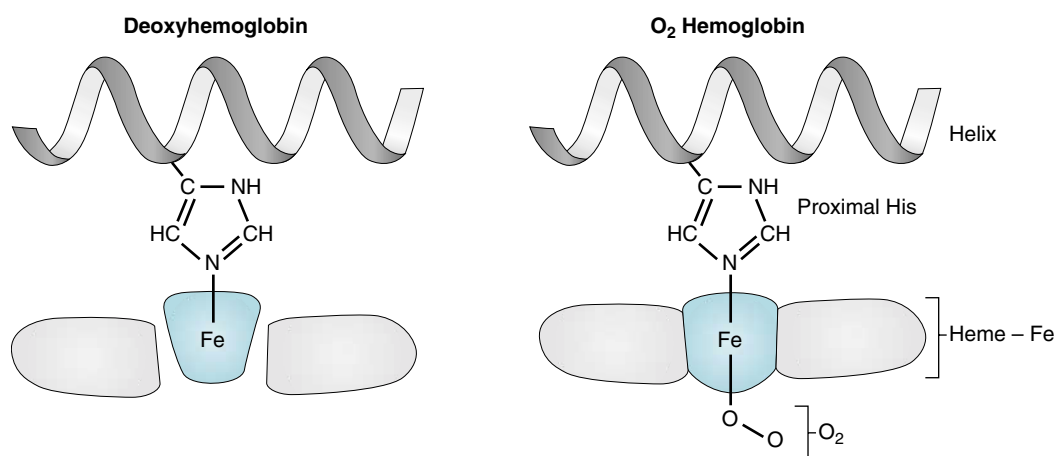


Fig. 7.14. Oxygen binding to the Fe^{2+} of heme in myoglobin. A histidine residue called the proximal histidine binds to the Fe^{2+} on one side of the porphyrin ring; O_2 binds to Fe^{2+} on the other side. O_2 binding causes a conformational change that pulls the Fe^{2+} back into the plane of the ring. As the proximal histidine moves, it moves the helix that contains it.

ring, it pulls the Fe^{2+} above the plane of the ring. When oxygen binds on the other side of the ring, it pulls the Fe^{2+} back into the plane of the ring. The pull of O_2 binding moves the proximal histidine toward the porphyrin ring, which moves the helix containing the proximal histidine. This conformational change has no effect on the function of myoglobin. However, in hemoglobin, the movement of one helix leads to the movement of other helices in that subunit, including one in a corner of the subunit that is in contact with a different subunit through salt bridges. The loss of these salt bridges then induces conformational changes in all other subunits, and all four subunits may change in a concerted manner from their original conformation to a new conformation.

B. Cooperativity of O_2 Binding in Hemoglobin

The cooperativity in oxygen binding in hemoglobin comes from conformational changes in tertiary structure that takes place when O_2 binds (Fig. 7.15). The conformational change of hemoglobin is usually described as changing from a T (tense) state with low affinity for O_2 to an R (relaxed) state with a high affinity for O_2 . Breaking the salt bridges in the contacts between subunits is an energy-requiring

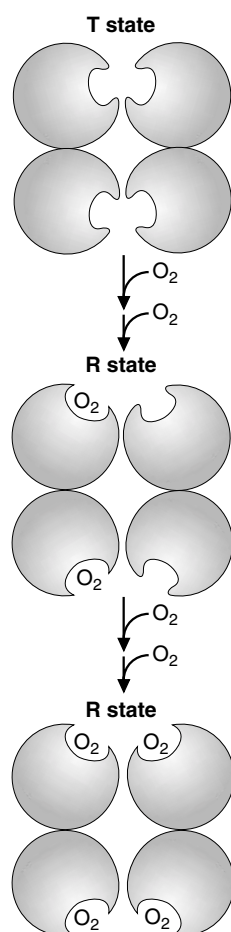


Fig. 7.15. Equilibria for binding of O_2 molecules to hemoglobin according to the concerted model of Monod, Wyman, and Changeux. Hemoglobin exists in two alternate conformations, the T (tense) state with a low affinity for O_2 and the R (relaxed) state with a higher affinity. In the T subunits, the binding sites are hindered, and in the R state the binding sites are open. Each successive addition of O_2 shifts the equilibrium further toward the R state. Because the conformation of all the subunits can change when O_2 binds to one subunit, oxygen binding is said to follow the concerted model. Most of the molecules change to the R state when 2 O_2 molecules have bound.



Amy Lloyd's serum protein electrophoresis indicated the presence of a sharp narrow peak or homogeneous "spike" in the characteristic γ -globulin zone known as an M protein (monoclonal protein) component. A narrow peak of spike in electrophoresis, which separates proteins according to charge distribution of the side chains, suggests an elevation of proteins with a similar or identical structure. Subsequently, it was shown that Amy Lloyd's immunoglobulin M-component was composed of a single homogeneous type of immunoglobulin (just one amino acid sequence in the N-terminal variable region). Thus, the M protein was produced by a single clone of antibody-secreting cells (cells that all arose from proliferation of one cell) in the bone marrow (called plasma cell dyscrasia). In amyloidosis, substitutions of particular amino acids in the light chain variable region are thought to destabilize its native conformation, resulting in fibrillogenesis.

In AL amyloidosis, amyloid is formed from degradation products of the λ or κ light chains that deposit most frequently in the extracellular matrix of the kidney and the heart but also may deposit in the tongue. In other types of amyloidosis, the amyloid arises from other proteins and deposits in a characteristic organ. For example, the amyloid associated with chronic inflammatory conditions, such as tuberculosis or rheumatoid arthritis, is derived from an acute phase serum protein called serum amyloid A that is produced by the liver in response to inflammation. It deposits most frequently in the kidney, and cardiac involvement is rare.

process and, consequently, the binding rate for the first oxygen is very low. When the next oxygen binds, many of the hemoglobin molecules containing one O_2 will already have all four subunits in the R state, and therefore the rate of binding is much higher. With two O_2 molecules bound, an even higher percentage of the hemoglobin molecules will have all four subunits in the R state. This phenomenon, known as positive cooperativity, is responsible for the sigmoidal oxygen saturation curve of hemoglobin (see Fig. 7.12).

VIII. STRUCTURE-FUNCTION RELATIONSHIPS IN IMMUNOGLOBULINS

The immunoglobulins (or antibodies) are one line of defense against invasion of the body by foreign organisms. In this capacity, they function by binding to ligands called antigens on the invading organisms, thereby initiating the process by which these organisms are inactivated or destroyed.

Immunoglobulins all have a similar structure; each antibody molecule contains two identical small polypeptide chains (the light or L chains) and two identical large polypeptide chains (the heavy or H chains) (Fig. 7.16). The chains are joined to each other by disulfide bonds.

The body has five major classes of immunoglobulins. The most abundant immunoglobulins in human blood are the γ -globulins, which belong to the IgG class. The γ -globulins have approximately 220 amino acids in their light chains and 440 in their heavy chains. Like most serum proteins, they have attached oligosaccharides that participate in targeting the protein for clearance from the blood. Both the light and heavy chains consist of domains known as the immunoglobulin fold, which is a collapsed β -barrel made from a number of β -sheets (Fig. 7.17).

Both the light and heavy chains contain regions termed variable (V) and constant (C) regions. The variable regions of the L and H chains (V_L and V_H , respectively) interact to produce a single antigen binding site at each branch of the Y-shaped molecule. Each population (clone) of B cells produces an antibody with a different amino acid composition in the variable region that is complementary to the structure of the antigen eliciting the response. The K_d of antibodies for their specific antigens is extremely small and varies from approximately 10^{-7} to 10^{-11} M. The antigen thus binds very tightly with almost no tendency to dissociate and can be removed from circulation as the antigen-antibody complex is ingested by macrophages. The constant domains that form the Fc part of the antibody are important for binding of the antigen-antibody complex to phagocytic cells for clearance and for other aspects of the immune response.

IX. PROTEIN FOLDING

Although the peptide bonds in a protein are rigid, flexibility around the other bonds in the peptide backbone allow an enormous number of possible conformations for each protein. However, every molecule of the same protein folds into the same stable three-dimensional structure. This shape is known as the native conformation.



The tight binding affinity of immunoglobulins for their specific antigen makes them useful for the measurement of small amounts of other compounds in various radioimmunoassays. The principle of the radioimmunoassay is that the immunoglobulin will specifically bind the compound being measured, and an additional component of the system that is labeled with a radioactive chemical will bind the immunoglobulin. The complex is then separated from the solution and the bound radioactivity measured. The different isozymes of CK and Tn-T used to track **Ann Jeina's** MI are measured with a type of radioimmunoassay by using antibodies specific for each isozyme. This method is much faster than the laborious separation of isozymes by electrophoresis. Radioimmunoassays are also useful for measuring the small amounts of hormones present in the blood for diagnosis of endocrine diseases.

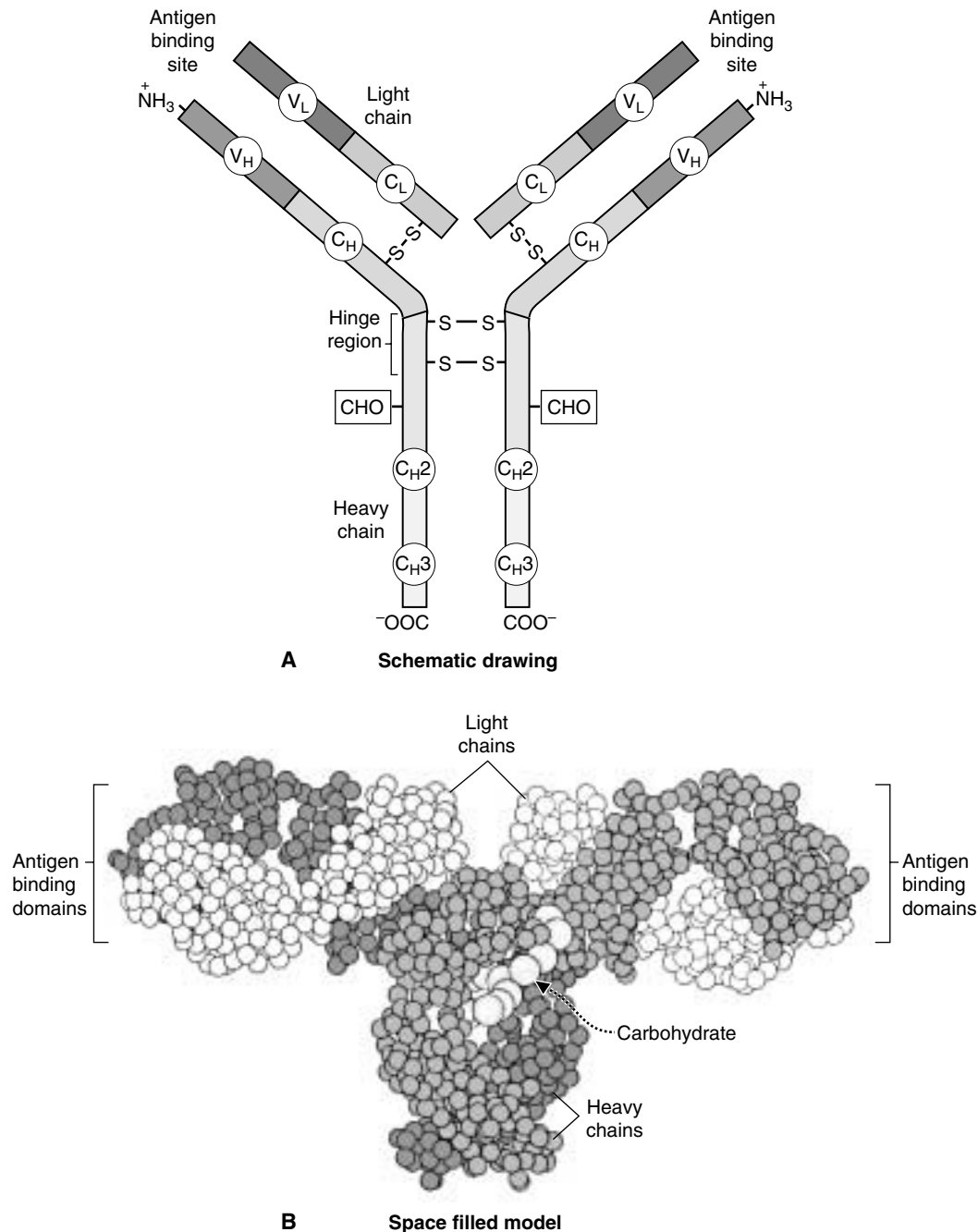


Fig. 7.16. Structure of immunoglobulins. **A.** Each IgG molecule contains two light (L) and two heavy (H) chains joined by disulfide bonds. Each light chain contains two domains, a variable domain (V_L) and a region of constant amino acid sequence (C_L). Each heavy chain has four domains: one variable domain (V_H) and three constant domains (C_H). The conformation of the constant domain contains the β -sheets that are called the immunoglobulin fold. The variable domains are specific for the antigen that is bound, whereas the constant regions are the same for all antibody molecules of a given class. Carbohydrate (CHO) is bound as indicated within the constant region of the heavy chains (CH). The hinge region allows flexibility when the molecule binds antigen. **B.** In the space-filled model, the light chains are light in color and the heavy chains are two different shades of gray. (Modified from Silverton EW, Navia MA, Davies DR, et al. Proc Natl Acad Sci, Three-dimensional structure of an intact human immunoglobulin USA 1977;11:5142)



Very little difference is seen in the energy state of the native conformation and a number of other stable conformations that a protein might assume. It appears that the prion protein, the cause of mad cow disease, is a normal cellular protein that has refolded into a different stable conformation with a lower energy state than its normal functional conformation (discussed under Biochemical Comments, later). If misfolded proteins do not precipitate into aggregates, they can be degraded in the cell by proteolytic reactions, or even refolded.

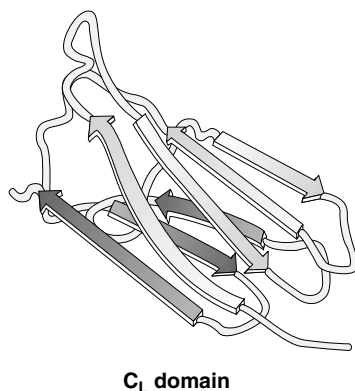
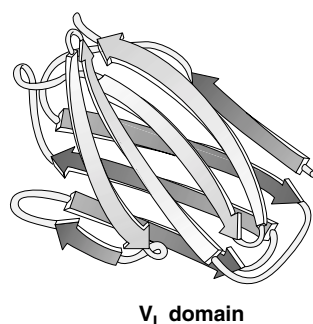


Fig. 7.17. Structure of the V_L and C_L domains of IgG. Layers of antiparallel β -sheets are stacked in these domains, which have been referred to as collapsed β -barrels. The antigen binds between the V_H and V_L immunoglobulin folds, and NOT in the barrel. The C_L domain is also called the immunoglobulin fold. (Top modified from Richardson JS. *Adv Protein Chem*. The anatomy and taxonomy of protein structure 1981;34:167; bottom reprinted in part with permission from Edmundson AB, et al. *Biochemistry* 1975;14:3954. © 1975 American Chemical Society.)

A. Primary Structure Determines Folding

The primary structure of a protein determines its three-dimensional conformation. More specifically, the sequence of amino acid side chains dictates the fold pattern of the three-dimensional structure and the assembly of subunits into quaternary structure. Under certain conditions, denatured proteins can refold into their native conformation, regaining their original function. Proteins can be denatured with organic solvents such as urea that disrupt hydrophobic interactions and convert the protein to a soluble random coil. Many simple single-subunit proteins like ribonuclease that are denatured in this way spontaneously refold into their native conformation if carefully brought back to physiologic conditions. Even complex multi-subunit proteins containing bound cofactors can sometimes spontaneously renature under the right conditions. Thus, the primary structure essentially specifies the folding pattern.

In the cell, not all proteins fold into their native conformation on their own. As the protein folds and refolds while it is searching for its native low energy state, it passes through many high-energy conformations that slow the process (called kinetic barriers). These kinetic barriers can be overcome by heat shock proteins, which use energy provided by ATP hydrolysis to assist in the folding process (Fig. 7.18). Heat shock proteins were named for the fact that their synthesis in bacteria increased when the temperature was suddenly raised. They are

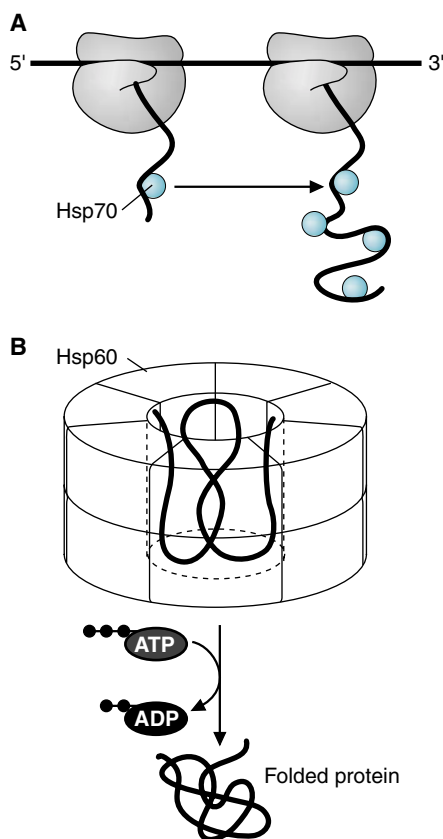


Fig. 7.18. Role of heat shock proteins in folding. A. The Hsp70 family of proteins prevent folding of the nascent chain and promote unfolding. The ATPase domain of the protein has the actin fold. B. The Hsp60 class of protein has a barrel shape into which the protein fits. It acts as a template, binding and rebinding portions of the unfolded protein until folding is completed. It hydrolyzes many ATP bonds to provide energy for the process.

present in human cells as different families of proteins with different activities. For example, the hsp70 proteins bind to nascent polypeptide chains as their synthesis is being completed to keep the uncompleted chains from folding prematurely. They also unfold proteins prior to their insertion through the membrane of mitochondria and other organelles. The multi-subunit barrel-shaped hsp60 family of proteins is called chaperonins. The unfolded protein fits into the barrel cavity that excludes water and serves as a template for the folding process. The hydrolysis of several ATP molecules is used to overcome the energy barriers to reaching the native conformation.

A *cis-trans* isomerase and a disulfide isomerase also participate in folding. The *cis-trans* isomerase converts a *trans* peptide bond preceding a proline into the *cis* conformation, which is well suited for making hairpin turns. The disulfide isomerase breaks and reforms disulfide bonds between the -SH groups of two cysteine residues in transient structures formed during the folding process. After the protein has folded, cysteine-SH groups in close contact in the tertiary structure can react to form the final disulfide bonds.

B. Protein Denaturation

1. DENATURATION THROUGH NONENZYMATIC MODIFICATION OF PROTEINS

Amino acids on proteins can undergo a wide range of chemical modifications that are not catalyzed by enzymes, such as nonenzymatic glycosylation or oxidation. Such modifications usually lead to a loss of function and denaturation of the protein, sometimes to a form that cannot be degraded in the cell. In nonenzymatic glycosylation, glucose that is present in blood, or in interstitial or intracellular fluid, binds to an exposed amino group on a protein (Fig. 7.19). The two-step process forms an irreversibly glycosylated protein. Proteins that turn over very slowly in the body, such as collagen or hemoglobin, exist with a significant fraction present in the glycosylated form. Because the reaction is nonenzymatic, the rate of glycosylation is proportionate to the concentration of glucose present, and individuals with hyperglycemia have much higher levels of glycosylated proteins than individuals with normal blood glucose levels. Collagen and other glycosylated proteins in tissues are further modified by nonenzymatic oxidation and form additional cross-links. The net result is the formation of large protein aggregates referred to as AGEs (advanced glycosylation end-products). AGE is a meaningful acronym because AGEs accumulate with age, even in individuals with normal blood glucose levels.

2. PROTEIN DENATURATION BY TEMPERATURE, pH, AND SOLVENT

Proteins can be denatured by changes of pH, temperature, or solvent that disrupt ionic, hydrogen, and hydrophobic bonds. At a low pH, ionic bonds and hydrogen bonds formed by carboxylate groups would be disrupted; at a very alkaline pH, hydrogen and ionic bonds formed by the basic amino acids would be disrupted. Thus, the pH of the body must be maintained within a range compatible with three-dimensional structure. Temperature increases vibrational and rotational energies in the bonds, thereby affecting the energy balance that goes into making a stable three-dimensional conformation.

Hydrophobic molecules can also denature proteins by disturbing hydrophobic interactions in the protein. For example, long-chain fatty acids can inhibit many enzyme-catalyzed reactions by binding nonspecifically to hydrophobic pockets in proteins and disrupting hydrophobic interactions. Thus, long-chain fatty acids and other highly hydrophobic molecules have their own binding proteins in the cell.



Di Abietes' physician used her glycosylated hemoglobin levels, specifically the HbA_{1c} fraction, to determine whether she had sustained hyperglycemia over a long period. The rate of irreversible nonenzymatic glycosylation of hemoglobin and other proteins is directly proportional to the glucose concentration to which they are exposed over the last 4 months. The danger of sustained hyperglycemia is that, over time, many proteins become glycosylated and subsequently oxidized, affecting their solubility and ability to function. The glycosylation of collagen in the heart, for example, is believed to result in a cardiomyopathy in patients with chronic uncontrolled diabetes mellitus. In contrast, glycosylation of hemoglobin has little effect on its function.



Proteins are denatured in the gastric juice of the stomach, which has a pH of 1 to 2. Although this pH cannot break peptide bonds, disruption of the native conformation makes the protein a better substrate for digestive enzymes.



Thermal denaturation is often illustrated by the process of cooking an egg. In the presence of heat, the protein albumin converts from its native translucent state to a denatured white precipitate. Protein precipitates can sometimes be dissolved by amphipathic agents such as urea, guanidine HCl, or SDS (sodium dodecylsulfate) that form extensive hydrogen bonds and hydrophobic interactions with the protein.

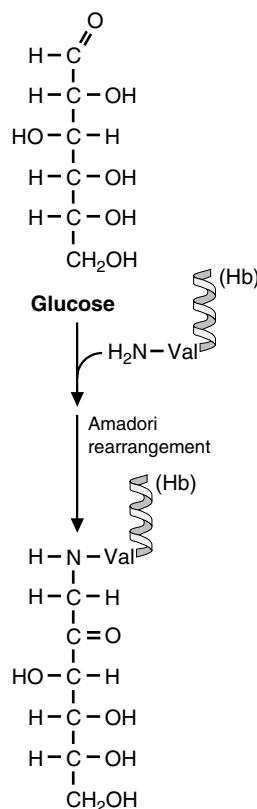


Fig. 7.19. Nonenzymatic glycosylation of hemoglobin. Glucose forms a Schiff base with the N-terminal amino group of the protein, which rearranges to form a stable glycosylated product. Similar nonenzymatic glycosylation reactions occur on other proteins.

CLINICAL COMMENTS

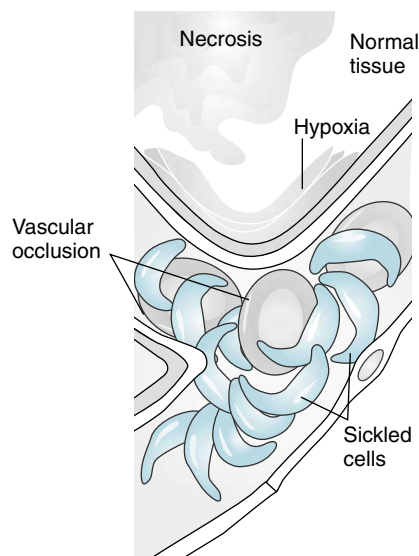


Fig. 7.20 Sickled red blood cells occlude a blood vessel, causing hypoxia (low O_2 in cells) and necrosis (cell death).



Will Sichel. Will Sichel continues to experience severe low back and lower extremity pain for many hours after admission. The diffuse pains of sickle cell crises are believed to result from occlusion of small vessels in a variety of tissues, thereby depriving cells of oxygen and causing ischemic or anoxic damage to the tissues. In a sickle cell crisis, long hemoglobin polymers form, causing the red blood cells to become distorted and change from a biconcave disc to an irregular shape, such as a sickle (for which the disease was named) or a stellate structure (Fig. 7.20). The aggregating Hb polymers damage the cellular membrane and promote aggregation of membrane proteins, increased permeability of the cell, and dehydration. Surface charge and antigens of red blood cells are carried on the transmembrane proteins glycophorin and Band 3 (the erythrocyte anion exchange channel, see Chapter 10). Hemoglobin S binds tightly to the cytoplasmic portion of band 3, contributing to further polymer aggregation and uneven distribution of negative charge on the sickle cell surface. As a result, the affected cells adhere to endothelial cells in capillaries, occluding the vessel and decreasing blood flow to the distal tissues. The subsequent hypoxia in these tissues causes cellular damage and even death.

The sickled cells are sequestered and destroyed mainly by phagocytic cells, particularly those in the spleen. An anemia results as the number of circulating red blood cells decreases and bilirubin levels rise in the blood as hemoglobin is degraded.

After a few days of treatment, **Will Sichel's** crisis was resolved. In the future, should Will suffer a cerebrovascular accident as a consequence of vascular occlusion or have recurrent life-threatening episodes, a course of long-term maintenance blood transfusions to prevent repeated sickle crises may be indicated. Iron chelation would have to accompany such a program to prevent or delay the development of iron overload. Although a few individuals with this disease have survived into the sixth decade, mean survival is probably into the fourth decade. Death usually results from renal failure or cardiopulmonary disease.



Anne Jeina. Mrs. Jeina's diagnosis of an acute myocardial infarction (MI) was based partly on measurements of CK-MB, myoglobin, and cTN-T (the cardiac isozyme of troponin-T, a subunit of the regulatory protein troponin). Early diagnosis is critical for a decision on the type of therapeutic intervention to be used. Of these proteins, myoglobin appears in the blood most rapidly. However, its levels are relatively nonspecific for cardiac injury because the amino acid sequences of cardiac and skeletal muscle myoglobins are identical. Myoglobin measurements do have a very high negative predictive value within the 2- to 6-hour period after the onset of symptoms (i.e., if myoglobin is not elevated, a myocardial infarction did not occur). In contrast, serum cardiac troponin-T is a relatively late, but highly specific, marker of myocardial injury. It is typically detected in an acute MI within 3 to 5 hours after onset of symptoms, is positive in most cases within 8 hours, and approaches 100% sensitivity at 10 to 12 hours. It remains elevated for 5 to 10 days.

Mrs. Jeina stayed in the hospital until she had been free of chest pain for 5 days. She was discharged on a low-fat diet and was asked to participate in the hospital patient exercise program for patients recovering from a recent heart attack. She was scheduled for regular examinations by her physician.



Amy Lloyd. Amy Lloyd has AL amyloidosis, which is characterized by deposition of amyloid fibers derived principally from the variable region of λ or κ immunoglobulin light chains. Increased amounts of the fragments of the light chains called Bence-Jones proteins appeared in her urine. Fibril deposition in the extracellular matrix of her kidney glomeruli has resulted in mild renal failure. Deposition of amyloid in the extracellular matrix of her heart muscle resulted in the cardiac arrhythmia seen on an electrocardiogram. In addition to other signs of right-sided heart failure, she had peripheral edema. The loss of weight may have been caused by infiltrations of amyloid in the gastrointestinal tract or by constipation and diarrhea resulting from involvement of the autonomic nervous system. Treatment may be directed against the plasma cell proliferation, or against the symptomatic results of organ dysfunction.

During Amy Lloyd's evaluation, she developed a cardiac arrhythmia that was refractory to treatment. The extensive amyloid deposits in her heart had disrupted conduction of electrical impulses in the heart muscle, ultimately resulting in cardiac arrest. On autopsy, amyloid deposits were found within the heart, tongue, liver, adipose tissue, and every organ examined except the central nervous system, which had been protected by the blood-brain barrier.



Dianne Abietes. Di Abietes' HbA_{1c} of 8.5% was just above the normal range (5.8–7.2% of total hemoglobin), and her physician decided not to alter her insulin treatment plan. Glycosylation is a nonenzymatic reaction that occurs with a rate directly proportionate to the concentration of glucose in the blood. In the normal range of blood glucose concentrations (approximately 80–140 mg/dL, depending on time after a meal), 6 to 7% of the hemoglobin is glycosylated to form HbA_{1c}. Hemoglobin turns over in the blood as red blood cells are phagocytosed and their hemoglobin degraded and new red blood cells are derived from retic-



Troponin is a heterotrimeric protein involved in the regulation of striated and cardiac muscle contraction.

Most troponin in the cell is bound to the actin–tropomyosin complex in the muscle fibril. The three subunits of troponin consist of troponin-C, troponin-T, and troponin-I, each with a specific function in the regulatory process. Troponin-T and troponin-I exist as different isoforms in cardiac and skeletal muscle (sequences with a different amino acid composition), thus allowing the development of specific antibodies against each form. As a consequence, either cardiac troponin-T or cardiac troponin-I may be rapidly measured in blood samples by immunoassay with a good degree of specificity.



Four minor components of adult hemoglobin (HbA) result from posttranslational, nonenzymatic glycosylation of different amino acid residues (HbA_{1a1}, HbA_{1a2}, HbA_{1b1}, and HbA_{1c}). In HbA_{1c}, the fraction that is usually measured, the glycosylation occurs on an N-terminal valine.



Prion diseases are categorized as transmissible spongiform encephalopathies, which are neurodegenerative diseases characterized by spongiform degeneration and astrocytic gliosis in the central nervous system. Frequently, protein aggregates and amyloid plaques are seen. These aggregates are resistant to proteolytic degradation.



Familial prion diseases are caused by point mutations in the gene encoding the Pr protein (point mutations are changes in one base in the DNA nucleotide sequence). The diseases have a variety of names related to the different mutations and the clinical syndrome (e.g., Gertsmann-Straussler-Scheinker disease and familial Creutzfeldt-Jakob disease). Familial Creutzfeldt-Jakob disease (fCJD) arises from an inherited mutation and has an autosomal dominant pedigree. It typically presents in the fourth decade of life. The mutation lowers the activation energy for refolding and the prion proteins fold into the PrP^{Sc} conformation more readily. It is estimated that the rate of generating prion disease by refolding of PrP^c in the normal cell is approximately 3,000 to 4,000 years. Lowering of the activation energy for refolding by mutation presumably decreases this time to the observed 30- to 40-year prodromal period. Sporadic Creutzfeldt-Jakob disease may arise from somatic cell mutation or rare spontaneous refoldings that initiate a cascade of refolding into the PrP^{Sc} conformation. The sporadic form of the disease accounts for 85% of all cases of CJD.

ulocytes. The average lifespan of a red blood cell is 120 days. Thus, the extent of hemoglobin glycosylation is a direct reflection of the average serum glucose concentration to which the cell has been exposed over its 120-day lifespan. **Di Abietes'** elevated Hb_{A1c} indicates that her blood glucose levels have been elevated on an average over the preceding 6 weeks to 4 months. An increase of Di's insulin dosage would decrease her hyperglycemia but increase her risk of hypoglycemic events.

BIOCHEMICAL COMMENTS



Protein Misfolding and Prions. Prion proteins are believed to cause a neurodegenerative disease by acting as a template to misfold other cellular prion proteins into a form that cannot be degraded. The word *prion* stands for proteinaceous infectious agent. The prion diseases can be acquired either through infection (mad-cow disease), or from sporadic or inherited mutations (e.g., Creutzfeldt-Jakob disease). Although the infectious prion diseases represent a small proportion of human cases, their link to mad cow disease in the United Kingdom (new variant CJD), to growth hormone inoculations in the United States and France (iatrogenic or “doctor-induced” CJD), and to ritualistic cannibalism in the Fore tribespeople (Kuru) have received the most publicity.

The prion protein is normally found in the brain and is encoded by a gene that is a normal component of the human genome. The disease-causing form of the prion protein has the same amino acid composition but is folded into a different conformation that aggregates into multimeric protein complexes resistant to proteolytic degradation (Fig. 7.21). The normal conformation of the prion protein has been designated PrP^c and the disease-causing form PrP^{Sc} (sc for the prion disease scrapies in sheep). Although PrP^{Sc} and PrP^c have the same amino acid composition, the PrP^{Sc} conformer is substantially enriched in β -sheet structure compared with the normal PrP^c conformer, which has little or no β -sheet structure and is approximately 40% α -helix. This difference favors the aggregation of PrP^{Sc} into multimeric complexes. These two conformations presumably have similar energy levels. Fortunately, spontaneous refolding of PrP proteins into the PrP^{Sc} conformation is prevented by a large activation energy barrier that makes this conversion extremely slow. Thus, very few molecules of PrP^{Sc} are normally formed during a lifetime.

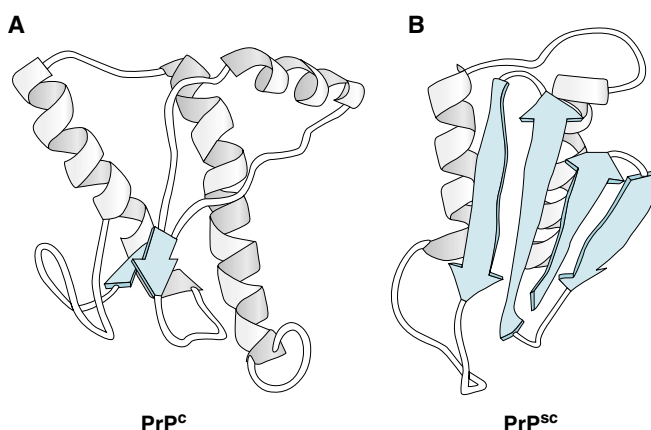


Fig. 7.21. The conformation of PrP^c and PrP^{Sc}. The prion proteins have two domains, an N-terminal region that binds four Cu²⁺ per chain, and a C-terminal region. In PrP^c, the C-terminal regions contain three substantial helices and two 3-residue β strands joined by two to three hydrogen bonds (approximately 40% α -helix and almost no β -sheet structure). It exists as a monomer. In PrP^{Sc}, the C-terminal region is folded into an extensive β -sheet. The overall structure is approximately 40 to 50% β -sheet, 20 to 30% α -helices. This conformation promotes aggregation.

The infectious disease occurs with the ingestion of PrP^{Sc} dimers in which the prion protein is already folded into the high β structure. These PrP^{Sc} proteins are thought to act as a template to lower the activation energy barrier for the conformational change, causing native proteins to refold into the PrP^{Sc} conformation much more rapidly (much like the role of chaperonins). The refolding initiates a cascade as each new PrP^{Sc} formed acts as a template for the refolding of other molecules. As the number of PrP^{Sc} molecules increase in the cell, they aggregate into a multimeric assembly that is resistant to proteolytic digestion.

Suggested References

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REVIEW QUESTIONS—CHAPTER 7

- Which of the following characterize α -helix regions of proteins?
 - They all have the same primary structure.
 - They are formed principally by hydrogen bonds between a carbonyl oxygen atom in one peptide bond and the amide hydrogen from a different peptide bond.
 - They are formed principally by hydrogen bonds between a carbonyl atom in one peptide bond and the hydrogen atoms on the side chain of another amino acid.
 - They are formed by hydrogen bonding between two adjacent amino acids in the primary sequence.
 - They require a high content of proline and glycine.
- Which of the following is a characteristic of globular proteins?
 - Hydrophilic amino acids tend to be on the inside.
 - Hydrophobic amino acids tend to be on the outside.
 - Tertiary structure is formed by hydrophobic and electrostatic interactions between amino acids, and by hydrogen bonds between amino acids and between amino acids and water.
 - Secondary structures are formed principally by hydrophobic interactions between amino acids.
 - Covalent disulfide bonds are necessary to hold the protein in a rigid conformation.
- A protein has one transmembrane domain composed entirely of α -helical secondary structure. Which of the following amino acids would you expect to find in the transmembrane domain?
 - Proline
 - Glutamate
 - Lysine
 - Leucine
 - Arginine
- Autopsies of patients with Alzheimer's disease show protein aggregates called neurofibrillary tangles and neuritic plaques in various regions of the brain. These plaques exhibit the characteristic staining of amyloid. Which of the following structural features is the most likely characteristic of at least one protein in these plaques?
 - A high content of β -pleated sheet structure
 - A high content of α -helical structure
 - A high content of random coils
 - Disulfide bond crosslinks between polypeptide chains
 - A low-energy native conformation

5. While studying a novel pathway in a remote species of bacteria, you discover a new globular protein that phosphorylates a substrate, using ATP as the phosphate donor. This protein most likely contains which of the following structures?
- (A) An actin fold
 - (B) An immunoglobulin fold
 - (C) A nucleotide binding fold
 - (D) A globin fold
 - (E) A β -barrel

8 Enzymes as Catalysts

Enzymes are proteins that act as **catalysts**, compounds that increase the rate of chemical reactions (Fig. 8.1). Enzyme catalysts bind reactants (substrates), convert them to products, and release the products. Although enzymes may be modified during their participation in this reaction sequence, they return to their original form at the end. In addition to increasing the speed of reactions, enzymes provide a means for regulating the rate of metabolic pathways in the body. This chapter describes the properties of enzymes that allow them to function as catalysts. The next chapter explains the mechanisms of enzyme regulation.

Enzyme binding sites. An enzyme binds the **substrates** of the reaction and converts them to **products**. The substrates are bound to specific **substrate binding sites** on the enzyme through interactions with the amino acid residues of the enzyme. The spatial geometry required for all the interactions between the substrate and the enzyme makes each enzyme **selective for its substrates** and ensures that only **specific products** are formed.

Active catalytic site. The substrate binding sites overlap in the **active catalytic site** of the enzyme, the region of the enzyme where the reaction occurs. Within the catalytic site, **functional groups** provided by **coenzymes**, **tightly bound metals**, and, of course, **amino acid residues** of the enzyme, participate in catalysis.

Activation energy and the transition state. The functional groups in the catalytic site of the enzyme activate the substrates and decrease the energy needed to form the high-energy intermediate stage of the reaction known as the **transition state complex**. Some of the catalytic strategies employed by enzymes, such as **general acid-base catalysis**, formation of **covalent intermediates**, and **stabilization of the transition state**, are illustrated by **chymotrypsin**.

pH and temperature profiles. Enzymes have a functional pH range determined by the pK_a s of functional groups in the active site and the interactions required for three-dimensional structure. Non-denaturing increases of temperature increase the reaction rate.

Mechanism-based inhibitors. The effectiveness of many **drugs** and **toxins** depends on their ability to inhibit an enzyme. The strongest inhibitors are **covalent inhibitors**, compounds that form covalent bonds with a reactive group in the enzyme active site, or **transition state analogues** that mimic the transition state complex.

Enzyme names. Most enzyme names end in “ase.” Enzymes usually have both a common name and a systematic classification that includes a name and an Enzyme Commission (EC) number.

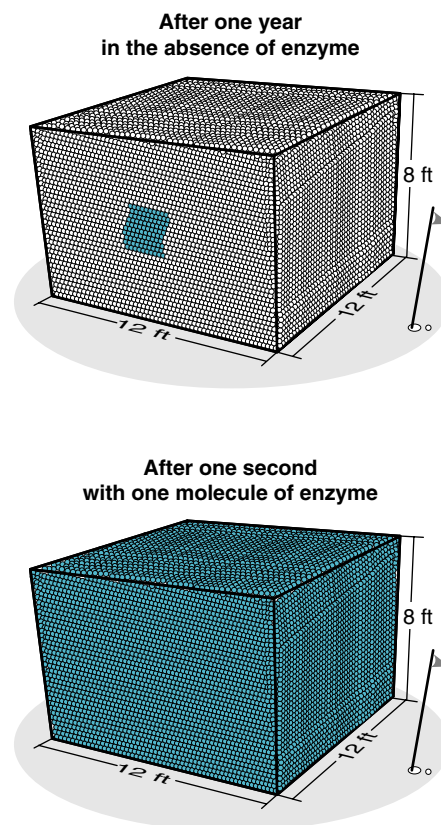


Fig. 8.1. Catalytic power of enzymes. Many enzymes increase the rate of a chemical reaction by a factor of 10^{11} or higher. To appreciate an increase in reaction rate by this order of magnitude, consider a room-sized box of golf balls that “react” by releasing energy and turning blue. The $12\text{ ft} \times 12\text{ ft} \times 8\text{ ft}$ box contains 380,000 golf balls. If the rate of the reaction in the absence of enzyme were 100 golf balls per year, the presence of 1 molecule of enzyme would turn the entire box of golf balls blue in 1 second (assuming a 10^{11} increase in reaction rate).



Most, if not all, of the tissues and organs in the body are adversely affected by chronic ingestion of excessive amounts of alcohol, including the liver, pancreas, heart, reproductive organs, central nervous system, and the fetus. Some of the effects of alcohol ingestion, such as the psychotropic effects on the brain or inhibition of vitamin transport, are direct effects caused by ethanol itself. However, many of the acute and chronic pathophysiological effects of alcohol relate to the pathways of ethanol metabolism (see Chapter 25).



THE WAITING ROOM



A year after recovering from salicylate poisoning (see Chapter 4), **Dennis “the Menace” Veere** was playing in his grandfather’s basement. Dennis drank an unknown amount of the insecticide malathion, which is sometimes used for killing fruit flies and other insects (Fig. 8.2). Sometime later when he was not feeling well, Dennis told his grandfather what he had done. Mr. Veere retrieved the bottle and rushed Dennis to the emergency room of the local hospital. On the way, Dennis vomited repeatedly and complained of abdominal cramps. At the hospital, he began salivating and had an uncontrollable defecation.

In the emergency room, physicians passed a nasogastric tube for stomach lavage, started intravenous fluids, and recorded vital signs. Dennis’s pulse rate was 48 beats per minute (slow), and his blood pressure was 78/48 mm Hg (low). The physicians noted involuntary twitching of the muscles in his extremities.



Lotta Topaigne was diagnosed with acute gouty arthritis involving her right great toe (see Chapter 5). The presence of insoluble urate crystals within the joint space confirmed the diagnosis. Several weeks after her acute gout attack subsided, Ms. Topaigne was started on allopurinol therapy in an oral dose of 150 mg twice per day.



Al Martini, a 44-year-old man who has been an alcoholic for the past 5 years, had a markedly diminished appetite for food. One weekend he became unusually irritable and confused after drinking two fifths of scotch and eating very little. His landlady convinced him to visit his doctor. Physical examination indicated a heart rate of 104 beats/min. His blood pressure was slightly low, and he was in early congestive heart failure. He was poorly oriented to time, place, and person.

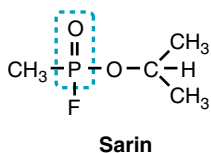
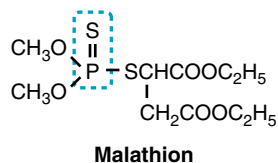


Fig. 8.2. Organophosphorous compounds. Malathion and parathion are organophosphorous insecticides. Nausea, coma, convulsions, respiratory failure, and death have resulted from the use of parathion by farmers who have gotten it on their skin. Malathion is similar in structure to parathion, but not nearly as toxic. The nerve gas Sarin, another organophosphorous compound, was used in a terrorist attack in a Japanese subway.

I. THE ENZYME-CATALYZED REACTION

Enzymes, in general, provide speed, specificity, and regulatory control to reactions in the body. Enzymes are usually proteins that act as catalysts, compounds that increase the rate of chemical reactions. Enzyme-catalyzed reactions have three basic steps:

- (1) binding of substrate: $\text{E} + \text{S} \leftrightarrow \text{ES}$
- (2) conversion of bound substrate to bound product: $\text{ES} \leftrightarrow \text{EP}$
- (3) release of product: $\text{EP} \leftrightarrow \text{E} + \text{P}$

An enzyme binds the substrates of the reaction it catalyzes and brings them together at the right orientation to react. The enzyme then participates in the making and breaking of bonds required for product formation, releases the products, and returns to its original state once the reaction is completed.

Enzymes do not invent new reactions; they simply make reactions occur faster. The catalytic power of an enzyme (the rate of the catalyzed reaction divided by the rate of the uncatalyzed reaction) is usually in the range of 10^6 to 10^{14} . Without the catalytic power of enzymes, reactions such as those involved in nerve conduction, heart contraction, and digestion of food would occur too slowly for life to exist.

Each enzyme usually catalyzes a specific biochemical reaction. The ability of an enzyme to select just one substrate and distinguish this substrate from a group of very similar compounds is referred to as specificity (Fig. 8.3). The enzyme converts

this substrate to just one product. The specificity, as well as the speed, of enzyme-catalyzed reactions result from the unique sequence of specific amino acids that form the three-dimensional structure of the enzyme.

A. The Active Site

To catalyze a chemical reaction, the enzyme forms an enzyme–substrate complex in its active catalytic site (Fig. 8.4). The active site is usually a cleft or crevice in the enzyme formed by one or more regions of the polypeptide chain. Within the active site, cofactors and functional groups from the polypeptide chain participate in transforming the bound substrate molecules into products.

Initially, the substrate molecules bind to their substrate binding sites, also called the substrate recognition sites (see Fig. 8.4B). The three-dimensional arrangement of binding sites in a crevice of the enzyme allows the reacting portions of the substrates to approach each other from the appropriate angles. The proximity of the bound substrate molecules and their precise orientation toward each other contribute to the catalytic power of the enzyme.

The active site also contains functional groups that directly participate in the reaction (see Fig. 8.4C). The functional groups are donated by the polypeptide chain, or by bound cofactors (metals or complex organic molecules called coenzymes). As the substrate binds, it induces conformational changes in the enzyme that promote further interactions between the substrate molecules and the enzyme functional groups. (For example, a coenzyme might form a covalent intermediate with the substrate, or an amino acid side chain might abstract a proton from the reacting substrate.) The activated substrates and the enzyme form a transition state complex, an unstable high-energy complex with a strained electronic configuration that is intermediate between substrate and product. Additional bonds with the enzyme stabilize the transition state complex and decrease the energy required for its formation.

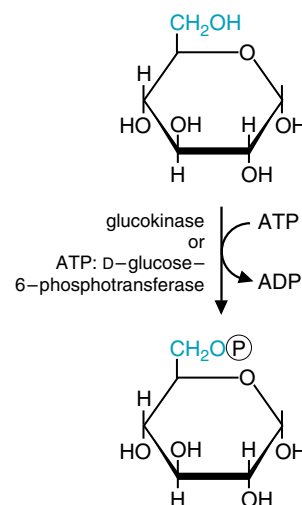


Fig. 8.3. Reaction catalyzed by glucokinase, an example of enzyme reaction specificity. Glucokinase catalyzes the transfer of a phosphate from ATP to carbon 6 of glucose. It cannot rapidly transfer a phosphate from other nucleotides to glucose, or from ATP to closely related sugars such as galactose, or from ATP to any other carbon on glucose. The only products formed are glucose 6-phosphate and ADP.

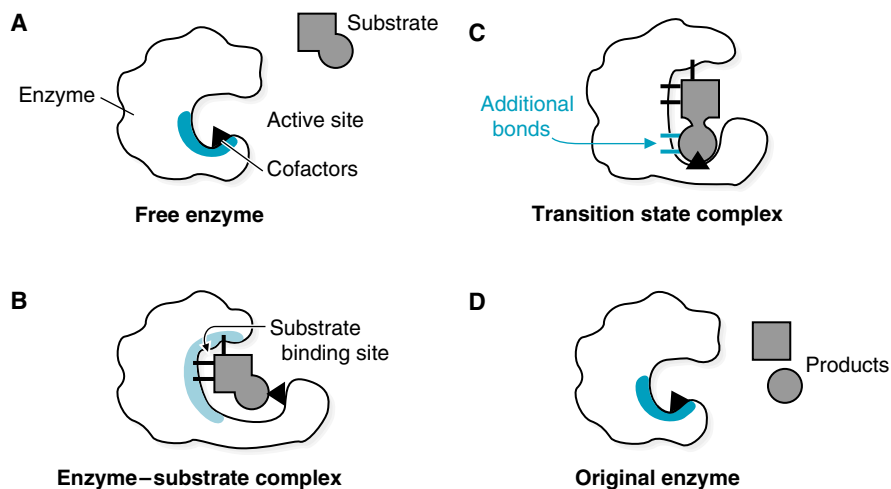


Fig. 8.4. Reaction in the enzyme active catalytic site. **A.** The enzyme contains an active catalytic site, shown in dark blue, with a region or domain where the substrate binds. The active site also may contain cofactors, nonprotein components that assist in catalysis. **B.** The substrate forms bonds with amino acid residues in the substrate binding site, shown in light blue. Substrate binding induces a conformational change in the active site. **C.** Functional groups of amino acid residues and cofactors in the active site participate in forming the transition state complex, which is stabilized by additional noncovalent bonds with the enzyme, shown in blue. **D.** As the products of the reaction dissociate, the enzyme returns to its original conformation.

The transition state complex decomposes to products, which dissociate from the enzyme (see Fig. 8.4D). The enzyme generally returns to its original form. The free enzyme then binds another set of substrates, and repeats the process.

B. Substrate Binding Sites

Enzyme specificity (the enzyme's ability to react with just one substrate) results from the three-dimensional arrangement of specific amino acid residues in the enzyme that form binding sites for the substrates and activate the substrates during the course of the reaction. The “lock-and-key” and the “induced-fit” models for substrate binding describe two aspects of the binding interaction between the enzyme and substrate.

1. LOCK-AND-KEY MODEL FOR SUBSTRATE BINDING

The substrate binding site contains amino acid residues arranged in a complementary three-dimensional surface that “recognizes” the substrate and binds it through multiple hydrophobic interactions, electrostatic interactions, or hydrogen bonds (Fig. 8.5). The amino acid residues that bind the substrate can come from very different parts of the linear amino acid sequence of the enzyme, as seen in glucokinase. The binding of compounds with a structure that differs from the substrate even to a small degree may be prevented by steric hindrance and charge-repulsion. In the lock-and-key model, the complementarity between the substrate and its binding site is compared to that of a key fitting into a rigid lock.

2. “INDUCED FIT” MODEL FOR SUBSTRATE BINDING

Complementarity between the substrate and the binding site is only part of the picture. As the substrate binds, enzymes undergo a conformational change (“induced fit”) that repositions the side chains of the amino acids in the active site and increases the number of binding interactions (see Fig. 8.4). The induced fit model for substrate bind-

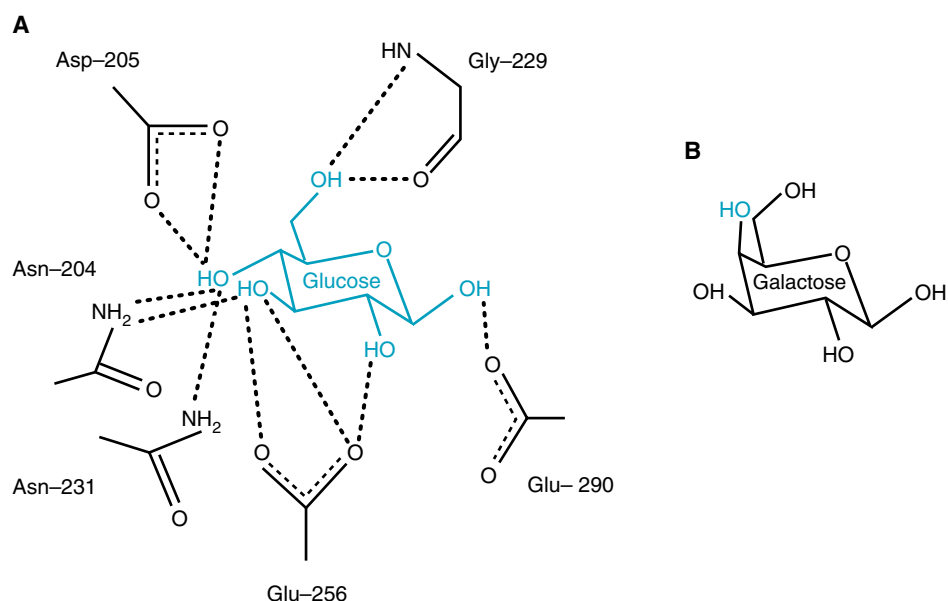


Fig. 8.5. Glucose binding site in glucokinase. A. Glucose, shown in blue, is held in its binding site by multiple hydrogen bonds between each hydroxyl group and polar amino acids from different regions of the enzyme amino acid sequence in the actin fold (see Chapter 7). The position of the amino acid residue in the linear sequence is given by its number. The multiple interactions enable glucose to induce large conformational changes in the enzyme. (Modified from Pilkis SJ, Weber IT, Harisson RW, Bell GI. *J Biol Chem.* Glucokinase : structural analysis of a protein involved in susceptibility to diabetes 1994;21925–21928.) B. Enzyme specificity is illustrated by the comparison of galactose and glucose. Galactose differs from glucose only in the position of the -OH group shown in blue. However, it is not phosphorylated at a significant rate by the enzyme. Cells therefore require a separate galactokinase for the metabolism of galactose.

ing recognizes that the substrate binding site is not a rigid “lock” but rather a dynamic surface created by the flexible overall three-dimensional structure of the enzyme.

The function of the conformational change induced by substrate binding, the induced fit, is usually to reposition functional groups in the active site in a way that promotes the reaction, improves the binding site of a cosubstrate, or activates an adjacent subunit through cooperativity. For example, consider the large conformational changes that occur in the actin fold of glucokinase when glucose binds. The induced fit involves changes in the conformation of the whole enzyme that close the cleft of the fold, thereby improving the binding site for ATP, and excluding water (which might interfere with the reaction) from the active site (Fig. 8.6). Thus, the multiple interactions between the substrate and the enzyme in the catalytic site serve both for substrate recognition and for initiating the next stage of the reaction, formation of the transition state complex.

C. The Transition State Complex

For a reaction to occur, the substrates undergoing the reaction need to be activated. If the energy levels of a substrate are plotted as the substrate is progressively converted to product, the curve will show a maximum energy level that is higher than that of either the substrate or the product (Fig. 8.7). This high energy level occurs at the

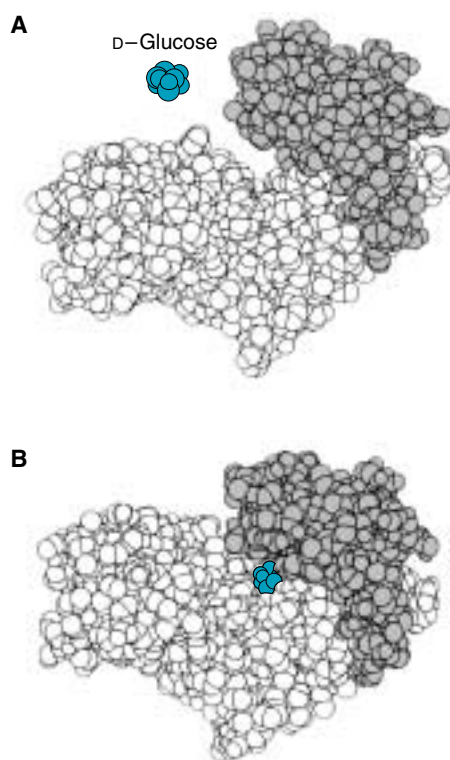


Fig. 8.6. Conformational change resulting from the binding of glucose to hexokinase. (The figure is actually yeast hexokinase, which is more similar to human glucokinase than it is to the other human hexokinase isozymes). The shaded and unshaded areas show the two domains (four subdomains) that form the actin fold with its ATP-binding cleft. **A.** Free enzyme. **B.** With glucose bound, the cleft closes, forming the ATP binding site. The closure of the cleft when glucose binds to hexokinase (or human glucokinase) is one of the largest “induced fits” known. The combination of secondary structures in the actin fold that give hexokinase the flexibility required for this shift are discussed in Chapter 7, section IV.B.1., “The Actin Fold.” (From Bennett WS, Steitz TA. J Mol Biol. Structure of a complex between yeast hexokinase A and glucose II. Detailed comparisons of conformation and active site configuration with the native hexokinase B. 1980;211–230.)



Because the transition state complex binds more tightly to the enzyme than does the substrate, compounds that resemble its electronic and three-dimensional surface (transition state analogs) are more potent inhibitors of an enzyme than are substrate analogs. Consequently, a drug developed as a transition state analog would be highly specific for the enzyme it is designed to inhibit. However, transition state analogs are highly unstable when not bound to the enzyme, and would have great difficulty making it from the digestive tract or injection site to the site of action. Some of the approaches in drug design that are being used to deal with the instability problem include: designing drugs that are almost transition state analogs but have a stable modification; designing a pro-drug that is converted to a transition state analog at the site of action; using the transition state analog to design a complementary antibody.



Abzymes (catalytic antibodies) are made as antibodies against analogs of the transition state complex. They thus have an arrangement of amino acid side chains in their variable regions that is similar to the active site of the enzyme in the transition state. Consequently, they can act as artificial enzymes. For example, abzymes have been developed against analogs of the transition state complex of cocaine esterase, the enzyme that degrades cocaine in the body. These abzymes have esterase activity, and monthly injections of the abzyme drug can be used to rapidly destroy cocaine in the blood, thereby decreasing the dependence of addicted individuals. (See Chapter 7 for antibody structure.)



Hydrolysis is the use of water to lyse (break) a bond. Proteolysis is the hydrolysis of a peptide bond in a protein, a reaction catalyzed by enzymes called proteases.

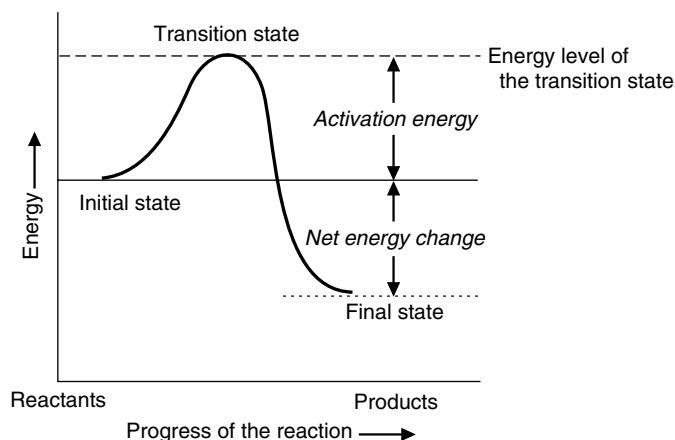


Fig. 8.7. Energy diagram showing the energy levels of the substrates as they progress toward products in the absence of enzyme. The substrates must pass through the high-energy transition state during the reaction. Although a favorable loss of energy occurs during the reaction, the rate of the reaction is slowed by the energy barrier to forming the transition state. The energy barrier is referred to as the activation energy.

transition state. For some enzyme-catalyzed reactions, the transition state is a condition in which bonds in the substrate are maximally strained. For other enzyme-catalyzed reactions, the electronic configuration of the substrate becomes very strained and unstable as it enters the transition state. The highest energy level corresponds to the most unstable substrate configuration, and the condition in which the changing substrate molecule is most tightly bound to participating functional groups in the enzyme. The difference in energy between the substrate and the transition state complex is called the activation energy.

According to transition state theory, the overall rate of the reaction is determined by the number of molecules acquiring the activation energy necessary to form the transition state complex. Enzymes increase the rate of the reaction by decreasing this activation energy. They use various catalytic strategies, such as electronic stabilization of the transition state complex or acid-base catalysis, to obtain this decrease.

Once the transition state complex is formed, it can collapse back to substrates or decompose to form products. The enzyme does not change the initial energy level of the substrates or the final energy level of the products.

II. CATALYTIC MECHANISM OF CHYMOTRYPSIN

The enzyme chymotrypsin provides a good example of the strategies and amino acid side chains used by enzymes to lower the amount of activation energy required. Chymotrypsin is a digestive enzyme released into the intestine that catalyzes the hydrolysis of specific peptide bonds in denatured proteins. It is a member of the serine protease superfamily, enzymes that use a serine in the active site to form a covalent intermediate during proteolysis. In the overall hydrolysis reaction, an OH^- from water is added to the carbonyl carbon of the peptide bond, and an H^+ to the N, thereby cleaving the bond (Fig. 8.8). The bond that is cleaved is called the scissile bond.

A. The Reaction in the Absence of Enzyme

In the reaction carried out in the absence of enzyme, the negatively charged hydroxyl group of water attacks the carbonyl carbon, which carries a partial positive charge. An unstable oxyanion tetrahedral transition state complex is formed in which the oxygen atom carries a full negative charge. The rate of the chemical reaction in the absence of chymotrypsin is slow because there are too few OH^-

molecules in H_2O with enough energy to form the transition state complex and too few OH^- molecules colliding with the substrate at the right orientation.

B. Catalytic Strategies in the Reaction Catalyzed by Chymotrypsin

In the reaction catalyzed by chymotrypsin, the same oxyanion intermediate is formed by using the hydroxyl group of a serine residue for the attack instead of a free hydroxyl anion. The rate of the chymotrypsin-catalyzed reaction is faster because functional groups in the enzyme active site activate the attacking hydroxyl group, stabilize the oxyanion transition state complexes, form a covalent intermediate, and destabilize the leaving group. The reaction takes place in two stages: (a) cleavage of the peptide bond in the denatured substrate protein and formation of a covalent acyl-enzyme intermediate (Fig. 8.9, steps 1–5), and (b) hydrolysis of the acyl-enzyme intermediate to release the remaining portion of the substrate protein (Fig. 8.9, steps 6–9). The names of the catalytic strategies employed in the various steps are in *italics* in the following paragraphs.

1. SPECIFICITY OF BINDING TO CHYMOTRYPSIN

Chymotrypsin hydrolyzes the peptide bond on the carbonyl side of a phenylalanine, tyrosine, or tryptophan in a denatured protein. The substrate recognition site consists of a hydrophobic binding pocket that holds the hydrophobic amino acid contributing the carbonyl group of the scissile bond (see Fig. 8.9, Step 1). The substrate protein must be denatured to fit into the pocket and be held rigidly in place by glycines in the enzyme peptide backbone. Scissile bond specificity is also provided by the subsequent steps of the reaction, such as moving serine 195 into attacking position (*proximity and orientation*).

2. FORMATION OF THE ACYL-ENZYME INTERMEDIATE IN CHYMOTRYPSIN

In the first stage of the reaction, the peptide bond of the denatured protein substrate is cleaved as an active site serine hydroxyl group attacks the carbonyl carbon of the scissile bond (*nucleophilic catalysis—a nucleophile is a chemical group that is attracted to the positively charged nucleus*) (Fig. 8.9, Step 2). Aspartate and histidine cooperate in converting this hydroxyl group (with a partial negative charge on the oxygen) into a better nucleophilic attacking group by giving it a more negative charge. An active site histidine acts as a base and abstracts a proton from the serine hydroxyl (*acid-base catalysis*). The protonated histidine is stabilized by the negative charge of a nearby aspartate.

The aspartate-histidine-serine combination, referred to as the catalytic triad, is an example of cooperative interactions between amino acid residues in the active site. The strong nucleophilic attacking group created by this charge-relay system has the same general effect on reaction rate as increasing the concentration of hydroxyl ions available for collision in the uncatalyzed reaction.

In the next step of the reaction sequence, an oxyanion tetrahedral transition state complex is formed that is stabilized by hydrogen bonds with $-\text{NH}$ groups in the peptide backbone (Fig. 8.9, Step 3). The original view of the way enzymes form transition state complexes was that they stretched the bonds or distorted the bond angles of the reacting substrates. However, most transition state complexes, such as the oxyanion tetrahedral complex, are better described as showing “electronic strain,” an electrostatic surface that would be highly improbable if it were not stabilized by bonds with functional groups on the enzyme. *Stabilization of the transition state complex* lowers its energy level and increases the number of molecules that reach this energy level.

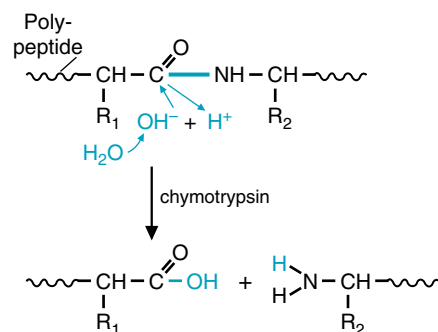
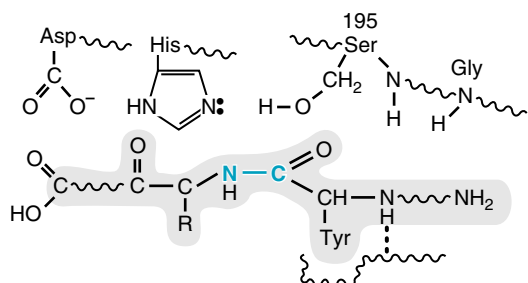
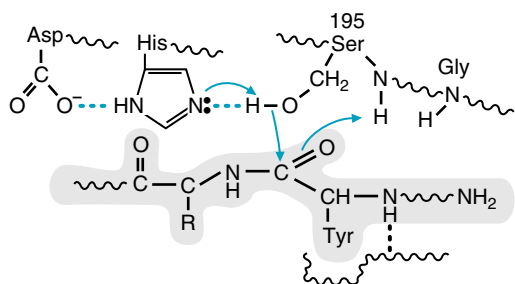
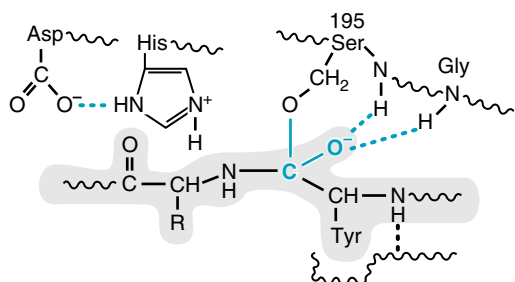
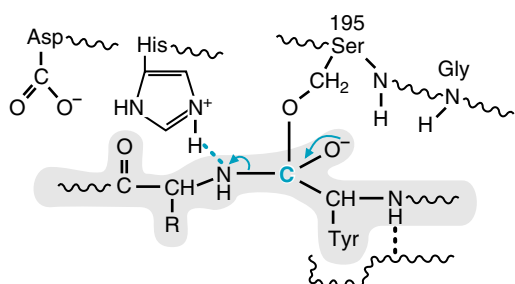
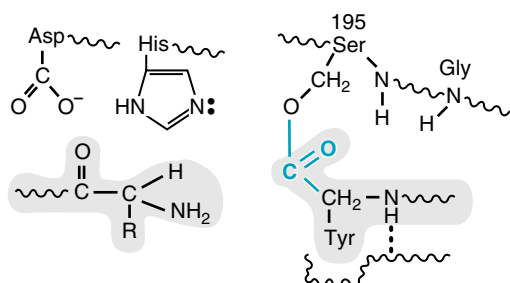
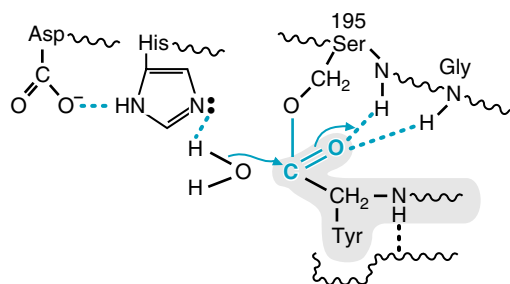
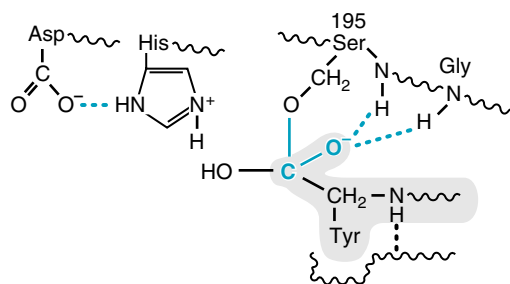
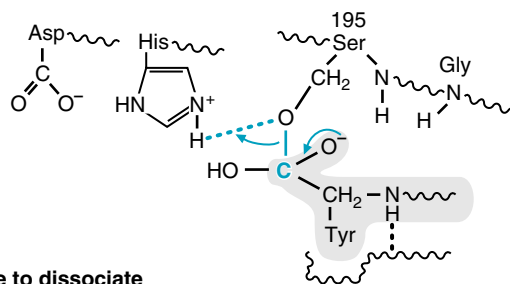
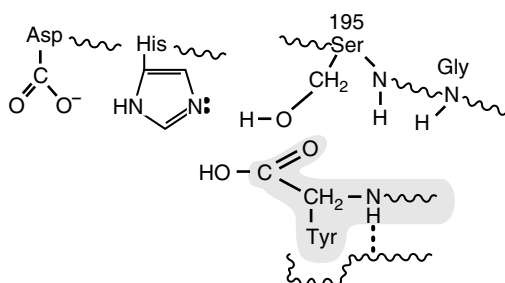


Fig. 8.8. Chymotrypsin hydrolyzes certain peptide bonds in proteins. The scissile bond is shown in blue. The carbonyl carbon, which carries a partial positive charge, is attacked by a hydroxyl group from water. An unstable tetrahedral oxyanion intermediate is formed, which is the transition state complex. As the electrons return to the carbonyl carbon, it becomes a carboxylic acid, and the remaining proton from water adds to the leaving group to form an amine.

Q: In the stomach, gastric acid decreases the pH to 1 to 2 to denature proteins through disruption of hydrogen bonding. The protease in the stomach, pepsin, is a member of the aspartate protease superfamily, enzymes that use two aspartate residues in the active site for acid-base catalysis of the peptide bond. Why can they not use histidine like chymotrypsin?

Serine proteases in blood coagulation. The use of an active site serine to cleave a peptide bond is common in the proteolytic enzymes of blood coagulation as well as those of digestion. Blood clots are formed from fibrin, a protein present in the blood as the inactive precursor, fibrinogen. The serine protease thrombin cleaves a peptide bond in fibrinogen to form active fibrin. Thrombin has the same aspartate-histidine-serine catalytic triad found in chymotrypsin and works in essentially the same way. Thrombin is also present as an inactive precursor, prothrombin, which is itself activated through proteolytic cleavage by another blood coagulation serine protease.

1. Substrate binding**2. Histidine activates serine for nucleophilic attack****3. The oxyanion tetrahedral intermediate is stabilized by hydrogen bonds****4. Cleavage of the peptide bond****5. The covalent acyl-enzyme intermediate****6. Water attacks the carbonyl carbon****7. Second oxyanion tetrahedral intermediate****8. Acid catalysis breaks the acyl-enzyme covalent bond****9. The product is free to dissociate**

Subsequently, the serine in the active site forms a full covalent bond with the carbon of the carbonyl group as the peptide bond is cleaved (*covalent catalysis*). The formation of a stable covalent intermediate is a catalytic strategy employed by many enzymes and often involves serine or cysteine residues. The covalent intermediate is subsequently hydrolyzed (*acid-base catalysis*). The dissociating products of an enzyme-catalyzed reaction are often “destabilized” by some degree of charge repulsion in the active site. In the case of chymotrypsin, the amino group formed after peptide bond cleavage is destabilized or “uncomfortable” in the presence of the active site histidine (*destabilization of developing product*).

3. HYDROLYSIS OF THE ACYL-CHYMOTRYPSIN INTERMEDIATE

The next sequence of events hydrolyzes the acyl-enzyme intermediate to release the bound carbonyl-side peptide (Fig. 8.9, Steps 6-9). The active site histidine activates water to form an OH^- for a nucleophilic attack, resulting in a second oxyanion transition state complex. When the histidine adds the proton back to serine, the reaction is complete, and the product dissociates.

C. Energy Diagram in the Presence of Chymotrypsin

The number of steps in real enzymatic reactions results in a multi-bump energy diagram (Fig. 8.10). At the initial stage of the reaction, a dip occurs because energy is provided by formation of the initial multiple weak bonds between the substrate and enzyme. As the reaction progresses, the curve rises because additional energy is required for formation of the transition state complex. This energy is provided by the subsequent steps in the reaction replacing the initial weak bonds with progressively tighter bonds. Semi-stable covalent intermediates of the reaction have lower energy levels than do the transition state complexes, and are present in the reaction diagram as dips in the energy curve. The final transition state complex has the highest energy level in the reaction and is therefore the most unstable state. It can collapse back to substrates or decompose to form products.

III. FUNCTIONAL GROUPS IN CATALYSIS

The catalytic strategies employed by chymotrypsin to increase the reaction rate are common to many enzymes. One of these catalytic strategies, proximity and orientation, is an intrinsic feature of substrate binding and part of the catalytic mechanism of all enzymes. All enzymes also stabilize the transition state by electrostatic interactions, but not all enzymes form covalent intermediates.

Great variety occurs in the functional groups employed by different enzymes to carry out these catalytic strategies. Some enzymes, such as chymotrypsin, rely on



To participate in general acid-base catalysis, the amino acid side chain must be able to abstract a proton at one stage of the reaction, and donate it back at another. Histidine (pK_a 6.3) would be protonated at this low pH, and could not abstract a proton from a potential nucleophile. However, aspartic acid, with a (pK_a of about 2) can release protons at a pH of 2. The two aspartates work together to activate water through the removal of a proton to form the hydroxyl nucleophile.

Fig. 8.9. Catalytic mechanism of chymotrypsin. The substrate (a denatured protein) is in the shaded area. **1.** As the substrate protein binds to the active site, serine-195 and a histidine (his57) are moved closer together and at the right orientation for the nitrogen electrons on histidine to attract the hydrogen of serine. Without this change of conformation on substrate binding, the catalytic triad cannot form. **2.** Histidine serves as a general base catalyst as it abstracts a proton from the serine, increasing the nucleophilicity of the serine-oxygen, which attacks the carbonyl carbon. **3.** The electrons of the carbonyl group form the oxyanion tetrahedral intermediate. The oxyanion is stabilized by the N-H groups of serine-195 and glycine in the chymotrypsin peptide backbone. **4.** The amide nitrogen in the peptide bond is stabilized by interaction with the histidine proton. Here the histidine acts as a general acid catalyst. As the electrons of the carbon-nitrogen peptide

bond withdraw into the nitrogen, the electrons of the carboxyanion return to the substrate carbonyl carbon, resulting in cleavage of the peptide bond. **5.** The cleavage of the peptide bond results in formation of the covalent acyl-enzyme intermediate, and the amide half of the cleaved protein dissociates. **6.** The nucleophilic attack by H_2O on the carbonyl carbon is activated by histidine, whose nitrogen electrons attract a proton from water. **7.** The second tetrahedral oxyanion intermediate (the transition state complex) is formed. It is again stabilized by hydrogen bonds with the peptide backbone bonds of glycine and serine. **8.** As the histidine proton is donated to the electrons of the bond between the serine oxygen and the substrate carbonyl group, the electrons from the oxyanion return to the substrate carbon to form the carboxylic acid, and the acyl-enzyme bond is broken. **9.** The enzyme, as it releases substrate, returns to its original state.

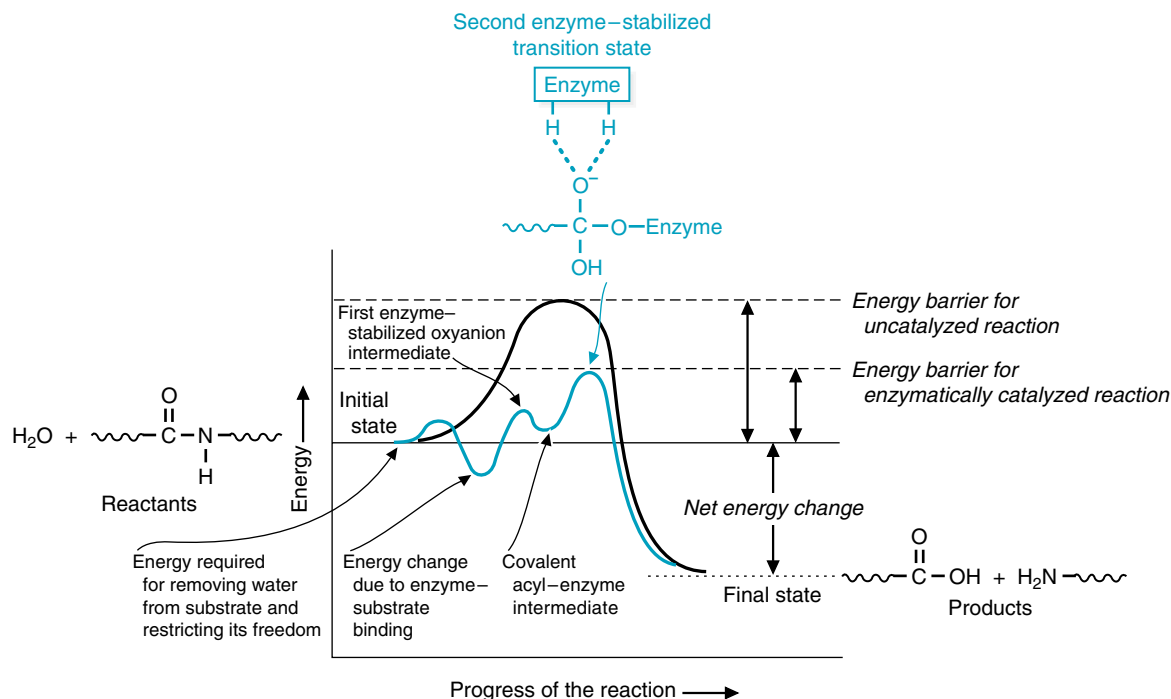


Fig. 8.10. A postulated energy diagram for the reaction catalyzed by chymotrypsin. In the presence of enzyme (blue); in the absence of enzyme (black). The energy barrier to the transition state is lowered in the enzyme-catalyzed reaction by the formation of additional bonds between the substrate and enzyme in the transition state complex. The energy is provided by substrate binding to the enzyme. The enzyme does not, however, change the energy levels of the substrate or product.



Because most vitamins function as coenzymes, the symptoms of vitamin deficiencies reflect the loss of specific enzyme activities dependent on the coenzyme form of the vitamin. Thus, drugs and toxins that inhibit proteins required for coenzyme synthesis (e.g., vitamin transport proteins or biosynthetic enzymes) can cause the symptoms of a vitamin deficiency. This type of deficiency is called a functional deficiency, whereas an inadequate intake is called a dietary deficiency.

Most coenzymes are tightly bound to their enzymes and do not dissociate during the course of the reaction. However, a functional or dietary vitamin deficiency that decreases the level of a coenzyme will result in the presence of the apoenzyme in cells (an enzyme devoid of cofactor).

Ethanol is an “antivitamin” that decreases the cellular content of almost every coenzyme. For example, ethanol inhibits the absorption of thiamine, and acetaldehyde produced from ethanol oxidation displaces pyridoxal phosphate from its protein binding sites, thereby accelerating its degradation.

amino acid residues within the active site. Other enzymes increase their repertoire by employing cofactors to provide a functional group with the right size, shape, and properties. Cofactors are nonprotein compounds that participate in the catalytic process. They are generally divided into three categories: coenzymes, metal ions (e.g., Fe^{2+} , Mg^{2+} , or Zn^{2+}), and metallocoenzymes (similar to the Fe^{2+} -heme in hemoglobin, see Chapter 7).

A. Functional Groups on Amino Acid Side Chains

Almost all of the polar amino acids participate directly in catalysis in one or more enzymes (Table 8.1). Serine, cysteine, lysine, and histidine can participate in covalent

Table 8.1. Some Functional Groups in the Active Site

Function of Amino Acid	Enzyme Example
<i>Covalent intermediates</i>	
Cysteine-SH	Glyceraldehyde 3-phosphate dehydrogenase
Serine-OH	Acetylcholinesterase
Lysine-NH ₂	Aldolase
Histidine-NH	Phosphoglucosmutase
<i>Acid-base catalysis</i>	
Histidine-NH	Chymotrypsin
Aspartate-COOH	Pepsin
<i>Stabilization of anion formed during the reaction</i>	
Peptide backbone-NH	Chymotrypsin
Arginine-NH	Carboxypeptidase A
Serine-OH	Alcohol dehydrogenase
<i>Stabilization of cation formed during the reaction</i>	
Aspartate-COO ⁻	Lysozyme

catalysis. Histidine, because it has a pK_a that can donate and accept a proton at neutral pH, often participates in acid-base catalysis. Most of the polar amino acid side chains are nucleophilic and participate in nucleophilic catalysis by stabilizing more positively charged groups that develop during the reaction.

B. Coenzymes in Catalysis

Coenzymes are complex nonprotein organic molecules that participate in catalysis by providing functional groups, much like the amino acid side chains. In the human, they are usually (but not always) synthesized from vitamins. Each coenzyme is involved in catalyzing a specific type of reaction for a class of substrates with certain structural features. Coenzymes can be divided into two general classes: activation-transfer coenzymes and oxidation-reduction coenzymes.

1. ACTIVATION-TRANSFER COENZYMES

The activation-transfer coenzymes usually participate directly in catalysis by forming a covalent bond with a portion of the substrate; the tightly held substrate moiety is then activated for transfer, addition of water, or some other reaction. The portion of the coenzyme that forms a covalent bond with the substrate is its functional group. A separate portion of the coenzyme binds tightly to the enzyme.

Thiamine pyrophosphate provides a good illustration of the manner in which coenzymes participate in catalysis (Fig. 8.11). It is synthesized in human cells from the vitamin thiamine by the addition of a pyrophosphate. This pyrophosphate provides negatively charged oxygen atoms that chelate Mg^{2+} , which then binds tightly to the enzyme. The functional group that extends into the active site is the reactive carbon atom with a dissociable proton (see Fig. 8.11). In all of the enzymes that use thiamine pyrophosphate, this reactive thiamine carbon forms a covalent bond with a substrate keto group while cleaving the adjacent carbon-carbon bond. However, each thiamine-containing enzyme catalyzes the cleavage of a different substrate (or group of substrates with very closely related structures).

Coenzymes have very little activity in the absence of the enzyme and very little specificity. The enzyme provides specificity, proximity, and orientation in the substrate recognition site, as well as other functional groups for stabilization of the transition state, acid-base catalysis, etc. For example, thiamine is made into a better nucleophilic attacking group by a basic amino acid residue in the enzyme that removes the dissociable proton (EnzB: in Fig. 8.11), thereby generating a negatively charged thiamine carbon anion. Later in the reaction, the enzyme returns the proton.

Coenzyme A (CoA), biotin, and pyridoxal phosphate are also activation-transfer coenzymes synthesized from vitamins. CoA (CoASH), which is synthesized from the vitamin pantothenate, contains an adenosine 3', 5'-bisphosphate which binds reversibly, but tightly, to a site on an enzyme (Fig. 8.12A). Its functional group, a sulfhydryl group at the other end of the molecule, is a nucleophile that always



Nucleophiles carry full or partial negative charges (like the oxygen atom in the serine -OH) or have a nitrogen that can act as an electron-donating group by virtue of its two unpaired electrons. Covalent catalysis and acid-base catalysis are carried out by nucleophilic groups. Electrophiles carry full or partial positive charges (e.g., the peptide backbone-NH was used as an electrophilic group in chymotrypsin). In general, nucleophilic and electrophilic catalysis occur when the respective nucleophilic or electrophilic groups on the enzyme stabilize substrate groups of the opposite polarity that develop during the reaction.



Q: Although coenzymes look as though they should be able to catalyze reactions autonomously (on their own), they have almost no catalytic power when not bound to the enzyme. Why?



Many alcoholics such as **Al Martini** develop thiamine deficiency because alcohol inhibits the transport of thiamine through the intestinal mucosal cells. In the body, thiamine is converted to thiamine pyrophosphate (TPP). TPP acts as a coenzyme in the decarboxylation of α -keto acids such as pyruvate and α -ketoglutarate (see Fig. 8.11) and in the utilization of pentose phosphates in the pentose phosphate pathway. As a result of thiamine deficiency, the oxidation of α -keto acids is impaired. Dysfunction occurs in the central and peripheral nervous system, the cardiovascular system, and other organs.



Most coenzymes, such as functional groups on the enzyme amino acids, are regenerated during the course of the reaction. However, CoASH and a few of the oxidation-reduction coenzymes are transformed during the reaction into products that dissociate from the enzyme at the end of the reaction (e.g., CoASH is converted to an acyl CoA derivative, and NAD^+ is reduced to NADH). These dissociating coenzymes are nonetheless classified as coenzymes rather than substrates because they are common to so many reactions, the original form is regenerated by subsequent reactions in a metabolic pathway, they are synthesized from vitamins, and the amount of coenzyme in the cell is nearly constant.



For a substrate to react with a coenzyme, it must collide with a coenzyme at exactly the right angle. The probability of the substrate and coenzyme in free solution colliding in exactly the right place at the exactly right angle is very small. In addition to providing this proximity and orientation, enzymes contribute in other ways, such as activating the coenzyme by abstracting a proton (e.g., thiamine-pyrophosphate and coenzyme A) or polarizing the substrate to make it more susceptible to nucleophilic attack.

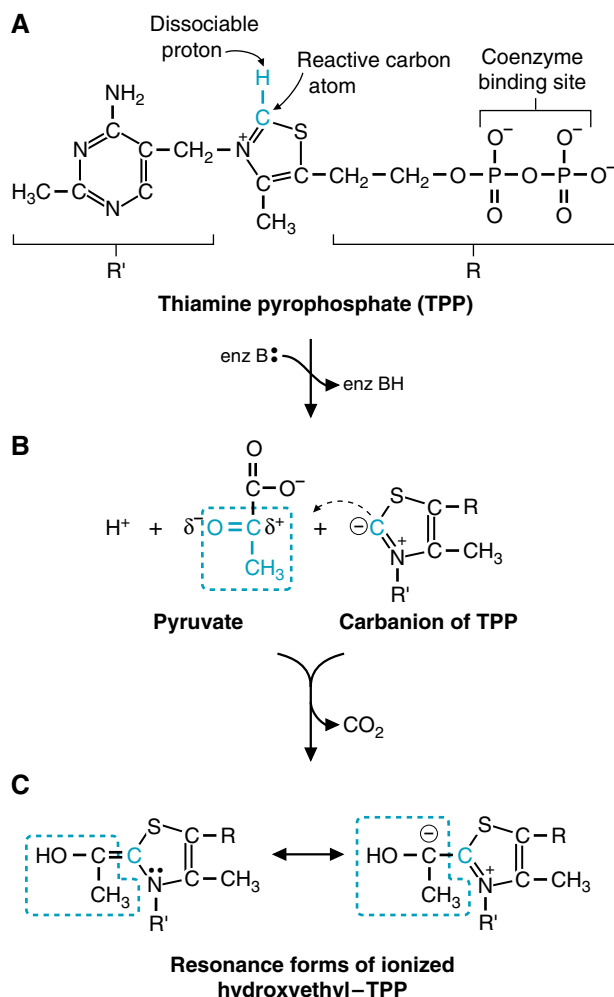


Fig. 8.11. The role of the functional group of thiamine pyrophosphate (the reactive carbon shown in blue) in formation of a covalent intermediate. **A.** A base on the enzyme (**B**) abstracts a proton from thiamine, creating a carbanion (general acid-base catalysis). **B.** The carbanion is a strong nucleophile and attacks the partially positively charged keto group on the substrate. **C.** A covalent intermediate is formed, which is stabilized by resonance forms. The uncharged intermediate is the stabilized transition state complex.

attacks carbonyl groups and forms acyl thioesters (in fact, the “A” in CoA stands for the acyl group that becomes attached).

Biotin, which does not contain a phosphate group, is covalently bound to a lysine in enzymes called carboxylases (see Fig. 8.12B). Its functional group is a nitrogen atom that covalently binds a CO_2 group in an energy-requiring reaction. This bound CO_2 group is activated for addition to another molecule. In the human, biotin functions only in carboxylation reactions.

Pyridoxal phosphate is synthesized from the vitamin pyridoxine, which is also called vitamin B_6 (Fig. 8.13). The reactive aldehyde group usually functions in enzyme-catalyzed reactions by forming a covalent bond with the amino groups on amino acids. The positively charged ring nitrogen withdraws electrons from a bond in the bound amino acid, resulting in cleavage of that bond. The enzyme participates by removing protons from the substrate and by keeping the amino acid and the pyridoxal group in a single plane to facilitate shuttling of electrons.

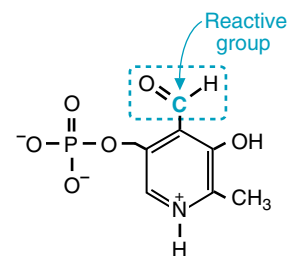
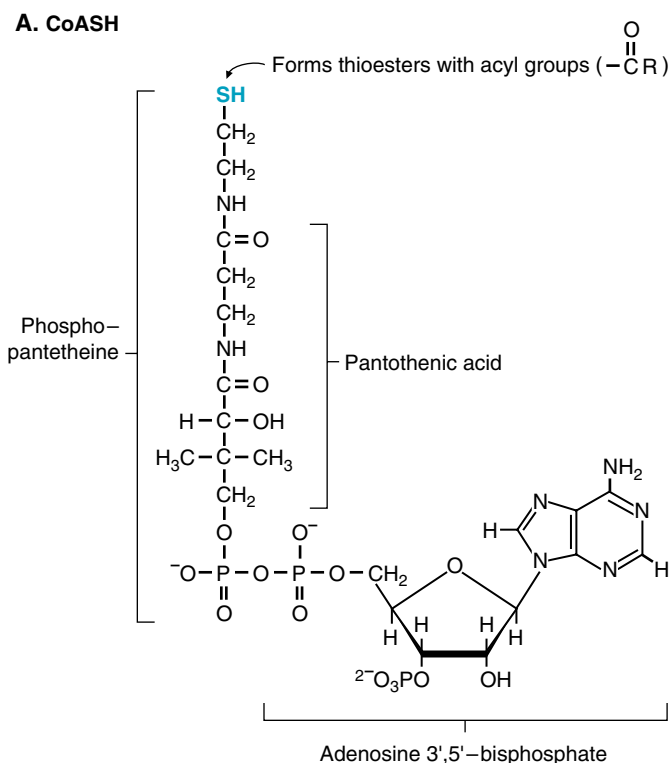


Fig. 8.13 Reactive sites of pyridoxal phosphate. The functional group of pyridoxal phosphate is a reactive aldehyde (shown in blue) that forms a covalent intermediate with amino groups of amino acids (a Schiff base). The positively charged pyridine ring is a strong electron-withdrawing group that can pull electrons into it (electrophilic catalysis).

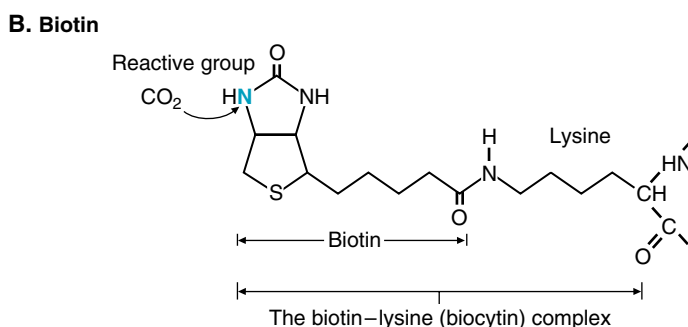


Fig. 8.12. CoA and biotin, activation-transfer coenzymes. A. Coenzyme A (CoA or CoASH) and phosphopantetheine are synthesized from the vitamin pantothenate (pantothenic acid). The active sulfhydryl group, shown in blue, binds to acyl groups (e.g., acetyl, succinyl, or fatty acyl) to form thioesters. B. Biotin activates and transfers CO₂ to compounds in carboxylation reactions. The reactive N is shown in blue. Biotin is covalently attached to a lysine residue in the carboxylase enzyme.

These coenzymes illustrate three features all activation-transfer coenzymes have in common: (1) a specific chemical group involved in binding to the enzyme, (2) a separate and different functional or reactive group that participates directly in the catalysis of one type of reaction by forming a covalent bond with the substrate, and (3) dependence on the enzyme for additional specificity of substrate and additional catalytic power.

2. OXIDATION-REDUCTION CCOENZYMES

A large number of coenzymes are involved in oxidation-reduction reactions catalyzed by enzymes categorized as oxidoreductases. Some coenzymes, such as



In general, coenzymes are nonprotein organic cofactors that participate in reactions. They may be covalently bound to enzymes (like biotin), dissociate during deficiency (thiamine-pyrophosphate), or freely dissociate at the end of the reaction (CoASH). Cofactors that are covalently or very tightly bound to nonenzyme proteins are usually called prosthetic groups. A prosthetic group, such as the heme in hemoglobin, usually does not dissociate from a protein until the protein is degraded.



When a compound is oxidized, it loses electrons. As a result, the oxidized carbon has fewer H atoms or gains an O atom. The reduction of a compound is the gain of electrons, which shows in its structure as the gain of H, or loss of O. In the oxidation of lactate to pyruvate (see Fig. 8.14), lactate loses two electrons as a hydride ion, and a proton (H^+) is released; NAD^+ , which accepts the hydride ion, is reduced to NADH. The carbon atom with the keto group is now at a higher oxidation state because both of the electrons in bonds between carbon and oxygen are counted as belonging to oxygen, whereas the two electrons in the C-H bond are shared equally between carbon and hydrogen.

The catalysis of oxidation-reduction reactions is carried out by a class of enzymes called oxidoreductases. A subclass of oxidoreductases is given the common name dehydrogenases (such as lactate dehydrogenase), because they transfer hydrogen (hydrogen atoms or hydride atoms) from the substrate to an electron-accepting coenzyme, such as NAD^+ .

nicotinamide adenine dinucleotide (NAD^+) and flavin adenine dinucleotide (FAD), can transfer electrons together with hydrogen and have unique roles in the generation of ATP from the oxidation of fuels. Other oxidation-reduction coenzymes work with metals to transfer single electrons to oxygen. Vitamin E and vitamin C (ascorbic acid) are oxidation-reduction coenzymes that can act as antioxidants and protect against oxygen free radical injury. The different functions of oxidation-reduction coenzymes in metabolic pathways are explained in Chapters 19 through 22.

Oxidation-reduction coenzymes follow the same principles as activation-transfer coenzymes, except that they do not form covalent bonds with the substrate. Each coenzyme has a unique functional group that accepts and donates electrons and is specific for the form of electrons it transfers (e.g., hydride ions, hydrogen atoms, oxygen). A different portion of the coenzyme binds the enzyme. Like activation-transfer coenzymes, oxidation-reduction coenzymes are not good catalysts without participation from amino acid side chains on the enzyme.

The enzyme lactate dehydrogenase, which catalyzes the transfer of electrons from lactate to NAD^+ , illustrates these principles (Fig. 8.14). The coenzyme nicotinamide adenine dinucleotide (NAD^+) is synthesized from the vitamin niacin (which forms the nicotinamide ring), and from ATP (which contributes an AMP). The ADP portion of the molecule binds tightly to the enzyme and causes conformational

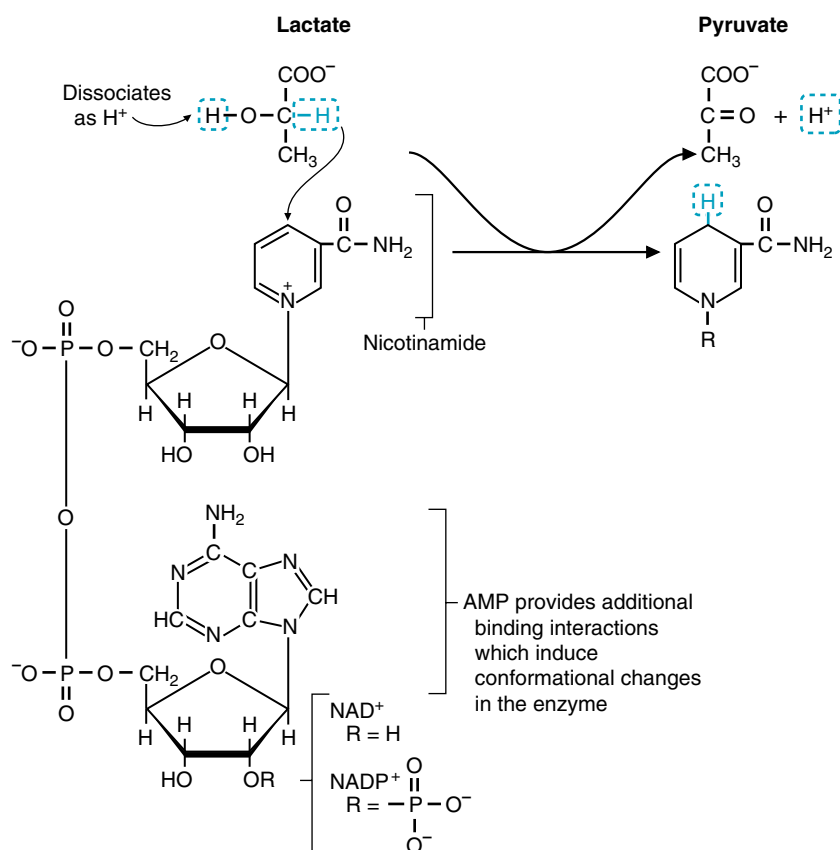


Fig. 8.14 The coenzyme NAD^+ accepting a hydride ion, shown in blue, from lactate. NAD^+ -dependent dehydrogenases catalyze the transfer of a hydride ion (H^-) from a carbon to NAD^+ in oxidation reactions such as the oxidation of alcohols to ketones or aldehydes to acids. The positively charged pyridine ring nitrogen of NAD^+ increases the electrophilicity of the carbon opposite it in the ring. This carbon then accepts the negatively charged hydride ion. The proton from the alcohol group is released into water. NADP functions by the same mechanism, but it is usually involved in pathways of reductive synthesis.

changes in the enzyme. The functional group of NAD^+ is the carbon on the nicotinamide ring opposite the positively charged nitrogen. This carbon atom accepts the hydride ion (a hydrogen atom that has two electrons) transferred from a specific carbon atom on the substrate. The H^+ from the substrate alcohol (OH) group then dissociates, and a keto group ($\text{C}=\text{O}$) is formed. One of the roles of the enzyme is to contribute a histidine nitrogen that can bind the dissociable proton on lactate, thereby making it easier for NAD^+ to pull off the other hydrogen with both electrons. Finally, NADH dissociates.

C. Metal Ions in Catalysis

Metal ions, which have a positive charge, contribute to the catalytic process by acting as electrophiles (electron-attracting groups). They assist in binding of the substrate, or they stabilize developing anions in the reaction. They can also accept and donate electrons in oxidation-reduction reactions.

The ability of certain metals to bind multiple ligands in their coordination sphere enables them to participate in binding substrates or coenzymes to enzymes. For example, Mg^{2+} plays a role in the binding of the negatively charged phosphate groups of thiamine pyrophosphate to anionic or basic amino acids in the enzyme (see Fig. 8.11). The phosphate groups of ATP are usually bound to enzymes through Mg^{2+} chelation.

The metals of some enzymes bind anionic substrates or intermediates of the reaction to alter their charge distribution, thereby contributing to catalytic power. The enzyme alcohol dehydrogenase, which transfers electrons from ethanol to NAD^+ to generate acetaldehyde and NADH , illustrates this role (Fig. 8.15). In the active site of alcohol dehydrogenase, an activated serine pulls a proton off the ethanol $-\text{OH}$ group, leaving a negative charge on the oxygen that is stabilized by zinc. This electronic configuration allows the transfer of a hydride ion to NAD^+ . Zinc is essentially fulfilling the same function in alcohol dehydrogenase that histidine fulfills in lactate dehydrogenase.

D. Noncatalytic Roles of Cofactors

Cofactors sometimes play a noncatalytic structural role in certain enzymes, binding different regions of the enzyme together to form the tertiary structure. They also can serve as substrates that are cleaved during the reaction.

IV. OPTIMAL pH AND TEMPERATURE

If the activity of most enzymes is plotted as a function of the pH of the reaction, an increase of reaction rate is usually observed as the pH goes from a very acidic level to the physiologic range; a decrease of reaction rate occurs as the pH goes from the physiologic range to a very basic range (Fig. 8.16). The shape of this curve in the acid region usually reflects the ionization of specific functional groups in the active site (or in the substrate) by the increase of pH, and the more general formation of hydrogen bonds important for the overall conformation of the enzyme. The loss of activity on the basic side usually reflects the inappropriate ionization of amino acid residues in the enzyme.



The parietal cells of the stomach secrete HCl into the lumen of the stomach, resulting in a pH between 1 and 2. This strongly acidic environment is capable of irreversibly denaturing most proteins by protonating amino acids, thereby preventing the hydrogen bond formation necessary for tertiary structure. Many of the peptide bonds in proteins would not be accessible to digestive proteases unless the protein was denatured. Pepsin, a digestive protease present in the stomach, is an exceptional enzyme because its pH optimum is approximately 1.6 and it is active in the acidic environment of the stomach. As denatured dietary proteins pass into the intestinal lumen, the pH of the gastric juice is raised above 6 by secretion of bicarbonate from the exocrine pancreas. At this higher pH, chymotrypsin and other proteases from the pancreas can act on the denatured proteins.



In humans, most of ingested ethanol is oxidized to acetaldehyde in the liver by alcohol dehydrogenase (ADH):



ADH is active as a dimer, with an active site containing zinc present in each subunit. The human has at least seven genes that encode isozymes of ADH, each with a slightly different range of specificities for the alcohols it oxidizes.

The acetaldehyde produced from ethanol is highly reactive, toxic, and immunogenic. In **Al Martini** and other patients with chronic alcoholism, acetaldehyde is responsible for much of the liver injury associated with chronic alcoholism

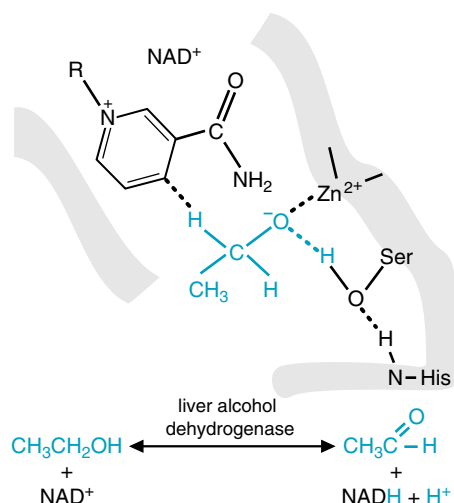


Fig. 8.15. Liver alcohol dehydrogenase catalyzes the oxidation of ethanol (shown in blue) to acetaldehyde. The active site of liver alcohol dehydrogenase contains a bound zinc atom, and a serine side chain $-\text{OH}$ and a histidine nitrogen that participate in the reaction. The histidine pulls an H^+ off the active site serine, which pulls the H^+ off of the substrate $-\text{OH}$ group, leaving the oxygen with a negative charge that is stabilized by zinc.

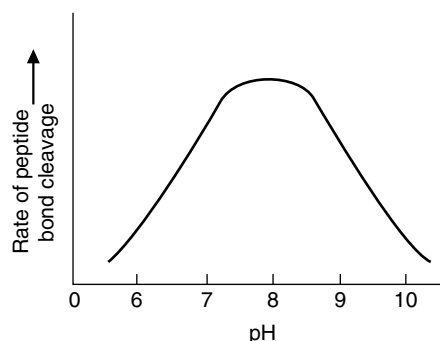


Fig. 8.16. pH profile of an enzyme. The rate of the reaction increases as the pH increases from 6 to 7.4. The exact shape of the curve depends on the protonation state of active site amino acid residues or on the hydrogen bonding required for maintenance of three-dimensional structure in the enzyme. For the enzyme shown in the figure, the increase of reaction rate corresponds to deprotonation of the active site histidine. At a pH above 8.5, deprotonation of an amino-terminal -NH_3^+ alters the conformation at the active site and the activity decreases. Other enzymes might have a lower pH maximum, a broader peak, or retain their activity in the basic side of the curve.

Most human enzymes function optimally at a temperature of approximately 37°C . An increase of temperature from 0°C to 37°C increases the rate of the reaction by increasing the vibrational energy of the substrates. The maximum activity for most human enzymes occurs near 37°C because denaturation (loss of secondary and tertiary structure) occurs at higher temperatures.

V. MECHANISM-BASED INHIBITORS

Inhibitors are compounds that decrease the rate of an enzymatic reaction. Mechanism-based inhibitors mimic or participate in an intermediate step of the catalytic reaction. The term includes transition state analogs and compounds that can react irreversibly with functional groups in the active site.



The symptoms experienced by **Dennis Veere** resulted from inhibition of acetylcholinesterase. Acetylcholinesterase cleaves the neurotransmitter acetylcholine to acetate and choline in the postsynaptic terminal, thereby terminating the transmission of the neural signal (see Fig. 8.17). Malathion is metabolized in the liver to a toxic derivative (malaoxon) that binds to the active site serine in acetylcholinesterase and other enzymes, an action similar to that of diisopropylfluorophosphate. As a result, acetylcholine accumulates and overstimulates the autonomic nervous system (the involuntary nervous system, including heart, blood vessels, glands), thereby accounting for Dennis's vomiting, abdominal cramps, salivation, and sweating. Acetylcholine is also a neurotransmitter for the somatic motor nervous system, where its accumulation resulted in Dennis's involuntary muscle twitching (muscle fasciculations).

Acetylcholine is also a neurotransmitter for the somatic motor nervous system, where its accumulation resulted in Dennis's involuntary muscle twitching (muscle fasciculations).

A. Covalent Inhibitors

Covalent inhibitors form covalent or extremely tight bonds with functional groups in the active catalytic site. These functional groups are activated by their interactions with other amino acid residues, and are therefore far more likely to be targeted by drugs and toxins than amino acid residues outside the active site.

The lethal compound diisopropyl phosphofluoridate (DFP, or diisopropylfluorophosphate) is an organophosphorus compound that served as a prototype for the development of the nerve gas Sarin and other organophosphorus toxins, such as the insecticides malathion and parathion (Fig. 8.17). DFP exerts its toxic effect by forming a covalent intermediate in the active site of acetylcholinesterase, thereby preventing the enzyme from degrading the neurotransmitter acetylcholine. Once the covalent bond is formed, the inhibition by DFP is essentially irreversible, and activity can only be recovered as new enzyme is synthesized. DFP also inhibits many other enzymes that use serine for hydrolytic cleavage, but the inhibition is not as lethal.

Aspirin (acetylsalicylic acid) provides an example of a pharmacologic drug that exerts its effect through the covalent acetylation of an active site serine in the enzyme prostaglandin endoperoxide synthase (cyclooxygenase). Aspirin resembles a portion of the prostaglandin precursor that is a physiologic substrate for the enzyme.

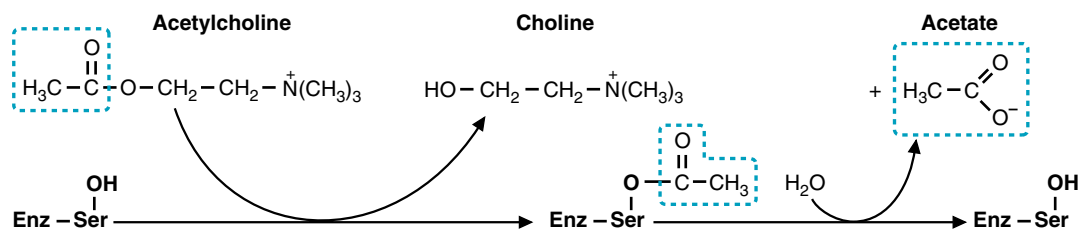
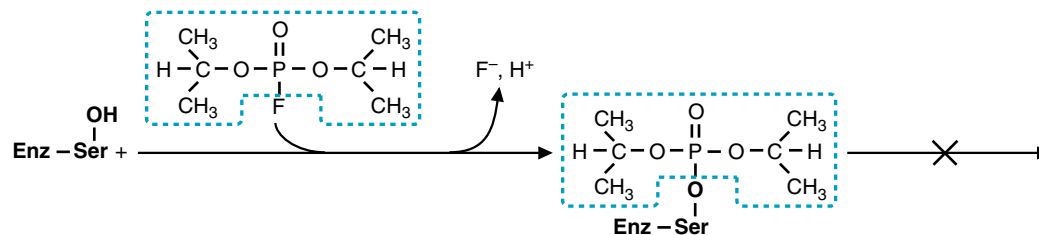
A. Normal reaction of acetylcholinesterase**B. Reaction with organophosphorus inhibitors**

Fig. 8.17 A. Acetylcholinesterase normally catalyzes inactivation of the neurotransmitter acetylcholine in a hydrolysis reaction. The active site serine forms a covalent intermediate with a portion of the substrate during the course of the reaction. B. Diisopropyl phosphorofluoridate (DFP), the ancestor of current organophosphorus nerve gases and pesticides, inactivates acetylcholinesterase by forming a covalent complex with the active site serine that cannot be hydrolysed by water. The result is that the enzyme cannot carry out its normal reaction, and acetylcholine accumulates.

B. Transition State Analogs and Compounds that Resemble Intermediate Stages of the Reaction

Transition state analogs are extremely potent and specific inhibitors of enzymes because they bind so much more tightly to the enzyme than do substrates or products. Drugs cannot be designed that precisely mimic the transition state because of its highly unstable structure. However, substrates undergo progressive changes in their overall electrostatic structure during the formation of a transition state complex, and effective drugs often resemble an intermediate stage of the reaction more closely than they resemble the substrate. Medical literature often refers to such compounds as substrate analogs, even though they bind more tightly than substrates.

1. PENICILLIN

The antibiotic penicillin is a transition state analog that binds very tightly to glycopeptidyl transferase, an enzyme required by bacteria for synthesis of the cell wall (Fig. 8.18). Glycopeptidyl transferase catalyzes a partial reaction with penicillin that covalently attaches penicillin to its own active site serine. The reaction is favored by the strong resemblance between the peptide bond in the β -lactam ring of penicillin and the transition state complex of the natural transpeptidation reaction. Active site inhibitors such as penicillin that undergo partial reaction to form irreversible inhibitors in the active site are sometimes termed “suicide inhibitors.”

2. ALLOPURINOL

Allopurinol, a drug used to treat gout, decreases urate production by inhibiting xanthine oxidase. This inhibition provides an example of an enzyme that commits suicide by converting a drug to a transition state analog. The normal physiologic

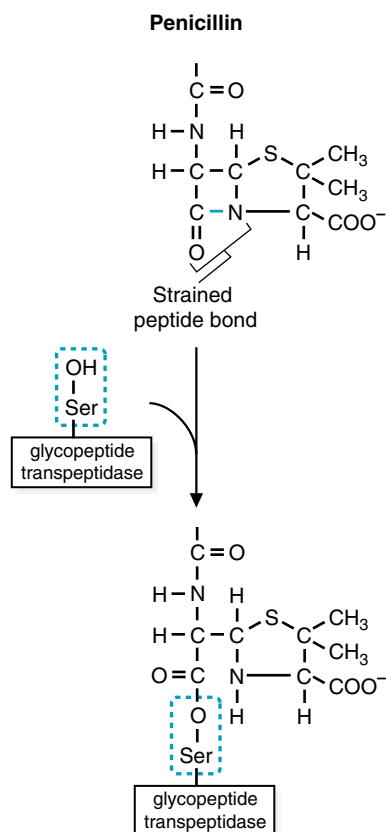


Fig. 8.18 The antibiotic penicillin inhibits the bacterial enzyme glycopeptide transpeptidase. The transpeptidase is a serine protease involved in cross-linking components of bacterial cell walls and is essential for bacterial growth and survival. It normally cleaves the peptide bond between two D-alanine residues in a polypeptide. Penicillin contains a strained peptide bond within the β -lactam ring that resembles the transition state of the normal cleavage reaction, and thus penicillin binds very readily in the enzyme active site. As the bacterial enzyme attempts to cleave this penicillin peptide bond, penicillin becomes irreversibly covalently attached to the enzyme's active site serine, thereby inactivating the enzyme.



Lotta Topaigne is being treated with allopurinol for gout, which is caused by an accumulation of sodium urate crystals in joints and joint fluid, particularly in the ankle and great toe. Allopurinol is a suicide inhibitor of the enzyme xanthine oxidase, which is involved in the degradation of purine nucleotides AMP and GMP to uric acid (urate). Although hypoxanthine levels increase in the presence of allopurinol, hypoxanthine does not participate in urate crystal formation and precipitation at this concentration. It is excreted in the urine.

function of xanthine oxidase is the oxidation of hypoxanthine to xanthine and xanthine to uric acid (urate) in the pathway for degradation of purines (Fig. 8.19). The enzyme contains a molybdenum–sulfide (Mo-S) complex that binds the substrates and transfers the electrons required for the oxidation reactions. Xanthine oxidase oxidizes the drug allopurinol to oxypurinol, a compound that binds very tightly to a molybdenum–sulfide complex in the active site. As a result, the enzyme has committed suicide and is unable to carry out its normal function, the generation of uric acid (urate).

D. Heavy Metals

Heavy metal toxicity is caused by tight binding of a metal such as mercury (Hg), lead (Pb), aluminum (Al), or iron (Fe), to a functional group in an enzyme. Heavy metals are relatively nonspecific for the enzymes they inhibit, particularly if the metal is associated with high dose toxicity. Mercury, for example, binds to so many enzymes, often at reactive sulfhydryl groups in the active site, that it has been difficult to determine which of the inhibited enzymes is responsible for mercury toxicity. Lead provides an example of a metal that inhibits through replacing the normal functional metal in an enzyme. Its developmental and neurologic toxicity may be caused by its

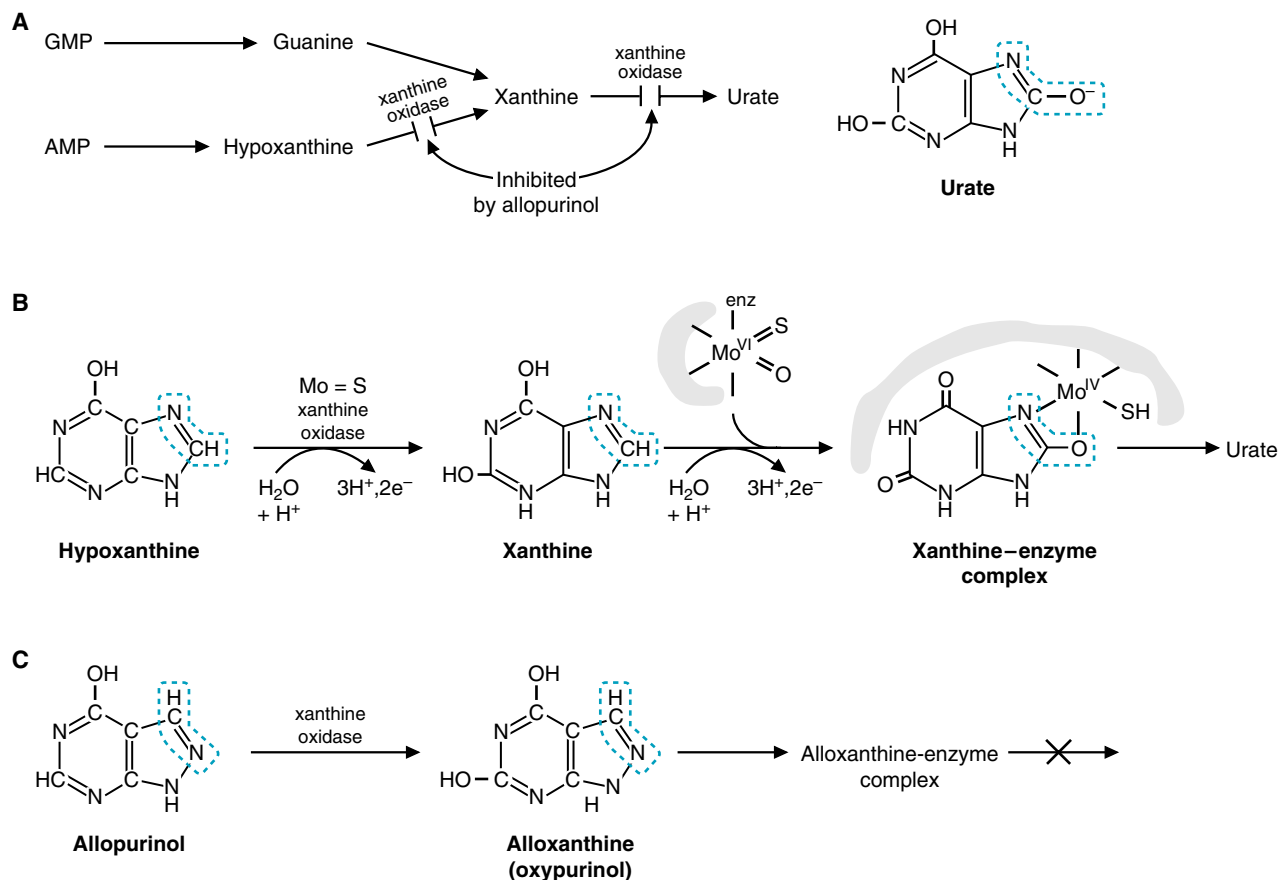


Fig. 8.19. Allopurinol is a suicide inhibitor of xanthine oxidase. **A.** Xanthine oxidase catalyzes the oxidation of hypoxanthine to xanthine, and xanthine to uric acid (urate) in the pathway for degradation of purine nucleotides. **B.** The oxidations are performed by a molybdenum-oxo-sulfide coordination complex in the active site that complexes with the group being oxidized. Oxygen is donated from water. The enzyme can work either as an oxidase (O_2 accepts the 2e^- and is reduced to H_2O_2) or as a dehydrogenase (NAD^+ accepts the 2e^- and is reduced to NADH). **C.** Xanthine oxidase is able to perform the first oxidation step and convert allopurinol to alloxanthine (oxypurinol). As a result, the enzyme has committed suicide; the oxypurinol remains bound in the molybdenum coordination sphere, where it prevents the next step of the reaction.

ability to replace Ca^{2+} in two regulatory proteins important in the central nervous system and other tissues, Ca^{2+} -calmodulin and protein kinase C.

CLINICAL COMMENTS



Dennis Veere. Dennis Veere survived his malathion intoxication because he had ingested only a small amount of the chemical, vomited shortly after the agent was ingested, and was rapidly treated in the emergency room. Lethal doses of oral malathion are estimated at 1 g/kg of body weight for humans. Emergency room physicians used a drug (oxime) to reactivate the acetylcholinesterase in Dennis before the aged complex formed. They also used intravenous atropine, an anticholinergic (antimuscarinic) agent, to antagonize the action of the excessive amounts of acetylcholine accumulating in cholinergic receptors throughout his body.

After several days of intravenous therapy, the signs and symptoms of acetylcholine excess abated, and therapy was slowly withdrawn. Dennis made an uneventful recovery.



Once ingested, the liver converts malathion to the toxic reactive compound, malaoxon, by replacing the sulfur with an oxygen. Malaoxon then binds to the active site of acetylcholinesterase and reacts to form the covalent intermediate. Unlike the complex formed between diisopropylfluorophosphate and acetylcholinesterase, this initial acylenzyme intermediate is reversible. However, with time, the enzyme-inhibitor complex “ages” (dealkylation of the inhibitor and enzyme modification) to form an irreversible complex.



At low concentrations of ethanol, liver alcohol dehydrogenase is the major route of ethanol oxidation to acetaldehyde, a highly toxic chemical. Acetaldehyde not only damages the liver, it can enter the blood and potentially damage the heart and other tissues. At low ethanol intakes, much of the acetaldehyde produced is safely oxidized to acetate in the liver by acetaldehyde dehydrogenases.



Enzymes have been systematically classified by the Enzyme Commission of the International Union of Biochemists and Molecular Biologists into six major groups according to the type of reaction catalysed. The major groups are further subdivided so that each enzyme has an individual Enzyme Commission number. Most enzymes have systematic names, reflecting their activities. In addition, most enzymes have common or trivial names, which are shorter, reflect the activity of the enzyme in a less standardized way, and may not end in “ase”. For example, glucokinase (common name) has the systematic name of ATP:D-hexose 6-phosphotransferase, and its enzyme commission number is EC 2.7.1.2. The “2” is the Enzyme Commission (EC) number of the general class (transferase), followed by a period and “7”, the number of the subclass for transfer of phosphorus-containing groups. The “1” denotes transfer to an alcohol acceptor, and the final “2” is the specific number of the enzyme.



Xanthine oxidase uses molybdenum for electron transfer to O_2 . This enzyme accounts for the human dietary requirement for molybdenum, which is classified as a trace mineral.



Lotta Topaigne. Within several days of starting allopurinol therapy, Ms. Topaigne’s serum uric acid level began to decrease. Several weeks later, the level in her blood was normal. However, while Lotta was adapting to allopurinol therapy, she experienced a mild gout attack, which was treated with a low dose of colchicine (see Chapter 10).



Al Martini. Al Martini was admitted to the hospital after intravenous thiamine was initiated at a dose of 100 mg/day (compared with an RDA of 1.4 mg/day). His congestive heart failure was believed to be the result, in part, of the cardiomyopathy (heart muscle dysfunction) of acute thiamine deficiency known as beriberi heart disease. This nutritional cardiac disorder and the peripheral nerve dysfunction usually respond to thiamine replacement. However, an alcoholic cardiomyopathy can also occur in well-nourished patients with adequate thiamine levels. Exactly how ethanol, or its toxic metabolite acetaldehyde, causes alcoholic cardiomyopathy in the absence of thiamine deficiency is unknown.

BIOCHEMICAL COMMENTS



Basic Reactions and Classes of Enzymes. In the following chapters of the text, students will be introduced to a wide variety of reaction pathways and enzyme names. Although it may seem that the number of reactions is infinite, many of these reactions are similar and occur frequently in different pathways. Recognition of the type of reaction can aid in remembering the pathways and enzyme names, thereby reducing the amount of memorization required. You may wish to use this section for reference as you go through your first biochemical pathways.

The Enzyme Commission has divided the basic reaction types and the enzymes catalyzing them into six broad numbered classes: (1) oxidoreductases, (2) transferases, (3) hydrolases, (4) lyases, (5) isomerases, and (6) ligases. Each broad class of enzymes includes subsets of enzymes with a systematic name and a common name (e.g., dehydrogenases and kinases).



Oxidoreductases. Oxidation-reduction reactions are very common in biochemical pathways and are catalyzed by a broad class of enzymes called oxidoreductases. Whenever an oxidation-reduction reaction occurs, at least one substrate gains electrons and becomes reduced, and another substrate loses electrons and becomes oxidized. One subset of reactions is catalyzed by *dehydrogenases*, which accept and donate electrons in the form of hydride ions (H^-) or hydrogen atoms. Usually an electron-transferring coenzyme, such as $NAD^+/NADH$, acts as an electron donor or acceptor (e.g., see Fig. 8.14 and Fig. 8.15).

In another subset of reactions, O_2 donates either one or both of its oxygen atoms to an acceptor (for example, see xanthine oxidase, Fig. 8.19). When this occurs, O_2 becomes reduced, and an electron donor is oxidized. Enzymes participating in reactions with O_2 are called *hydroxylases* and *oxidases* when one oxygen atom is incorporated into a substrate and the other oxygen atom into water, or both atoms are incorporated into water. They are called *oxygenases* when both atoms of oxygen are incorporated into the acceptor. Most hydroxylases and oxidases require metal ions, such as Fe^{2+} , for electron transfer.



Transferases. Transferases catalyze group transfer reactions—the transfer of a functional group from one molecule to another. If the transferred group is a high-energy phosphate (as shown in Fig. 8.3), the enzyme is a *kinase*; if the transferred group is a carbohydrate residue, the enzyme is a

glycosyltransferase; if it is a fatty acyl group, the enzyme is an *acyltransferase*. A common feature of these reactions is that the group being transferred exists as a good leaving group on the donor molecule.

Another subset of group transfer reactions consists of transaminations (Fig. 8.20). In this type of reaction, the nitrogen group from an amino acid is donated to an α -keto acid, forming a new amino acid and the α -keto acid corresponding to the donor amino acid. Enzymes catalyzing this last type of reaction are called *transaminases* or *aminotransferases*. The coenzyme pyridoxal phosphate is required for all transaminases (see Fig. 8.13).

When the physiologically important aspect of the reaction is the compound synthesized, the transferase may be called a *synthase*. For example, the enzyme commonly called glycogen synthase transfers a glucosyl residue from UDP-glucose to the end of a glycogen molecule. Its systematic name is UDP-glucose-glycogen glycosyltransferase.



Hydrolases. In hydrolysis reactions, C-O, C-N, or C-S bonds are cleaved by the addition of H_2O in the form of OH^- and H^+ to the atoms forming the bond (see, for example, Fig. 8.8). The enzyme class names specify the group being cleaved (e.g., the enzyme commonly named chymotrypsin is a protease, a hydrolase that cleaves peptide bonds in proteins).



Lyases. The lyase class of enzymes consists of a diverse group of enzymes cleaving C-C, C-O, and C-N bonds by means other than hydrolysis or oxidation. Some of the enzymes catalyzing C-C bond cleavage are called *aldolases*, *decarboxylases* (when carbon dioxide is released from a substrate), and *thiolases* (when the sulfur-containing nucleophile of cysteine or CoASH is used to break a carbon-carbon bond) (Fig. 8.21). The structures amenable to carbon-carbon bond cleavage usually require a carbonyl carbon that can act as an electron sink to stabilize the carbanion transiently formed when the carbon-carbon bond breaks.

This broad class of enzymes also includes dehydratases and many synthases. *Dehydratases* remove the elements of water from two adjacent carbon-carbon bonds to form a double bond. Certain enzymes in this group, such as certain group transferases, are commonly called synthases when the physiologically important direction of the reaction favors the formation of a carbon-carbon bond (e.g., citrate synthase).

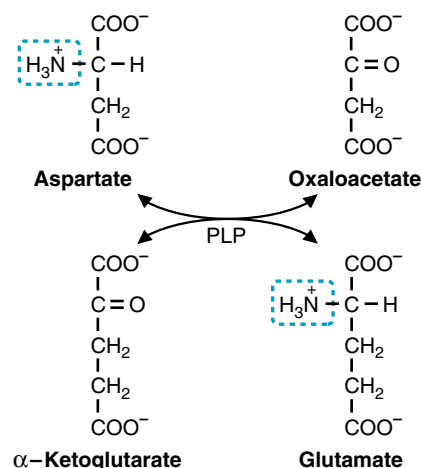


Fig. 8.20. Transamination reactions. Pyridoxal phosphate (PLP) on aspartate aminotransferase transfers an amino group from aspartate to the α -keto acid (α -ketoglutarate) to form a new amino acid (glutamate). The enzyme used to be called glutamate-oxaloacetate transaminase.

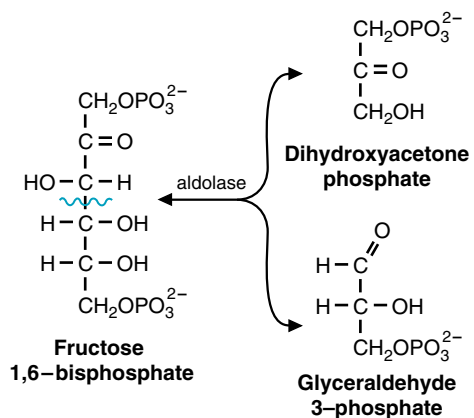


Fig. 8.21. Aldolases catalyze carbon-carbon cleavage in reactions that are usually reversible. In glycolysis, the enzyme fructose 1,6-bisphosphate aldolase cleaves a carbon-carbon bond in fructose 1,6-bisphosphate. Aldolases have a lysine epsilon amino group in the active site that participates in the reaction.

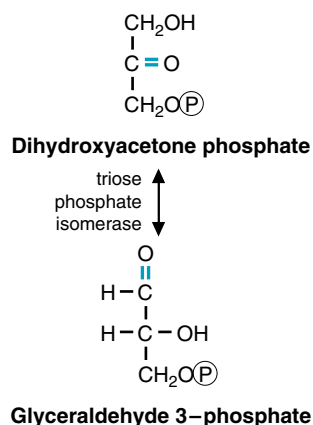


Fig. 8.22. Isomerases rearrange atoms within a molecule. In the pathway of glycolysis, triose phosphate isomerase converts dihydroxyacetone phosphate to glyceraldehydes 3-phosphate by rearranging hydrogen atoms. No other substrates or products of the reaction exist.



Isomerases. Many biochemical reactions simply rearrange the existing atoms of a molecule, that is, create isomers of the starting material (Fig. 8.22). Enzymes rearranging the bond structure of a compound are called *isomerases*, whereas enzymes catalyzing movement of a phosphate from one atom to another are called *mutases*.



Ligases. Ligases synthesize C-C, C-S, C-O, and C-N bonds in reactions coupled to the cleavage of a high-energy phosphate bond in ATP or another nucleotide. *Carboxylases*, for example, add CO_2 to another compound in a reaction requiring ATP cleavage to provide energy (see Fig. 8.12B). Most carboxylases require the coenzyme biotin. Other ligases are named *synthetases* (e.g., fatty acyl CoA synthetase). Synthetases differ from the synthases mentioned under “lyases” and “group transferases” in that synthetases derive the energy for new bond formation from cleavage of high-energy phosphate bonds, and synthases use a different source of energy.

Suggested References

Dressler D, Potter H. Discovering enzymes. Scientific American Library. New York: W.H. Freeman, 1991.
 Fersht A. Structure and Mechanism in Protein Science. New York: W.H. Freeman, 1999.
 EPA has a good website for information on organophosphate compounds at www.epa.gov/pesticides.
 The ENZYME database (<http://www.expasy.ch/enzyme/>) provides basic information about specific enzymes. It is based on the recommendations of the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology. In this database, you can obtain the EC number of any enzyme, its recommended name, alternative names, cofactors, and human diseases associated with the enzyme. You can trace the enzyme from its name to the reaction it catalyses—to the metabolic pathway(s) in which it participates—to the large and intricate Boehringer Mannheim Biochemical Pathways Wallchart. You also can search the database by metabolites or pathways.



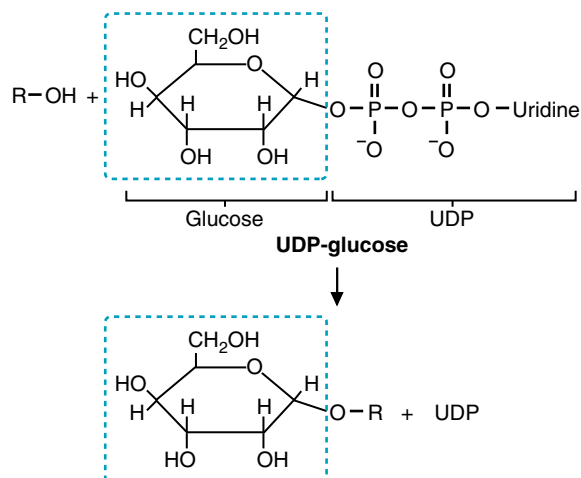
REVIEW QUESTIONS—CHAPTER 8

Questions below cover material from Chapters 6 and 7, as well as Chapter 8 (including Biochemical Comments).

1. A patient was born with a congenital mutation in an enzyme, severely affecting its ability to bind an activation-transfer coenzyme. As a consequence,
 - (A) the enzyme would be unable to bind the substrate of the reaction.
 - (B) the enzyme would be unable to form the transition state complex.
 - (C) the enzyme would normally use a different activation-transfer coenzyme.
 - (D) the enzyme would normally substitute the functional group of an active site amino acid residue for the coenzyme.
 - (E) the reaction could be carried out by the free coenzyme, provided the diet carried an adequate amount of its vitamin precursor.
2. An individual had a congenital mutation in glucokinase in which a proline was substituted for a leucine on a surface helix far from the active site, but within the hinge region of the actin fold. This mutation would be expected to
 - (A) have no effect on the rate of the reaction because it is not in the active site.
 - (B) have no effect on the rate of the reaction because proline and leucine are both nonpolar amino acids.
 - (C) have no effect on the number of substrate molecules reaching the transition state.
 - (D) probably affect the binding of ATP or a subsequent step in the reaction sequence.
 - (E) probably cause the reaction to proceed through an alternate mechanism.

3. A patient developed a bacterial overgrowth in his intestine that decreased the pH of the luminal contents from their normal pH of approximately 6.5 down to 5.5. This decrease of pH is likely to
- (A) denature proteins reaching the intestine with their native structure intact.
 - (B) disrupt hydrogen bonding essential for maintenance of tertiary structure.
 - (C) inhibit intestinal enzymes dependent on histidine for acid-base catalysis.
 - (D) inhibit intestinal enzymes dependent on an active site lysine for binding substrate.
 - (E) have little effect on hydrolases.

Questions 4 and 5 refer to the reaction shown below:



4. The type of reaction shown above fits into which of the following classifications?
- (A) Group transfer
 - (B) Isomerization
 - (C) Carbon-carbon bond breaking
 - (D) Carbon-carbon bond formation
 - (E) Oxidation-reduction
5. The type of enzyme catalyzing this reaction is a
- (A) kinase
 - (B) dehydrogenase
 - (C) glycosyltransferase
 - (D) transaminase
 - (E) isomerase

9 Regulation of Enzymes

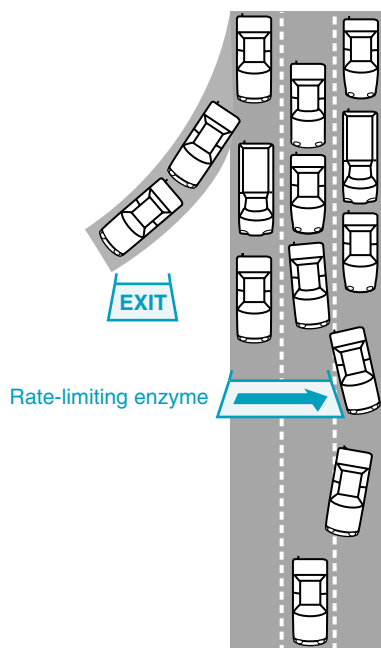


Fig. 9.1. The flux of substrates down a metabolic pathway is analogous to cars traveling down a highway. The rate-limiting enzyme is the portion of the highway that is narrowed to one lane by a highway barrier. This single portion of the highway limits the rate at which cars can arrive at their final destination miles later. Cars will back up before the barrier (similar to the increase in concentration of a precursor when a rate-limiting enzyme is inhibited). Some cars may exit and take an alternate route (similar to precursors entering another metabolic pathway.) Moving the barrier just a little to open an additional lane is like activating a rate-limiting enzyme; it increases flow through the entire length of the pathway.

In the body, thousands of diverse enzymes are regulated to fulfil their individual functions without waste of dietary components. Thus, with changes in our physiologic state, time of eating, environment, diet, or age, the rates of some enzymes must increase and others decrease. In this chapter, we describe the mechanisms for regulation of enzyme activity and the strategies employed to regulate the metabolic pathways in which they reside.

Regulation matches function. Changes in the rate of a metabolic pathway occur because at least one enzyme in that pathway, the **regulatory enzyme**, has been **activated** or **inhibited**, or the amount of enzyme has increased or decreased. Regulatory enzymes usually catalyze the **rate-limiting**, or slowest, step in the pathway, so that increasing or decreasing their rate changes the rate of the entire pathway (Fig. 9.1). The mechanisms used to regulate the rate-limiting enzyme in a pathway reflect the function of the pathway.

Substrate concentration. The rate of all enzymes is dependent on substrate concentration. Enzymes exhibit **saturation kinetics**; their rate increases with increasing substrate concentration $[S]$, but reaches a maximum velocity (V_{max}) when the enzyme is saturated with substrate. For many enzymes, the **Michaelis-Menten equation** describes the relationship between v_i (the initial velocity of a reaction), $[S]$, V_{max} , and the K_m (the substrate concentration at which $v_i = 1/2 V_{max}$).

Reversible inhibition. Enzymes are reversibly inhibited by **structural analogs** and **products**. These inhibitors are classified as **competitive**, **noncompetitive**, or **uncompetitive**, depending on their effect on formation of the **enzyme-substrate complex**.

Allosteric enzymes. **Allosteric activators** or **inhibitors** are compounds that bind at sites other than the active catalytic site and regulate the enzyme through **conformational changes** affecting the catalytic site.

Covalent modification. Enzyme activity also may be regulated by a covalent modification, such as **phosphorylation** of a serine, threonine, or tyrosine residue by a **protein kinase**.

Protein-protein interactions. Enzyme activity can be modulated through the reversible binding of a **modulator protein**, such as Ca^{2+} **calmodulin**. **Monomeric G proteins** (GTP-binding proteins) activate target proteins through reversible binding.

Zymogen cleavage. Some enzymes are synthesized as inactive precursors, called **zymogens**, that are activated by **proteolysis** (e.g., the digestive enzyme chymotrypsin).

Changes in enzyme concentration. The concentration of an enzyme can be regulated by changes in the rate of enzyme synthesis (e.g., induction of gene transcription) or the rate of degradation.

Regulation of metabolic pathways. The regulatory mechanisms for the rate-limiting enzyme of a pathway always reflects the **function of the pathway in a particular tissue**. In **feedback regulation**, the end product of a pathway directly or indirectly controls its own rate of synthesis; in **feedforward regulation**, the

substrate controls the rate of the pathway. **Biosynthetic** and **degradative** pathways are controlled through different but **complementary regulation**. Pathways are also regulated through **compartmentation** of enzymes.



THE WAITING ROOM



Al Martini is a 44-year-old man who has been an alcoholic for the past 5 years. He was recently admitted to the hospital for congestive heart failure (see Chapter 8). After being released from the hospital, he continued to drink. One night he arrived at a friend's house at 7:00 P.M. Between his arrival and 11:00 P.M., he drank four beers and five martinis (for a total ethanol consumption of 9.5 oz). His friends encouraged him to stay an additional hour and drink coffee to sober up. Nevertheless, he ran his car off the road on his way home. He was taken to the emergency room of the local hospital and arrested for driving under the influence of alcohol. His blood alcohol concentration at the time of his arrest was 240 mg/dL, compared with the legal limit of ethanol for driving of 80 mg/dL.



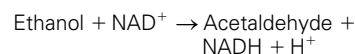
Ann O'Rexia, a 23-year old woman, 5 feet 7 inches tall, is being treated for anorexia nervosa (see Chapters 1–3). She has been gaining weight, and is now back to 99 lb from a low of 85 lb. Her blood glucose is still below normal (fasting blood glucose of 72 mg/dL, compared with a normal range of 80–100 mg/dL). She complains to her physician that she feels tired when she jogs, and she is concerned that the “extra weight” she has gained is slowing her down.



We will generally be using the pathways of fuel oxidation to illustrate the role of various mechanisms of enzyme regulation in metabolic pathways. Chapters 1 through 3 provide an overview of the names and functions of these pathways, including the TCA cycle, glycolysis, glycogen synthesis, glycogenolysis, and fatty acid oxidation.



Al Martini was not able to clear his blood ethanol rapidly enough to stay within the legal limit for driving. Ethanol is cleared from the blood at about 1/2 ounce/hr (15 mg/dL per hour). Liver metabolism accounts for more than 90% of ethanol clearance from the blood. The major route of ethanol metabolism in the liver is the enzyme liver alcohol dehydrogenase (ADH), which oxidizes ethanol to acetaldehyde with generation of NADH.



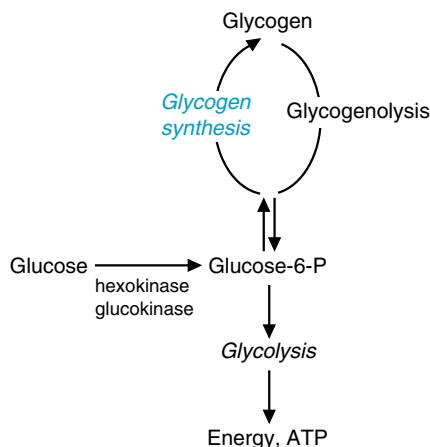
The multienzyme complex MEOS (**m**icrosomal **e**thanol **o**xidizing **s**ystem), which is also called cytochrome P450-2E1, provides an additional route for ethanol oxidation to acetaldehyde in the liver.

Although the regulation of metabolic pathways is an exceedingly complex subject, dealt with in most of the subsequent chapters of this text, a number of common themes are involved. Physiologic regulation of a metabolic pathway depends on the ability to alter flux through the pathway by activating the enzyme catalyzing the rate-limiting step in the pathway (see Fig. 9.1). The type of regulation employed always reflects the function of the pathway and the need for that pathway in a particular tissue or cell type. Pathways producing a necessary product are usually feedback-regulated through a mechanism directly or indirectly involving the concentration of product (e.g., allosteric inhibition or induction/repression of enzyme synthesis). The concentration of product signals when enough of the product has been synthesized. Storage and toxic disposal pathways are usually regulated directly or indirectly through a feedforward mechanism reflecting the availability of precursor. Regulatory enzymes are often tissue-specific isozymes whose properties reflect the different functions of a pathway in particular tissues. Pathways are also regulated through compartmentation, collection of enzymes with a common function within a particular organelle or at a specific site in the cell.

The mechanisms employed to regulate enzymes have been organized into three general categories: regulation by compounds that bind reversibly in the active site (including dependence of velocity on substrate concentration and product levels); regulation by changing the conformation of the active site (including allosteric regulators, covalent modification, protein–protein interactions, and zymogen cleavage); and regulation by changing the concentration of enzyme (enzyme synthesis and degradation).



One of the fuels used by **Ann O' Rexia's** skeletal muscles for jogging is glucose, which is converted to glucose 6-phosphate (glucose 6-P) by the enzymes hexokinase (HK) and glucokinase (GK). Glucose 6-P is metabolized in the pathway of glycolysis to generate ATP. This pathway is feedback regulated, so that as her muscles use ATP, the rate of glycolysis will increase to generate more ATP.



When she is resting, her muscles and liver will convert glucose 6-phosphate to glycogen (a fuel storage pathway, shown in blue). Glycogen synthesis is feed-forward regulated by the supply of glucose and by insulin and other hormones that signal glucose availability.

Glycogenolysis (glycogen degradation) is activated during exercise to supply additional glucose 6-P for glycolysis. Unless Ann consumes sufficient calories, her glycogen stores will not be replenished after exercise, and she will tire easily.

I. REGULATION BY SUBSTRATE AND PRODUCT CONCENTRATION

A. Velocity and Substrate Concentration

The velocity of all enzymes is dependent on the concentration of substrate. This dependence is reflected in conditions such as starvation, in which a number of pathways are deprived of substrate. In contrast, storage pathways (e.g., glucose conversion to glycogen in the liver) and toxic waste disposal pathways (e.g., the urea cycle, which prevents NH_4^+ toxicity by converting NH_4^+ to urea) are normally regulated to speed up when more substrate is available. In the following sections, we use the Michaelis-Menten equation to describe the response of an enzyme to changes in substrate concentration and use glucokinase to illustrate the role of substrate supply in regulation of enzyme activity.



The Michaelis-Menten equation relates the initial velocity of the reaction (v_i) to the concentration of enzyme substrate complexes (ES). This equation is derived for a reaction in which a single substrate, S, is converted to a single product, P. The enzyme (E) and S associate to form ES with the rate constant of k_1 . The complex dissociates with the rate constant of k_2 , or is converted to P with the rate constant k_3 . Under conditions in which $[S] \gg [E]$, [P] is negligible, and the rate of conversion of ES to an enzyme-product complex is very fast, $v_i = k_3 [ES]$. The concentration of ES is a fraction of E_T , the total amount of enzyme present as ES and E.

Therefore,

$$v_i = k_3[ES] = \frac{k_3[ET][S]}{K_m + [S]}$$

Where $K_m = (k_2 + k_3)/k_1$. Substitution of $V_{\max}$ for $k_3 [ET]$ gives the Michaelis-Menten equation (see Equation 9.1)

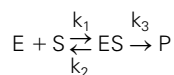
1. MICHAELIS-MENTEN EQUATION

The equations of enzyme kinetics provide a quantitative way of describing the dependence of enzyme rate on substrate concentration. The simplest of these equations, the Michaelis-Menten equation, relates the initial velocity (v_i) to the concentration of substrate [S] and the two parameters K_m and $V_{\max}$ (Equation 9.1) The $V_{\max}$ of the enzyme is the maximal velocity that can be achieved at an infinite concentration of substrate, and the K_m of the enzyme for a substrate is the concentration of substrate required to reach $\frac{1}{2} V_{\max}$. The Michaelis-Menten model of enzyme kinetics applies to a simple reaction in which the enzyme and substrate form an enzyme-substrate complex (ES) that can dissociate back to the free enzyme and substrate. The initial velocity of product formation, v_i , is proportionate to the concentration of enzyme-substrate complexes [ES]. As substrate concentration is increased, the concentration of enzyme-substrate complexes increases, and the reaction rate increases proportionately.

The graph of the Michaelis-Menten equation (v_i as a function of substrate concentration) is a rectangular hyperbola that approaches a finite limit, $V_{\max}$, as the fraction of total enzyme present as enzyme-substrate complex increases (Fig. 9.2).

Equation 9.1. The Michaelis-Menten equation:

For the reaction



the Michaelis-Menten equation is given by

$$v_i = \frac{V_{\max}[S]}{K_m + [S]}$$

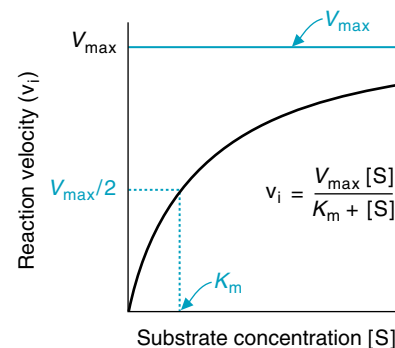
where $K_m = (k_2 + k_3)/k_1$ and $V_{\max} = k_3[ET]$ 

Fig. 9.2. A graph of the Michaelis-Menten equation. $V_{\max}$ (solid blue line) is the initial velocity extrapolated to infinite $[S]$. K_m (dashed blue line) is the concentration of S at which $v_i = V_{\max}/2$.

At a hypothetical infinitely high substrate concentration, all of the enzyme molecules contain bound substrate, and the reaction rate is at $V_{\max}$. The approach to the finite limit of $V_{\max}$ is called saturation kinetics because velocity cannot increase any further once the enzyme is saturated with substrate. Saturation kinetics is a characteristic property of all rate processes dependent on the binding of a compound to a protein.

The K_m of the enzyme for a substrate is defined as the concentration of substrate at which v_i equals $1/2 V_{\max}$. The velocity of an enzyme is most sensitive to changes in substrate concentration over a concentration range below its K_m (see Fig. 9.2). At substrate concentrations less than $1/10$ th of the K_m , a doubling of substrate concentration nearly doubles the velocity of the reaction; at substrate concentrations 10 times the K_m , doubling the substrate concentration has little effect on the velocity.

The K_m of an enzyme for a substrate is related to the dissociation constant, K_d , which is the rate of substrate release divided by the rate of substrate binding. For example, a genetic mutation that decreases the rate of substrate binding to the enzyme decreases the affinity of the enzyme for the substrate and increases the K_d and K_m of the enzyme for that substrate. The higher the K_m , the higher is the substrate concentration required to reach $1/2 V_{\max}$.

2. HEXOKINASE ISOZYMES HAVE DIFFERENT K_m VALUES FOR GLUCOSE

A comparison between the isozymes of hexokinase found in red blood cells and in the liver illustrates the significance of the K_m of an enzyme for its substrate. Hexokinase catalyses the first step in glucose metabolism in most cells, the transfer of a phosphate from ATP to glucose to form glucose 6-phosphate. Glucose 6-phosphate may then be metabolized in glycolysis, which generates energy in the form of ATP, or converted to glycogen, a storage polymer of glucose. Hexokinase I, the isozyme in red blood cells (erythrocytes), has a K_m for glucose of approximately 0.05 mM (Fig. 9.3). The isozyme of hexokinase, called glucokinase, which is found in the liver and pancreas, has a much higher K_m of approximately 5 to 6 mM. The red blood cell is totally dependent on glucose metabolism to meet its needs for ATP. At the low K_m of the erythrocyte hexokinase, blood glucose could fall drastically below its normal fasting level of approximately 5 mM, and the red blood cell could still phosphorylate glucose at rates near $V_{\max}$. The liver, however, stores large amounts of “excess” glucose as glycogen or converts it to fat. Because glucokinase has a K_m of approximately 5 mM, the rate of glucose phosphorylation in the liver will tend to increase as blood glucose increases after a high-carbohydrate meal, and decrease as blood glucose levels fall. The high K_m of hepatic glucokinase thus promotes the storage of glucose as liver glycogen or as fat, but only when glucose is in excess supply.



MODY. Patients with maturity onset diabetes of the young (MODY) have a rare genetic form of diabetes mellitus in which the amount of insulin being secreted from the pancreas is too low, resulting in hyperglycemia. The disease is caused by mutations in the gene for pancreatic glucokinase (a closely related isozyme of liver glucokinase) that affect its kinetic properties (K_m or $V_{\max}$). Glucokinase is part of the mechanism controlling release of insulin from the pancreas. A decreased activity of glucokinase results in lower insulin secretion for a given blood glucose level.



Q: As Ann O'Rexia eats a high carbohydrate meal, her blood glucose will rise to approximately 20 mM in the portal vein, and much of the glucose from her carbohydrate meal will enter the liver. How will the activity of glucokinase in the liver change as glucose is increased from 4 mM to 20 mM? (Hint: Calculate v_i as a fraction of $V_{\max}$ for both conditions, using a K_m for glucose of 5 mM and the Michaelis-Menten equation).

A: Glucokinase, which has a high K_m for glucose, phosphorylates glucose to glucose 6-phosphate about twice as fast after a carbohydrate meal than during fasting. Substitute the values for S and K_m into the Michaelis-Menten equation. The initial velocity will be 0.44 times V_{max} when blood glucose is at 4 mM and about 0.80 times V_{max} when blood glucose is at 20 mM. In the liver, glucose 6-phosphate is a precursor for both glycogen and fat synthesis. Thus, these storage pathways are partially regulated through a direct effect of substrate supply. They are also partially regulated through an increase of insulin and a decrease of glucagon, two hormones that signal the supply of dietary fuel.

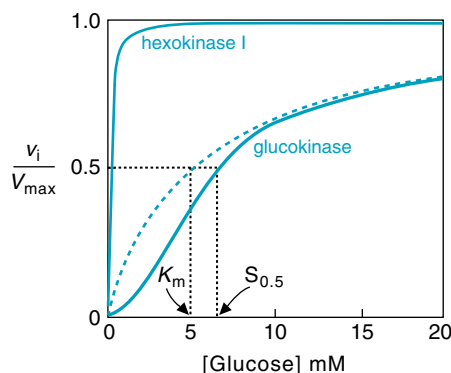


Fig. 9.3. A comparison between hexokinase I and glucokinase. The initial velocity (v_i) as a fraction of V_{max} is graphed as a function of glucose concentration. The plot for glucokinase (heavy blue line) is slightly sigmoidal (S-shaped), possibly because the rate of an intermediate step in the reaction is so slow that the enzyme does not follow Michaelis-Menten kinetics. The dashed blue line has been derived from the Michaelis-Menten equation fitted to the data for concentrations of glucose above 5 mM. For S-shaped curves, the concentration of substrate required to reach half V_{max} , or half-saturation, is sometimes called the $S_{0.5}$ or $K_{0.5}$, rather than K_m . At $v_i/V_{max} = 0.5$, the K_m is 5 mM, and the $S_{0.5}$ is 6.7 mM.

B: The use of V_{max} in the medical literature to describe the maximal rate at which a certain amount of tissue converts substrate to product can be confusing. The best way to describe an increase in enzyme activity in a tissue is to say that the maximal capacity of the tissue has increased. In contrast, the term k_{cat} has been developed to clearly describe the speed at which an enzyme can catalyze a reaction under conditions of saturating substrate concentration. The rate constant k_{cat} , the turnover number of the enzyme, has the units of min^{-1} (micromoles of product formed per minute divided by the micromoles of active site).

3. VELOCITY AND ENZYME CONCENTRATION

The rate of a reaction is directly proportional to the concentration of enzyme; if you double the amount of enzyme, you will double the amount of product produced per minute, whether you are at low or at saturating concentrations of substrate. This important relationship between velocity and enzyme concentration is not immediately apparent in the Michaelis-Menten equation because the concentration of total enzyme present (E_t) has been incorporated into the term V_{max} (that is, V_{max} is equal to the rate constant k_3 times E_t .) However, V_{max} is most often expressed as product produced per minute per milligram of enzyme and is meant to reflect a property of the enzyme that is not dependent on its concentration.

4. MULTISUBSTRATE REACTIONS

Most enzymes have more than one substrate, and the substrate binding sites overlap in the catalytic (active) site. When an enzyme has more than one substrate, the sequence of substrate binding and product release affect the rate equation. As a

C: The liver alcohol dehydrogenase most active in oxidizing ethanol has a very low K_m for ethanol of approximately 0.04 mM, and is at over 99% of its V_{max} at the legal limit of blood alcohol concentration for driving (80 mg/dL or about 17 mM). In contrast, the MEOS isozyme most active toward ethanol has a K_m of approximately 11 mM. Thus, MEOS makes a greater contribution to ethanol oxidation and clearance from the blood at higher ethanol levels than lower ones. Liver damage, such as cirrhosis, results partly from toxic byproducts of ethanol oxidation generated by MEOS. **Al Martini**, who has blood alcohol levels of 240 mg/dL (approximately 52 mM), is drinking enough to potentially cause liver damage, as well as his car accident and arrest for driving under the influence of alcohol. The various isozymes and polymorphisms of alcohol dehydrogenase and MEOS are discussed in more detail in Chapter 25.

consequence, an apparent value of K_m ($K_{m,app}$) depends on the concentration of cosubstrate or product present.

5. RATES OF ENZYME-CATALYZED REACTIONS IN THE CELL

Equations for the initial velocity of an enzyme-catalyzed reaction, such as the Michaelis-Menten equation, can provide useful parameters for describing or comparing enzymes. However, many multisubstrate enzymes, such as glucokinase, have kinetic patterns that do not fit the Michaelis-Menten model (or do so under non-physiologic conditions). The Michaelis-Menten model is also inapplicable to enzymes present in a higher concentration than their substrates. Nonetheless, the term “ K_m ” is still used for these enzymes to describe the approximate concentration of substrate at which velocity equals $\frac{1}{2} V_{max}$.

B. Reversible Inhibition within the Active Site

One of the ways of altering enzyme activity is through compounds binding in the active site. If these compounds are not part of the normal reaction, they inhibit the enzyme. An inhibitor of an enzyme is defined as a compound that decreases the velocity of the reaction by binding to the enzyme. It is a reversible inhibitor if it is not covalently bound to the enzyme and can dissociate at a significant rate. Reversible inhibitors are generally classified as competitive, noncompetitive, or uncompetitive with respect to their relationship to a substrate of the enzyme. In most reactions, the products of the reaction are reversible inhibitors of the enzyme producing them.

1. COMPETITIVE INHIBITION

A competitive inhibitor “competes” with a substrate for binding at the enzyme’s substrate recognition site and therefore is usually a close structural analog of the substrate (Fig. 9.4). An increase of substrate concentration can overcome competitive inhibition; when the substrate concentration is increased to a sufficiently high level, the substrate binding sites are occupied by substrate, and inhibitor molecules cannot bind. Competitive inhibitors, therefore, increase the apparent K_m

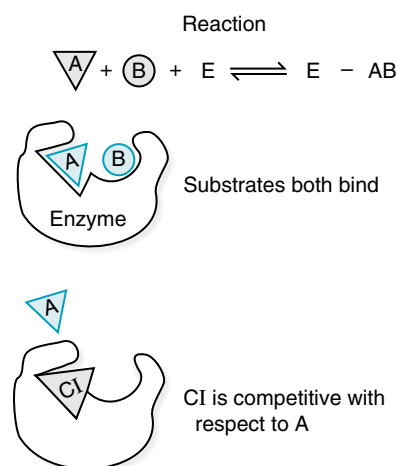


Fig. 9.4. Competitive inhibition with respect to substrate A. A and B are substrates for the reaction forming the enzyme substrate complex (E-AB). The enzyme has separate binding sites for each substrate, which overlap in the active site. The competitive inhibitor (CI) competes for the binding site of A, the substrate it most closely resembles.



Some of **Al Martini's** problems have arisen from product inhibition of liver alcohol dehydrogenase by NADH. As ethanol is oxidized in liver cells, NAD^+ is reduced to NADH and the NADH/ NAD^+ ratio rises. NADH is an inhibitor of alcohol dehydrogenase, competitive with respect to NAD^+ , so the increased NADH/ NAD^+ ratio slows the rate of ethanol oxidation and ethanol clearance from the blood.

NADH is also a product inhibitor of enzymes in the pathway that oxidizes fatty acids. Consequently, these fatty acids accumulate in the liver, eventually contributing to the alcoholic fatty liver.

of the enzyme ($K_{m,app}$) because they raise the concentration of substrate necessary to saturate the enzyme. They have no effect on V_{max} .

2. NONCOMPETITIVE AND UNCOMPETITIVE INHIBITION

If an inhibitor does not compete with a substrate for its binding site, the inhibitor is either a noncompetitive or uncompetitive inhibitor with respect to that particular substrate (Fig. 9.5). To illustrate noncompetitive inhibition, consider a multisubstrate reaction in which substrates A and B react to form a product. An inhibitor (NI) that is a structural analog of substrate B would fit into substrate B's binding site, but the inhibitor would be a noncompetitive inhibitor with regard to the other substrate, substrate A. An increase of A will not prevent the inhibitor from binding to substrate B's binding site. The inhibitor, in effect, lowers the concentration of the active enzyme and therefore changes the V_{max} of the enzyme. If the inhibitor has absolutely no effect on the binding of substrate A, it will not change the K_m for A (a pure noncompetitive inhibitor).

Some inhibitors, such as metals, might not bind at either substrate recognition site. In this case, the inhibitor would be noncompetitive with respect to both substrates.

An inhibitor that is uncompetitive with respect to a substrate will bind only to enzyme containing that substrate. Suppose, for example, that in Figure 9.5 an inhibitor that is a structural analog of B and binds to the B site could only bind to an enzyme that contains A. That inhibitor would be called uncompetitive with respect to A. It would decrease both the V_{max} of the enzyme and its apparent K_m for A.

3. SIMPLE PRODUCT INHIBITION IN METABOLIC PATHWAYS

All products are reversible inhibitors of the enzymes that produce them and may be competitive, noncompetitive, or uncompetitive relative to a particular substrate. Simple product inhibition, a decrease in the rate of an enzyme caused by the accumulation of its own product, plays an important role in metabolic pathways; it prevents one enzyme in a sequence of reactions from generating a product faster than it can be used by the next enzyme in that sequence.



Product inhibition of hexokinase by glucose 6-phosphate conserves blood glucose for tissues needing it. Tissues take up glucose from the blood and phosphorylate it to glucose 6-phosphate, which can then enter a number of different pathways (including glycolysis and glycogen synthesis). As these pathways become more active, glucose 6-phosphate concentration decreases, and the rate of hexokinase increases. When these pathways are less active, glucose 6-phosphate concentration increases, hexokinase is inhibited, and glucose remains in the blood for other tissues.

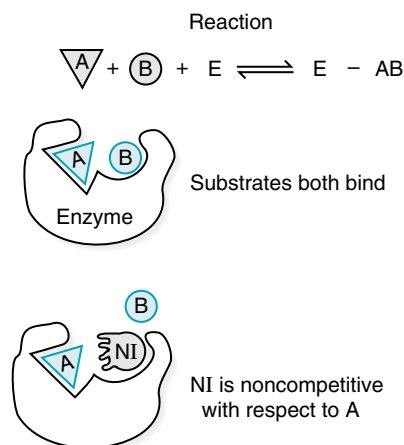


Fig. 9.5. NI is a noncompetitive inhibitor with respect to substrate A. A can still bind to its binding site in the presence of NI. However, NI is competitive with respect to B because it binds to the B binding site. In contrast, an inhibitor that is uncompetitive with respect to A might also resemble B, but it could only bind to the B site after A is bound.

III. REGULATION THROUGH CONFORMATIONAL CHANGES

In substrate response and product inhibition, the rate of the enzyme is affected principally by the binding of a substrate or a product within the catalytic site. Most rate-limiting enzymes are also controlled through regulatory mechanisms that change the conformation of the enzyme in a way that affects the catalytic site. These regulatory mechanisms include: (1) allosteric activation and inhibition; (2) phosphorylation or other covalent modification; (3) protein–protein interactions between regulatory and catalytic subunits, or between two proteins; and (4) proteolytic cleavage. These types of regulation can rapidly change an enzyme from an inactive form to a fully active conformation.

In the sections below, we describe the general characteristics of these regulatory mechanisms and illustrate the first three with glycogen phosphorylase, glycogen phosphorylase kinase, and protein kinase A.

A. Conformational Changes in Allosteric Enzymes

Allosteric activators and inhibitors (allosteric effectors) are compounds that bind to the allosteric site (a site separate from the catalytic site) and cause a conformational change that affects the affinity of the enzyme for the substrate. Usually an allosteric enzyme has multiple interacting subunits that can exist in active and inactive conformations, and the allosteric effector promotes or hinders conversion from one conformation to another.

1. COOPERATIVITY IN SUBSTRATE BINDING TO ALLOSTERIC ENZYMES

Allosteric enzymes usually contain two or more subunits and exhibit positive cooperativity; the binding of substrate to one subunit facilitates the binding of substrate to another subunit (Fig. 9.6). The first substrate molecule has difficulty in binding to the enzyme because all of the subunits are in the conformation with a low affinity for substrate (the taut “T” conformation) (see Chapter 7, section VII.B.). The first substrate molecule to bind changes its own subunit and at least one adjacent subunit to the high-affinity conformation (the relaxed “R” state.) In the example of the tetramer hemoglobin, discussed in Chapter 7, the change in one subunit facilitated changes in all four subunits, and the molecule generally changed to the new conformation in a concerted fashion. However, most allosteric enzymes follow a more stepwise (sequential) progression through intermediate stages (see Fig. 9.6)

2. ALLOSTERIC ACTIVATORS AND INHIBITORS

Allosteric enzymes bind activators at the allosteric site, a site physically separate from the catalytic site. The binding of an allosteric activator changes the conformation of the catalytic site in a way that increases the affinity of the enzyme for the substrate.

In general, activators of allosteric enzymes bind more tightly to the high-affinity R state of the enzyme than the T state (i.e., the allosteric site is open only in the R enzyme) (Fig. 9.7). Thus, the activators increase the amount of enzyme in the active state, thereby facilitating substrate binding in their own and other subunits. In contrast, allosteric inhibitors bind more tightly to the T state, so either substrate concentration or activator concentration must be increased to overcome the effects of the allosteric inhibitor. Allosteric inhibitors might have their own binding site on the enzyme, or they might compete with the substrate at the active site and prevent cooperativity. Thus, the term “allosteric inhibitor” is more generally applied to any inhibitor of an allosteric enzyme.

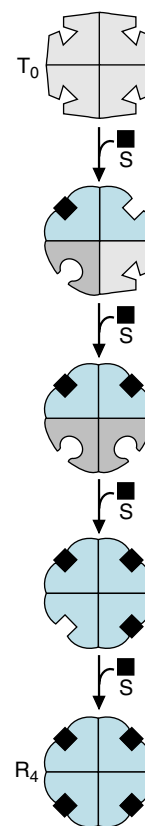


Fig. 9.6. A sequential model for an allosteric enzyme. The sequential model is actually the preferred path from the T_0 (taut, with 0 substrate bound) low-affinity conformation to the R_4 (relaxed, with four substrate molecules bound) conformation, taken from an array of all possible equilibrium conformations that differ by the conformation of only one subunit. The final result is a stepwise path in which intermediate conformations exist, and subunits may change conformations independently, depending on their geometric relationship to the subunits already containing bound substrate.

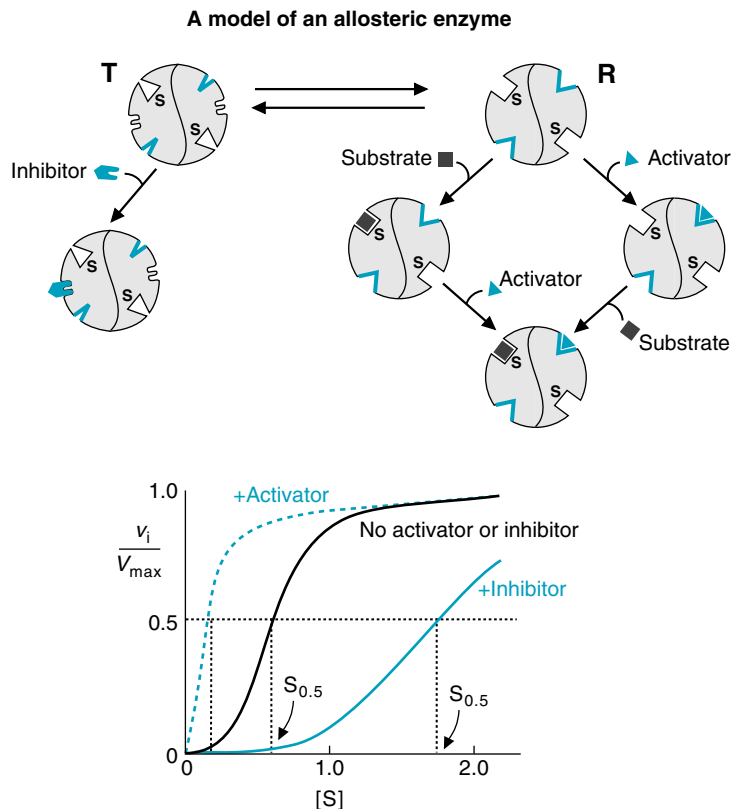


Fig. 9.7. Activators and inhibitors of an allosteric enzyme (simplified model). This enzyme has two identical subunits, each containing three binding sites: one for the substrate (s), one for the allosteric activator (blue triangle), and one for the allosteric inhibitor (two-pronged shape). The enzyme has two conformations, a relaxed active conformation (R) and an inactive conformation (T). The activator binds only to its activator site when the enzyme is in the R configuration. The inhibitor binding site is open only when the enzyme is in the T state.

A plot of velocity ($v_i/V_{\max}$) versus substrate concentration reveals that binding of the substrate at its binding site stabilizes the active conformation so that the second substrate binds more readily, resulting in an S (sigmoidal)-shaped curve. The graph of $v_i/V_{\max}$ becomes hyperbolic in the presence of activator (which stabilizes the high-affinity R form), and more sigmoidal with a higher $S_{0.5}$ in the presence of inhibitor (which stabilizes the low-affinity form).



Some of the rate-limiting enzymes in the pathways of fuel oxidation (e.g., muscle glycogen phosphorylase in glycogenolysis, phosphofructokinase-1 in glycolysis and isocitrate dehydrogenase in the TCA cycle) are allosteric enzymes regulated by changes in the concentration of ADP or AMP, which are allosteric activators. The function of fuel oxidation pathways is the generation of ATP. When the concentration of ATP in a muscle cell begins to decrease, ADP and AMP increase; ADP activates isocitrate dehydrogenase, and AMP activates glycogen phosphorylase and phosphofructokinase-1. The response is very fast, and small changes in the concentration of activator can cause large changes in the rate of the reaction.

In the absence of activator, a plot of velocity versus substrate concentration for an allosteric enzyme usually results in a sigmoid or S-shaped curve (rather than the rectangular hyperbola of Michaelis-Menten enzymes) as the successive binding of substrate molecules activates additional subunits (see Fig. 9.7). In plots of velocity versus substrate concentration, the effect of an allosteric activator generally makes the sigmoidal S-shaped curve more like the rectangular hyperbola, with a substantial decrease in the $S_{0.5}$ (K_m) of the enzyme, because the activator changes all of the subunits to the high-affinity state. Such allosteric effectors are “K effectors”; they change the K_m but not the $V_{\max}$ of the enzyme. An allosteric inhibitor makes it more difficult for substrate or activators to convert the subunits to the most active conformation, and therefore inhibitors generally shift the curve to the right, either increasing the $S_{0.5}$ alone, or increasing it together with a decrease in the $V_{\max}$.

3. ALLOSTERIC ENZYMES IN METABOLIC PATHWAYS

Regulation of enzymes by allosteric effectors provides several advantages over other methods of regulation. Allosteric inhibitors usually have a much stronger

effect on enzyme velocity than competitive, noncompetitive, and uncompetitive inhibitors in the active catalytic site. Because allosteric effectors do not occupy the catalytic site, they may function as activators. Thus, allosteric enzymes are not limited to regulation through inhibition. Furthermore, the allosteric effector need not bear any resemblance to substrate or product of the enzyme. Finally, the effect of an allosteric effector is rapid, occurring as soon as its concentration changes in the cell. These features of allosteric enzymes are often essential for feedback regulation of metabolic pathways by endproducts of the pathway or by signal molecules that coordinate multiple pathways.

B. Conformational Changes from Covalent Modification

1. PHOSPHORYLATION

The activity of many enzymes is regulated through phosphorylation by a protein kinase or dephosphorylation by a protein phosphatase (Fig. 9.8). Serine/threonine protein kinases transfer a phosphate from ATP to the hydroxyl group of a specific serine (and sometimes threonine) on the target enzyme; tyrosine kinases transfer a phosphate to the hydroxyl group of a specific tyrosine residue. Phosphate is a bulky, negatively charged residue that interacts with other nearby amino acid residues of the protein to create a conformational change at the catalytic site. The conformational change makes certain enzymes more active and other enzymes less active. The effect is reversed by a specific protein phosphatase that removes the phosphate by hydrolysis.

2. MUSCLE GLYCOGEN PHOSPHORYLASE

Muscle glycogen phosphorylase, the rate-limiting enzyme in the pathway of glycogen degradation, degrades glycogen to glucose 1-phosphate. It is regulated by the allosteric activator AMP, which increases in the cell as ATP is used for muscular contraction (Fig. 9.9). Thus, a rapid increase in the rate of glycogen degradation to glucose 1-phosphate is achieved when an increase of AMP signals that more fuel is needed for ATP generation in the glycolytic pathway.

Glycogen phosphorylase also can be activated through phosphorylation by glycogen phosphorylase kinase. Either phosphorylation or AMP binding can change the enzyme to the same fully active conformation. The phosphate is removed by protein

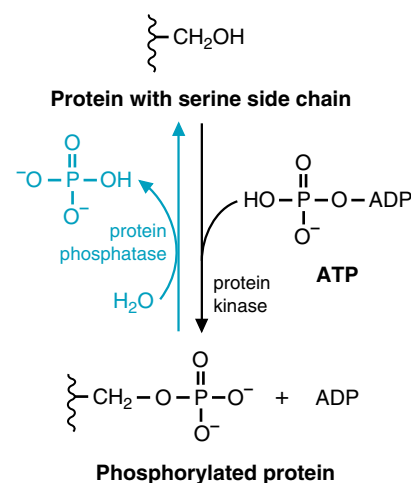


Fig. 9.8. Protein kinases and protein phosphatases.



When **Ann O'Rexia** begins to jog, AMP activates her muscle glycogen phosphorylase, which degrades glycogen to glucose 1-phosphate. This compound is converted to glucose 6-phosphate, which feeds into the glycolytic pathway to generate ATP for muscle contraction. As she continues to jog, her adrenaline (epinephrine) levels rise, producing the signal that activates glycogen phosphorylase kinase. This enzyme phosphorylates glycogen phosphorylase, causing it to become even more active than with AMP alone (see Fig. 9.9).

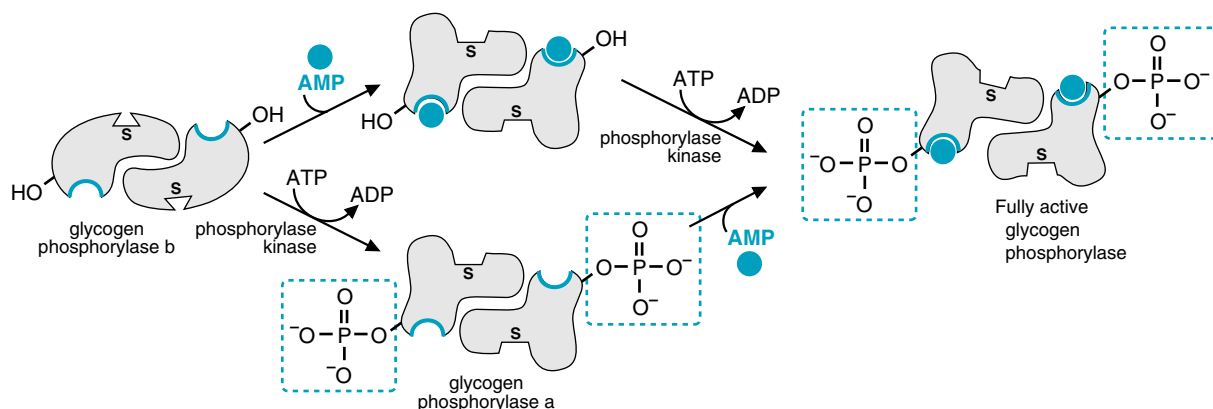


Fig. 9.9. Activation of muscle glycogen phosphorylase by AMP and by phosphorylation. Muscle glycogen phosphorylase is composed of two identical subunits. The substrate binding sites in the active catalytic site are denoted by S. AMP binds to the allosteric site, a site separate from the active catalytic site. Glycogen phosphorylase kinase can transfer a phosphate from ATP to one serine residue in each subunit. Either phosphorylation or binding of AMP causes a change in the active site that increases the activity of the enzyme. The first event at one subunit facilitates the subsequent events that convert the enzyme to the fully active form.

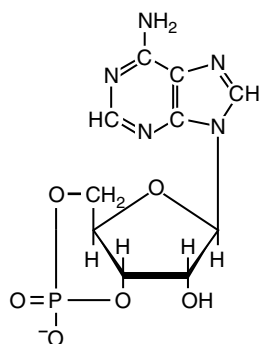


Fig. 9.10. Structure of cAMP (3',5' -cyclic AMP.) The phosphate group is attached to hydroxyl groups on both the 3rd (3') and 5th (5') carbons of ribose, forming a cycle.

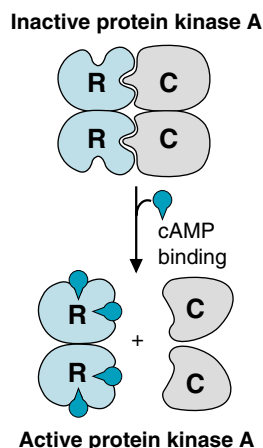


Fig. 9.11. Protein kinase A. When the regulatory subunits (R) of protein kinase A bind the allosteric activator, cAMP, they dissociate from the enzyme, thereby releasing active catalytic subunits (C).

phosphatase-1. Glycogen phosphorylase kinase links the activation of muscle glycogen phosphorylase to changes in the level of the hormone adrenaline in the blood. It is regulated through phosphorylation by protein kinase A and by activation of Ca^{2+} -calmodulin (a modulator protein) during contraction.

3. PROTEIN KINASE A

Some protein kinases, called dedicated protein kinases, are tightly bound to a single protein and regulate only the protein to which they are tightly bound. However, other protein kinases and protein phosphatases will simultaneously regulate a number of rate-limiting enzymes in a cell to achieve a coordinated response. For example, protein kinase A, a serine/threonine protein kinase, phosphorylates a number of enzymes that regulate different metabolic pathways. One of these enzymes is glycogen phosphorylase kinase (see Fig. 9.9).

Protein kinase A provides a means for hormones to control metabolic pathways. Adrenaline and many other hormones increase the intracellular concentration of the allosteric regulator 3', 5'-cyclic AMP (cAMP), which is referred to as a hormonal second messenger (Fig. 9.10). cAMP binds to regulatory subunits of protein kinase A, which dissociate and release the activated catalytic subunits (Fig. 9.11). Dissociation of inhibitory regulatory subunits is a common theme in enzyme regulation. The active catalytic subunits phosphorylate glycogen phosphorylase and other enzymes at serine residues.

In the example shown in Figure 9.9, adrenaline indirectly increases cAMP, which activates protein kinase A, which phosphorylates phosphorylase kinase, which phosphorylates glycogen phosphorylase. The sequence of events in which one kinase phosphorylates another kinase is called a phosphorylation cascade. Because each stage of the phosphorylation cascade is associated with one enzyme molecule activating many enzyme molecules, the initial activating event is greatly amplified.

4. OTHER COVALENT MODIFICATIONS

A number of proteins are covalently modified by the addition of groups such as acetyl, ADP-ribose, or lipid moieties (see Chapter 6). These modifications may directly activate or inhibit the enzyme. However, they also may modify the ability of the enzyme to interact with other proteins or to reach its correct location in the cell.

C. Conformational Changes from Protein-Protein Interactions

Changes in the conformation of the active site also can be regulated by direct protein-protein interaction. This type of regulation is illustrated by Ca^{2+} -calmodulin and small (monomeric) G proteins.

1. THE CALCIUM-CALMODULIN FAMILY OF MODULATOR PROTEINS

Modulator proteins bind to other proteins and regulate their activity by causing a conformational change at the catalytic site or by blocking the catalytic site (steric hindrance). They are protein allosteric effectors that can either activate or inhibit the enzyme or protein to which they bind.

Ca^{2+} -calmodulin is an example of a dissociable modulator protein that binds to a number of different proteins and regulates their function. It also exists in the cytosol and functions as a Ca^{2+} binding protein (Fig. 9.12). The center of the symmetric molecule is a hinge region that bends as Ca^{2+} -calmodulin folds over the protein it is regulating.

One of the enzymes activated by Ca^{2+} -calmodulin is muscle glycogen phosphorylase kinase, which is also activated by protein kinase A (see Fig. 9.9). When a

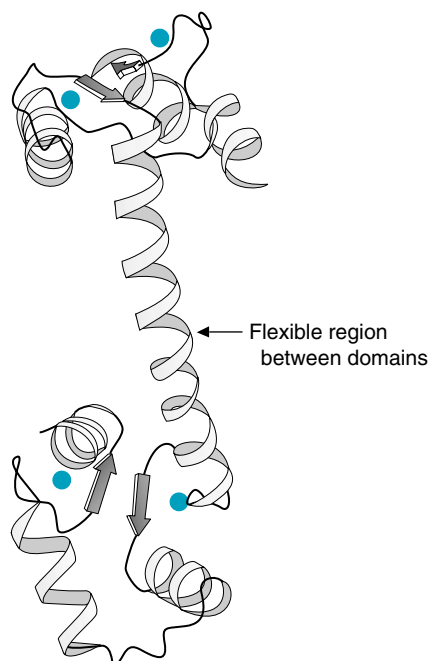


Fig. 9.12. Calcium-calmodulin has four binding sites for calcium (shown in blue). Each calcium forms a multiligand coordination sphere by simultaneously binding several amino acid residues on calmodulin. Thus, it can create large conformational changes in proteins when it binds. Calmodulin has a flexible region in the middle connecting the two domains.

neural impulse triggers Ca^{2+} release from the sarcoplasmic reticulum, Ca^{2+} binds to the calmodulin subunit of muscle glycogen phosphorylase kinase, which undergoes a conformational change. This activated kinase then phosphorylates glycogen phosphorylase, ultimately increasing the generation of ATP to supply energy for muscle contraction. Simultaneously, Ca^{2+} binds to troponin-C, a member of the Ca^{2+} -calmodulin superfamily that serves as a nondissociable regulatory subunit of troponin, a regulator of muscle contraction. Calcium binding to troponin prepares the muscle for contraction. Thus, the supply of energy for contraction is activated simultaneously with the contraction machinery.

2. SMALL (MONOMERIC) G PROTEINS REGULATE THROUGH CONFORMATIONAL CHANGES

The masters of regulation through reversible protein association in the cell are the monomeric G proteins, small single-subunit proteins that bind and hydrolyze GTP. GTP (guanosine triphosphate) is a purine nucleotide that, like ATP, contains high-energy phosphoanhydride bonds that release energy when hydrolyzed. When G proteins bind GTP, their conformation changes so that they can bind to a target protein, which is then either activated or inhibited in carrying out its function (Fig. 9.13, step 1).

G proteins are said to possess an internal clock because they are GTPases that slowly hydrolyze their own bound GTP to GDP and phosphate. As they hydrolyze GTP, their conformation changes and the complex they have formed with the target protein disassembles (see Fig. 9.13, step 2). The bound GDP on the inactive G protein is eventually replaced by GTP, and the process can begin again (see Fig. 9.13, step 3).

The activity of many G proteins is regulated by accessory proteins (GAPs, GEFs, and GDIs), which may, in turn, be regulated by allosteric effectors. GAPs (GTPase activating proteins) increase the rate of GTP hydrolysis by the G protein, and therefore the rate of dissociation of the G protein-target protein complex (see Fig. 9.13,

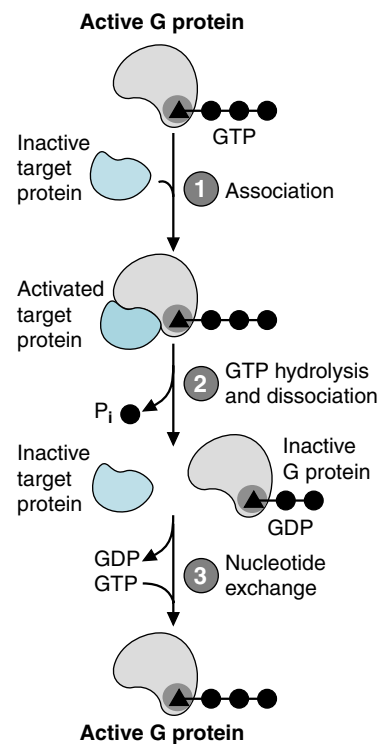


Fig. 9.13. Monomeric G proteins. (1) When GTP is bound, the conformation of the G protein allows it to bind target proteins, which are then activated. (2) The G protein hydrolyzes a phosphate from GTP to form GDP, which changes the G-protein conformation and causes it to dissociate from the target protein. (3) GDP is exchanged for GTP, which reactivates the G protein.

step 2). When a GEF protein (guanine nucleotide exchange factor) binds to a G-protein, it increases the rate of GTP exchange for a bound GDP, and therefore activates the G-protein (see Fig. 9.13, step 3). GDI proteins (GDP dissociation inhibitor) bind to the GDP-G protein complex and inhibit dissociation of GDP, thereby keeping the G protein inactive.

The Ras superfamily of small G proteins is divided into five families: Ras, Rho, Arf, Rab, and Ran. These monomeric G proteins play major roles in the regulation of growth, morphogenesis, cell motility, axonal guidance, cytokinesis, and trafficking through the Golgi, nucleus, and endosomes. They are generally bound to a lipid membrane through a lipid anchor, such as a myristoyl group or farnesyl group, and regulate the assembly and activity of protein complexes at these sites. The small G protein Ras, for example, is involved in regulation of cellular proliferation by a number of hormones called growth factors (Fig. 9.14). It is attached to the plasma membrane by a farnesyl group (see Chapter 6, section IV.B.). The activity of Ras is regulated by a guanine nucleotide exchange protein called SOS (son of sevenless). When SOS is in its active conformation, it binds to Ras, thereby activating dissociation of GDP and binding of GTP. When Ras binds GTP, it is activated, allowing it to bind and activate a protein kinase called Raf. The net effect will be the activation of transcription of certain genes. (Rho, Arf, Rab, and Ran are illustrated in Chapter 10, and the function of Ras is discussed in greater detail in Chapter 11).

D. Proteolytic Cleavage

Although many enzymes undergo some cleavage during synthesis, others enter lysosomes, secretory vesicles or are secreted as proenzymes, which are precursor proteins that must undergo proteolytic cleavage to become fully functional. Unlike most other forms of regulation, proteolytic cleavage is irreversible.

The precursor proteins of proteases (enzymes that cleave specific peptide bonds) are called zymogens. To denote the inactive zymogen form of an enzyme, the name is modified by addition of the suffix “ogen” or the prefix “pro.” The synthesis of zymogens as inactive precursors prevents them from cleaving proteins prematurely at their sites of synthesis or secretion. Chymotrypsinogen, for example, is stored in vesicles within pancreatic cells until secreted into ducts leading to the intestinal lumen. In the digestive tract, chymotrypsinogen is converted to chymotrypsin by the proteolytic enzyme trypsin, which cleaves off a small peptide from the N-terminal region (and two internal peptides). This cleavage activates chymotrypsin by causing



Most of the proteases involved in blood clotting are zymogens, such as fibrinogen and prothrombin, which circulate in blood in the inactive form. They are cleaved to the active form (fibrin and thrombin, respectively) by other proteases, which have been activated by their attachment to the site of injury in a blood vessel wall. Thus, clots form at the site of injury and not randomly in circulation.

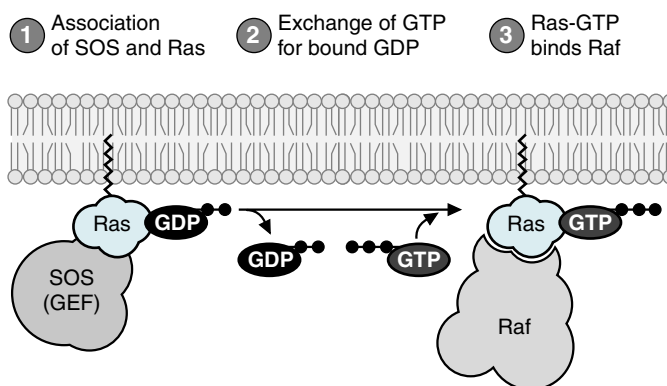


Fig. 9.14. The monomeric G protein Ras. When SOS is activated, it binds to Ras, a monomeric G protein anchored to the plasma membrane. SOS is a guanine nucleotide exchange protein that activates the exchange of GTP for bound GDP on Ras. Activated Ras containing GTP binds the target enzyme Raf, thereby activating it.

a conformational change in the spacing of amino acid residues around the binding site for the denatured protein substrate and around the catalytic site.

IV. REGULATION THROUGH CHANGES IN AMOUNT OF ENZYME

Tissues continuously adjust the rate at which different proteins are synthesized to vary the amount of different enzymes present. The expression for $V_{\max}$ in the Michaelis-Menten equation incorporates the concept that the rate of a reaction is proportional to the amount of enzyme present. Thus, the maximal capacity of a tissue can change with increased protein synthesis, or with increased protein degradation.

A. Regulated Enzyme Synthesis

Protein synthesis begins with the process of gene transcription, transcribing the genetic code for that protein from DNA into messenger RNA. The code in messenger RNA is then translated into the primary amino acid sequence of the protein. Generally the rate of enzyme synthesis is regulated by increasing or decreasing the rate of gene transcription, processes generally referred to as induction (increase) and repression (decrease). However, the rate of enzyme synthesis is sometimes regulated through stabilization of the messenger RNA. (These processes are covered in Section Three). Compared with the more immediate types of regulation discussed above, regulation by means of induction/repression of enzyme synthesis is usually slow in the human, occurring over hours to days.

B. Regulated Protein Degradation

The content of an enzyme in the cell can be altered through selective regulated degradation as well as through regulated synthesis. Although all proteins in the cell can be degraded with a characteristic half-life within lysosomes, protein degradation via two specialized systems, proteasomes and caspases, is highly selective and regulated. Protein degradation is dealt with in more detail in Chapter 37.

V. REGULATION OF METABOLIC PATHWAYS

The different means of regulating enzyme activity described above are used to control metabolic pathways, cellular events, and physiologic processes to match the body's requirements. Although many metabolic pathways are present in the body, a few common themes or principles are involved in their regulation. Of course, the overriding principle is: **Regulation of a pathway matches its function.**

A. Principles of Pathway Regulation

Metabolic pathways are a series of sequential reactions in which the product of one reaction is the substrate of the next reaction (Fig. 9.15). Each step or reaction is usually catalyzed by a separate enzyme. The enzymes of a pathway have a common function—conversion of substrate to the final endproducts of the pathway. A pathway also may have a branchpoint at which an intermediate becomes the precursor for another pathway.

1. REGULATION OCCURS AT THE RATE-LIMITING STEP

Pathways are principally regulated at one key enzyme, the regulatory enzyme, which catalyzes the rate-limiting step in the pathway. This is the slowest step and is usually not readily reversible. Thus, changes in the rate-limiting step can influence flux through the rest of the pathway (see Fig. 9.1). The rate-limiting step is usually the first committed step in a pathway, or a reaction that is related to, or influenced



The maximal capacity of MEOS (cytochrome P450-2E1) is increased in the liver with continued ingestion of ethanol through a mechanism involving induction of gene transcription. Thus, **Al Mar-tini** has a higher capacity to oxidize ethanol to acetaldehyde than a naive drinker (a person not previously subjected to alcohol). Nevertheless, the persistence of his elevated blood alcohol level shows he has saturated his capacity for ethanol oxidation ($V_{\max}$ ed out). Once his enzymes are operating near $V_{\max}$, any additional ethanol he drinks will not appreciably increase the rate of ethanol clearance from his blood.



During fasting or infective stress, protein degradation in skeletal muscle is activated to increase the supply of amino acids in the blood for gluconeogenesis, or for the synthesis of antibodies and other component of the immune response. Under these conditions, synthesis of ubiquitin, a protein that targets proteins for degradation in proteasomes, is increased by the steroid hormone cortisol.

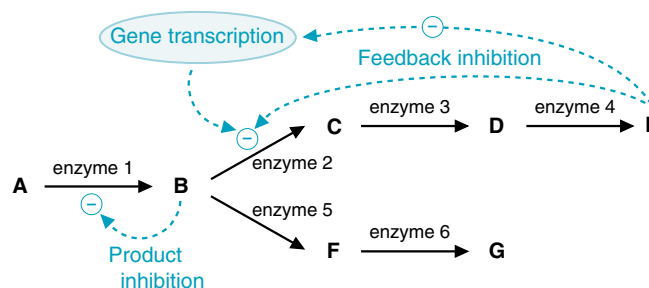


Fig. 9.15. A common pattern for feedback inhibition of metabolic pathways. The letters represent compounds formed from different enzymes in the reaction pathway. Compound B is at a metabolic branchpoint: it can go down one pathway to E or down an alternate pathway to G. The endproduct of the pathway, E, might control its own synthesis by allosterically inhibiting enzyme 2, the first committed step of the pathway, or inhibiting transcription of the gene for enzyme 2. As a result of the feedback inhibition, B accumulates and more B enters the pathway for conversion to G, which could be a storage, or disposal pathway. In this hypothetical pathway, B is a product inhibitor of enzyme 1, competitive with respect to A. Precursor A might induce the synthesis of enzyme 1, which would allow more A to go to G.



The pathways of energy production must be regulated by a mechanism that can respond rapidly to requirements for more ATP, such as the allosteric regulation of glycogen phosphorylase by AMP. However, storage pathways or biosynthetic pathways can be regulated by a mechanism that responds more slowly to changing conditions. For example, cholesterol partially feedback regulates its own rate of synthesis by decreasing transcription of the gene for the rate-limiting enzyme (HMG-CoA reductase). The enzyme concentration of a tissue may change even more slowly in response to developmental changes.



When **Ann O'Rexia** jogs, the increased use of ATP for muscle contraction results in an increase of AMP, which allosterically activates both the allosteric enzyme phosphofructokinase-1, the rate-limiting enzyme of glycolysis, and glycogen phosphorylase, the rate-limiting enzyme of glycogenolysis. This is an example of *feedback regulation* by the ATP/AMP ratio. Unfortunately, her low caloric consumption has not allowed *feed-forward activation* of the rate-limiting enzymes in her fuel storage pathways, and she has very low glycogen stores. Consequently, she has inadequate fuel stores to supply the increased energy demands of exercise.

by, the first committed step. Additional regulated enzymes occur after each metabolic branchpoint to direct flow into the branch. (e.g., in Fig. 9.15, feedback inhibition of enzyme 2 results in accumulation of B, which enzyme 5 then uses for synthesis of compound G). Inhibition of the rate-limiting enzyme in a pathway usually leads to accumulation of the pathway precursor.

2. FEEDBACK REGULATION

Feedback regulation refers to a situation in which the endproduct of a pathway controls its own rate of synthesis (see Fig. 9.15). Feedback regulation usually involves allosteric regulation of the rate-limiting enzyme by the endproduct of a pathway (or a compound that reflects changes in the concentration of the endproduct). The endproduct of a pathway may also control its own synthesis by inducing or repressing the gene for transcription of the rate-limiting enzyme in the pathway. This type of regulation is much slower to respond to changing conditions than allosteric regulation.

3. FEED-FORWARD REGULATION

Certain pathways, such as those involved in the disposal of toxic compounds, are feed-forward regulated. Feed-forward regulation may occur through an increased supply of substrate to an enzyme with a high K_m , allosteric activation of a rate-limiting enzyme through a compound related to substrate supply, substrate-related induction of gene transcription (e.g., induction of cytochrome P450-2E1 by ethanol), or increased concentration of a hormone that stimulates a storage pathway by controlling enzyme phosphorylation state.

4. TISSUE ISOZYMES OF REGULATORY PROTEINS

The human body is composed of a number of different cell types that perform specific functions unique to that cell type and synthesize only the proteins consistent with their functions. Because regulation matches function, regulatory enzymes of pathways usually exist as tissue-specific isozymes with somewhat different regulatory properties unique to their function in different cell types. For example, hexokinase and glucokinase are tissue-specific isozymes with different kinetic properties.

5. COUNTER-REGULATION OF OPPOSING PATHWAYS

A pathway for the synthesis of a compound usually has one or more enzymatic steps that differ from the pathway for degradation of that compound. A biosynthetic pathway can therefore have a different regulatory enzyme than the opposing degradative pathway, and one pathway can be activated while the other is inhibited (e.g., glycogen synthesis is activated while glycogen degradation is inhibited).

6. SUBSTRATE CHANNELING THROUGH COMPARTMENTATION

In the cell, compartmentation of enzymes into multienzyme complexes or organelles provides a means of regulation, either because the compartment provides unique conditions or because it limits or channels access of the enzymes to substrates. Enzymes or pathways with a common function are often assembled into organelles. For example, enzymes of the TCA cycle are all located within the mitochondrion. The enzymes catalyze sequential reactions, and the product of one reaction is the substrate for the next reaction. The concentration of the pathway intermediates remains much higher within the mitochondrion than in the surrounding cellular cytoplasm.

Another type of compartmentation involves the assembly of enzymes catalyzing sequential reactions into multi-enzyme complexes so that intermediates of the pathway can be directly transferred from the active site on one enzyme to the active site on another enzyme, thereby preventing loss of energy and information.

7. LEVELS OF COMPLEXITY

You may have noticed by now that regulation of metabolic pathways in the human is exceedingly complex; this might be called the second principle of metabolic regulation. As you study different pathways in the subsequent chapters of the text, it may help to develop diagrams such as Fig. 9.15 to keep track of the function and rationale behind different regulatory interactions.



The different isozymes of hexokinase (e.g., hexokinase I and glucokinase) are tissue-specific isozymes that arose through gene duplication. Glucokinase, the low-affinity enzyme found in liver, is a single polypeptide chain with a molecular weight of 55 kDa that contains one active catalytic site. The hexokinases found in erythrocytes, skeletal muscles, and most other tissues are 110 kDa and are essentially two mutated glucokinase molecules synthesized as one polypeptide chain. However, only one catalytic site is functional. All of the tissue-specific hexokinases but glucokinase have K_m s for glucose that are less than 0.2 mM.



An example of a multi-enzyme complex is provided by MEOS (microsomal ethanol oxidizing system), which is composed of two different subunits with different enzyme activities. One subunit transfers electrons from NADPH to a cytochrome Fe-heme group on the 2nd subunit, which then transfers the electrons to O_2 .

CLINICAL COMMENTS



Al Martini. In the Emergency Room, **Al Martini** was evaluated for head injuries. From the physical examination and blood alcohol levels, it was determined that his mental state resulted from his alcohol consumption. Although his chronic ethanol consumption had increased his level of MEOS (and, therefore, rate of ethanol oxidation in his liver), his excessive drinking resulted in a blood alcohol level greater than the legal limit of 80 mg/dL. He suffered bruises and contusions but was otherwise uninjured. He left in the custody of the police officer.



Ann O'Rexia. Ann O'Rexia's physician explained that she had inadequate fuel stores for her exercise program. To jog, her muscles require an increased rate of fuel oxidation to generate the ATP for muscle contraction. The fuels used by muscles for exercise include glucose from muscle glycogen, fatty acids from adipose tissue triacylglycerols, and blood glucose supplied by liver glycogen. These fuel stores were depleted during her prolonged bout of starvation. In addition, starvation resulted in the loss of muscle mass as muscle protein was being degraded to supply amino acids for other processes, including gluconeogenesis (the synthesis of glucose from amino acids and other noncarbohydrate precursors). Therefore, Ann will need to increase her caloric consumption to rebuild her fuel stores. Her physician helped her calculate the



The hormone epinephrine (released during stress and exercise) and glucagon (released during fasting) activate the synthesis of cAMP in a number of tissues. cAMP activates protein kinase A. Because protein kinase A is able to phosphorylate key regulatory enzymes in many pathways, these pathways can be coordinately regulated. In muscle, for example, glycogen degradation is activated while glycogen synthesis is inhibited. At the same time, fatty acid release from adipose tissue is activated to provide more fuel for muscle. The regulation of glycolysis, glycogen metabolism, and other pathways of metabolism is much more complex than we have illustrated here and is discussed in many subsequent chapters of this text.

additional amount of calories her jogging program will need, and they discussed which foods she would eat to meet these increased caloric requirements. He also helped her visualize the increase of weight as an increase in strength.

BIOCHEMICAL COMMENTS



The Lineweaver-Burk transformation. The K_m and V_{max} for an enzyme can be visually determined from a plot of $1/v_i$ versus $1/S$, called a Lineweaver-Burk or a double reciprocal plot. The reciprocal of both sides of the Michaelis-Menten equation generates an equation that has the form of a straight line, $y = mx + b$ (Fig. 9.16). K_m and V_{max} are equal to the reciprocals of the intercepts on the abscissa and ordinate, respectively. Although double reciprocal plots are often used to illustrate certain features of enzyme reactions, they are not directly used for the determination of K_m and V_{max} values by researchers.

For the reaction in which an enzyme forms a complex with both substrates, the K_m for one substrate can vary with the concentration of cosubstrate (Fig. 9.17). At each constant concentration of cosubstrate, the plot of $1/v_i$ vs $1/[S]$ is a straight line. To obtain V_{max} , the graph must be extrapolated to saturating concentrations of both substrates, which is equivalent to the intersection point of these lines for different cosubstrate concentrations.

Lineweaver-Burk plots provide a good illustration of competitive inhibition and pure noncompetitive inhibition (Fig. 9.18). In competitive inhibition, plots of $1/v$ vs $1/[S]$ at a series of inhibitor concentrations intersect on the ordinate. Thus, at infinite substrate concentration, or $1/[S] = 0$, there is no effect of the inhibitor. In pure noncompetitive inhibition, the inhibitor decreases the velocity even when $[S]$ has been extrapolated to an infinite concentration. However, if the inhibitor has no effect on the binding of the substrate, the K_m is the same for every concentration of inhibitor, and the lines intersect on the abscissa.

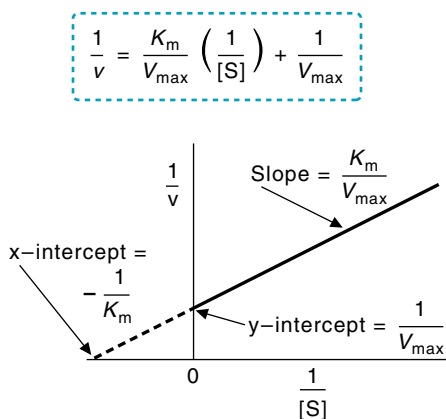
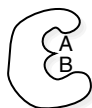


Fig. 9.16. The Lineweaver-Burk transformation (shown in blue) for the Michaelis-Menten equation converts it to a straight line of the form $y = mx + b$. When $[S]$ is infinite, $1/[S] = 0$, and the line crosses the ordinate (y-axis) at $1/v = 1/V_{max}$. The slope of the line is K_m/V_{max} . Where the line intersects the abscissa (x-axis), $1/[S] = -1/K_m$.

For the reaction: $A + B \longrightarrow C + D$



When the enzyme forms a complex with both substrates

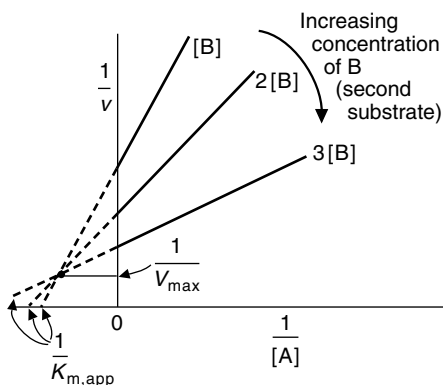


Fig. 9.17. A Lineweaver-Burk plot for a two-substrate reaction in which A and B are converted to products. In the graph, $1/[A]$ is plotted against $1/v$ for three different concentrations of the cosubstrate, $[B]$, $2[B]$, and $3[B]$. As the concentration of B is increased, the intersection on the abscissa, equal to $1/K_{m,app}$ is increased. The “app” represents “apparent”, as the $K_{m,app}$ is the K_m at whatever concentration of cosubstrate, inhibitor, or other factor is present during the experiment.

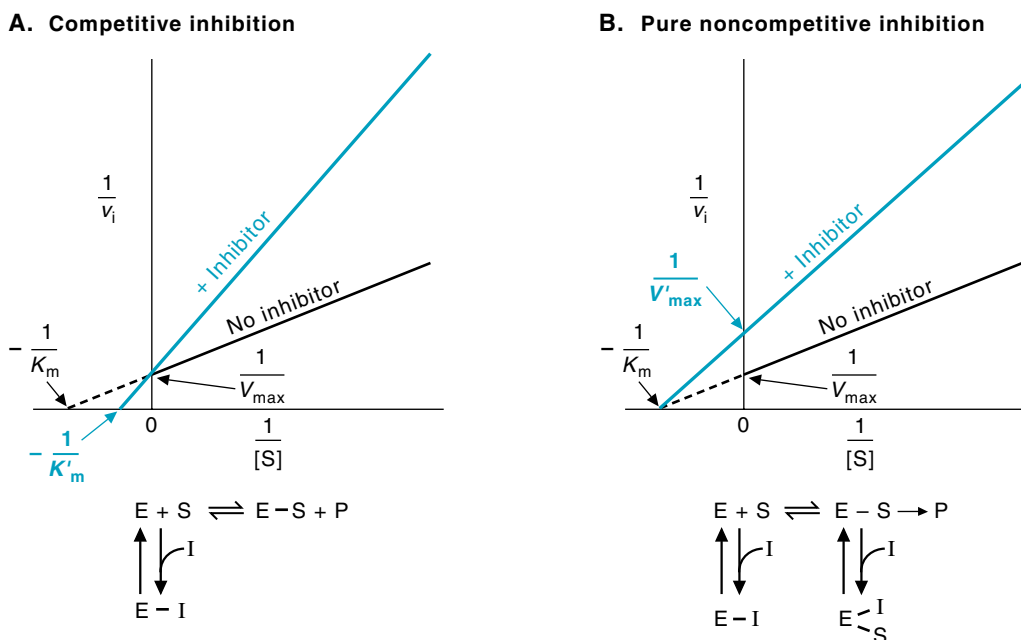


Fig. 9.18. Lineweaver-Burk plots of competitive and pure noncompetitive inhibition. A. $1/v_i$ versus $1/[S]$ in the presence of a competitive inhibitor. The competitive inhibitor alters the intersection on the abscissa. The new intersection is $1/K_{m,app}$ (also called $1/K'_m$). A competitive inhibitor does not affect V_{max} . B. $1/v_i$ versus $1/[S]$ in the presence of a pure noncompetitive inhibitor. The noncompetitive inhibitor alters the intersection on the ordinate, $1/V_{max,app}$ or $1/V'_{max}$, but does not affect $1/K_m$. A pure noncompetitive inhibitor binds to E and ES with the same affinity. If the inhibitor has different affinities for E and ES, the lines will intersect above or below the abscissa, and the noncompetitive inhibitor will change both the K'_m and the V'_m .



REVIEW QUESTIONS—CHAPTER 9

- Which of the following describes a characteristic feature of an enzyme obeying Michaelis-Menten kinetics?
 - The enzyme velocity is at $\frac{1}{2}$ the maximal rate when 100% of the enzyme molecules contain bound substrate.
 - The enzyme velocity is at $\frac{1}{2}$ the maximal rate when 50% of the enzyme molecules contain bound substrate.
 - The enzyme velocity is at its maximal rate when 50% of the enzyme molecules contain bound substrate.
 - The enzyme velocity is at its maximal rate when all of the substrate molecules in solution are bound by the enzyme.
 - The velocity of the reaction is independent of the concentration of enzyme.
- The pancreatic glucokinase of a patient with MODY had a mutation replacing a leucine with a proline. The result was that the K_m for glucose was decreased from a normal value of 6 mM to a value of 2.2 mM, and the V_{max} was changed from 93 units/mg protein to 0.2 units/mg protein. Which of the following best describes the patient's glucokinase compared with the normal enzyme?
 - The patient's enzyme requires a lower concentration of glucose to reach $\frac{1}{2} V_{max}$.
 - The patient's enzyme is faster than the normal enzyme at concentrations of glucose below 2.2 mM.
 - The patient's enzyme is faster than the normal enzyme at concentrations of glucose above 2.2 mM.
 - At near saturating glucose concentration, the patient would need 90 to 100 times more enzyme than normal to achieve normal rates of glucose phosphorylation.
 - As blood glucose levels increase after a meal from a fasting value of 5 mM to 10 mM, the rate of the patient's enzyme will increase more than the rate of the normal enzyme.

3. Methanol (CH_3OH) is converted by alcohol dehydrogenases to formaldehyde (CHO), a compound that is highly toxic in the human. Patients who have ingested toxic levels of methanol are sometimes treated with ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) to inhibit methanol oxidation by alcohol dehydrogenase. Which of the following statements provides the best rationale for this treatment?
- (A) Ethanol is a structural analog of methanol, and might therefore be an effective noncompetitive inhibitor.
 - (B) Ethanol is a structural analog of methanol that would be expected to compete with methanol for its binding site on the enzyme.
 - (C) Ethanol would be expected to alter the V_{max} of alcohol dehydrogenase for the oxidation of methanol to formaldehyde.
 - (D) Ethanol would be an effective inhibitor of methanol oxidation regardless of the concentration of methanol.
 - (E) Ethanol would be expected to inhibit the enzyme by binding to the formaldehyde binding site on the enzyme, even though it cannot bind at the substrate binding site for methanol.
4. Which of the following describes a characteristic of most allosteric enzymes?
- (A) They are composed of single subunits.
 - (B) In the absence of effectors, they generally follow Michaelis-Menten kinetics.
 - (C) They show cooperativity in substrate binding.
 - (D) They have allosteric activators that bind in the catalytic site.
 - (E) They have irreversible allosteric inhibitors that bind at allosteric sites.
5. A rate-limiting enzyme catalyzes the first step in the conversion of a toxic metabolite to a urinary excretion product. Which of the following mechanisms for regulating this enzyme would provide the most protection to the body?
- (A) The product of the pathway should be an allosteric inhibitor of the rate-limiting enzyme.
 - (B) The product of the pathway should act through gene transcription to decrease synthesis of the enzyme.
 - (C) The toxin should act through gene transcription to increase synthesis of the enzyme.
 - (D) The enzyme should have a high K_m for the toxin.
 - (E) The product of the first enzyme should allosterically activate the subsequent enzyme in the pathway.

10 Relationship between Cell Biology and Biochemistry

The basic unit of a living organism is the cell. In the human, each tissue is composed of similar cell types, which differ from those in other tissues. The diversity of cell types serves the function of the tissue and organs in which they reside, and each cell type has unique structural features that reflect its role. In spite of their diversity in structure, human cell types have certain architectural features in common, such as the plasma membrane, membranes around the nucleus and organelles, and a cytoskeleton (Fig. 10.1). In this chapter, we review some of the chemical characteristics of these common features, the functions of organelles, and the transport systems for compounds into cells and between organelles.

Plasma membrane. The cell membrane is a **lipid bilayer** that serves as a selective barrier; it restricts the entry and exit of compounds. Within the plasma membrane, different **integral proteins** facilitate the **transport** of specific compounds by



The cells of humans and other animals are eukaryotes (eu, good; karyon, nucleus) because the genetic material is organized into a membrane-enclosed nucleus. In contrast, bacteria are prokaryotes (pro, before; karyon, nucleus); they do not contain nuclei or other organelles found in eukaryotic cells.

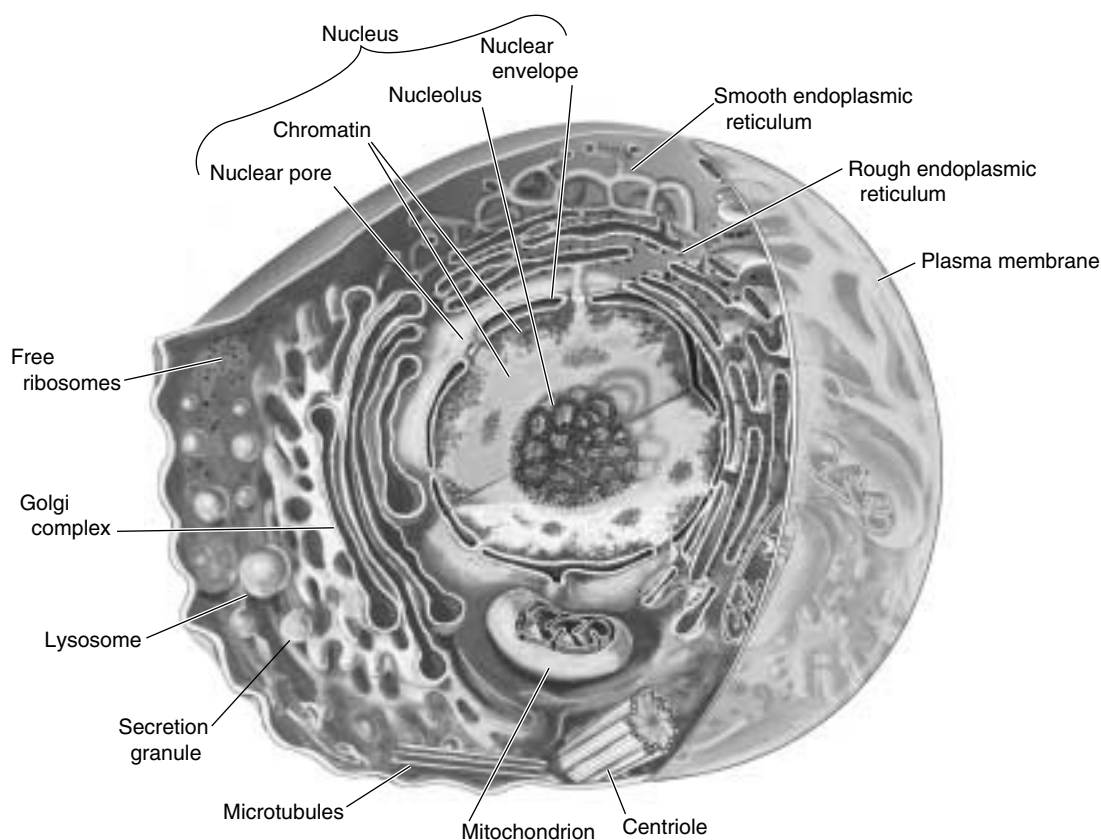


Fig. 10.1. Common components of human cells.



The cytoplasm of the cell is the portion of the cell between the cell membrane and the nucleus. Mitochondria, lysosomes and peroxisomes are referred to as cytoplasmic organelles. The Golgi and the endoplasmic reticulum are referred to as cytoplasmic membrane systems. The plasma membrane can be gently disrupted by detergents or shear stress without damage to the other membrane systems. When a suspension that has been treated this way is centrifuged for a long period of time (100,000*g* for 1 hour), the organelles and membrane systems will collect at the bottom of the tube. The remaining clear liquid of soluble enzymes, cofactors, and metabolites is the cytosol.

*energy-requiring active transport, facilitated diffusion, or by forming pores or gated-channels. The plasma membrane is supported by a **membrane skeleton** composed of proteins.*

Organelles and cytoplasmic membrane systems. Most organelles within the cell are compartments surrounded by a membrane system that restricts exchange of compounds and information with other compartments (see Fig. 10.1). In general, each organelle has unique functions that are served by the enzymes and other compounds it contains, or the environment it maintains. **Lysosomes** contain **hydrolytic enzymes** that degrade proteins and other large molecules. The **nucleus** contains the genetic material and carries out **gene replication** and **transcription of DNA**, the first step of protein synthesis. The last phase of protein synthesis occurs on **ribosomes**. For certain proteins, the ribosomes become attached to the complex membrane system called the **endoplasmic reticulum**; for other proteins, synthesis is completed on ribosomes that remain in the cytoplasm. The **endoplasmic reticulum** is also involved in lipid synthesis and transport of molecules to the Golgi. The **Golgi** forms vesicles for transport of molecules to the plasma membrane and other membrane systems, and for secretion. **Mitochondria** are organelles of **fuel oxidation** and **ATP generation**. **Peroxisomes** contain many enzymes that use or produce **hydrogen peroxide**. The **cytosol** is the intracellular compartment free of organelles and membrane systems.

Cytoskeleton. The **cytoskeleton** is a flexible fibrous protein support system that maintains the geometry of the cell, fixes the position of organelles, and moves compounds within the cell or the cell itself. It is composed principally of **actin microfilaments, intermediate filaments, tubulin microtubules, and their attached proteins**.



THE WAITING ROOM



Al Martini had been drinking heavily when he drove his car off the road and was taken to the hospital emergency room (see Chapters 8 and 9). Although he suffered only minor injuries, his driving license was suspended.



Two years after **Dennis “the Menace” Veere** successfully recovered from his malathion poisoning, he visited his grandfather, **Percy Veere**. Mr. Veere took Dennis with him to a picnic at the shore, where they ate steamed crabs. Later that night, Dennis experienced episodes of vomiting and watery diarrhea, and Mr. Veere rushed him to the hospital emergency room. Dennis’s hands and feet were cold, he appeared severely dehydrated, and he was approaching hypovolemic shock (a severe drop in blood pressure). He was diagnosed with cholera, caused by the bacteria *Vibrio cholerae*.



Before **Lotta Topaigne** was treated with allopurinol (see Chapter 8), her physician administered colchicine (acetyltrimethylcolchicinic acid) for the acute attack of gout affecting her great toe. After taking a high dose of colchicine divided over several-hour intervals, the throbbing pain in her toe had abated significantly. The redness and swelling also seemed to have lessened slightly.



V. cholerae epidemics are associated with unsanitary conditions affecting the drinking water supply and are rare in the United States. However, these bacteria grow well under the alkaline conditions found in seawater and attach to chitin in shellfish. Thus, sporadic cases occur in the southeast United States associated with the ingestion of contaminated shellfish.



Cell lysis, the breaking of the cell membrane and release of cell contents, occurs when the continuity of the cell membrane is disrupted.

I. COMPARTMENTATION IN CELLS

Membranes are lipid structures that separate the contents of the compartment they surround from its environment. An outer plasma membrane separates the cell from the

external aqueous environment. Organelles (such as the nucleus, mitochondria, lysosomes, and peroxisomes) are also surrounded by a membrane system that separates the internal compartment of the organelle from the cytosol. The function of these membranes is to collect or concentrate enzymes and other molecules serving a common function into a compartment with a localized environment. The transporters and receptors in each membrane system control this localized environment and communication of the cell or organelle with the surrounding milieu.

The following sections describe various organelles and membrane systems found in most human cells and outline the relationship between their properties and function. Each organelle has different enzymes and carries out different general functions. For example, the nucleus contains the enzymes for DNA and RNA synthesis.

Not all cells in the human are alike. Different cell types differ quantitatively in their organelle content, or their organelles may contain vastly different amounts of a particular enzyme, *consistent with the function of the cell*. For example, liver mitochondria contain a key enzyme for synthesizing ketone bodies, but they lack a key enzyme for their use. The reverse is true in muscle mitochondria. Thus, the enzymic content of the organelles varies somewhat from cell type to cell type.

II. PLASMA MEMBRANE

A. Structure of the Plasma Membrane

All mammalian cells are enclosed by a plasma membrane composed of a lipid bilayer (two layers) containing embedded proteins (Fig. 10.2). The membranes are continuous and sealed so that the hydrophobic lipid bilayer selectively restricts the exchange of polar compounds between the external fluid and the intracellular compartment. The membrane is referred to as a fluid mosaic because it consists of a mosaic of proteins and lipid molecules that can, for the most part, move laterally in the plane of the membrane. The proteins are classified as integral proteins, which span the cell membrane, or peripheral proteins, which are attached to the membrane surface through electrostatic bonds to lipids or integral proteins. Many of the proteins and lipids on the external leaflet contain covalently bound carbohydrate chains and therefore are glycoproteins and glycolipids. This layer of carbohydrate on the outer surface of the cell is called the glycocalyx.

1. LIPIDS IN THE PLASMA MEMBRANE

Each layer of the plasma membrane lipid bilayer is formed primarily by phospholipids, which are arranged with their hydrophilic head groups facing the aqueous medium and their fatty acyl tails forming a hydrophobic membrane core (see Fig. 10.2). The principle phospholipids in the membrane are the glycerol lipids phosphatidylcholine, phosphatidylethanolamine, and phosphatidylserine and the sphingolipid sphingomyelin (Fig. 10.3). The lipid composition varies among different cell types, with phosphatidylcholine being the major plasma membrane lipid in most cell types and sphingolipids the most variable.

The lipid composition of the bilayer is asymmetric, with a higher content of phosphatidylcholine and sphingomyelin in the outer leaflet and a higher content of phosphatidylserine and phosphatidylethanolamine in the inner leaflet. Phosphatidylserine contains a net negative charge that contributes to the membrane potential and might be important for binding positively charged molecules within the cell. Phosphatidylinositol, which is found only in the inner membrane, functions in the transfer of information from hormones and neurotransmitters across the cell membrane into the cell (Fig. 10.4).



Bacteria are single cells surrounded by a cell membrane and a cell wall exterior to the membrane.

They are prokaryotes, which do not contain nuclei or other organelles (i.e. membrane-surrounded subcellular structures) found in eukaryotic cells. Nonetheless, bacteria carry out many similar metabolic pathways, with the enzymes located in either the intracellular compartment or the cell membrane.

The *Vibrio cholerae* responsible for **Dennis Veere's** cholera are gram-negative bacteria. Their plasma membrane is surrounded by a thin cell wall composed of a protein-polysaccharide structure called peptidoglycan and an outer membrane. In contrast, gram-positive bacteria have a plasma membrane and a thick peptidoglycan cell wall that retains the Gram stain. *Vibrio* grow best under aerobic conditions, but also can grow under low oxygen conditions. They possess enzymes similar to those in human cells for glycolysis, the TCA cycle, and oxidative phosphorylation. They have a low tolerance for acid, which partially accounts for their presence in slightly basic seawater and shellfish.



The variable carbohydrate components of the glycolipids on the cell surface function as cell recognition markers. For example, the A, B, or O blood groups are determined by the carbohydrate composition of the glycolipids. Cell surface glycolipids may also serve as binding sites for viruses and bacterial toxins before penetrating the cell. For example, the cholera AB toxin binds to GM₁-gangliosides on the surface of the intestinal epithelial cells. The toxin is then endocytosed in caveolae (invaginations or "caves" that can form in specific regions of the membrane).



One of the bacterial toxins secreted by *Clostridium perfringens*, the bacteria that cause gas gangrene, is a lipase that hydrolyzes phosphocholine from phosphatidylcholine and from sphingomyelin. The resulting lysis of the cell membrane releases intracellular contents that provide the bacteria with nutrients for rapid growth. These bacteria are strict anaerobes and grow only in the absence of oxygen. As their toxins lyse membranes in the endothelial cells of blood vessels, the capillaries are destroyed, and the bacteria are protected from oxygen transported by the red blood cells. They are also protected from antibiotics and components of the immune system carried in the blood.

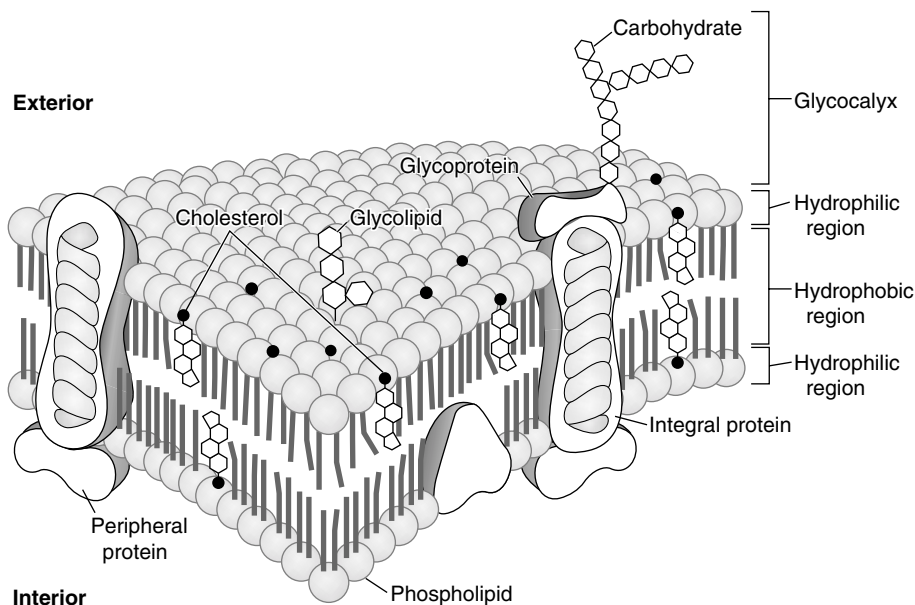


Fig. 10.2. Basic structure of an animal cell membrane.

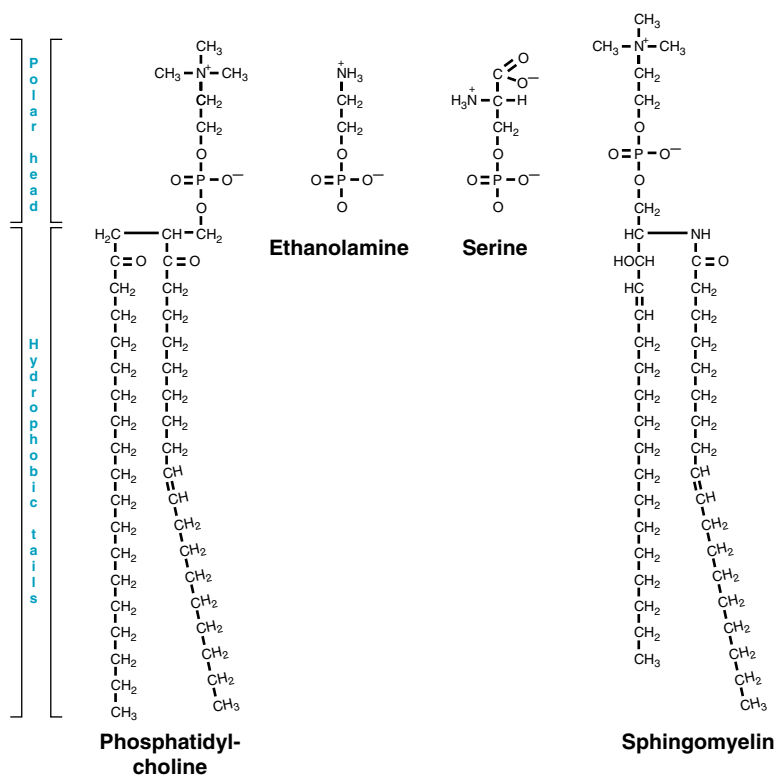


Fig. 10.3. Common phospholipids in the mammalian cell membrane. The polar head groups shown for ethanolamine and serine replace the choline in phosphatidylcholine to form phosphatidylethanolamine and phosphatidylserine, respectively. Phosphatidylcholine, phosphatidylethanolamine, and phosphatidylserine are phosphoacylglycerols. In contrast, sphingomyelin does not contain the glycerol backbone but has a sphingosine backbone and is a sphingolipid.

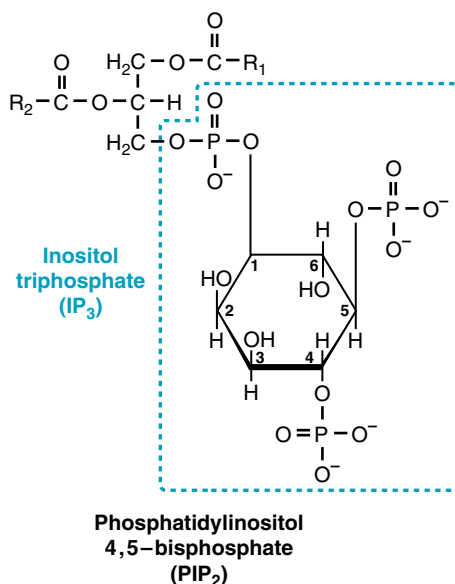


Fig. 10.4. Phosphatidylinositol bisphosphate (PIP₂). R1 and R2 are fatty acyl chains. The portion of PIP₂ that becomes inositol triphosphate, the polar head group extending into the cytosol, is shown in blue.

Cholesterol, which is interspersed between the phospholipids, maintains membrane fluidity. In the phosphoacylglycerols, unsaturated fatty acid chains bent into the *cis* conformation form a pocket for cholesterol, which binds with its hydroxyl group in the external hydrophilic region of the membrane and its hydrophobic steroid nucleus in the hydrophobic membrane core (Fig. 10.5). The presence of cholesterol and the *cis* unsaturated fatty acids in the membrane prevent the hydrophobic chains from packing too closely together. As a consequence, lipid and protein molecules that are not bound to external or internal structural proteins can rotate and move laterally in the plane of the leaflet. This movement enables the plasma membrane to partition between daughter cells during cell division, to



Al Martini is suffering from both short-term and long-term effects of ethanol on his central nervous system. Data support the theory that the short-term effects of ethanol on the brain partially arise from an increase in membrane fluidity caused when ethanol intercalates between the membrane lipids. The changes in membrane fluidity may affect proteins that span the membrane (integral proteins), such as ion channels and receptors for neurotransmitters involved in conducting the nerve impulse.

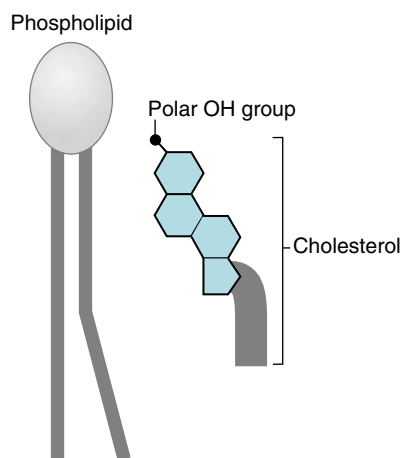


Fig. 10.5. Cholesterol in the plasma membrane. The polar hydroxyl group of cholesterol is oriented toward the surface. The hydrocarbon tail and the steroid nucleus (blue) lie in the hydrophobic core. A *cis* double bond in the fatty acyl chain of a phospholipid bends the chain to create a hydrophobic binding site for cholesterol.

deform as cells pass through capillaries, and to form and fuse with vesicle membranes. The fluidity of the membrane is partially determined by the unsaturated fatty acid content of the diet.

The composition of the membrane is dynamic. Sections of membrane form buds that pinch off into vesicles and membrane vesicles formed in the Golgi and elsewhere bring new and recycled components back to the membrane. Individual fatty acyl chains turn over as they are hydrolyzed from the lipids and replaced, and enzymes called flippases transfer lipids between leaflets.

2. PROTEINS IN THE PLASMA MEMBRANE

The integral proteins contain transmembrane domains with hydrophobic amino acid side chains that interact with the hydrophobic portions of the lipids to seal the membrane (see Fig. 10.2). Hydrophilic regions of the proteins protrude into the aqueous medium on both sides of the membrane. Many of these proteins function as either channels or transporters for the movement of compounds across the membrane, as receptors for the binding of hormones and neurotransmitters, or as structural proteins (Fig. 10.6).

Peripheral membrane proteins, which were originally defined as those proteins that can be released from the membrane by ionic solvents, are bound through weak electrostatic interactions with the polar head groups of lipids or with integral proteins. One of the best-characterized classes of peripheral proteins is the spectrin family of proteins, which are bound to the intracellular membrane surface and provide mechanical support for the membrane. Spectrin is bound to actin, which together form a structure that is called the inner membrane skeleton or the cortical skeleton (see Fig. 10.6).

A third classification of membrane proteins consists of lipid-anchored proteins bound to the inner or outer surface of the membrane. The glycosylphosphatidylinositol (GPI) anchor is a covalently attached lipid that anchors proteins to the



Two of the prominent integral proteins in the red blood cell membrane are glycophorin, which provides an external negative charge that repels other cells, and band 3, which is a channel for bicarbonate and chloride exchange. The transport of bicarbonate into the red blood cell in exchange for chloride helps to carry the bicarbonate to the lungs, where it is expired as CO_2 .



All cells contain an inner membrane skeleton of spectrin-like proteins. Red blood cell spectrin was the first member of the spectrin family described. The protein dystrophin present in skeletal muscle cells is a member of the spectrin family. Genetic defects in the dystrophin gene are responsible for Duchenne's and Becker's muscular dystrophies.

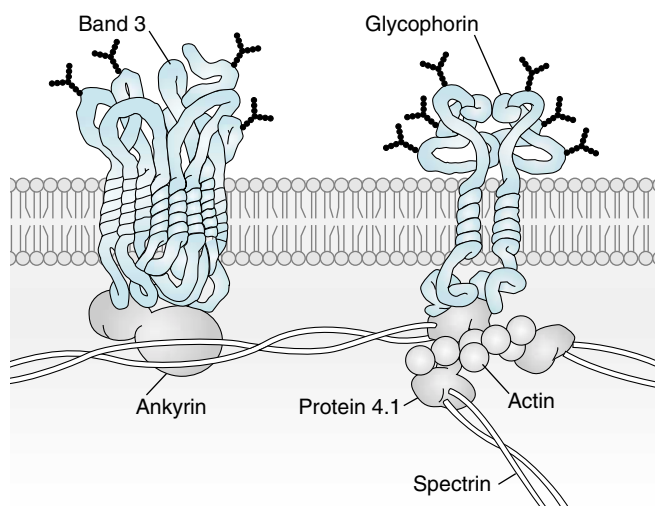


Fig. 10.6. Proteins in the red blood cell membrane. The proteins named Band 3 (the bicarbonate-chloride exchange transporter) and glycophorin contain nonpolar α -helical segments spanning the lipid bilayer. These proteins contain a large number of polar and charged hydrophilic amino acids in the intracellular and extracellular domains. On the inside of the cell, they are attached to peripheral proteins constituting the inner membrane skeleton. Band 3 is connected to spectrin filaments via the protein ankyrin. Glycophorin is connected to short actin filaments and spectrin via protein 4.1.

external surface of the membrane (Fig.10.7). A number of proteins involved in hormonal regulation are anchored to the internal surface of the membrane through palmityl (C16) or myristyl (C14) fatty acyl groups or through geranylgeranyl (C20) or farnesyl (C15) isoprenyl groups (see Ras, Chapter 9, Fig. 9.14, or Chapter 6, Fig. 6.14). However, many integral proteins also contain attached lipid groups to increase their stability in the membrane.

3. THE GLYCOCALYX OF THE PLASMA MEMBRANE

Some of the proteins and lipids on the external surface of the membrane contain short chains of carbohydrates (oligosaccharides) that extend into the aqueous medium. Carbohydrates therefore constitute 2 to 10% of the weight of plasma membranes. This hydrophilic carbohydrate layer, called the glycocalyx, protects the cell against digestion and restricts the uptake of hydrophobic compounds.

The glycoproteins generally contain branched oligosaccharide chains of approximately 15 sugar residues that are attached through N-glycosidic bonds to the amide nitrogen of an asparagine side chain (N-glycosidic linkage), or through a glycosidic bond to the oxygen of serine (O-glycoproteins). The membrane glycolipids are usually galactosides or cerebroside. Specific carbohydrate chains on the glycolipids serve as cell recognition molecules (see Chapter 5 for structures of classes of compounds).

B. Transport of Molecules across the Plasma Membrane

Membranes form hydrophobic barriers around cells to control the internal environment by restricting the entry and exit of molecules. As a consequence, cells require transport systems to permit entry of small polar compounds that they need (e.g., glucose) to concentrate compounds inside the cell (e.g., K^+) and to expel other



The prion protein, present in neuronal membranes, provides an example of a protein attached to the membrane through a GPI anchor. This is the protein that develops an altered pathogenic conformation in both mad cow disease and Creutzfeldt-Jakob disease (see Chapter 7, Biochemical Comments).

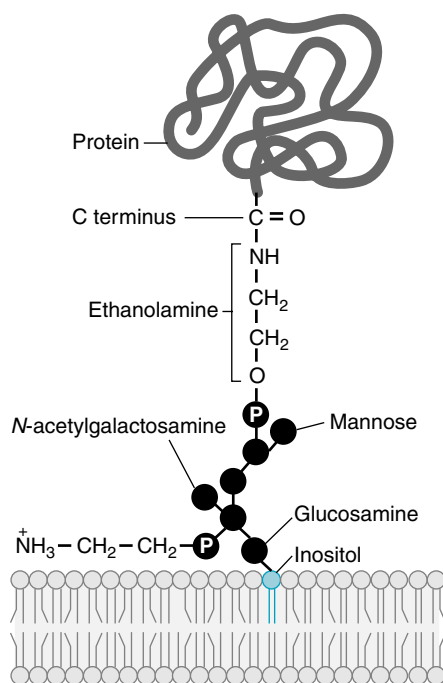


Fig. 10.7. The glycosylphosphatidylinositol glycan anchor (GPI). The carboxy terminus of the protein is attached to phosphoethanolamine, which is bound to a branched oligosaccharide that is attached to the inositol portion of phosphatidylinositol. The hydrophobic fatty acyl chains of the phosphatidylinositol portion are bound in the hydrophobic core of the membrane.



Dennis Veere has become dehydrated because he has lost so much water through vomiting and diarrhea (see Chapter 4). Cholera toxin increases the efflux of sodium and chloride ions from his intestinal mucosal cells into the intestinal lumen. The increase of water in his stools results from the passive transfer of water from inside the cell and body fluids, where it is in high concentration (i.e., intracellular Na^+ and Cl^- concentrations are low), to the intestinal lumen and bowel, where water is in lower concentration (relative to high Na^+ and Cl^-). The watery diarrhea is also high in K^+ ions and bicarbonate. All of the signs and symptoms of cholera generally derive from this fluid loss.

compounds (e.g., Ca^{2+} and Na^+). The transport systems for small organic molecules and inorganic ions fall into four categories: simple diffusion through the lipid bilayer or through a large pore; facilitative diffusion; gated channels; and active transport pumps (Fig. 10.8). These transport mechanisms are classified as passive if energy is not required, or active if energy is required. The energy is often provided by the hydrolysis of ATP.

In addition to these mechanisms for the transport of small individual molecules, cells engage in endocytosis. The plasma membrane extends or invaginates to surround a particle, a foreign cell, or extracellular fluid, which then closes into a vesicle that is released into the cytoplasm (see Fig. 10.8).

1. SIMPLE DIFFUSION

Gases such as O_2 and CO_2 and lipid-soluble substances (such as steroid hormones) can cross membranes by simple diffusion (see Fig. 10.8). In simple diffusion (free diffusion), molecules move by engaging in random collisions with other like molecules. There is a net movement from a region of high concentration to a region of low concentration because molecules keep bumping into each other where their concentration is highest. Energy is not required for diffusion, and compounds that are uncharged eventually reach the same concentrations on both sides of the membrane.

Water is considered to diffuse through membranes by unspecific movement through ion channels, pores, or around proteins embedded in the lipids. Certain cells (e.g., renal tubule cells) also contain large protein pores, called aquaporins, which permit a high rate of water flow from a region of a high water concentration (low solute concentration) to one of low water concentration (high solute concentration).

2. FACILITATIVE DIFFUSION THROUGH BINDING TO TRANSPORTER PROTEINS

Facilitative diffusion requires that the transported molecule bind to a specific carrier or transport protein in the membrane (Fig. 10.9). The transporter protein

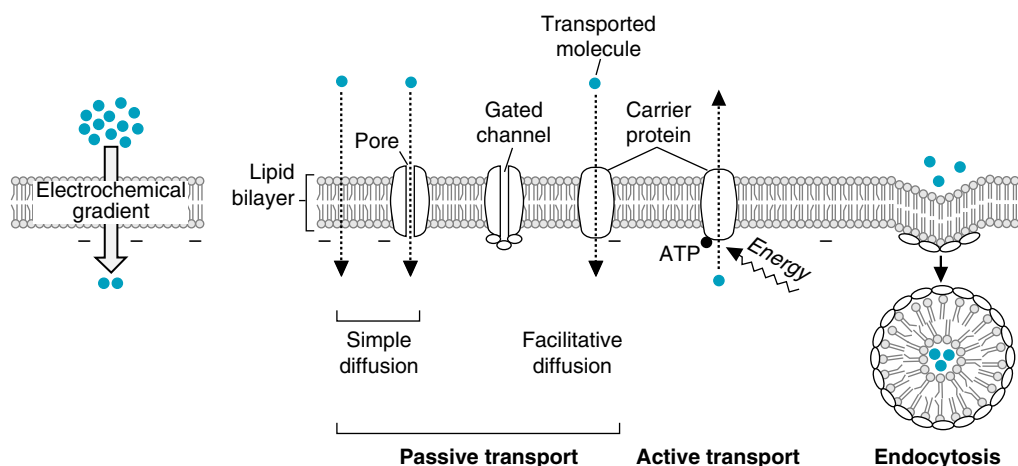


Fig. 10.8. Common types of transport mechanisms for human cells. The electrochemical gradient consists of the concentration gradient of the compound and the distribution of charge on the membrane, which affects the transport of charged ions such as Cl^- . Both protein amino acid residues and lipid polar head groups contribute to the net negative charge on the inside of the membrane. Generally, the diffusion of uncharged molecules (passive transport) is net movement from a region of high concentration to a low concentration, and active transport (energy-requiring) is net movement from a region of low concentration to one of high concentration.

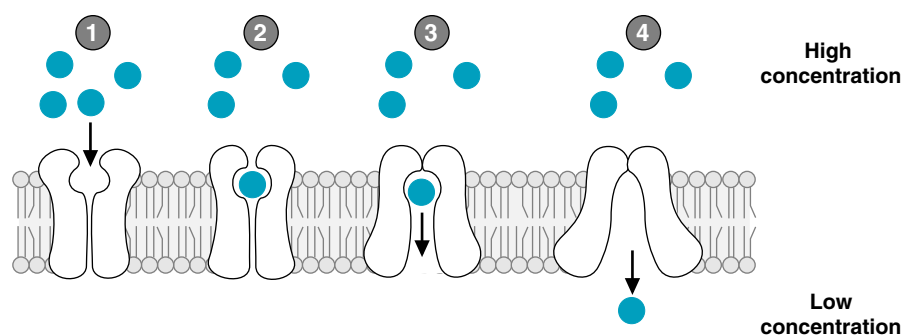


Fig. 10.9. Facilitative transport. Although the molecule being transported must bind to the protein transporter, the mechanism is passive diffusion, and the molecule moves from a region of high concentration to one of low concentration. “Passive” refers to the lack of an energy requirement for the transport.



All of the cells in the body have facilitative glucose transporters that transport glucose across the plasma membrane down an electrochemical (concentration) gradient as it is rapidly metabolized in the cell. In muscle and adipose tissue, insulin increases the content of facilitative glucose transporters in the cell membrane, thus increasing the ability of these tissues to take up glucose. Patients with type 1 diabetes mellitus, who do not produce insulin (e.g., **Di Abietes**, see Chapter 7), have a decreased ability to transport glucose into these tissues, thereby contributing to hyperglycemia (high blood glucose).

then undergoes a conformational change that allows the transported molecule to be released on the other side of the membrane. Although the transported molecules are bound to proteins, the transport process is still classified as diffusion because energy is not required, and the compound equilibrates (achieves a balance of concentration and charge) on both sides of the membrane.

Transporter proteins, like enzymes, exhibit saturation kinetics; when all the binding sites on all of the transporter proteins in the membrane are occupied, the system is saturated and the rate of transport reaches a plateau (the maximum velocity). By analogy to enzymes, the concentration of a transported compound required to reach $\frac{1}{2}$ the maximum velocity is often called the K_m (Fig. 10.10). Facilitative transporters are similar to enzymes with respect to two additional features: they are relatively specific for the compounds they bind and they can be inhibited by compounds that block their binding sites or change their conformation.

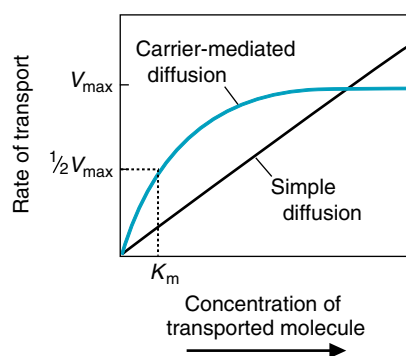


Fig. 10.10. Saturation kinetics of transporter proteins. When a compound must bind to a protein to be transported across a membrane, the velocity of transport depends on the amount of compound bound. It reaches a maximum rate when the compound’s concentration is raised so high that all of the transporter binding sites are occupied. The curve is a rectangular hyperbola that approaches V_{max} at infinite substrate concentration, identical to that of Michaelis-Menten enzymes. The K_m of transport is the concentration of compound required for $\frac{1}{2} V_{max}$. In contrast, simple diffusion of a compound does not require its binding to a protein, and the rate of transport increases linearly with increasing concentration of the compound.



The cystic fibrosis transmembrane conductance regulator (CFTR) was named for its role in cystic fibrosis.

A mutation in the gene encoding its transmembrane subunits results in dried mucus accumulation in the airways and pancreatic ducts.

The CFTR is also involved in the dehydration experienced by cholera patients such as **Dennis Veere**. In intestinal mucosal cells, cholera A toxin indirectly activates phosphorylation of the regulatory domain of CFTR by protein kinase A. Thus, the channel stays open and Cl^- and H_2O flow from the cell into the intestinal lumen, resulting in dehydration.



Protein-mediated transport systems, whether facilitative or active, are classified as antiports if they specifically exchange compounds of similar charge across a membrane; they are called symports or cotransporters if they simultaneously transport two molecules across the membrane in the same direction. Band 3 in the red blood cell membrane, which exchanges chloride ion for bicarbonate, provides an example of an antiport.

3. GATED CHANNELS IN PLASMA MEMBRANES

In gated channels, transmembrane proteins form a pore for ions that is either opened or closed in response to a stimulus: voltage changes across the membrane (voltage-gated channels), the binding of a compound (ligand-gated channels), or a regulatory change in the intracellular domain (phosphorylation-gated and pressure-gated channels). For example, the conduction of a nerve impulse along the axon depends on the passive flux of Na^+ ions through a voltage-gated channel that is opened by depolarization of the membrane. CFTR (cystic fibrosis transmembrane conductance regulator) is a Cl^- channel that provides an example of a ligand-gated channel regulated through phosphorylation (phosphorylation-gated) (Fig. 10.11). CFTR is a member of the ABC (adenine nucleotide binding cassette, or ATP binding cassette) superfamily of transport proteins. It has two transmembrane domains that form a closed channel, each connected to an ATP binding site, and a regulatory domain that sits in front of the channel. When the regulatory domain is phosphorylated by a kinase, its conformation changes and it moves away from the ATP binding domains. As ATP binds and is hydrolyzed, the transmembrane domains change conformation and open the channel, and chloride ions diffuse through. As the conformation reverts back to its original form, the channel closes.

Transport through a ligand-gated channel is considered diffusion, although ATP is involved, because only a few ATP molecules are being used to open and close the channel through which many, many chloride ions diffuse. However, the distinction between ligand-gated channels and facilitative transporters is not always as clear. Many gated channels show saturation kinetics at very high concentrations of the compounds being transported.

4. ACTIVE TRANSPORT REQUIRES ENERGY AND TRANSPORTER PROTEINS

Both active transport and facilitative transport are mediated by protein transporters (carriers) in the membrane. However, in facilitative transport, the compound is transported down an electrochemical gradient (the balance of concentration and charge across a membrane), usually from a high concentration to a low concentration, to equilibrate between the two sides of the membrane. In active transport, energy is used to concentrate the compound on one side of the membrane. If energy is directly applied to the transporter (e.g., ATP hydrolysis by Na^+, K^+ -ATPase), the transport is called primary active transport; if energy is used to establish an ion gradient (e.g., the Na^+ gradient), and the gradient is used to concentrate another compound, the transport is called secondary active transport.

The Na^+, K^+ -ATPase spans the plasma membrane, much like a gated pore, with a binding site for 3 Na^+ ions open to the intracellular side (Fig. 10.12). Energy from

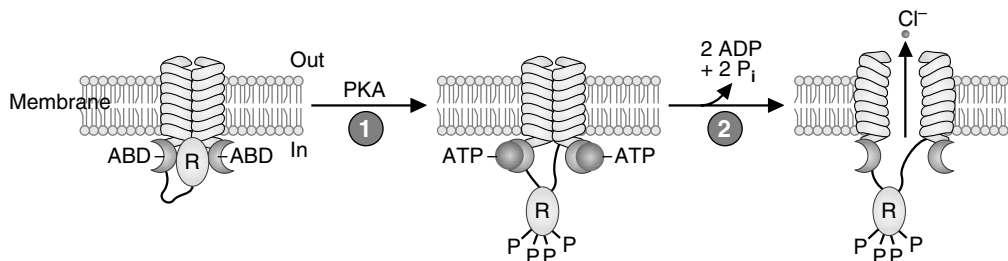


Fig. 10.11. CFTR, a ligand-gated channel controlled by phosphorylation. Two intracellular binding domains control opening of the channel, an adenine nucleotide binding domain (ABD) and a regulatory domain (R). **1** Phosphorylation of the regulatory subunit by protein kinase A causes a conformational change that allows ATP to bind to the adenine nucleotide binding domain (ABD). **2** Hydrolysis of bound ATP opens the channel so that chloride ions can diffuse through.

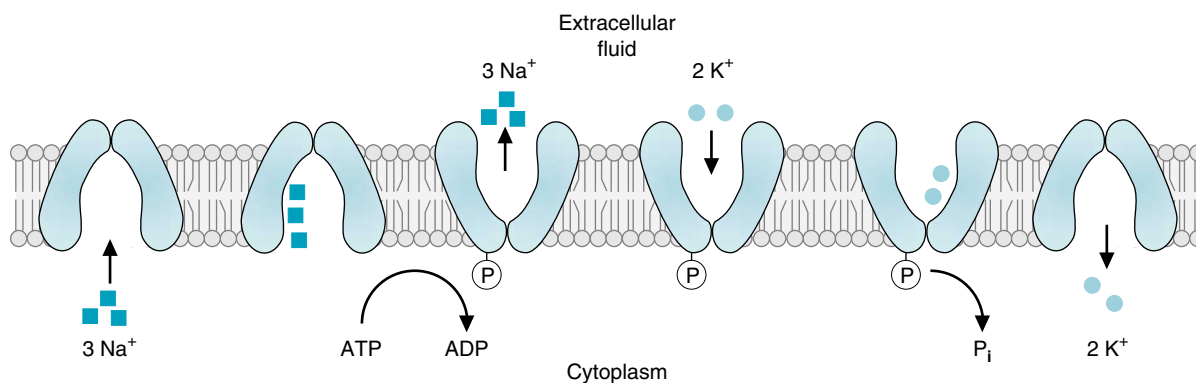


Fig. 10.12. Active transport by Na⁺,K⁺-ATPase. Three sodium ions bind to the transporter protein on the cytoplasmic side of the membrane. When ATP is hydrolyzed to ADP, the carrier protein is phosphorylated and undergoes a change in conformation that causes the sodium ions to be released into the extracellular fluid. Two potassium ions then bind on the extracellular side. Dephosphorylation of the carrier protein produces another conformational change, and the potassium ions are released on the inside of the cell membrane. The transporter protein then resumes its original conformation, ready to bind more sodium ions.

ATP hydrolysis is used to phosphorylate an internal domain and change the transporters' conformation so that bound Na⁺ ions are released to the outside, and two external K⁺ ions bind. K⁺ binding triggers hydrolysis of the bound phosphate group and a return to the original conformation, accompanied by release of K⁺ ions inside the cell. As a consequence, cells are able to maintain a much lower intracellular Na⁺ concentration and much higher intracellular K⁺ ion concentration than present in the external fluid.

The Na⁺ gradient, which is maintained by primary active transport, is used to power the transport of glucose, amino acids, and many other compounds into the cell through secondary active transport. An example is provided by the transport of glucose into cells of the intestinal epithelium in conjunction with Na⁺ ions (Fig. 10.13).

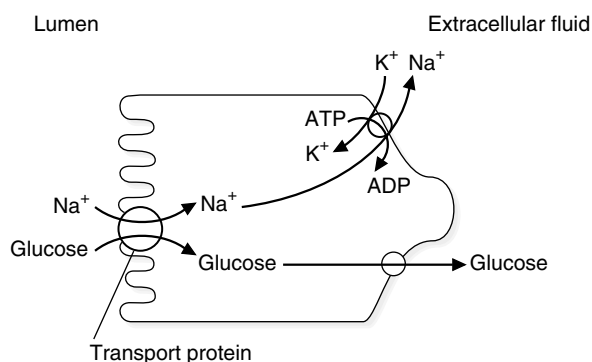


Fig. 10.13. Secondary active transport of glucose by the Na⁺-glucose cotransporter. One sodium ion binds to the carrier protein in the luminal membrane, stimulating the binding of glucose. After a conformational change, the protein releases Na⁺ and glucose into the cell and returns to its original conformation. Na⁺,K⁺-ATPase in the basolateral membrane pumps Na⁺ against its concentration gradient into the extracellular fluid. Thus, the Na⁺ concentration in the cell is low, and Na⁺ moves from the lumen down its concentration gradient into the cell and is pumped against its gradient into the extracellular fluid. Glucose, consequently, moves against its concentration gradient from the lumen into the cell by traveling on the same carrier as Na⁺. Glucose then passes down its concentration gradient into the extracellular fluid on a passive transporter protein.



The Ca²⁺-ATPase, a calcium pump, uses a mechanism similar to that of Na⁺,K⁺-ATPase to maintain intracellular Ca²⁺ concentration below 10⁻⁷ M in spite of the high extracellular concentration of 10⁻³ M. This transporter is inhibited by binding of the regulatory protein calmodulin. When the intracellular Ca²⁺ concentration increases, Ca²⁺ binds to calmodulin, which dissociates from the transporter, thereby activating it to pump Ca²⁺ out of the cell (see Chapter 9 for the structure of calmodulin). High levels of intracellular Ca²⁺ are associated with irreversible progression from cell injury to cell death.



The dehydration of cholera is often treated with an oral rehydration solution containing Na^+ , K^+ , and glucose or a diet of rice (which contains glucose and amino acids). Glucose is absorbed from the intestinal lumen via the Na^+ -dependent glucose cotransporters, which cotransport Na^+ into the cells together with glucose. Many amino acids are also absorbed by Na^+ -dependent cotransport. With the return of Na^+ to the cytoplasm, water efflux from the cell into the intestinal lumen decreases.



The vitamin folate provides an example of a compound transported into cells by caveolae, which form around the occupied folate receptor. In contrast, endocytosis of many compounds such as membrane hormone receptors occurs through clathrin-coated pits. The receptors are targeted for these pits by adaptor proteins that bind to a specific amino acid sequence in the receptor.

These cells create a gradient in Na^+ and then use this gradient to drive the transport of glucose from the intestinal lumen into the cell against its concentration gradient.

D. Vesicular Transport across the Plasma Membrane

Vesicular transport occurs when a membrane completely surrounds a compound, particle, or cell and encloses it into a vesicle. When the vesicle fuses with another membrane system, the entrapped compounds are released. Endocytosis refers to vesicular transport into the cell, and exocytosis to transport out of the cell. Endocytosis is further classified as phagocytosis if the vesicle forms around particulate matter (such as whole bacterial cells or metals and dyes from a tattoo), and pinocytosis if the vesicle forms around fluid containing dispersed molecules. Receptor-mediated endocytosis is the name given to the formation of clathrin-coated vesicles that mediate the internalization of membrane-bound receptors in vesicles coated on the intracellular side with subunits of the protein clathrin (Fig. 10.14). Potocytosis is the name given to endocytosis that occurs via caveolae (small invaginations or “caves”), which are regions of the cell membrane with a unique lipid and protein composition (including the protein caveolin-1).

III. LYSOSOMES

Lysosomes are the intracellular organelles of digestion enclosed by a single membrane that prevents the release of its digestive enzymes into the cytosol. They are central to a wide variety of body functions that involve elimination of unwanted material and recycling their components, including destruction of

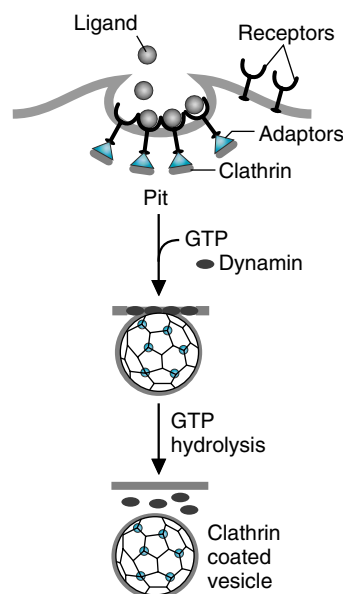


Fig. 10.14. Formation of a clathrin-coated vesicle. Ligands entering the cell through receptor-mediated endocytosis bind to receptors that cluster in an area of the membrane. Adaptor proteins bind to the receptor tails and to the clathrin molecules to enclose the budding membrane in a cage-like clathrin coat.

Molecules of a monomeric G protein called dynamin (from the Rab family) constrict the neck of the vesicle and pinch it off from the membrane as GTP is hydrolyzed.

infectious bacteria and yeast, recovery from injury, tissue remodeling, involution of tissues during development, and normal turnover of cells and organelles.

A. Lysosomal Hydrolases

The lysosomal digestive enzymes include nucleases, phosphatases, glycosidases, esterases, and proteases called cathepsins (Fig. 10.15). These enzymes are all hydrolases, enzymes that cleave amide, ester, and other bonds through the addition of water. Many of the products of lysosomal digestion, such as the amino acids, return to the cytosol. Lysosomes are therefore involved in recycling compounds.

Most of these lysosomal hydrolases have their highest activity near a pH of approximately 5.5 (the pH optimum). The intralysosomal pH is maintained near 5.5 principally by v -ATPases (vesicular ATPases), which actively pump protons into the lysosome. The cytosol and other cellular compartments have a pH nearer 7.2 and are therefore protected from escaped lysosomal hydrolases.

B. Endocytosis, Phagocytosis, and Autophagy

Lysosomes are formed from digestive vesicles called endosomes, which are involved in receptor-mediated endocytosis. They also participate in digestion of foreign cells acquired through phagocytosis and the digestion of internal contents in the process of autophagocytosis.

1. RECEPTOR-MEDIATED ENDOCYTOSIS

Lysosomes are involved in the digestion of compounds brought into the cells in endocytotic clathrin-coated vesicles formed by the plasma membrane (Fig. 10.16). These vesicles fuse to form multivesicular bodies called early endosomes. The early endosomes mature into late endosomes as they recycle clathrin, lipids, and other



Lysosomal storage diseases. Genetic defects in lysosomal enzymes, or in proteins such as the mannose 6-phosphate receptors required for targeting the enzymes to the lysosome, lead to an abnormal accumulation of undigested material in lysosomes that may be converted to residual bodies. The accumulation may be so extensive that normal cellular function is compromised, particularly in neuronal cells. Genetic diseases such as the Tay-Sachs disease (an accumulation of partially digested gangliosides in lysosomes), and Pompe's disease (an accumulation of glycogen particles in lysosomes) are caused by the absence or deficiency of specific lysosomal enzymes.

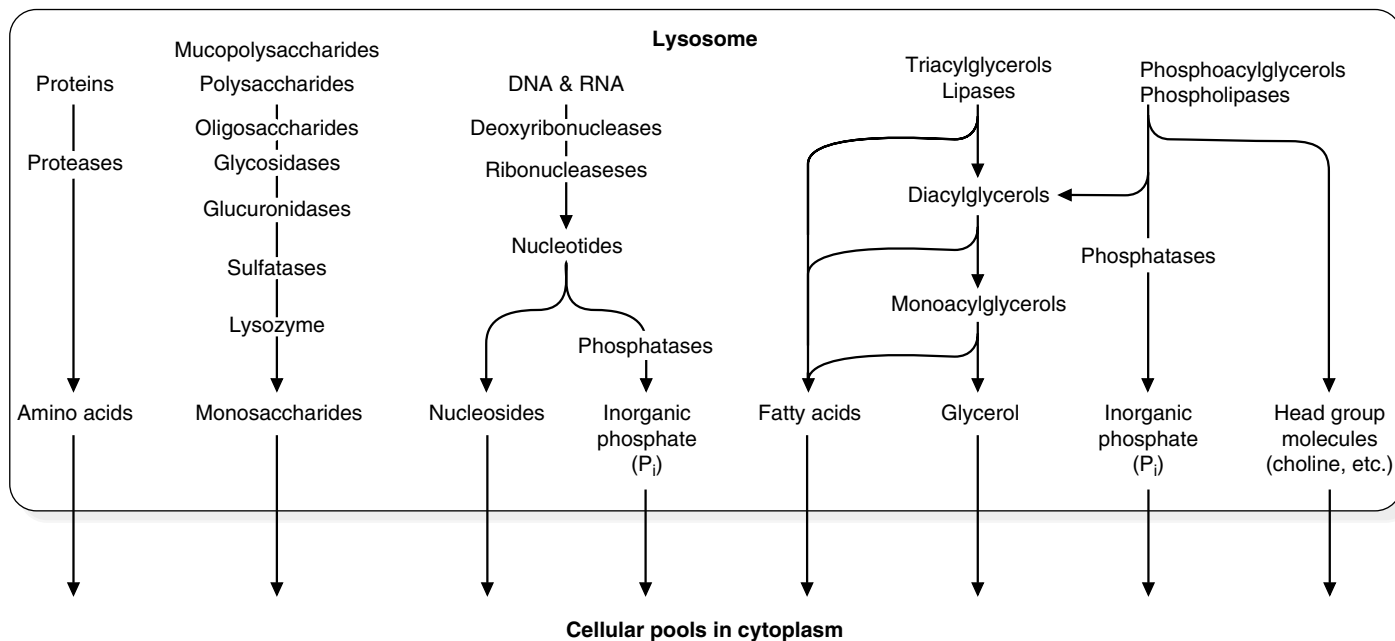


Fig. 10.15. Lysosomal reactions. Most lysosomal enzymes are hydrolases, which cleave peptide, ester, and glycosidic bonds by adding the components of water across the bond. These enzymes are active at the acidic pH of the lysosome and inactive if accidentally released into the cytosol.

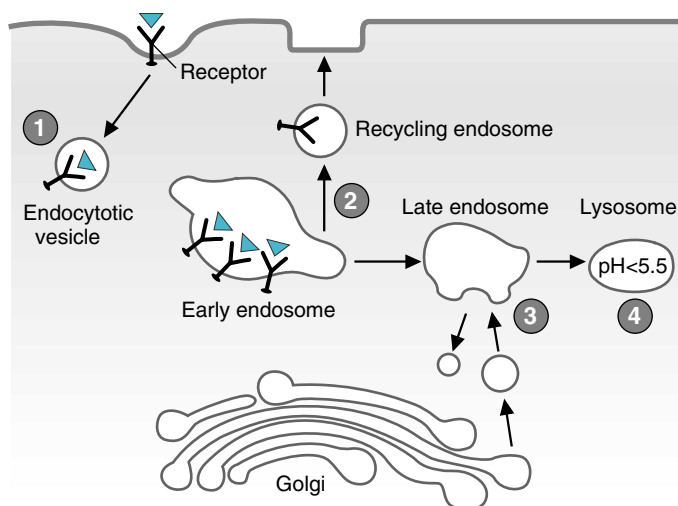


Fig. 10.16. Lysosomes in receptor-mediated endocytosis via clathrin-coated pits. **1** Endocytotic vesicles fuse to form early endosomes. **2** Vesicle contents are sorted, and receptors, clathrin, and lipids are sent back to the plasma membrane. **3** Transport vesicles from the trans-Golgi carry lysosomal hydrolases to the late endosome. **4** Lysosomes containing concentrated hydrolases digest proteins and other components acquired from endocytotic vesicles.



The elevated level of uric acid in **Lotta Topaigne's** blood led to the deposition of monosodium urate crystals in the joint space (synovial fluid) of her right great toe, resulting in podagra (painful great toe). Neutrophils, the mediators of the acute inflammation that followed, attempted to phagocytose the urate crystals. The engulfed urate crystals were deposited in the late endosomes and lysosomes of the neutrophil. Because urate crystals are particles that cannot be degraded by any of the lysosomal acid hydrolases, their accumulation caused lysis of the lysosomal membranes, followed by cell lysis and release of lysosomal enzymes into the joint space. The urate crystals also resulted in release of chemical mediators of inflammation that recruited other cells into the area. This further amplified the acute inflammatory reaction in the tissues of the joint capsule (synovitis), leading to the extremely painful swelling of acute gouty arthritis.



Phagocytosis and autophagy are part of the normal turnover of body components, such as degradation of cells that have a shorter lifespan than the whole organism and remodeling of tissues during pregnancy. For example, phagocytes, located mainly in the spleen and liver, remove approximately 3×10^{11} red blood cells from the circulation each day. During pregnancy, breast tissue is remodeled to develop the capacity for lactation; after weaning of an infant, the lactating breast returns to its original state (involution).

membrane components back to the plasma membrane in vesicles called recycling endosomes. The late endosomes mature into lysosomes as they progressively accumulate newly synthesized acid hydrolases and vesicular proton pumps brought to them in clathrin-coated vesicles from the Golgi. Thus, lysosomes do not acquire their full digestive power until after sorting of membrane lipids and proteins for recycling.

Within the Golgi, enzymes are targeted for endosomes (and eventually lysosomes) by addition of mannose 6-phosphate residues that bind to mannose 6-phosphate receptor proteins in the Golgi membrane. The mannose 6-phosphate receptors together with their bound acid hydrolases are incorporated into the clathrin-coated Golgi transport vesicles and released. The transport vesicles lose their clathrin coat and then fuse with the late endosomal membrane. The acidity of the endosome releases the acid hydrolases from the receptors into the vesicle lumen. The receptors are eventually recycled back to the Golgi.

2. PHAGOCYTOSIS AND AUTOPHAGY

One of the major roles of lysosomes is phagocytosis (Fig. 10.17). Neutrophils and macrophages, the major phagocytic cells, devour pathogenic microorganisms and clean up wound debris and dead cells, thus aiding in repair. As bacteria or other particles are enclosed into clathrin-coated pits in the plasma membrane, these vesicles bud off to form intracellular phagosomes. The phagosomes fuse with lysosomes, where the acidity and digestive enzymes destroy the contents. Pinocytotic vesicles also may fuse with lysosomes.

In autophagy (self-eating), intracellular components such as organelles or glycogen particles are surrounded by a membrane derived from ER vesicles, forming an autophagosome. The autophagosome fuses with a lysosome, and the contents of the phagolysosome are digested by lysosomal enzymes. Organelles usually turn over much more rapidly than the cells in which they reside (e.g., approximately four mitochondria in each liver cell are degraded per hour). Cells that are damaged but still viable recover, in part, by using autophagy to eliminate damaged components.

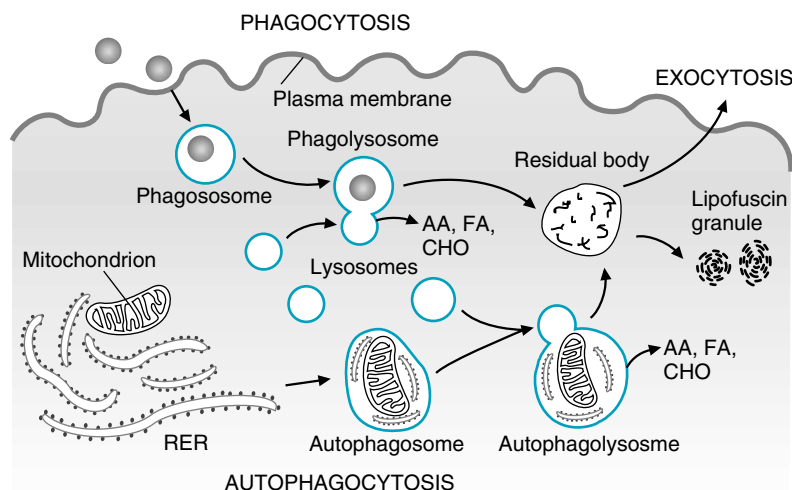


Fig. 10.17. Phagocytosis and autophagy. Cells and large particles are phagocytosed. The phagosomes fuse with lysosomes to form phagolysosomes. Recyclable amino acids (AA), fatty acids (FA), and carbohydrates (CHO) are released into the cytosol. Autophagosomes are formed in the ER as the cell digests mitochondria and its own large particles. These also merge with lysosomes. Undigested material may remain in the lysosomes to form residual bodies, which are either extruded (exocytosis) or remain in the cell as lipofuscin granules.

If a significant amount of undigestible material remains within the lysosome after the digestion process is completed, the lysosome is called a residual body. Depending on the cell type, residual bodies may be expelled (exocytosis) or remain indefinitely in the cell as lipofuscin granules that accumulate with age.

IV. MITOCHONDRIA

Mitochondria contain most of the enzymes for the pathways of fuel oxidation and oxidative phosphorylation and thus generate most of the ATP required by mammalian cells. Each mitochondrion is surrounded by two membranes, an outer membrane and an inner membrane, separating the mitochondrial matrix from the cytosol (Fig. 10.18). The inner membrane forms invaginations known as cristae containing the electron transport chain and ATP synthase. Most of the enzymes for the TCA cycle and other pathways for oxidation are located in the mitochondrial matrix, the compartment enclosed by the inner mitochondrial membrane. (The TCA cycle and electron transport chain are described in more detail in Chapters 20 and 21.)

The inner mitochondrial membrane is highly impermeable, and the proton gradient that is built up across this membrane during oxidative phosphorylation is essential for ATP generation from ADP and phosphate. The transport of ions occurs principally through facilitative transporters in a type of secondary active transport powered by the proton gradient established by the electron transport chain. The outer membrane contains pores made from proteins called porins and is permeable to molecules with a molecular weight up to about 1000 g/mole.

Mitochondria can replicate by division; however, most of their proteins must be imported from the cytosol. Mitochondria contain a small amount of DNA, which encodes for only 13 different subunits of proteins involved in oxidative phosphorylation. Most of the enzymes and proteins in mitochondria are encoded by nuclear DNA and synthesized on cytoplasmic ribosomes. They are imported

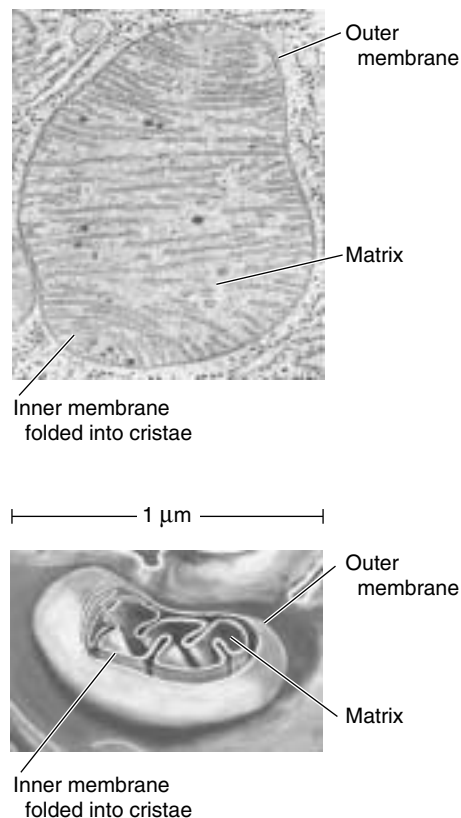


Fig. 10.18. Mitochondrion. Electron micrograph (top); three-dimensional drawing (bottom).



Mitochondrial diseases. Mitochondria contain DNA and can reproduce by replicating their DNA and then dividing in half. Although nuclear DNA encodes most of the enzymes found in mitochondria, mitochondrial DNA encodes some of the subunits of the electron transport chain proteins and ATP synthase. Mutations in mitochondrial DNA result in a number of genetic diseases that affect skeletal muscle, neuronal, and renal tissues. They are implicated in aging.

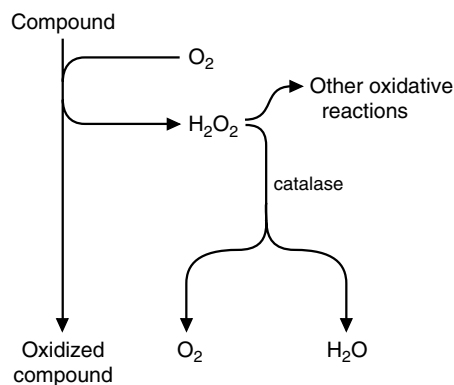


Fig. 10.19. Types of reactions in peroxisomes.



Peroxisomal Diseases. Peroxisomal diseases are caused by mutations affecting either the synthesis of functional peroxisomal enzymes or their incorporation into peroxisomes. For example, adrenoleukodystrophy probably involves a mutation that decreases the content of a transporter in the peroxisomal membrane. Zellweger's syndrome is caused by the failure to complete the synthesis of peroxisomes.

through membrane pores by a receptor-mediated process involving members of the heat shock family of proteins.

V. PEROXISOMES

Peroxisomes are cytoplasmic organelles, similar in size to lysosomes, that are involved in oxidative reactions using molecular oxygen (Fig. 10.19). These reactions produce the toxic chemical hydrogen peroxide (H_2O_2), which is subsequently used or degraded within the peroxisome by catalase and other enzymes. Peroxisomes function in the oxidation of very long chain fatty acids (containing 20 or more carbons) to shorter chain fatty acids, the conversion of cholesterol to bile acids, and the synthesis of ether lipids called plasmalogens. They are bounded by a single membrane.

Like mitochondria, peroxisomes can replicate by division. However, they are dependent on the import of proteins to function. They contain no DNA.

VI. NUCLEUS

The largest of the subcellular organelles of animal cells is the nucleus (Fig. 10.20). Most of the genetic material of the cell is located in the chromosomes of the nucleus, which are composed of DNA, an equal weight of small, positively charged proteins called histones, and a variable amount of other proteins. This nucleoprotein

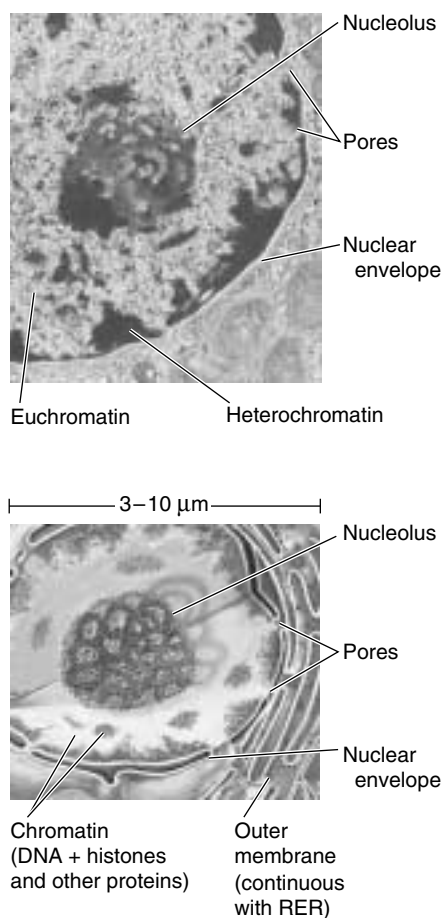


Fig. 10.20. Nucleus. Electron micrograph (top); three-dimensional drawing (bottom).

complex is called chromatin. The nucleolus, a substructure of the nucleus, is the site of rRNA transcription and processing, and of ribosome assembly. Replication, transcription, translation, and the regulation of these processes are the major focus of the molecular biology section of this text (see Section Three).

The nucleus is separated from the rest of the cell (the cytoplasm) by the nuclear envelope, which consists of two membranes joined at nuclear pores. The outer nuclear membrane is continuous with the rough endoplasmic reticulum. To convert the genetic code of the DNA into the primary sequence of a protein, DNA is transcribed into RNA, which is modified and edited into mRNA. The mRNA travels through the nuclear pores into the cytoplasm, where it is translated into the primary sequence of a protein on ribosomes (Fig. 10.21). Ribosomes, which are generated in the nucleolus, also must travel through nuclear pores to the cytoplasm. Conversely, proteins required for replication, transcription, and other processes pass into the nucleus through these pores. Thus, transport through the pore is specific for the molecule and the direction of transport.

Specificity and direction of travel through the nuclear pore (import vs. export) is dictated by binding proteins (importins vs. exportins), by a small GTP protein called Ran, and by the location of the regulatory protein, RanGAP (GTPase activating protein) only on the cytoplasmic side (Fig. 10.22). Proteins transported into the nucleus have a nuclear localization signal that causes them to bind to one of the subunits of cytosolic proteins called importins. The other subunit of the importin molecule binds to cytoplasmic filaments attached to the outer ring of the nuclear pore. As the importin-protein complex enters the nucleus, the small GTP-binding protein Ran binds to an importin subunit, causing release of the transported protein into the nucleus. The Ran-importin complex is returned to the cytosol, where RanGAP (GTPase activating protein) activates hydrolysis of bound GTP to GDP and phosphate. The energy released by GTP hydrolysis changes the conformation of Ran and the complex dissociates. The free importin can then bind another protein.

Table 10.1 Monomeric G Proteins in the Ras Superfamily

G-Protein Family	Function	Some Family Members	Location and Membrane Attachment Site
Ras	Regulator of gene expression and cell growth, found in mutated oncogenic forms in many human tumors	H-Ras, K-Ras, N-Ras, Ral A, Rad, Rap, Rit,	Anchored to plasma membrane by farnesyl, palmitoyl, or other lipid groups
Rho	Controls organization of actin cytoskeleton and gene expression (F-actin bundling, myosin filament assembly)	Rho (A-E), Cdc42, Rac (1-3)	Anchored to plasma membrane by lipids, and translocates to cytosol
Arf/Sar	Assembly of coatmer-coated vesicles (COPI and COPII) for vesicular trafficking pathways originating in the Golgi	Arf (1-6), Sar 1a,1b; Arl (1-7)	Arf is anchored to vesicular membranes by myristyl groups, but Sar is anchored by the protein itself.
Rab	Targeting of vesicles involved in secretory and endocytotic pathways and formation of v-SNARE-t-SNARE complexes	Dynamin, Rab (11-33)	Anchored to lipid membranes with geranylgeranyl (C20 isoprenoid) groups and other lipids
Ran	Transport through nuclear pore complexes.	Ran	Not anchored to lipid membrane. Found in cytosol and nucleus



The Ras family of monomeric G proteins.

Ras and Ran belong to a superfamily of proteins called small G proteins (also called small GTP-binding proteins, small GTPases, or monomeric G proteins: see Chapter 9, section III.C.2). These proteins function as timing regulators for a variety of cell functions. They are referred to as “small” because they are composed of a single subunit with a weight of 20 to 40 kDa, and they are called GTPases because they slowly hydrolyze bound GTP. When small G proteins contain bound GTP, they bind to and activate their target proteins. As their bound GTP is hydrolyzed to GDP and Pi, their conformation changes dramatically, and they dissociate from the target protein. They thus serve as “automatic clocks” that shut themselves off. Many of the monomeric GTP binding proteins are regulated by GAPs (GTPase activating proteins), GEFs (GTP exchange proteins which stimulate GDP dissociation and GTP binding), or GDIs (GDP-dissociation inhibitory proteins). (See Chapter 9 for a more complete discussion of these regulators). The function of each of the five major classes of Ras monomeric G proteins in cell biology is summarized in Table 10.1.

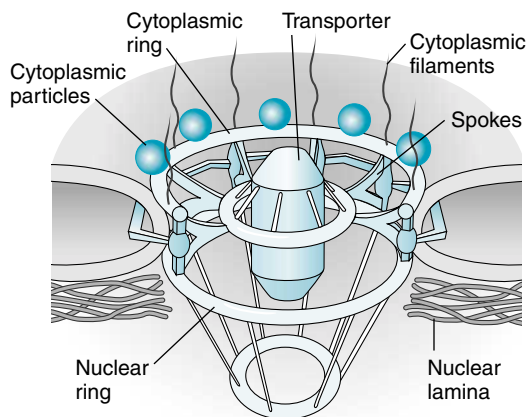


Fig. 10.21. The nuclear pore complex. The approximately 100 different polypeptide chains of the nuclear pore complex form an assembly of 8 spokes attached to two ring structures (a cytoplasmic ring in the outer nuclear membrane and a nuclear ring through the inner membrane) with a transporter “plug” in the center. Small molecules, ions, and proteins with less than a 50-kDa mass passively diffuse through the pore in either direction. However, RNAs and most proteins are too large to diffuse through, and are actively transported in a process that requires energy, is selective for the molecule transported, is unidirectional, and can be regulated.

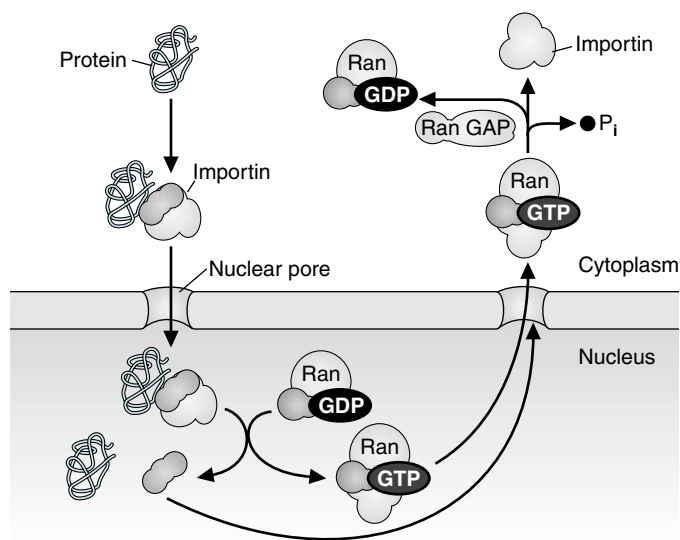


Fig. 10.22. Nuclear import. Proteins with the nuclear localization signal bind to importins, which carry them through the nuclear pore into the nucleus. The monomeric G protein Ran containing bound GTP binds to one of the subunits of importin. This causes dissociation of the importin subunits and release of the imported protein in the nucleus. The Ran-importin complex exits a nuclear pore. On the cytoplasmic side, a RanGAP (GTPase activating protein) activates the hydrolysis of GTP to GDP, which causes dissociation of the complex. RanGDP is subsequently returned to the nucleus, where an accessory protein activates dissociation of GDP and association of GTP.

RNAs are transported from the nucleus to the cytoplasm as ribonucleoproteins, which are targeted for export by a specific amino acid sequence called the nuclear export signal. The nucleoprotein forms a complex with additional proteins called exportins and with Ran. This complex is transported through the pore to the cytoplasm, where RanGAP activates hydrolysis of the bound GTP. In the absence of GTP, the complex dissociates with the release of RNA into the cytoplasm, and the exportins and Ran are transported back to the nucleus.

VII. ENDOPLASMIC RETICULUM

The endoplasmic reticulum (ER) is a network of membranous tubules within the cell consisting of smooth endoplasmic reticulum (SER), which lacks ribosomes, and rough endoplasmic reticulum (RER), which is studded with ribosomes (Fig. 10.23). The SER has a number of functions. It contains enzymes for the synthesis of many lipids, such as triacylglycerols and phospholipids. It also contains the cytochrome P450 oxidative enzymes involved in metabolism of drugs and toxic chemicals such as ethanol and the synthesis of hydrophobic molecules such as steroid hormones. Glycogen is stored in regions of liver cells that are rich in SER.

The RER is involved in the synthesis of certain proteins. Ribosomes attached to the membranes of the RER give them their “rough” appearance. Proteins produced on these ribosomes enter the lumen of the RER, travel to the Golgi complex in vesicles, and are subsequently either secreted from the cell, sequestered within membrane-enclosed organelles such as lysosomes, or embedded in the plasma membrane. Posttranslational modifications of these proteins, such as the initiation of N-linked glycosylation and the addition of GPI anchors, occur in the RER. In contrast, proteins encoded by the nucleus and found in the cytosol, peroxisomes, or mitochondria are synthesized on free ribosomes in the cytosol and are seldom modified by the attachment of oligosaccharides.



Chronic ingestion of ethanol has increased the content of MEOS, the microsomal ethanol oxidizing system, in **Al Martini's** liver. MEOS is a cytochrome P450 enzyme that catalyzes the conversion of ethanol, NADPH and O_2 to acetaldehyde, $NADP^+$, and $2 H_2O$ (see Chapter 9). The adjective *microsomal* is a term derived from experimental cell biology that is sometimes used for processes occurring in the ER. When cells are lysed in the laboratory, the ER is fragmented into vesicles called *microsomes*, which can be isolated by centrifugation. Microsomes, as such, are not actually present in cells.

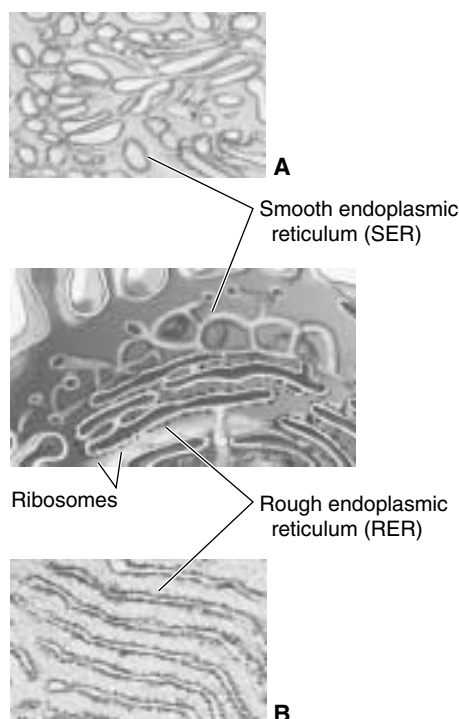


Fig. 10.23 A. Smooth endoplasmic reticulum. B. Rough endoplasmic reticulum. A and B are electron micrographs. A three-dimensional drawing is in the middle.

VIII. GOLGI COMPLEX

The Golgi complex is involved in modifying proteins produced in the RER and in sorting and distributing these proteins to the lysosomes, secretory vesicles, or the plasma membrane. It consists of a curved stack of flattened vesicles in the cytoplasm that is generally divided into three compartments: the *cis*-Golgi network, which is often convex and faces the nucleus; the *medial* Golgi stacks; and the *trans* Golgi network, which often faces the plasma membrane (Fig. 10.24).

Proteins are transported to and from the Golgi in at least three kinds of vesicles: coatamer-coated COP I vesicles, coatamer-coated COP II vesicles, and clathrin-coated vesicles (see Fig. 10.24). Proteins produced on the RER travel in COP II vesicles to an endoplasmic reticulum-Golgi intermediate compartment (ERGIC), and then to the *cis*-Golgi network, where they enter the lumen. Here N-linked oligosaccharide chains that were added to proteins in the RER are modified, and O-linked oligosaccharides are added. COP I vesicles recycle material from the Golgi back to the ER and possibly transfer material from the Golgi to other sites.

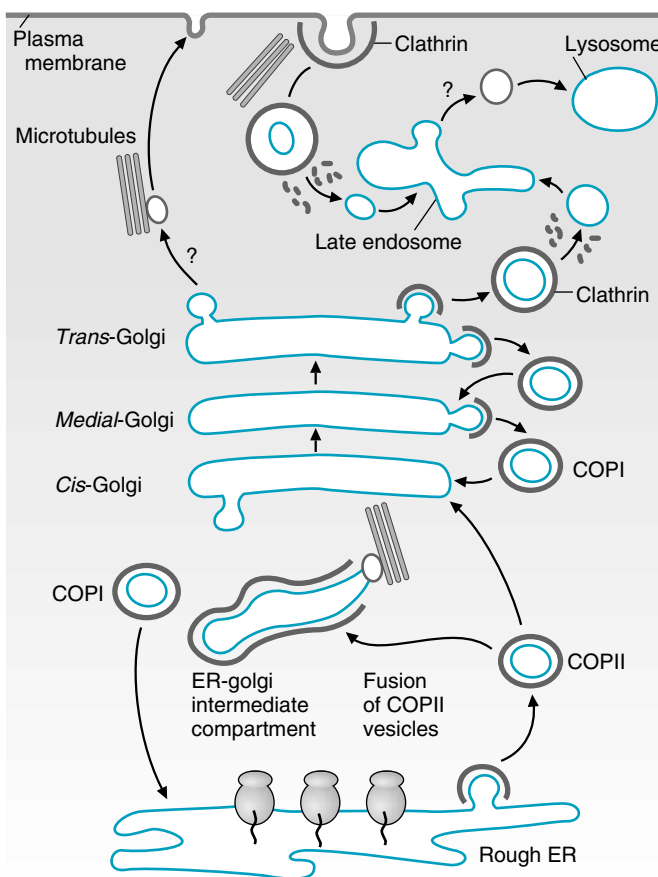


Fig. 10.24. Vesicular transport to and from the Golgi complex. COP II vesicles (coatamer-coated) form in the rough ER and move to the Golgi. COP I vesicles generally go from the trans to the cis Golgi to the ER. Vesicles that go to late endosomes (eventually lysosomes) from the Golgi or the plasma membrane are clathrin-coated. Less is known about exocytotic vesicles. Vesicle transport, as well as transport of organelles and secretory proteins, occurs along microtubules (structures formed from the protein tubulin).

Vesicles released from the *trans* face of the Golgi complex travel to endosomes as clathrin-coated vesicles.

COP vesicles are coated with a complex composed of coatamer proteins (COP), an Arf family monomeric G protein that mediates vesicle assembly, and other proteins (Fig. 10.25). COP I vesicles contain the monomeric G protein Arf (ADP-ribosylating factor), and COP II vesicles contain the monomeric G protein Sar (another member of the Arf family). In both types of vesicles, hydrolysis of GTP causes dissociation of the G-protein and disassembly of the vesicle coat. The vesicle components are then recycled. Glycoproteins or glycolipids once anchored in the membrane of the vesicle remain in the plasma membrane when the vesicular and plasma membranes fuse.

Vesicles that have lost their coats are ready to fuse with the target membrane. The vesicle membranes contain proteins called v-SNARES (vesicle-SNARES) (see Fig. 10.25). Each type of v-SNARE is able to recognize and bind to its complementary t-SNARE (target SNARE) on the target membrane, thus ensuring that



The monomeric G protein Arf was named for its contribution to the pathogenesis of cholera and not for its normal function in the assembly of COP I vesicles. However, it is also required for the transport of *V. cholerae* A-toxin. The cholera toxin is endocytosed in caveolae vesicles that subsequently merge with lysosomes (or are transformed into lysosomes), where the acidic pH contributes to activation of the toxin. As the toxin is transported through the Golgi and ER, it is further processed and activated. Arf forms a complex with the A-toxin that promotes its travel between compartments. The A-toxin is actually an ADP-ribosylase (an enzyme that cleaves NAD and attaches the ADP portion to a protein) (see Chapter 6, Fig. 6.14), and hence, Arf became known as the ADP-ribosylating factor. The ADP-ribosylation of proteins regulating the CFTR chloride channel leads to **Dennis Veere's** dehydration and diarrhea.

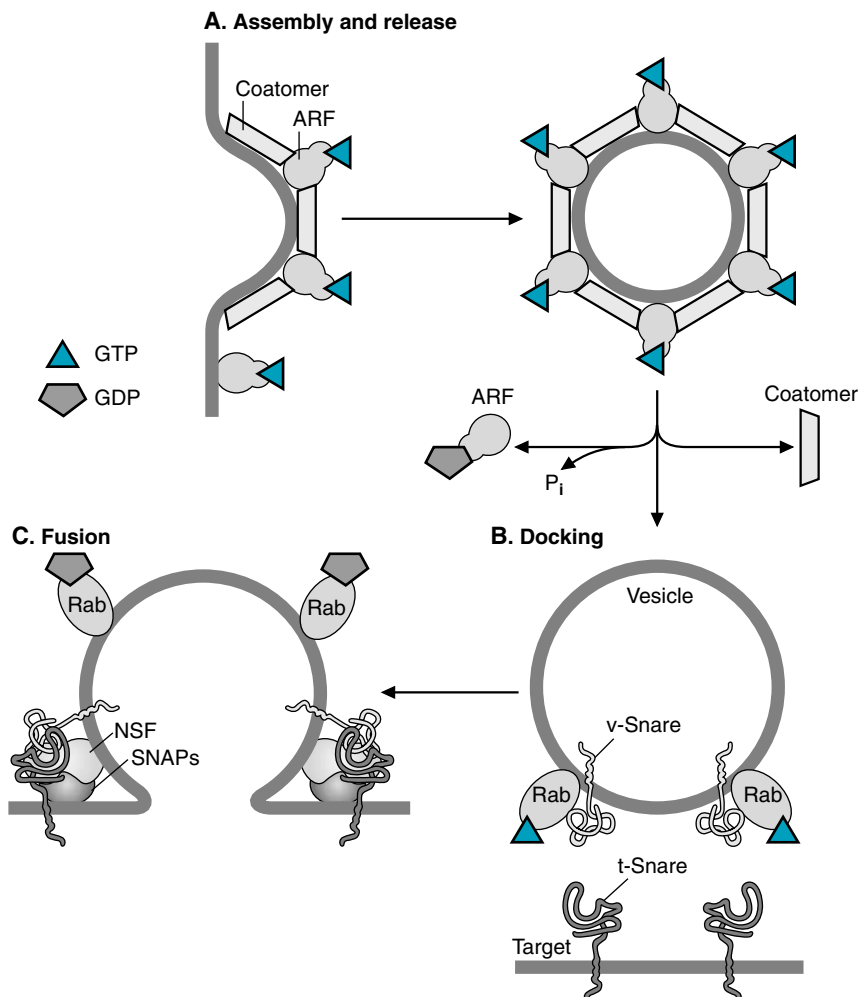


Fig. 10.25. Transport in COP-coated vesicles. **A.** Assembly and release. Arf with bound GTP assembles a region of the trans-Golgi membrane containing receptors for the protein cargo and coatamers. As GTP is hydrolyzed to GDP, the coat is released. **B.** Docking. The small G protein Rab assists in docking. v-Snares in the vesicle membrane recognize complementary t-Snares in the target membrane. **C.** Fusion. NSF and SNAP are fusion proteins.



Secretory vesicles. The hormone insulin is synthesized as a prohormone, proinsulin, which is incorporated into secretory vesicles. These vesicles contain a protease that is activated by the acidic pH of the secretory vesicle. It cleaves proinsulin into the A, B, and C chains (see Fig. 6.13).



Lotta Topaigne was given colchicine, a drug that is frequently used to treat gout. One of its actions is to prevent phagocytic activity by binding to dimers of the α and β subunits of tubulin. When the tubulin dimer–colchicine complexes bind to microtubules, further polymerization of the microtubules is inhibited, depolymerization predominates, and the microtubules disassemble. Microtubules are necessary for vesicular movement of urate crystals during phagocytosis and release of mediators that activate the inflammatory response. Thus, colchicine diminishes the inflammatory response, swelling and pain caused by formation of urate crystals.

the vesicle contents are delivered to the right location. Several additional proteins are required for fusion of the vesicle with the target membrane, including Rab (another monomeric G protein) and two additional proteins called SNAP (soluble NSF attachment proteins) and NSF (N-ethylmaleimide sensitive factor).

Exocytotic vesicles release proteins into the extracellular space after fusion of the vesicular and plasma cell membranes. Exocytotic vesicles containing hormones also may contain proteases that cleave the prohormone at a specific site and v -ATPases that acidify the vesicle and activate the protease (similar to lysosomes).

IX. CYTOSKELETON

The structure of the cell, the shape of the cell surface, and the arrangement of subcellular organelles is organized by three major protein components: microtubules composed of tubulin, which move and position organelles and vesicles; thin filaments composed of actin, which form a cytoskeleton, and intermediate filaments composed of different fibrous proteins. Actin and tubulin, which are involved in cell movement, are dynamic structures composed of continuously associating and dissociating globular subunits. Intermediate filaments, which play a structural role, are composed of stable fibrous proteins that turn over more slowly.

A. Microtubules

Microtubules, cylindrical tubes composed of tubulin subunits, are present in all nucleated cells and the platelets in blood (Fig. 10.26). They are responsible for the positioning of organelles in the cell cytoplasm and the movement of vesicles, including phagocytic vesicles, exocytotic vesicles, and the transport vesicles between the ER, Golgi, and endosomes (see Fig. 10.24). They also form the spindle apparatus for cell division. The microtubule network (the minus end) begins in the nucleus at the centriole and extends outward to the plasma membrane (usually the plus end). Microtubule-associated proteins (MAPs) attach microtubules to other cellular components, and can determine cell shape and polarity.

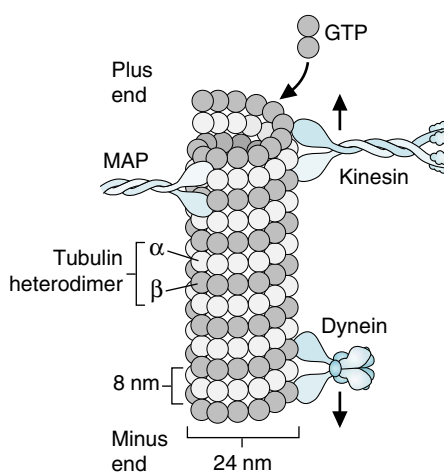


Fig. 10.26. Microtubules composed of $\alpha\beta$ tubulin heterodimers. MAP, microtubule-associated protein. These proteins project outward to attach the microtubules to other cellular components. The microtubule grows by the addition of $\alpha\beta$ dimers containing bound GTP to the plus end of the polymer. Kinesin and dyneins are motor proteins that transport cargo (e.g., vesicles) along the microtubule.

Motor proteins called kinesins and cytoplasmic dyneins use ATP energy to move cargo along the microtubules. Kinesins moves molecules, vesicles, and organelles toward the plus end of microtubules, usually toward the plasma membrane. Cytoplasmic dyneins are huge proteins that move vesicles and organelles to the minus end, generally toward the nucleus. They are also involved in the positioning of the Golgi complex and the movement of chromosomes during mitosis.

Microtubules consist of polymerized arrays of α and β tubulin dimers that form 13 protofilaments organized around a hollow core (see Fig. 10.26). Three different tubulin polypeptides (α , β , and γ) of similar amino acid composition are encoded by related genes; α and β dimers polymerize to form most microtubules, and γ -tubulin is found only in the centrosome. Tubulin dimers composed of one α and one β subunit bind GTP, which creates a conformational change in the dimer that favors addition of dimers to the tubulin polymer. The dimers can add to and dissociate from both ends of the tubulin, but the end to which they add more rapidly (the plus end) has a net rate of growth, and the end to which they add more slowly (the minus end) has a net rate of loss. As GTP is hydrolyzed to GDP, the binding of tubulin subunits is weakened, resulting in their dissociation (dynamic instability). Thus, the net rate and direction of growth is dictated by the fastest growing end of the microtubule.

B. Actin Filaments

Actin filaments form a network controlling the shape of the cell and movement of the cell surface, thereby allowing cells to move, divide, engulf particles, and contract. Actin is present in all living cells. The actin polymer, called F-actin, is composed of a helical arrangement of globular G-actin subunits (Fig. 10.27). Within the polymer, each G-actin subunit contains a bound ATP or ADP that

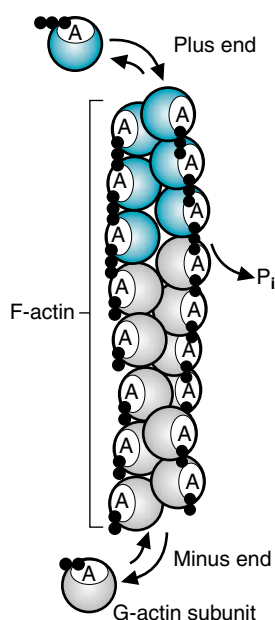


Fig. 10.27. Actin filaments. The polymer F-actin is assembled from G-actin subunits containing bound ATP. While bound, the ATP is slowly hydrolyzed to ADP. The conformational change shifts the equilibrium so that dissociation of the G-actin subunits is favorable at the minus end of the polymer. Once dissociated, the actin subunits exchange ADP for ATP, which may again associate with the actin polymer. At the plus end of the molecule, association is favored over dissociation.



A variety of human cells have cilia and flagella, hairlike projections from the surface that have a stroke-like motion. These projections contain a flexible organized array of microtubules. Fluid or mucus is propelled over the surface of ciliated epithelial cells by the coordinated beating of cilia. A sperm cell swims by means of a flagellum.



Colchicine has a narrow therapeutic index (i.e., the amount of drug that produces the desirable therapeutic effect is not much lower than the amount that produces an adverse effect). Its therapeutic effect depends on inhibiting tubulin synthesis in neutrophils, but it can also prevent tubulin synthesis (and, thus, cell division and other cellular processes) in other cells. Fortunately, neutrophils concentrate colchicine, so they are affected at lower intakes than other cell types. Neutrophils lack the transport protein P-glycoprotein, a member of the ABC cassette family (which includes the CFTR channel). In most other cell types, P-glycoprotein exports chemicals such as colchicine, thus preventing their accumulation.

holds the actin fold into a closed conformation (see Chapter 7). The actin polymer is dynamic. New subunits of G-actin containing ATP continuously combine with the assembled F-actin polymer at the plus end. As F-actin elongates, bound ATP is hydrolyzed to ADP, so that most of the polymer contains G-actin-ADP subunits. The conformation of ADP-actin favors dissociation from the minus end of the polymer; thus, the polymer is capable of lengthening from the plus end. This directional growth can account for certain types of cell movement and shape changes: the formation of pseudopodia that surround other cells during phagocytosis, the migration of cells in the developing embryo, or the movement of white blood cells through tissues.

Actin polymers form the thin filaments (also called microfilaments) in the cell that are organized into compact ordered bundles or loose network arrays by cross-linking proteins. Short actin filaments bind to the cross-linking protein spectrin to form the cortical actin skeleton network (see Fig. 10.6). In muscle cells, long actin filaments combine with thick filaments, composed of the protein myosin, to produce muscle contraction. The assembly of G-actin subunits into polymers, bundling of fibers, and attachments of actin to spectrin and to the plasma membrane proteins and organelles, are mediated by a number of actin-binding proteins and G-proteins from the Rho family.

C. Intermediate Filaments

Intermediate filaments (IF) are composed of fibrous protein polymers that provide structural support to membranes of the cells and scaffolding for attachment of other cellular components. Each IF subunit is composed of a long rod-like α -helical core containing globular spacing domains, and globular N- and C-terminal domains. The α -helical segments of two subunits coil around each other to form a coiled coil, and then combine with another dimer coil to form a tetramer. Depending on the type of filament, the dimers may be either hetero- or homo-dimers. The tetramers join end-to-end to form protofilaments and approximately eight protofilaments combine to form filaments (Fig. 10.28). Filament assembly is partially controlled through phosphorylation.

In contrast to actin thin filaments, the 50 or so different types of intermediate filaments are each composed of a different protein having the same general structure described above (Table 10.2). Some of the intermediate filaments, such as the nuclear lamins, are common to all cell types. These filaments provide a lattice-like support network attached to the inner nuclear membrane. Other intermediate filaments are specific for types of cells (e.g., epithelial cells have cytokeratins, and neurons have neurofilaments). These provide an internal network that helps to support the shape and resilience of the cell.

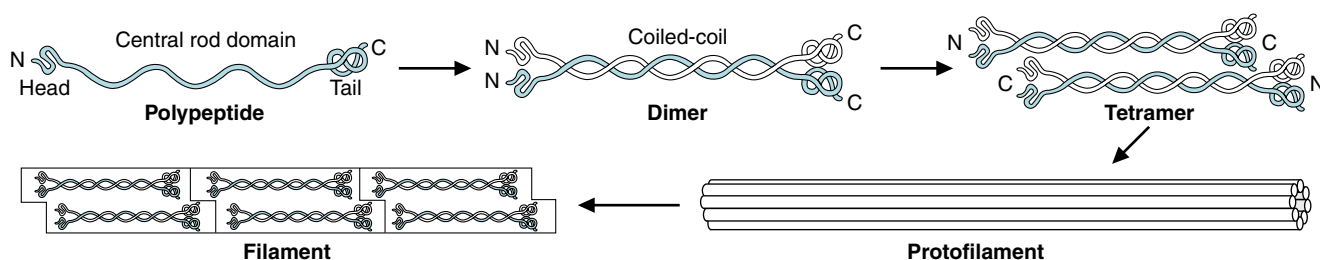


Fig. 10.28. Formation of a cytokeratin filament. The central rod of the keratin monomer is principally α -helical structure. A specific acidic keratin monomer combines with a specific basic keratin monomer to form a heterodimer coil (a coiled coil structure). Two dimers combine in antiparallel fashion to form a tetramer, and the tetramers combine head-to-tail to form protofilaments. Approximately eight protofilaments combine to form a filament. The filament is thicker than actin filaments (called thin filaments or microfilaments) and thinner than microtubules (thick tubes) and is therefore called an intermediate filament.

Table 10.2 Intermediate Filaments

Type	Protein	Location
Type I	Acidic keratins	Different epithelial cells express different keratins.
Type II	Basic keratins	Epithelial cells. Each cell type has a specific acidic and a specific basic keratin.
Type III	Vimentin-like	Cells of mesenchymal origin
	Vimentin	Muscle
	Desmin	Glial cells in the central nervous system
	Glial fibrillary acid protein	
Type IV	Neurofilaments (NF-L, NF-M, NF-H)	Neurons, particularly long thin motor axons
Type V	Nuclear lamins A, B, and C	Forms a mesh-like network in the inner lining of cellular nuclear envelopes.
Type VI	Nestin	Stem cells of central nervous system

CLINICAL COMMENTS



Al Martini. Al Martini has been drinking for 5 years and has begun to exhibit mental and systemic effects of chronic alcohol consumption. In his brain, ethanol has altered the fluidity of neuronal lipids, causing changes in their response to neurotransmitters released from exocytotic vesicles. In his liver, increased levels of MEOS (CYP2E1) located in the smooth ER increased his rate of ethanol oxidation to acetaldehyde, a compound that is toxic to the cell. His liver also continues to oxidize ethanol to acetaldehyde through a cytosolic enzyme, liver alcohol dehydrogenase.

One of the toxic effects of acetaldehyde is inhibition of tubulin polymerization. Tubulin is used in the liver for secretion of very low-density lipoprotein (VLDL) particles containing newly synthesized triacylglycerols. As a result, these triacylglycerols accumulate in the liver, and he has begun to develop a fatty liver. Acetaldehyde may also damage protein components of the inner mitochondrial membrane and affect its ability to pump protons to the cytosol.



Lotta Topaigne had a rapid and gratifying clinical response to the hourly administration of colchicine. This drug diminishes phagocytosis and the subsequent release of the lysosomal enzymes that initiate the inflammatory response in synovial tissue.

The inflammatory response that causes the symptoms of an acute gout attack begins when neutrophils and macrophages ingest urate crystals. In neutrophils, urate activates the conversion of the polyunsaturated fatty acid arachidonic acid (present in membrane phospholipids) to leukotriene B₄. The release of this prostaglandin contributes to the pain. Colchicine, through its effect on tubulin, inhibits phagocytosis, leukotriene B₄ release, and recruitment and cell division of additional cells involved in inflammation. Colchicine also inhibits the tubulin-dependent release of histamine from mast cells. As a result, there was a rapid improvement in the pain and swelling in Lotta's great toe.

After the gout attack subsided, Ms. Topaigne was placed on allopurinol, a drug that inhibits urate production (see Chapter 8). During the next 6 months of allopurinol therapy, Ms. Topaigne's blood urate levels decreased. She did not have another gout attack during this time.

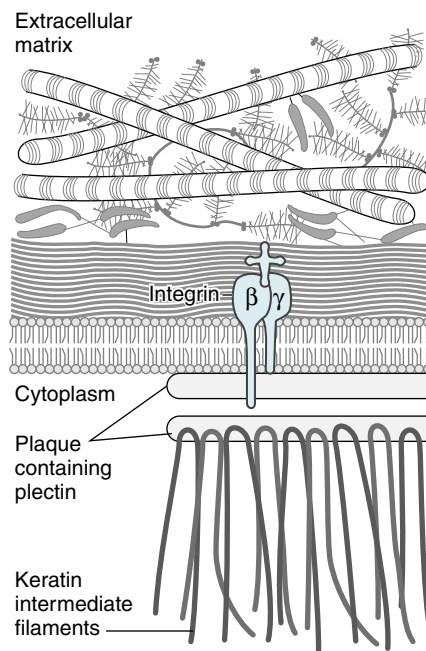


Dennis Veere. Dennis Veere was diagnosed with cholera. He was placed on intravenous rehydration therapy, followed by oral rehydration therapy with high glucose and Na⁺-containing fluids (to be continued in Chapter 11).

Vibrio cholerae secrete an A toxin that is processed and transported in the cell in conjunction with the monomeric G protein Arf (ADP-ribosylation factor). The A



In epidermolysis bullosa simplex, the skin blisters in response to a very slight mechanical stress. The familial form of the disease is generally caused by mutations in either of the two forms of keratin that constitute the keratin heterodimers of the basal layer of the epidermis. The weakened keratin cytoskeleton results in cytolysis when stress is applied. The disease also can be caused by mutations in plectin, the protein that attaches keratin to the membrane protein integrin in hemidesmosomes (a structure involved in attaching cells to the extracellular matrix).



toxin ADP-ribosylates the G_{α} subunit of a heterotrimeric G protein (a process discussed in Chapter 11.) The net result is activation of protein kinase A, which then phosphorylates the CFTR (cystic fibrosis transmembrane conductance regulator) chloride channel so that it remains permanently open. The subsequent efflux of chloride, sodium, and water into the bowel lumen is responsible for Dennis Veere's diarrhea and subsequent dehydration.

BIOCHEMICAL COMMENTS



The mitochondria of eukaryotic cells have many features in common with prokaryotic cells and, in fact, may have originated when primordial anaerobic eukaryotes engulfed ancient aerobic prokaryotes, establishing a symbiotic relationship. These prokaryotes provided eukaryotes with a more efficient mechanism for oxidizing fuels to obtain energy.



Differences between bacteria and human cells. Different species of bacteria have some common structural features that distinguish them from animal cells. They are single-cell organisms that are prokaryotes ("before nucleus"). Their genetic material (DNA) is concentrated in the central region of the cell called a nucleoid, rather than a nucleus, because it is not separated from the rest of the cellular contents by a membrane. Likewise, bacteria contain no cytoplasmic organelles defined by membranes. They do have a plasma membrane that encloses the cytoplasm. External to this membrane is a peptidoglycan cell wall composed of extensively cross-linked polysaccharides that form a protective shield on the surface of the cell.

Bacterial cells obtain nutrients from the medium on which they grow. Many of their metabolic pathways for fuel oxidation are similar to those in eukaryotes and generate NADH and ATP; however, individual steps in these pathways may use different coenzymes or very different enzymes for catalysis than do human cells. Like human cells, bacteria use intermediates of glycolysis and other basic degradative pathways to serve as precursors for biosynthetic pathways, and energy acquired from catabolic pathways is used in anabolic pathways. Aerobic bacteria, such as *Escherichia coli*, contain enzymes of the tricarboxylic acid (TCA) cycle and the components of the electron transport chain, which are located in the cell membrane. However, many bacteria are anaerobes and can function in the absence of oxygen.

Many of the metabolic differences between human cells and bacteria are related to their interactions with their environment. Some bacteria, such as *E. coli*, can adapt to adverse or changing conditions (high versus low O_2 tension or a single supply of nutrients from which to synthesize everything) by dramatic shifts in the genes that are transcribed. Other bacteria find a unique environmental niche where they do not have to compete with other bacteria for nutrients (e.g., *Lactobacilli* in yogurt are adapted to an acidic pH). In contrast, the human cells are adapted to interacting with blood and interstitial fluid, which provides a well-controlled pH, a constant nutrient supply, and a medium for communication between very distant cells. As a consequence of their constant environment, adult human cells seldom need to adapt (or can adapt) to widely fluctuating conditions through large variations in the genes transcribed. As a consequence of being organized into a multicellular organism, human cell types have been able to specialize in function, structure, and enzyme content.

Suggested References

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- Takai Y, Sasaki T, Matozaki T. Small GTP-binding proteins. *Physiol Rev* 2001;81:153–208. Although this is a lengthy review for medical students, it is very useful in identifying the roles of different G proteins.
- A number of extremely good cell and molecular biology textbooks have recently been published. Some of the most recent textbooks that we found helpful include:
- Karp G. *Cell and Molecular Biology: Concepts and Experiments*. 3rd Ed. New York: John Wiley and Sons, 2002.